

The Swords of Armageddon

U.S. Nuclear Weapons Development since 1945

VOLUME IV

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Release 01

October 1995

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VOLUME IV THERMONUCLEAR WEAPONS DEVELOPMENT, 1952 - SUMMER 1954

April 1952: The AEC Reports to the President

At the beginning of April, in its annual report to the President for 1951, the AEC summarized progress in thermonuclear weaponry during the preceding year and described plans for future testing; prospects for ultimate success were vastly improved over those of a year earlier:

Weapon tests at Eniwetok last spring and investigations since then have greatly improved the prospect for ultimately securing a practical hydrogen bomb.

.....

The Hydrogen Bomb Program. A year ago we reported that the inherent difficulties were clearer, the time required to carry out the program was believed to be longer, and the ultimate practical success was less certain than on January 31, 1950 when you issued the directive to determine the feasibility of a thermonuclear (hydrogen) weapon. Nor did we know a year ago the answer to the crucial question of how much tritium would be needed to ignite a mass of liquid deuterium.

Now we can report that, although the feasibility of a thermonuclear weapon remains to be established, important new approaches to the problem were born out (of) the GREENHOUSE tests last spring and subsequent investigations which appear to have greatly reduced the requirement for tritium and have improved the outlook and time-scale for ultimately achieving a practical thermonuclear weapon.

There now appear to be two promising weapon-design possibilities, the significance of which can be determined only by successive tests during the next two years. The first is a device employing liquid deuterium which will be tested at Eniwetok in the fall of 1952 and, if successful, is expected to provide a yield between one and 15 megatons of TNT, or roughly 50 to 750 times the 20-kiloton yield of the Nagasaki-type atomic bomb. This device is so large, however, it can

only establish principles; it is not a prototype of a practical weapon.

A second, more advanced design, using solid lithium deuteride, which might result in a practical and deliverable weapon, may be tested in the late summer of 1953 when sufficient lithium-6 is available. A plant to produce this material is being constructed at Oak Ridge.

.....

.... Operation GREENHOUSE at Eniwetok demonstrated the feasibility of igniting deuterium-tritium mixtures, as referred to above ...¹

Improvements in Computing Capabilities

The logjam in computing resources was rapidly breaking by the spring of 1952. The Institute for Advanced Studies at Princeton University released the design of John von Neumann's recently completed MANIAC computer, and two of these machines were built. One was installed at the Institute and the other at Los Alamos.

The new equipment made possible much more exact and far more extensive nuclear calculations than had previously been completed. Subsequently, there was a period in 1952 when the MANIAC, a Sperry-Rand UNIVAC (Universal Automatic Computer) in Philadelphia, and the National Bureau of Standards' SEAC (Standards Eastern Automatic Computer) in Washington, D. C. were all engaged essentially full time on LASL thermonuclear computations.

Calculations were now so extensive, for example, that in the course of running a Super problem on the Princeton MANIAC, the number of basic arithmetical steps performed was of the same order of magnitude as the total number of similar operations performed at Los Alamos, excluding those done on the Los Alamos MANIAC, from the lab's beginning in 1943 to the present.² The first large-scale hydrodynamic calculations for the MIKE device were begun on the Los Alamos MANIAC computer in March 1952.³

¹Atomic Energy Commission, Annual Report to the President for 1951, April 8, 1952, pp. 5, 6. Emphasis in original.

²Alexander, p. 18; AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 200; Mark, LA-5647-MS, p. 11.

³MANAGING NUCLEAR OPERATIONS, Ashton B. Carter, John D. Steinbruner, and Charles A. Zraket, eds., The Brookings

The operation of the thermonuclear test device was to proceed in successive stages: (1) explosion of a fission "principal bomb;" (2) propagation of the resulting radiation; (3) radiation implosion of the deuterium-fueled thermonuclear fuel capsule and its sparkplug; (4) thermonuclear detonation of the compressed capsule; and (5) fission of a surrounding natural or enriched uranium fusion tamper, resulting from its bombardment by the 14 MeV neutrons radiating from the compressed secondary capsule.

During the preceding six months, most of the effort of LASL's theoretical division had gone into calculating how well the second stage might be compressed. The results, obtained in part with the help of the National Bureau of Standards SEAC computer, indicated a usable compression of the liquid deuterium well in excess of the minimum compression required to sustain the thermonuclear reactions, as previously estimated by Dr. Teller.

Calculations by Project Matterhorn had also indicated that a sustainable reaction would require a high degree of compression.

Regardless of the outcome of the test, it was expected to provide valuable information on the propagation of radiation from the fission "trigger" to the imploding secondary; it would also give data on the fission properties of large quantities of uranium-238. In any event, whether the test was a success or a failure, it would help show what further tests were required.⁴

There were at this time two radically-different approaches proposed to testing the hydrogen device. The simplest, quickest, and most direct route was to test the entire system at once and see what happened. The other approach was to test separate subsystems within the device.

The first approach, while more daring and risky, would not clearly reveal why a device worked or failed. The separate subsystems testing was called "components testing" and its proponents argued that this approach, while more cautious and possibly taking more time, was a more scientific method:

That is, rather than proceeding as Los Alamos now is by the crash testing of a full-scale thermonuclear device, which, if

Institution, Washington, D.C., 1987, p. 66.

⁴Minutes of the Thirtieth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, April 27, 28, and 29, 1952, pp. 28, 29, 30.

it works, will not yield data on precisely why it worked, and, if it fails, will not show just why it failed, the Teller approach would take up specific components for individual testing.

For example, it was proposed, as soon as possible, to test a small alarm clock (sic) device at Las Vegas, reducing the reactive content of the thermonuclear component to a minimum, so that the yield would be held to a minimum.

In like fashion, propagation could be studied without studying propagation in a reactive material. This approach, according to Mr. X, would get to the efficient design of a weapon more rapidly than the other.⁵

Ironically, Edward Teller and some of the same individuals within the JCAE and DOD who favored the creation of a second weapons laboratory to counteract LASL's perceived conservatism also argued that components testing was a more logical and success-prone approach to a full-scale thermonuclear test. On April 17, Teller stated to members of the JCAE staff that

he maintained a skepticism on whether the (planned test) device will work at all and on what information can be obtained should it work or not work. His own approach is that of component testing. He would concentrate upon detailed study of radiation implosion [deleted]

He feels that it is the safest path to a well-designed deliverable weapon [deleted] He wants work to develop in parallel by his approach, starting immediately. It might be noted that this approach is especially interesting in view of the frequently heard comments [deleted] our thermonuclear weapon program would have to back up, reflect, and start all over again.⁶

⁵Memorandum to Mr. Borden dated April 7, 1952 from John S. Walker re Thermonuclear Program. "Las Vegas" is a reference to the Nevada Test Site; "Mr. X" may have been David Griggs, the chief scientist of the Air Force in 1951, and a friend of staff members of the JCAE.

⁶Memorandum to the File dated April 17, 1952 from John S. Walker, subject: Conversation with Dr. Edward Teller on the

Late in April, delays in engineering, fabricating, and procuring materials for the thermonuclear device cryogenic system and other technical apparatus forced postponement of the scheduled date of the Mike shot from October 1 to November 1, 1952.⁷

By the end of April, the GAC believed that future thermonuclear development work would probably overload LASL, certainly its Theoretical Division. Under this premise, it would make sense to ask the University of California Radiation Laboratory at Berkeley to help out with the test program. This program might be directed by Dr. Herbert York, reporting to LASL, and backed up by Dr. E. O. Lawrence. UCRL might also be encouraged to go ahead on a broader front of weapons research and development, as long as it did not "pirate" any LASL personnel, consultants, or subcontractors, and did not either interfere with or damage LASL weapons work.

The consensus of the GAC was that UCRL should not be asked to take over the entire thermonuclear program, but it should be encouraged to take an interest in the field, and, with "due consultation" of LASL, should be supported in following up any problem it would like to work on. Whether or not this type of activity would lead to a second independent weapons laboratory was not clear at this time.⁸

Earlier in the month, during the 30th meeting of the GAC, when the subject of how seriously technical progress might have been impeded by excessive secrecy, among points raised was that a rational policy of secrecy must be related to estimates of what was then known by the Soviets, even though information of what was known behind the Iron Curtain about the U.S. nuclear weapons program was very hard to come by. The Central Intelligence Agency, from time to time, prepared analyses of what might be concluded from information which had been "overtly" published in the U.S. and abroad.

evening of Tuesday, April 15, 1952. On June 19, 1952, Teller reiterated his doubts about whether or not the proposed LASL thermonuclear test device would work. (Memorandum for the File dated June 19, 1952 from Bill Borden and John Walker, subject: Second Laboratory -- Dr. Teller.)

⁷Letter dated April 25, 1952 from H. D. Smyth, Acting Chairman, USAEC, to Honorable Robert LeBaron, Chairman, MLC.

⁸Minutes of the Thirtieth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, April 27, 28, and 29, 1952, p. 33, 34, 35.

The U.S. was also severely handicapped by not being able to exchange technical information with the British. Given that British intelligence was at this time riddled with Soviet agents, this may not have been an altogether bad situation for the U.S.

Another opinion expressed was that DOD leaks about LASL's real achievements in weapon, including many articles by Hanson Baldwin in the New York Times, had been far more serious breaches of security than would be the revelation of much of the purely scientific information now classified as secret.⁹

At the beginning of May, construction was started on a lithium-6 separation plant at Oak Ridge. Construction time was estimated at one year, with a design output of at least one kilogram of lithium-6 per day when completed.¹⁰

At this time, AEC Chairman Gordon Dean was still strongly opposed to a second weapons lab:

Mr. Dean spoke several times of the second laboratory issue -- and with some bitterness, he made it plain that in his view a second laboratory is not necessary; that it would do more harm than good; and that -- confound it all -- why did we keep beating him on a second laboratory? He referred scathingly to the recent Air Force threat to set up a second laboratory under Air Force auspices if the (Atomic Energy) Commission does not proceed.

We had the impression that Mr. Dean was more emphatic and outspoken on the second laboratory than on any other point discussed during the entire interview.¹¹

In the middle of May, Major General Howard G. Bunker, an Air Force member of the MLC, stated that if a thermonuclear

⁹Minutes of the Thirtieth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, April 27, 28, and 29, 1952, p. 7, 24. For examples of Baldwin's disclosures, see CHINESE ATTITUDES TOWARD NUCLEAR WEAPONS: CHINA AND THE UNITED STATES DURING THE KOREAN WAR. Mark A. Ryan, M. E. Sharpe, Armonk, New York, 1989, pp. 42, 74, 75, 146, 147, and 263.

¹⁰Borden-Walker H-bomb chronology, p. 77.

¹¹Memorandum to Senator McMahon dated May 7, 1952 from Bill Borden and John Walker.

weapon would "cost" only two fission weapons, and if its weight could be held to a maximum of 15,000 lbs. so that it could be delivered by existing USAF aircraft, then the Air Force would probably prefer to see one-half of the strategic weapons stockpile composed of hydrogen bombs. He further estimated that the Air Force would probably prefer to have 2,000 to 3,000 thermonuclear weapons at its disposal.¹²

The Genesis of the Second Weapons Laboratory

Later in the month, the nascent second U.S. nuclear weapons laboratory, then proposed for Livermore, California, began to take shape as an offshoot of the venerable Lawrence Berkeley Laboratory (then the University of California Radiation Laboratory). Although its creation appeared to be more and more likely, Norris Bradbury and many LASL personnel -- who were fighting to maintain LASL's prerogatives and privileges -- continued to strongly oppose the new lab and to disavow any major assistance to it after its establishment.

The proposed lab's program was to complement LASL's development of thermonuclear weaponry; in Norris Bradbury's skeptical opinion, presented in a long, seven-page critique of the new lab and its possible future relations with LASL, the new lab would have to sink or swim without any help from LASL:

The Berkeley group (is) interested in establishing a laboratory whose primary mission is the experimental and theoretical exploration of thermonuclear components, mechanisms, and devices, particularly radiation implosion as related to its use primarily in thermonuclear devices, but with no specific limitation to this field. ...

The prime interest of the Berkeley administration in establishing the Livermore group is to undertake independent exploration in the radiation implosion field. Such work would be carried out in close collaboration with the Los Alamos Scientific Laboratory but not under supervision or control. Berkeley personnel feel that any directive to the new organization must be specific in the sense that the entry of the new Laboratory (sic) into this field must be both encouraged and

¹²Memorandum to the File dated May 14, 1952 from J. K. Mansfield, subject: Military Need for H-Weapons.

directed. ...

... there appears to be available a fairly clear philosophy under which a senior group (Ernest O. Lawrence, Donald Cooksey, Luis Alvarez, Herbert York) of demonstrated competence and enthusiasm is willing to establish and staff a new Laboratory in the thermonuclear weapons field.

Echoing previous AEC statements, Bradbury went on to state that the likelihood of the new lab attracting a large number of qualified and talented personnel "must be regarded as highly dubious at this time." He suggested that in order to properly transfer the present state of thermonuclear knowledge, new employees of the second lab should serve a one- or two-year residency at LASL.

Even as he proposed this residency program, Bradbury argued that time spent by LASL staff educating a second lab staff would detract from LASL's own efforts to continue to make timely progress in thermonuclear weapons development. Raising a security issue, the LASL director also noted that Dr. Hans Bethe had "recently expressed some concern regarding the wide dissemination of radiation implosion engineering concepts."

Bradbury also argued that since both LASL and the new lab were planning to conduct "component test" experiments during Operations UPSHOT and CASTLE, both of which would make large demands upon LASL's computing facilities and theoretical physicists, it was "unlikely that much time for analysis and study of Livermore proposals will be available at Los Alamos." He even suggested that LASL co-opt any promising Livermore innovations:

However, it is entirely reasonable to believe that the collaboration and sincerity of the two Laboratories will be sufficiently real that a really outstanding idea originating in the new group will be enthusiastically adopted at Los Alamos if the Livermore organization is, at the time, unable to push it through to adequate completion.

Nonetheless, Bradbury was sufficiently realistic about the opinion of some of Teller's allies that LASL was hindering the establishment of the Livermore lab to state that, regarding LASL review of Livermore test proposals, "the temper of the new Laboratory and some of the personnel it might attract could hardly accept Los Alamos review, in any event."

The LASL director also gave his discouraging opinion of field

testing problems that the Livermore lab might face; he seemed to think that these, in the aggregate, might be insurmountable, although these types of problems had all been overcome in the past by LASL:

The problem confronting the Livermore group in planning and carrying out thermonuclear component tests is not confined to theoretical planning, calculation, and evaluation ... There is the problem of obtaining and fabricating the special materials required (for) the tests. ... There is the problem of designing and constructing, calibrating and installing diagnostic test equipment. There is the problem of laboratory tests of components of the device, such as safety tests and the measurement of time sequences (transit and related times). There is the problem of field assembly of the device.

There is the problem of carrying out necessary but possibly uninteresting measurements in the test of the device, such as measurements of the performance of the primary fission bomb. The Livermore group must plan to carry out, or arrange to have carried out, these and similar activities. ... Los Alamos will probably find it impossible, without seriously upsetting its own development and test programs, to supply any appreciable aid to Livermore, apart, possibly, from supplying duplicate components and data which it makes or obtains for its own purposes. ...

In a foreshadowing of a complaint that would arise several years later, Bradbury noted that all of LASL's test plans were reviewed by the AEC, the GAC, and the MLC to guarantee the maximum scientific progress for a given amount of scarce and expensive fissionable and fusionable materials. In the recent past, LASL had been criticized by the GAC for conducting "too many experiments" without enough interpretation of their results.

Bradbury feared that if Livermore test proposals were not similarly reviewed in advance by all these organizations, "it will give rise to enormous irritation and loss of morale if the new Laboratory is given a free hand and the old one restricted." (Bradbury's fears here were groundless: all of the Livermore lab test plans were eventually reviewed by the same groups that also reviewed LASL's test plans.)

The new lab would also compete for limited available testing

and operating funds:

Moreover, the dollars spent on tests have been subject to increasing criticism by the Bureau of the Budget. It will be difficult to explain to Los Alamos that their test budget is decreased because the new laboratory must have a test budget. It will be only slightly less difficult to make any statement of the opposite nature (i.e., that test budgets are being increased) which is both encouraging and true.

The (AEC) must face the question of how to say to Los Alamos "Your funds will not be affected by the new Laboratory" when we continually face them with requests for economy in the face of Congressional budget limitations. The statement itself, unfortunately, is probably not true. The reason for its falsity may not be obvious at present, but will inevitably become so: an institution, once founded, never dies. If Livermore is established, it will continue because it will have acquired roots.

Funds available to the Commission may some day become limited. Perhaps they are even now. The Commission will then have face the problem of supporting both Los Alamos and Livermore. It will almost certainly do so by the perfectly normal method of cutting both budgets. .. The result of this cut (will be) that Los Alamos will have less funds with which to operate than had Livermore not been established. ...

Bradbury also suggested that requirements for special test device materials might place an undue burden on the AEC's production complex (Bradbury was right about this: later test series, simply by virtue of the number of individual tests, did in fact strain the AEC's ability to maintain weapons production and support test program demands; machinists and technicians who might be employed on as production or assembly line had to be reassigned to custom-build one-of-a-kind nuclear test devices or components):

Another problem which must be faced is the matter of supplying Livermore with materials, like special high explosive components for implosion warheads, to be used in thermonuclear tests, as well as uranium and plutonium incident to

exploratory experiments in the radiation implosion field. The Los Alamos Scientific Laboratory has always been willing to supply to other laboratories materials which it was in a special position to fabricate and which could be prepared without detriment to other laboratory programs.

However, the facilities of the Los Alamos Scientific Laboratory are largely geared to the demands of its own test programs, and the extent to which materials can be supplied elsewhere may be limited. It will be clearly inimical to the interests of the AEC to relegate Los Alamos to the role of "supplier" to the new Laboratory, yet it will be extremely difficult to defend the establishment, for example, of facilities for developing light material explosive lenses at Livermore.

If Livermore confines its explosives requests to stockpile or developed items, it is possible that its needs can be handled through Los Alamos. It will, however, be highly unlikely that Los Alamos can respond favorably to requests for development programs originating from Livermore in view of the Los Alamos Scientific Laboratory preoccupation with its own program. Probably it would be necessary to request Oak Ridge to make uranium components and Rocky Flats or Hanford to fabricate plutonium for Livermore.¹³

Soon after this letter was completed, Bradbury dispatched yet another missive, this one to the AEC Director of Military Application requesting that cryogenic hardware for the "emergency capability" liquid-fueled JUGHEAD bomb be developed as soon as possible, for both ground and air handling and transportation of the weapon.¹⁴ The JUGHEAD was to be a deliverable version of the liquid-fueled device

¹³"Observations on the Livermore Laboratory Proposal," N. E. Bradbury, May 21, 1952. One significant point about this paper is that Bradbury, who frequently referred in correspondence to Los Alamos as "the LASL," spelled out its name every time it was mentioned in this document, perhaps trying to emphasize its role or history.

¹⁴Borden-Walker H-Bomb Chronology, p. 82.

slated for test late in 1952.

By May 28, Dr. Edward Teller had submitted to the AEC a proposal for a series of five thermonuclear device component tests; two would take place in February 1953.¹⁵

At the end of the month, the chairman of the MLC wrote to President Truman and stated that not merely one, but several, types of hydrogen bombs could be constructed, with yields on the order of several megatons. A deliverable bomb was anticipated by 1954 at the latest.¹⁶

Within two weeks of Bradbury's lengthy denunciation of the second weapons lab, which had by now been designated as the University of California Radiation Laboratory at Livermore, or UCRL, Dr. Herbert York, who would be its first director, submitted his own plans for its immediate future:

Proposed Program for Project Whitney, UCRL

Basis of Proposed Program

This proposal is based on two assumptions, both of which are based on discussions between members of UCRL, AEC, and LASL. These discussions have indicated that there is essentially universal agreement among all concerned.

I. First Assumption. The charge of the AEC will be more or less as follows:

First, UCRL will be asked to support the LASL weapons program, particularly in the field of diagnostic experiments, with special reference to what has come to be known as components testing.

Second, UCRL will be encouraged to assemble personnel and facilities which will not only be of immediate assistance to the

¹⁵Memorandum to the File dated May 28, 1952 from John S. Walker, subject: Thermonuclear Program.

¹⁶Borden-Walker H-bomb chronology, p. 80; letter dated May 30, 1952 to President Truman from Senator Brien McMahon. The Truman-to-McMahon letter is reprinted in FOREIGN RELATIONS OF THE UNITED STATES 1952-1954, Volume II, National Security Affairs, Part 2, U.S. Government Printing Office, Washington, D.C., 1984, pp. 955-957.

currently envisaged test program but will also contribute to the longer-range development of new ideas.

I. Second Assumption. The diagnostic work will be along the following lines:

First, we will investigate methods for determining the fundamental parameters involved in thermonuclear weapons design, such as [deleted] velocity of radiation flow in channels, timing of hydrodynamic motions, temperatures, compressions, etc., and then we will carry out in the field experiments using the methods developed.

Second, we will make other less basic measurements requested by LASL.

York proposed a diagnostic planning group that would start out by studying the theory of the various types of thermonuclear weapons then known, and current explosion measurement methods (photographic, electronic, and radiochemical). Modern electronic computers, both digital and analog, and skilled operators, engineers, and programmers would be required to perform analyses of radiation flow and hydrodynamical motion.

York also noted that it might be possible, even at an early date, for the computers to be used to investigate possible new weapons designs, working on either LASL or UCRL problems.¹⁷

By June, a large number of facilities across the U.S. were engaged full-time in work on the Eniwetok test device. The University of California Radiation Laboratory was developing diagnostic methods and equipment for the test. A group at Princeton University (Project Matterhorn) was computing theoretical thermonuclear burning problems. The National Bureau of Standards in Boulder, Colorado had constructed for the AEC a new cryogenics plant and was producing liquid hydrogen.

The Arthur D. Little's Cambridge Corporation had designed and constructed large capacity mobile dewars to transport refrigerated liquid hydrogen. Ohio State University had installed and was maintaining a hydrogen liquefaction facility at Eniwetok atoll in preparation for the November test. ACF was engineering and constructing the non-nuclear components of the PANDA experiment, another name for the

¹⁷Attachment to letter dated June 3, 1952 to Dr. N. E. Bradbury, Los Alamos Scientific Laboratory, from Herbert F. York.

forthcoming November test at Eniwetok.¹⁸

By this time, plans were underway to employ UCRL scientists in diagnostic experiments during IVY. The AEC was by now receptive to suggestions of broader programs of thermonuclear research by UCRL. Soon afterwards, AEC Chairman Gordon Dean formally notified the regents of the University of California that the Commission would like UCRL to "undertake work in the thermonuclear field." (By the end of June, the UC regents would concur with the AEC in this wish, and would expand the existing AEC-UC contract to cover this new work.)¹⁹

Early in the month, the AEC advised the MLC that thermonuclear weapons development was occupying one-quarter to one-third of LASL's current manpower. Plans were being made to test-solid fueled-devices in the fall of 1953 during the proposed Operation CASTLE. As a result of an extensive and still-continuing program of machine calculations carried out on the National Bureau of Standards' SEAC computer, it now appeared possible that a solid-fueled version of the ALARM CLOCK could be built in the form of a bomb deliverable by existing large aircraft.

Availability of lithium-6 was a governing factor in this development, and steps were being taken to build a lithium isotope separation plant with enough production capacity to provide the lithium required both for the 1953 tests and for possible stockpiling of solid-fueled weapons, if the tests were successful.²⁰

At this time, AEC Chairman Gordon Dean reiterated his opposition to a second U.S. nuclear weapons lab:

As explained in the hearing before the Joint Committee (on Atomic Energy) last fall, the AEC is of the opinion that to establish a second weapons laboratory would dilute the

¹⁸Borden-Walker H-bomb chronology, p. 83; letter dated June 11, 1952 to Robert LeBaron, Chairman, MLC, from William Borden, Executive Director, Joint Committee on Atomic Energy, U.S. Congress; letter dated June 10, 1952 to Senator Brien McMahon, Chairman, MLC, from Gordon Dean, Chairman, USAEC.

¹⁹Memorandum dated April 29, 1954 to Chester G. Brinck, Assistant General Counsel, from Leonard A. Jacobvitz, Attorney, Office of Assistant General Counsel, subject: Participation of the University of California Radiation Laboratory in Weapons Development Program.

²⁰Letter dated June 10, 1952 to Senator Brien McMahon, Chairman, MLC, from Gordon Dean, Chairman, USAEC.

efforts of Los Alamos and would have no compensating advantages for many years. It is becoming increasingly clear that thermonuclear and fission weapons have become more closely related in design and planning and that separation of these two fields of activity would be disadvantageous. We are of the opinion that it will be more effective to bring additional efforts to bear on the increasingly important thermonuclear program ...²¹

On June 12, plans for a second weapons laboratory had received the support of the Secretary of Defense; the new facility might be located at Midway, near Chicago. Plans for the lab noted that while there was much weapons work to be done, including lightweight warheads for anti-aircraft missiles, the "most promising weapons work" was in the thermonuclear field. The plans reiterated in considerable detail a series of seven component tests, including both what should be done and how it could be coordinated with LASL. Also stated were seven specific and significant ways of conducting the testing. The lab might comprise from 500 to 800 persons.²²

A Proposal to Delay the Thermonuclear Test

Near the middle of the month, Norris Bradbury responded to a possibility raised by J. Robert Oppenheimer and others that the November thermonuclear test be postponed. Oppenheimer and his allies were hoping in the interest of arms control that if the shot were delayed until after the November 1952 presidential election, and if Democratic candidate Senator Adlai Stevenson were triumphant over General Dwight Eisenhower, Stevenson's Republican opponent, the thermonuclear test might be canceled completely.

A corollary to this idea held that if the U.S. did not test and develop the H-bomb, the Soviets would also not test their own version.²³ This was a fall-back position from the idea

²¹Letter dated June 10, 1952 to Senator Brien McMahon, Chairman, MLC, from Gordon Dean, Chairman, USAEC.

²²Memorandum for the Files dated June 12, 1952 from John S. Walker, subject: Thermonuclear Program -- Second Laboratory.

²³Memorandum from Bill Borden dated May 28, 1952 to Senator McMahon. See also Memorandum by the Panel of Consultants on Disarmament, reprinted in FOREIGN RELATIONS OF THE UNITED STATES 1952-1954, Volume II, National Security Affairs, Part 2, U.S. Government Printing Office, Washington, D.C., 1984,

that if the U.S. did not develop such a weapon, the Soviets would do likewise, and was more grounded in wishful thinking than current geopolitical reality.

In a letter to Oppenheimer, Bradbury laid out the lab's plans for the test (the SAUSAGE and equilibrium Super refer to the planned thermonuclear shot):

Hans Bethe has passed on to me a question you have raised regarding the feasibility of delaying the test of the equilibrium Super at Eniwetok as well as the problems which prompt this query. Needless to say, these problems are of paramount importance, and if useful results can come from a delay in our operations, we would make every effort to adjust ourselves to the required circumstances.

The situation as we see it presently is as follows. We have as yet encountered no theoretical or experimental obstacle which would suggest a delay in the present November 1 date (for the test). Troubles either here or at Eniwetok can naturally still occur and could force a postponement on their own account. The weather at Eniwetok may force postponement. Nevertheless, we would still use November 1 as a target date at this time.

If we were directed not to shoot before November 15, but permitted to shoot on the first available day thereafter, I believe no serious problems would result. As you know, the problem of further postponement arises out of the fact that we have another shot to get off (the 500 KT shot) and that we would like, if at all possible, to permit not only our own men but the Task Force (also) to be home by Christmas. For shots of the order of 500 kilotons, there appear to be on the average about five suitable (test) days in October, three in November, but only one or two in December.

We have explored the possibility of shooting the 500 kiloton test first. This, however, takes a chance of jeopardizing the SAUSAGE test by damaging structures such as our cryogenics buildings and

equipment, and we have regarded the latter test (to be of) primary importance. Indeed, if the atoll becomes too "hot" (Author's note: contaminated by radiation) it may be necessary to forego altogether the 500 kiloton test. We have chosen this alternative rather than take any chance whatsoever with the equilibrium Super test.

If the operation were to be postponed (for other than technical reasons at the site) beyond November 15, it would probably be necessary to postpone it altogether until perhaps March 15, 1953 (Author's note: winter and typhoon season in the Pacific would remove any chance for firing any later than early December 1952). This would, of course, interfere with Operation UPSHOT presently planned for that time at Nevada.

It would also permit rather little time between this operation and the CASTLE tests planned for the Fall (sic) of 1953 in which to make practical use of any results from the SAUSAGE test. It would further require our Test Division to go overseas twice in one year -- a task none too easy.

We have considered the possibility of scheduling UPSHOT in the Fall of this year in order to relieve the interference noted above. Of course, we have just recently postponed this operation (UPSHOT) because we felt that we would not be in a position, in the time after the series just concluded (Author's note: Operation TUMBLER-SNAPPER) to do a worthwhile job of planning. These same arguments still hold and are complicated by the fact that for some time at least we will have to proceed on the basis that the SAUSAGE test is, indeed, on November 1 until we are directed otherwise.

This may take some time, but in the interim, we would not be able to any definitive planning (for) UPSHOT. I would actually be concerned about any relaxation in the SAUSAGE schedule until all design work was done and the theorists had nothing further to offer the engineers for this test.

In addition, the UPSHOT test is planned to include some [deleted] component tests

which may be better designed after the SAUSAGE test. In at least one such test, there is a real question as to the availability of Li-6 of adequate enrichment for a Fall UPSHOT test, even were there time to design and fabricate such a test between now and then.

Finally, I would be quite concerned about a long delay (in the SAUSAGE test) from the point of view of the morale of the many Los Alamos and contractor personnel involved. Even though the objectives of such a postponement were very high, they could probably not be told what they were, and even if told, might feel that this could well have been thought of some time ago when we were being urged to "hurry, hurry."

In short, a directed postponement to a date not before November 15 would present no serious difficulty and such a (delay) may occur by itself. A directed postponement to a later date would present serious problems. My own hope is that the two individuals most likely to be concerned (Author's note: presidential candidates Stevenson and Eisenhower) could be apprised of the problem, along with the present incumbent (Author's note: President Truman), long before they would actually have to face dealing with it.²⁴

Also at this time, the AEC took several steps to strengthen the U.S. thermonuclear research effort. The AEC approved plans to raise LASL scientific and technical staff to a total of 3,050, including the payment of appropriate consultant fees to bring to and retain at LASL key persons, such as Hans Bethe, Edward Teller, John von Neumann, Enrico Fermi, and Victor Weisskopf, who might be induced to work either full- or part-time if their compensation were increased.

The AEC ordered continuance of Project Matterhorn at

²⁴Letter dated 11 June 1952 to Dr. J. Robert Oppenheimer, Director, Institute for Advanced Study, Princeton, New Jersey, from N. E. Bradbury, Director, LASL. Operation UPSHOT was originally to have included a test of the mixing of light materials (lithium) and heavy fissionable materials in a device named the "Mixmaster;" in addition, some neutron-radiation "preheating" experiments were to be conducted on enriched lithium in bomb casing mockups during one or more UPSHOT detonations.

Princeton, and the assignment of the University of California Radiation Laboratory to prepare instrumentation for tests of thermonuclear devices. UCRL was also to be encouraged to participate in component or phase testing of thermonuclear weapons.

A new process to produce Li-6 was to be perfected as soon as possible, and a new lithium enrichment facility would be constructed after the AEC and DOD agreed on the rate and scale of future thermonuclear weapons manufacture.²⁵

In the middle of June, the DOD furnished to the AEC a further statement of military requirements for thermonuclear weapons. In a letter dated June 13 from MLC Chairman Robert LeBaron to AEC Chairman Gordon Dean, LeBaron stated:

A military requirement exists for the development of thermonuclear weapons giving yields of one megaton upwards: a military requirement exists for the production of such weapons provided they are compatible in size, shape, and weight with delivery systems that will be available in 1954.²⁶

By now, even before the IVY tests, the GAC, in a complete about-face from its October 1949 attitude, was very optimistic about successful thermonuclear weapons:

Several atomic explosions involving thermonuclear reactions have now been conducted, showing the feasibility of using these reactions to increase the yield of fission weapons. Although the underlying scientific facts have long been known, it is for only about a year, since the spring of 1951, that we have promising and practical schemes for very large-scale thermonuclear weapons.

Even more than in the case of the first atomic bomb, certainty with regard to these schemes will have to await the outcome of

²⁵Memorandum dated June 10, 1952 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from Roy B. Snapp, Secretary, USAEC, Subject: Commission Action Regarding Strengthening the Thermonuclear Effort.

²⁶Borden-Walker H-bomb chronology, p. 83; letter dated 4 February 1964 to John Conway, Executive Director, JCAE, from W. J. Howard, Assistant to the Secretary of Defense (Atomic Energy).

actual test, and may at some stage involve temporary disappointment with regard to the performance of a specific weapons design.

Yet we think it very likely that the feasibility of weapons hundreds, and perhaps thousands of times more powerful than the first atomic bomb will be manifest within the next years, and that such weapons can be added to our arsenal in substantial numbers to the extent that there is a military need for them.²⁷

The first reactor at the DuPont plant at Savannah River was nearing completion in the summer of 1952, and was scheduled for completion by March 1953.²⁸ Four of the 24 dual-temperature heavy water units were undergoing preliminary testing.²⁹

At this time, the amount of tritium on hand in the U.S. was only on the order of a few tens of grams, far less than would have been required for even one "classical Super." Some tritium was being used for tests and some was eventually lost in handling or through radioactive decay into helium. This small amount was obviously inadequate for any future weapons program, including advanced gas-boosted bomb designs which might use it; however, it was believed sufficient for the UPSHOT tests in Nevada in 1953 and the CASTLE series in the Pacific in 1954.³⁰

Late in the month, the total yield of the IVY thermonuclear test was expected to be four to five megatons, and a reasonably possible maximum of 10 MT. Any yield over 10 MT was considered a remote possibility. Nonetheless, for purposes of predicting possible damage to installations at Eniwetok atoll caused by blast, thermal radiation, and water

²⁷Letter dated June 14, 1952 to The President, The White House, from J. R. Oppenheimer, Chairman, for the General Advisory Committee. The "several atomic explosions involving thermonuclear reactions" included the GREENHOUSE George and Item shots, and the Operation SNAPPER Dog shot in Nevada, which had studied boosting effects of deuterium gas.

²⁸Between 1944 and 1968, the AEC built and operated 14 production reactors at the Hanford and Savannah River sites, finally producing about 100 metric tons of plutonium, as well as much smaller quantities of tritium, U-233, and polonium.

²⁹Hewlett and Duncan, p. 586.

³⁰September 12, 1952 Walker memorandum, pp. 20-22.

waves, maximum yields up to 40 MT were considered.³¹

On July 1, the regents of the University of California accepted the AEC's recommendations for the development of a second thermonuclear weapons lab to be run under UC aegis. While the most immediate project for this lab would be diagnostic work for thermonuclear tests to be conducted by LASL, the new lab was to also "propose and undertake a broad thermonuclear research program." Project Whitney, as the new project was named, expected to develop a work force of 800 men by the end of June 1953 and hoped to be numerically half as large as LASL by the end of June 1954.

Nine days later, the RAND Corporation issued studies entitled "Thermal Radiation and Aircraft Escape Problems for Large-Yield Bombs" and "Implications of Potential Weapons Development for Strategic Bombing and Air Defense." These papers reviewed potential problems and problem solutions associated with both the delivery of hydrogen weapons to an enemy and the possible consequences to the U.S. of attack by such weapons.³²

In mid-August 1952, in spite of the forthcoming test, Edward Teller was still fearful that the U.S. might be lagging behind the Soviets in the race to the Super (this was not the case; the Soviet Union was far behind the U.S. and had not even yet discovered the utility of radiation implosion):

I believe that we have pursued the thermonuclear development throughout the past seven years at much too slow a rate; and even since the Presidential Directive (of January 1950) progress has been slower and certainly narrower than is consistent with national security.

Our only comfort seems to be that the Russians have not as yet given any evidence of possessing an effective thermonuclear weapon. (But) we may at the beginning of an arduous program and it is quite possible that the Russians have advanced much farther along the road than we have.³³

³¹Report dated June 24, 1952 to Commander, Joint Task Force 132, from Duncan Curry, Jr., Chief of Staff, Task Group 132.1, subject: Report of Evacuation Plans Conference, p. 1.

³²Borden-Walker H-Bomb Chronology, p. 86.

³³"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein,

Late in August, construction of the lithium-6 separation plant at Oak Ridge was proceeding on a very tight schedule. The AEC was gambling that a production plant could be built from the results of only laboratory experimentation on the separation process, without the more-cautious and normal procedure of building a pilot plant for small-scale lithium separation.³⁴

There was some concern at the JCAE at this time that Li-6 production might become a bottleneck to the entire U.S. thermonuclear weapons program, and that weapons might be stockpiled on the basis of their requirements for this substance, rather than on the basis of the most promising scientific model.

Other major design uncertainties were the effects of Taylor instabilities and the efficacy of radiation implosion.³⁵

By September, as Operation IVY drew near, both Carson Mark and Hans Bethe were confident that the principle of radiation implosion would be successful.³⁶

At this time, the new Livermore lab was just getting started; for the time being, it was being run out of the University of California Radiation Laboratory in Berkeley, California:

Project WHITNEY at Berkeley probably dates from July 1952, nine and one-third years after establishment of Los Alamos in April 1943. WHITNEY personnel are being sent to Los Alamos in twos and threes for conferences and library work. An assistant is stationed at Los Alamos to facilitate Berkeley men visiting Los Alamos.

The Berkeley laboratory has asked for one test at UPSHOT and one at CASTLE -- details

HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 No. 2 (1989), pp. 327, 328.

³⁴Memorandum to the Files dated August 20, 1952 from Bill Borden, Executive Director, JCAE.

³⁵Memorandum to the File dated August 25, 1952 from John S. Walker and J. Kenneth Mansfield, subject: Thermonuclear Program -- Dr. Edward L. Teller.

³⁶Memorandum to File dated September 12, 1952 from John S. Walker, subject: Status Report on the Thermonuclear Program, p. 3. (hereafter referred to as September 12, 1952 Walker memorandum.)

as yet unknown, but admittedly an ambitious schedule. All routine coordination between the two laboratories appears to be going very smoothly and cordially. On top levels there is speculation of considerable tension, but no more, apparently, than was to be expected.³⁷

At the 744th AEC meeting on September 8, Dr. Ernest O. Lawrence reviewed the proposed program for the new laboratory. Although the original purpose of the lab was to help LASL with diagnostic measurements in the Pacific, a more ambitious program now being envisioned included a major project involving promising new concepts.

Dr. Edward Teller described the lab's leading project for the next year or more, a program named RAMROD. Using rough sketches, Teller discussed the advantages of the system and the problems to be solved before the concept could be realized. He particularly noted that the project would not compete with LASL for scarce materials.

When asked if the project would achieve worthwhile results, Dr. Lawrence said that the RAMROD entailed some risk; however, the project as now contemplated did not involve an extremely large effort for a very long time and even a relative failure would still be profitable. Lawrence added that it was intended that Livermore should become the second major weapons laboratory.³⁸

By October, the new Livermore lab planned to fire two shots in the spring 1953 Nevada test series. The second test would consist of Teller's design of the so-called RAMROD.³⁹

In the fall of 1952, the attention of the nation's thermonuclear weapons designers at Los Alamos was once again focusing on an isolated atoll in the remote reaches of the central Pacific. On September 9, the AEC and DOD had jointly issued the following press release:

The Department of Defense and the Atomic Energy Commission today announced that Joint Task Force 132 will conduct tests in the autumn months of this year looking toward the development of atomic weapons.

³⁷September 12, 1952 Walker memorandum, pp. 11, 12.

³⁸Minutes of AEC Meeting No. 744, September 8, 1952.

³⁹Memorandum to the Files dated October 7, 1952 from Edward L. Heller, subject: Project Whitney.

The tests will be held at the Commission's Pacific Proving Grounds on Eniwetok Atoll of the Marshall Islands in the Pacific. They will be conducted under full security provisions of the Atomic Energy Act. Only official observers of the United States Government (sic) and members of the task force will be present at the tests. The organization of Joint Task Force 132 .. was announced February 18, 1952.⁴⁰

Operation IVY

The first large-scale proof test of the Teller-Ulam configuration was to be conducted during a special two-shot nuclear test series named Operation IVY, scheduled for Eniwetok atoll in the fall of 1952.⁴¹

A prototype MK 18 "Super Oralloy Bomb," a high-yield pure-fission brainchild of Dr. Theodore B. Taylor, would also be airdropped and detonated.⁴² The MK 18 had the highest all-

⁴⁰Joint DOD-AEC press release dated September 9, 1952 reprinted in FOREIGN RELATIONS OF THE UNITED STATES 1952-1954, Volume II, National Security Affairs, Part 2, U.S. Government Printing Office, Washington, D.C., 1984, p. 1025.

⁴¹Much of the following history, except where noted, is drawn from OPERATION IVY, REPORT OF COMMANDER, TASK GROUP 132.1, WT-608 (EX), extracted version, Pacific Proving Grounds, Joint Task Force 132, November 1952, pp. 7 - 13, 19 - 22, 25 - 27, 44, and 55 - 60, and OPERATION IVY, Final Report, Joint Task Force 132, 1952, Headquarters, Joint Task Force Seven, Washington, D. C., 9 January 1953. The name IVY was assigned to the test operation in May 1951; IVY was then scheduled for the spring of 1952. (Atomic Energy Commission, Memorandum for Information, Code Names for Proposed Tests, Note by Secretary, May 3, 1951.) Three months later, IVY was not expected to be conducted before the winter of 1952-53. (Letter dated August 15, 1951 from Sumner T. Pike, Acting Chairman, USAEC, to Honorable Robert LeBaron, Chairman, Military Liaison Committee to the AEC.)

⁴²McPhee, pp. 89, 90, 91; York, 83, 84, 85; Schneider, pp. 180, 181, 185, 188, 189, 190, 191; OPERATION IVY 1952, DNA 6036F, Defense Nuclear Agency, Washington, D.C., 1 December 1982, pp. 1, 18, 19, 199. The official DNA radiological history states (p. 1) that "The energy from KING was generated by the fission ... of plutonium atoms." This seems rather unlikely, given the preponderance of evidence from other sources that the MK 18 was uranium-fueled and the existence in all plutonium of the Pu-240 isotope which causes predetonation in large quantities

fission yield of any weapon then in the U.S. nuclear stockpile, and it would serve as an immediately-deployable backup in the event that the Teller-Ulam ideas did not work (in a similar fashion, a liquid-fueled thermonuclear bomb was fabricated in 1954 in case several solid-fueled test devices did not work; see "Operation CASTLE" below).

The MK 18 shot was designated "King" (for kilotons) and the full-scale thermonuclear test was designated "Mike" (for megatons).⁴³ These mnemonic names made it easy to distinguish between purposes of the tests.

King shot was expected to yield between 400 and 600 kilotons, with the most likely value around 500 KT, depending upon the efficiency of the weapon (LASL's T Division predicted 525 KT; W Division estimated a more-conservative 425 KT).⁴⁴ King was

of plutonium (500 Kt of fission yield is equivalent to the total fission of 25 kg, or 55 lbs., of plutonium; at 30% efficiency, the MK 18 would have required 183 lbs. of ^{239}Pu). On p. 199, DNA 6036F states "KING was a stockpile weapon modified to produce a large yield."

Further evidence of the all-U-235 composition of the MK 18 pit is provided in a description of radiochemical analysis of post-KING shot debris: there is no mention of Pu-239; only 15% of all KING fission occurred in nuclides other than U-235. (OPERATION IVY, FINAL REPORT, Joint Task Force 132, 1952, Headquarters, Joint Task Force Seven, Washington, D.C., 9 January 1953, p. A-2.)

⁴³SPECIAL WEAPONS EFFECTS ON AIRCRAFT, Julius King, Historical Branch, Office of Information Services, Wright Air Development Center (WADC), Air Research and Development Command (ARDC), July 1957, p. 51. The designation "King" was originally assigned to an airdrop planned for Operation GREENHOUSE.

⁴⁴Atomic Energy Commission, Progress Report to the Joint Committee, June through November 1951, Part III, Weapons, December 17, 1951, p. 21; Facts Bearing on the Problem and Discussion, Enclosure A to Report by the Commander Joint Task Force 132 to the Joint Chiefs of Staff on Armed Forces Participation in the 1952 Nuclear and Thermonuclear Experimental Device Tests, 1 December 1951; "Concept of TG 132.1 Forward Area Operations," undated, p. 1; Memorandum for Record dated November 19, 1952 from Lt. Col. William R. Sturges, Jr., Subject: Trip to Eniwetok Atoll, November 7, 1952 through November 18, 1952. Another estimate of King yield was 500 KT, $\pm 20\%$. (Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. A-1.)

to be a test of a high-efficiency air-dropped device using a modified fuzing system to permit a 1,500-foot burst height.⁴⁵

The King shot, like so many other U.S. Pacific and Nevada nuclear tests, was to be preceded by a full-scale rehearsal, called "KX" or "King X-Ray." "KX" was to be the airdrop sometime before the actual King shot of a device ballistically-identical to the King weapon, but its detonation would come from approximately 5,000 lbs. of high explosives with no nuclear components.⁴⁶

Purpose of IVY

The GREENHOUSE tests in the spring of 1951 had confirmed the theoretical predictions that a thermonuclear reaction could be initiated by a fission explosion. One of the major purposes of Operation GREENHOUSE had been to answer key questions relating to the possibility of developing thermonuclear explosives. The GREENHOUSE George and Item shots were important steps in the investigation of basic problems in the development of a fusion weapon. After GREENHOUSE, the design of a large-yield thermonuclear device had been immediately undertaken at Los Alamos Scientific Laboratory.

The yield expected from the test device was from two to six megatons, with the most likely yield around four megatons; however, a range of yields between 300 KT and 50 MT was encompassed, the probability being that yield would be on the low side if it were not near the most-likely predicted value.⁴⁷ The Mike shot was thus expected to be either a

⁴⁵Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, pp. 6, A-1; FINAL REPORT, JOINT TASK FORCE 132, OPERATION "IVY" 1952, p. 5.

⁴⁶Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, pp. 6, A-11.

⁴⁷"Concept of TG 132.1 Forward Area Operations," undated, p. 1; Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. A-1. By July 1952, the yield prediction for Mike was 5 to 10 MT with only a remote possibility for 40 MT. (Memorandum for Record dated July 3, 1952 from Lt. Col. R. P. Campbell, Jr., DMA/USAEC, Subject: Notes from the Commission's Briefing of July 1, 1952.) OPERATION IVY, 1952 (DNA 6036F), the official Defense Nuclear Agency radiological history of Operation IVY, states: "Although the novelty of the experiment made yield prediction difficult, the device designers expected a yield of at least 4 MT and perhaps as

spectacular success or a miserable failure.

By December 1951, the yield of the Mike test was expected to be so big that it might be necessary to erect blast-deflecting earthen berms around some of the more important structures on Eniwetok and Parry islands.⁴⁸ One estimate stated that "the Super will be fired on Elugelab, and the predicted yield is one to 10 megatons, with a fairly remote possibility of its going as high as 50 to 90 megatons." The most likely yield was believed to be around five megatons.⁴⁹

Anticipated shot effects were equally cataclysmic. Heavy fallout, mostly generated by the uranium-clad secondary in the device, was expected to be deposited as far as 700 miles away. The blast was expected to dig a crater 500 feet deep and a half-mile in diameter and lift as much 21 million cubic yards of water and coral into the air. Water waves caused by the blast were calculated to vary from 43 feet at a distance of 6,000 yards from ground zero to 15 feet at 15,000 yards and only six feet at a distance of 40,000 yards (about 23 miles).⁵⁰

Diagnostic experiments were designed and constructed to measure:

1. The alpha (neutron multiplication rate) of the primary

much as 10 MT, assuming a fusion reaction could be triggered" (p. 18). Even at the lower estimated yield, Mike would be by far the most powerful nuclear device ever detonated. Even bigger explosions were contemplated: "In November 1951, when the Air Force still was gravely concerned with the probable effects of 100 and 200-kiloton bombs, the Atomic Energy Commission had revealed plans to test a 20 to 60-megaton fusion bomb at Eniwetok in late 1952." (King, p. 50.)

⁴⁸Atomic Energy Commission, Progress Report to the Joint Committee, June through November 1951, Part III, Weapons, December 17, 1951, p. 21.

⁴⁹"Notes on Pogo Staff Meeting, 29 December 1951, Blast Measurements for IVY and SNAPPER," p. 2. The Pogo group, named after Walt Kelly's opossum character in the comic strip "Pogo," was a LASL test experiment planning group. (Santa Fe New Mexican news story dated 15 January 1956.)

⁵⁰Undated and unidentified personal notebook concerning Operation IVY. In June 1952, wave heights at Engebi, near Elugelab, were predicted to be up to 100 feet. (Report dated June 24, 1952 to Commander, Joint Task Force 132, from Duncan Curry, Jr., Chief of Staff, Task Group 132.1, subject: Report of Evacuation Plans Conference, p. 2.)

2. The time interval between fission and fusion phases to determine whether or not the primary behaved as predicted
3. The DD (deuterium-deuterium) burn characteristics
4. The speed of propagation of the DD reaction
5. The DD reaction temperature as indicated by 14 MeV neutrons⁵¹

The general types of experiments performed were basically the same as those conducted during Operation GREENHOUSE. Weapon diagnosticians used sophisticated techniques to follow the process that occurred during the device explosion. Detectors and collectors were run up to, and sometimes inside, the device casing so that the radiation being sampled could be directly channeled some distance away and be recorded by instruments designed to survive the ensuing blast.

To enhance its transport, radiation was conducted through pipes, which were often evacuated or filled with special gases, such as helium, argon or nitrogen, from the device to stations where recording instruments were located or where the data could be telemetered to a survivable recording station.⁵²

Much of the data to determine the behavior and efficiency of Mike's fusion and fission reactions would be collected in a concrete bunker nearly two miles away from the shot island. Telemetry transmitters in the bunker were connected to the device shelter (the shot cab) by a helium-filled tunnel, through which gamma rays and neutrons could pass with little attenuation.

More neutron energy and quantity data would be measured by the radiation induced by neutrons in selected threshold-detector materials, such as gold, tantalum, indium, iodine, and zirconium. For both Mike and King shots, many detector stations were arrayed in radial lines extending outward from

⁵¹OPERATION IVY, Projects 2.1 to 2.4 and Project 2.6, Measurements of Reaction History, Volume I -- Summary of Results, WT-620 (EX), Pacific Proving Grounds, November 1952, Joint Task Force 132, p. 9; Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. D-6; Operation IVY Pacific Proving Grounds November 1952, Report to the Scientific Director, Measurement of Reaction History, Volume II - Experiments, Results, and Analysis, Mike Shot, Part I, WT-621, E. H. Krause and Staff, Radiation Division, Naval Research Laboratory, Washington, D.C., June 1953, p. 11. Note that there is no mention of tritium-tritium or tritium-deuterium reactions in these measurements.

⁵²DNA 6036F, p. 24.

surface zero.

For King shot, the weapon's alpha and transit time were also measured, the latter by a transmitter located inside the bomb casing which radioed signals to recording equipment aboard two nearby airborne B-50 aircraft.

The objective of the Mike shot was to test, by actual detonation, the theory of design for a thermonuclear reaction on a large scale. The results of the shot could be used to design, test, produce, and stockpile thermonuclear weapons. To guarantee that the thermonuclear reaction would reach a large magnitude, the Mike device was essentially overdesigned: among other things, the steel walls of the external casing were a foot thick to contain the explosions within it as long as possible. This thick casing wall contributed greatly to the weight of the device.

All the diagnostic experiments were configured to measure certain specific reactions in an effort to confirm the Teller-Ulam principles that dictated the design of Mike. The liquid-fueled SAUSAGE -- the Mike shot test device -- was adaptable to weaponization, but it would require major redesign; the overall size and weight would have to be reduced and the cryogenics system simplified to make it a usable weapon. Even before IVY, a reduced-weight weaponized Mike device was expected to be ready for testing in the spring of 1954 (a cryogenic bomb was prepared for the CASTLE tests in early 1954).⁵³

Organization of Task Units and Test Site Selection

In mid-February 1951, before GREENHOUSE started, the AEC expected to conduct a follow-up thermonuclear test at Eniwetok in the spring of 1952.⁵⁴ In August 1951, Sumner T. Pike, then Acting Chairman of the U.S. Atomic Energy Commission, stated in letters that a thermonuclear device would probably not be ready for testing before late fall 1952 or early spring 1953, and most likely not before the winter of 1952-53.⁵⁵

However, in late October 1951, progress in research and

⁵³WT-608 (EX), pp. 7, 8.

⁵⁴AEC 334/4, 15 February 1951.

⁵⁵Letters dated August 15, 1951, from Sumner T. Pike, Acting Chairman, USAEC, Washington, D.C., to Senator Brian McMahon, Chairman, Joint Committee on Atomic Energy, U.S. Congress, and to Honorable Robert LeBaron, Chairman, Military Liaison Committee to the Atomic Energy Commission.

development activities had been such that a more realistic date for planning purposes appeared to be the fall of 1952, possibly October 1. In addition to the thermonuclear device to be tested at Operation IVY, the Director of the Los Alamos Scientific Laboratory issued orders to plan for the proof test of a high-yield (500 KT) MK 18 "Super Oralloy Bomb" fission weapon as a second IVY event.

In view of this October 1952 target date, the Panda Committee, composed of key personnel of the Los Alamos Scientific Laboratory and certain contractors, was organized in October 1951, just seven months after the discovery of the principle of radiation implosion, to consider the operational, design, fabrication, and planning problems related to the thermonuclear device to be tested at IVY.

Consideration was given to the possibility of testing this device in "dry" form, that is, with lithium-6 deuteride, but the lack of large amounts of Li^6D at this time because large-scale fabrication plants had not yet been designed or built dictated the use of liquid deuterium instead.

The King shot was required to extend the knowledge of the effects of high-yield weapons and to better study certain factors in the design of high-efficiency ${}_{92}\text{U}^{235}$ bombs.

Some of the major problems confronting the Panda Committee at this time included (1) suitable design and fabrication of the test device; (2) procurement of adequate and reliable sources of liquid hydrogen, deuterium, and nitrogen in the PPG, as well as safe and reliable means of transporting liquefied gases; and (3) the planning of the final assembly and control of the test device before and up to shot time.⁵⁶

As the Panda Committee met, the basic requirements for testing the thermonuclear device became apparent; namely, a large cryogenics plant, dewar transport facilities, a specialized shelter (cab) in which to assemble the device, and the use of the USS CURTISS as a preliminary assembly ship.

The choice of a specific island site for the detonation of a device with the predicted yield of Mike shot presented many problems, from both construction and experimental points of view. But by November 1951, after consideration and discussion, the island group at the north end of Eniwetok atoll from Elugelab clockwise to Bogon was decided on as the shot island group.

⁵⁶Moore and Bechanan, p. 259.

Elugelab would be the shot island, thus permitting the use of islands both west and east of it for instrumentation. Elugelab was also the northernmost island on the atoll, the farthest point from the base camps on Parry and Eniwetok islands 20 miles to the south-southeast.

The sequence for the two detonations was initially planned to be King shot, the high-yield MK 18 fission weapon airburst, followed by Mike shot. However, further study in late February 1952 proved that it would be best to fire Mike first. This would permit the greatest flexibility of operations and would not risk damage to Mike shot installations from King, nor would the King shot interfere with preparations for Mike.

King required relatively little instrumentation and did not require the massive precisely-aligned arrays used to collect data during Mike shot, and Mike was not expected to cause a great deal of damage to these installations. Therefore the Panda Committee recommended that the shot sequence for Operation IVY be changed to Mike shot followed by King shot. At this time the planned date for Mike shot was changed from October 1, 1952, to November 1, 1952. King shot would be fired within three weeks of Mike.

Large-scale nuclear weapons tests far from the continental U.S. assumed all the characteristics and logistical requirements and problems of major military war games or invasion exercises; indeed, for most of the tests, the seaborne command post was usually an amphibious force flagship. Hundreds of tons of test devices and experimental diagnostic, recording, and support materiel, along with thousands of people, had to be shuttled to the distant test site. Test devices and experiments had to be assembled and checked out on site.

Movements of sampling, monitoring, and photography flights and seaborne missions had to be rehearsed before each shot date. These activities necessitated the creation, assignment, and training of a major sea and airborne task force, similar in some respects to the U.S. Navy flotillas which fought the war in the Pacific against Japan in the mid-1940s.

Joint Task Force 132 (JTF 132) was the organization that conducted the IVY test series. Personnel representing the U.S. Army, Air Force, Navy, and Marine Corps; the U.S. Atomic Energy Commission; several other federal government agencies; the Los Alamos Scientific Laboratory; and many other civilian contractor agencies made up this organization. Commander JTF 132 reported directly to the Chairman of the Joint Chiefs of Staff, and he was also the designated AEC agent while conducting the tests.

The joint nature of this test organization, like all previous

and subsequent nuclear test task forces, resulted from the requirements of the Atomic Energy Act of 1946. This legislation placed atomic energy development under civilian control; however, the remoteness of the IVY series test site far out in the central Pacific Ocean dictated a military organization for physical security and technical and logistical support. The "police action" in Korea and Cold War tensions heightened the need for tight physical security.

The total number of personnel involved in JTF 132 was nearly 11,650, of which 9,350 were military and 2,300 were civilians. Most of the civilians and over 6,600 of the military personnel operated from Eniwetok atoll and from task force ships based there. The majority of the remaining military men were Air Force personnel who were based at Kwajalein atoll, over 400 miles southeast of Eniwetok.

IVY Task Units

JTF 132 included a number of subgroups, or task units, a few of which are noteworthy here.

Task Unit 2 of Joint Task Force 132 (TU 2 of JTF 132), Production, was to produce and supply liquefied gases in the amounts and at the times required by various users. The Mike shot experiment required large amounts of liquid hydrogen, liquid deuterium, and liquid nitrogen. From time to time, smaller amounts of compressed pure hydrogen and dry nitrogen were required for various purposes.

Some thought was initially given to converting a surplus U.S. Navy aircraft carrier into a floating cryogenic facility with a 200 liter per hour liquid hydrogen plant and a 400 liter per hour liquid nitrogen plant.

To provide these large amounts and varieties of liquefied gases, in addition to the floating facility, the existing GREENHOUSE cryogenics facility on Parry island at Eniwetok had to be expanded. The GREENHOUSE plant was completely rebuilt and enlarged so it could be used for the liquefaction of deuterium on a continuous basis at a rate much higher than was required for GREENHOUSE.

The nitrogen liquefaction plant was overhauled and used as a standby in case of malfunction of the deuterium plant. To supply electricity for the new liquid gas facilities, a 3,000 kilowatt power plant was installed in the cryogenics area. The Parry plant was to produce 200 liters per hour of liquid nitrogen.⁵⁷

⁵⁷Moore and Bechanan, p. 260.

By November 1951, the floating cryogenic plant idea was abandoned in favor of upgrading the Parry island facilities.

Specifically, TU 2 was to:

1. Complete necessary modifications and have the GREENHOUSE-era forward area hydrogen liquefier in operating condition by July 1952.
2. Install and be running by 1 September 1952 the National Bureau of Standards hydrogen liquefier.
3. Modify the forward area liquid nitrogen plant so that it could operate at maximum capacity.
4. Operate and maintain the gas liquefaction plants.
5. Procure and install a suitable liquid nitrogen storage facility, and furnish liquefied gases to all users upon request.⁵⁸

Task Unit 3 of JTF 132, Special Materials Facilities, was to operate and maintain the eight dewar-refrigerator combinations at Eniwetok atoll for Operation IVY beginning on 15 September 1952. This included moving the dewars and refrigerators throughout the atoll as required by the operation schedule. TU 3 operations at Eniwetok consisted of many fillings and transfers of the dewars in coordination with TU 2 and TU 4. The filled dewars were moved repeatedly between Parry and Elugelab.

Operation IVY involved the handling of liquid hydrogen in larger quantities than ever before encountered (liquid hydrogen is now handled routinely in large amounts for hydrogen-fueled rockets). Considerable forethought and planning were required to avoid accidents or explosions while moving or handling liquid hydrogen and deuterium.

Specifically, TU 3 was also to:

1. Establish the forward area facilities required by the Cambridge Corporation to move and operate dewars and refrigerators.

⁵⁸Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. 8.

2. Make necessary arrangements to move all equipment, both inside the U.S. and between the U.S. and Eniwetok atoll.
3. Work with TU 2 to develop and furnish both equipment and safe procedures to transfer liquefied gases and test dewars.
4. Deliver filled dewars to the Mike shot cab at the request of TU 4.⁵⁹

Task Unit 4 (TU 4), Mike Assembly, was to prepare the Mike device (the "SAUSAGE") for shipment and to reassemble it at Elugelab. Task Unit 4 comprised several groups and contractors, including personnel from American Car & Foundry (ACF, which had fabricated many of the Mike device's components), and CMR, GMX, and W Divisions of the Los Alamos Scientific Laboratory.

Each of these groups was responsible for various specific assembly operations. For example, ACF was responsible for the mechanical assembly of the SAUSAGE; CMR and GMX were responsible for the cryogenics system and the deuterium fusion fuel; and W-1 was responsible for overall device design and all nuclear components, including the fission trigger.⁶⁰

Specifically, TU 4 was to:

1. Coordinate Mike device component assembly activities at all points, for all groups and contractors concerned.
2. Arrange for shipment of Mike device components by ACF and both GMX and CMR divisions of LASL. In coordination with LASL's W-Division, TU 4 was to arrange for the shipment of all active (fissionable) materials.
3. Coordinate the storage of all units and

⁵⁹Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, pp. 8, 9.

⁶⁰WT-608 (EX), p. 57; AEC Memorandum ADWD-250 from Edward Teller to Norris Bradbury, Los Alamos Scientific Laboratory, March 24, 1951, Subject: Plan for Setting up a Separate Thermonuclear Division.

equipment aboard the USS CURTISS.

4. Provide jettisoning gear and instructions for its use, in the event that components of the Mike device had to be thrown overboard from the CURTISS in event of emergency, disaster, or hostile action against the ship to prevent their loss to, or recovery or capture, by enemy personnel.
5. Effect final assembly of the complete Mike unit.⁶¹

TU 5, King Assembly, was to assemble and deliver the King weapon and a special MK 18 test unit to the strike aircraft. A test drop of a dummy MK 18 loaded with high explosives, barometric and radar fuses, and a functioning nuclear safing system (but no nuclear material) would be conducted about a week before the detonation of the oralloy-loaded MK 18, to test the firing system and the unique safing system.⁶² The components for the dummy and live bombs were procured through the Sandia Corporation Production Department in Albuquerque, New Mexico.

Specifically, TU 5 was to:

1. Prepare, and arrange for the shipment of, both the King and King X-Ray assembly components and all accessory

⁶¹Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. 9.

⁶²According to Mark Bernard Schneider in NUCLEAR WEAPONS AND AMERICAN STRATEGY, 1945-1953, "The MK 18 contained a relatively unreliable mechanical safety mechanism which utilized aluminum chains as neutron absorbers" (p. 191). The MK 18 contained so much oralloy that it was always very near to "going critical" on its own, much like a nuclear reactor with its control rods pulled out. The MK 18 detonated during King shot yielded 500 KT; at 17 kt per kilogram (2.2 lb.) of oralloy, then the equivalent of 29.41 kg (64.7 lb.) of oralloy was expended. Assuming an efficiency of 50%, then the MK 18 pit contained about 129 lbs. of oralloy. At 30% efficiency, this figure rises to nearly 216 lbs. According to Ted Taylor in THE CURVE OF BINDING ENERGY (p. 90), the "Super Oralloy Bomb" contained "A lot" of fissionable material. The aluminum safing chains were pulled out of the weapon pit before the core was inserted in the automatic in-flight insertion mechanism. (See MK 18 history in Volume VI.)

equipment.

2. Coordinate the shipboard storage of all units and equipment.
3. Working with representatives of LASL's W-Division, effect final assembly of the King and King X-Ray units.
4. Deliver assembled devices and their associated accessory equipment to the bomb drop aircraft, and ensure that the aircraft crew and weaponeers have been properly instructed.⁶³

Original plans called for shipment of the King weapon, a dummy test bomb with no nuclear material, all spare parts, and the necessary handling equipment aboard the USS CURTISS. However, by April 1952 it became apparent that the LASL time schedule did not permit completion of the design, procurement of special materials, tooling, machining, and production of the MK 18 nuclear assembly before the CURTISS sailed. New plans were made to ship the King weapon by air to allow LASL more latitude in the time schedule; the dummy bomb could still go by sea.

The arming and firing of the Mike device were the responsibilities of the Firing Party, TU 6. This task unit was divided into an arming team and a firing team, which were functionally separate organizations with different missions. The arming team was to arm the Mike device, which included making all final connections, tests, adjustments, and calibrations necessary to ensure a successful detonation.

The firing team was to operate the firing control room aboard the USS ESTES (AGC-12) and to detonate the device at H-hour. The Firing Party set up specific operations according to orders and check lists, and also established procedures for emergencies such as a misfire, low-order nuclear detonation (with dispersal of plutonium and other contaminants), delays, or postponements. A similar firing team was used for King shot but only activated timing signals and maintained a voice time-and-countdown broadcast.

For the Mike shot, the master timing equipment was located on Elugelab island near the zero cab. Radio signals from the control room aboard the ESTES were used to start timing signals for experimental equipment and the sequence timer

⁶³Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. 9.

used to fire the device. This same radio system could be used to halt the shot at any time before zero hour if critical experiments or equipment failed to start when scheduled.

Two independent television channels between the shot island and the ship were used for telemetering. Two TV cameras were focused on identical indicator panels, which displayed information required by the firing party commander to determine whether or not the detonation should take place, including cryogenic monitoring systems showing liquid hydrogen pressure and flow rate and deuterium temperature and vacuum level in the secondary; and the statuses of timing, power-on, and arming and firing circuits.⁶⁴

The Mike Device

To simplify the test, and to make analyses of the results easier, the SAUSAGE was fueled with liquid deuterium, rather than the more-practical lithium-6 deuteride used in later thermonuclear weapon tests (for a while, liquid-fueled bombs, with all their encumbering cryogenic support equipment, were considered by the Air Force as "emergency capability" weapons, to be rushed into production and deployed as quickly as possible).

Very little was known about the behavior of lithium hydrides in thermonuclear environments, and the many complex reactions among lithium isotopes (see "Fusion Physics" under "Weapons Physics" in Volume I) would only further complicate what was already a very complex and theoretical experiment. The use of liquid deuterium would also produce a much higher yield, on a pound-for-pound basis, than lithium-6 deuteride.⁶⁵ Fissile

⁶⁴"Concept of TG 132.1 Forward Area Operations," undated, p. 8; Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. A-7.

⁶⁵The Mike device contained several dozens of liters of liquid deuterium; however, fusion efficiency was probably not much better than 40%, and eight megatons (nearly 80%) of Mike's total yield came from fission of plutonium and uranium-235.

If only 2.4 MT of Mike's 10.4 MT yield came from D-D fusion, it would have been equivalent to the complete "burning" of about 61.5 liters of liquid deuterium. Assuming that D-D fusion efficiency was 40%, then Mike's total deuterium load was around 154 liters. If the fusion efficiency was 50%, then the SAUSAGE may have contained as little as 123 liters of deuterium. These estimates of fusion efficiency are probably high: a problem with Mike was the "short lifetime" of liquid deuterium: it was quickly converted to gas by the heat of the

plutonium-239 was incorporated into the "sparkplug" of the fusion stage in Mike.⁶⁶

It took much more than just the Teller-Ulam concepts to build the SAUSAGE. Calculations made in connection with the design of Mike were started in June 1951 and completed a year later.⁶⁷ There were four major problems to be solved: whether or not a fusion reaction, once ignited in a compressed mass of deuterium, would propagate; whether or not radiation implosion of the secondary was feasible; how big a primary yield was required to adequately compress the secondary and "sparkplug;" and whether or not the fusion reaction could be inertially-contained long enough to generate an appreciable yield.⁶⁸

By the summer of 1951, results from machine calculations showed that the answers to the first two questions were probably "yes" and that the combination of radiation implosion and thermonuclear reaction would work.

The problem of the size of the primary turned out to be much simpler than had been anticipated. Because LASL had learned so much about fission weapon design and behavior between 1946 and 1951, the primary yield required to detonate the Mike secondary was within the yield range of fissile cores compatible with the MK 5 HE assembly, i.e., less than 100 KT. Results of GREENHOUSE studies on the efficiency of the TX-5

fissioning "sparkplug". TU 2 generated a total of 1,000 liters of liquid deuterium during the late summer and early fall of 1952 at Eniwetok. (Moore and Bechanan, p. 261.)

⁶⁶Morland, THE SECRET THAT EXPLODED, p. 123; J. Carson Mark interview, LOS ALAMOS SCIENCE, Volume 4 No. 7, Winter/Spring 1983, p. 37 (Mark refers to "plutonium behavior at liquid hydrogen temperatures"). An internal AEC memorandum dated October 30, 1952, from AEC Chairman Gordon Dean to James S. Lay, Executive Secretary, National Security Council, about approval for Operation IVY, states that the Mike device contained plutonium and 93.5%-enriched oralloy, most of which was probably in the TX-5 primary stage core. OPERATION IVY, Mike Shot Cursory Reports on Experimental Programs, William E. Ogle and John H. Lofland, Jr., 8 November 1952, Eniwetok, Marshall Islands, lists under illustrations for Program 2, Progress of the Nuclear Reaction, "Scope Trace: Spark Plug Gamma Channel" and "Spark Plug Neutron Channel."

⁶⁷Bethe, "Comments on the History Of the H-Bomb", p. 49; Mark, LA-5647-MS, p. 12.

⁶⁸Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM.

were available by the fall of 1951.⁶⁹

The question of sufficient confinement time, initially considered relatively easy to answer, turned out to be more difficult than expected. Major difficulties occurred in the theoretical design of Mike in March 1952; they were solved by LASL scientists three months later just in time to meet the scheduled November test date.

The Panda Committee decided very early that the Buffalo, New York facilities of ACF would be used to fabricate the major portions of the thermonuclear test device and that shakedown of the final assembly procedure would also be conducted in Buffalo.

At the outset of planning, the Panda Committee considered two basic designs which could be used in the construction of the Mike device. Dual programs of design and partial component manufacture were conducted until mid-January 1952, when the more feasible design of the two as chosen for testing during IVY.

Typical of the design problems which had to be addressed was the question of whether deuterium would be used in a liquid or solid state in the device. While a cryogenic, liquid-fueled design was favored for its ease of analysis of results (as opposed to its more difficult engineering, construction, and maintenance aspects), the possibility of using a "dry" form of deuterium, chemically compounded with lithium-6, was also studied. Theoretical considerations, and the lack of lithium isotope separation plants, made the liquid-fueled device the final choice.⁷⁰

Another engineering decision involved the internal shape and composition of the radiation channel, i.e., the "duct" through which x-rays would propagate to compress the complex secondary. After considerable study, LASL engineers decided to line the channel with lead and plastic, the lead to reduce the likelihood of a pressure pulse from the vaporizing steel

⁶⁹Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM. OPERATION IVY, Mike Shot Cursory Reports on Experimental Programs, William E. Ogle and John H. Lofland, Jr., 8 November 1952, Eniwetok, Marshall Islands, lists under illustrations for Program 2, Progress of the Nuclear Reaction, "Scope Trace: TX-5 Alpha."

⁷⁰For just over two weeks in January 1952, LASL briefly abandoned plans to use liquid hydrogen and liquid deuterium in the IVY tests; orders were given to disband TU 2. These considerations were soon overturned. (Moore and Bechanan, p. 261.)

outer casing that might damage the secondary, and the plastic to act as a plasma generator to implode the secondary and to also further smooth out any pressure pulse. The channel itself was finally cylindrical, straight-walled from one end to the other.⁷¹

The lead and plastic were vital in maintaining the integrity of the secondary while it was being compressed. A high atomic number casing liner would ablate non-uniformly and create pressure "spikes" that might chop up the secondary. A low atomic number liner would have a slower and more uniform ablation rate:

You cover the steel with lead. That would make it more opaque, so it couldn't give a pressure pulse that quickly. That at least would keep you from getting down to the steel. But even the lead, the surface of the lead, would blow off. So then we covered the lead with plastic -- with polyethylene. That was low (atomic number), just (carbon and hydrogen). From a time point of view, the radiation ionization of the heavy materials with all that shielding would be so late that it would (no) effect on the overall system.⁷²

The use of lead and plastic linings inside thermonuclear device casings was also carried over to the first solid-fueled, stockpiled U.S. "emergency capability" hydrogen bombs, namely, the TX-14, TX-16, and TX-17.⁷³

By January 1952, ACF was proceeding with the design of the on-site structure, the cab, to house the device at Eniwetok, and was beginning the casting and machining of the outer steel case. At the same time, modifications were begun on a building at Buffalo in which the device was to be assembled. By June, the device was nearing completion and final assembly procedures were being developed.⁷⁴

By the middle of 1952, roughly 75% of all the development

⁷¹Rhodes, DARK SUN, pp. 485-487, 492.

⁷²Rhodes, DARK SUN, p. 492.

⁷³Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1953 - 30 June 1953, p. 188.

⁷⁴Moore and Bechanan, pp. 259, 260.

work of the Los Alamos lab was on thermonuclear weapons and the preparation of the Mike device in particular. In June, the AEC approved a request by lab director Norris Bradbury to recruit new technical and scientific personnel to bring LASL's total of such persons to 3,050.⁷⁵ The new MANIAC computer and its sister machines around the country, which had only recently gone into operation, were indispensable in designing the Mike device.

In addition, the University of California Radiation Laboratory in Berkeley, California was requested by the AEC to design and build much of the instrumentation for thermonuclear device tests and to participate in component or phase testing of thermonuclear weapons.⁷⁶

Engineering of the SAUSAGE for test began in October 1951, before theoretical design was completed. This situation led to many changes in engineering design as the theoretical design evolved, putting quite a strain on the engineers at LASL. Preparation of experiments to observe and record Mike's test behavior was an equally difficult problem, since such an instantaneous explosion of the predicted magnitude had never occurred before.

As far as future weaponization was concerned, LASL realized immediately that Mike's bulk of over 80 tons was totally impractical for a usable weapon. Reduction in weight to about 20 tons, and to a diameter of 65 inches, appeared possible by reducing the volume of the thermonuclear reaction vessel (the secondary) with a corresponding reduction in yield; by using Li^6D instead of liquid deuterium for fuel,⁷⁷ and by engineering weapons with lighter casings (the Mike casing was made quite thick to provide sufficient confinement time and to raise yield);⁷⁸

⁷⁵Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, pp. 9, 10; Minutes of AEC Meeting No. 708, June 6, 1952.

⁷⁶Letter dated June 9, 1952 to Honorable William C. Foster, Deputy Secretary of Defense, The Pentagon, from Gordon Dean, Chairman, USAEC.

⁷⁷Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, pp. 11, 12.

⁷⁸Edward Teller allegedly told Luis Alvarez that Washington had been threatening to cut off funding for the H-bomb project if the Eniwetok test was unsuccessful; because of this, every conceivable method had been incorporated into the SAUSAGE design to make it more powerful. ("Mad About the Bomb," Gregg Herken, HARPER'S, December 1983, p. 52.)

The basic design of the thermonuclear test device was frozen by the Panda Committee on January 18.⁷⁹ Component testing of radiation implosion began at Los Alamos in April 1952.⁸⁰ Mike's theoretical design was completed by June, and on the morning of June 30, AEC Chairman Gordon E. Dean, Los Alamos Laboratory Director Norris E. Bradbury, and AEC Division of Military Application Director Colonel Kenneth E. Fields met with President Truman in the oval office of the White House in Washington, D.C. to explain how Mike would work, using a model of the device to illustrate their description.⁸¹

Minor changes in the mechanical design of the Mike device continued right up to the time of final assembly at Eniwetok; as late as the end of October, there was a last-minute change in the amount of plutonium and enriched uranium in the primary in order to lower the likelihood of failure from one specific cause.⁸²

The SAUSAGE was essentially a large thermos bottle, standing just over 20 feet high and measuring 80 inches in diameter.⁸³ The device, with its attached diagnostic tubes (located at the ends of each of its stages) and full load of liquid deuterium, weighed 82 tons.⁸⁴ The outer casing was constructed from a number of welded steel sections, fabricated by the American Car & Foundry Company in its factory at Buffalo, New

⁷⁹Rhodes, DARK SUN, p. 487.

⁸⁰Teller, COMMENTS ON BETHE'S HISTORY OF THE THERMONUCLEAR PROGRAM, p. 3.

⁸¹Hewlett and Duncan, p. 590; Shelton, p. 5-50.

⁸²Memorandum dated October 30, 1952 to Mr. James S. Lay, Executive Secretary, National Security Council, from Gordon Dean, USAEC Chairman. According to Norris Bradbury, "in getting ready for the Mike shot, we actually had to fly out an important modified piece a few days before the shot was going to take place." Bradbury stated that something like this was also done just before the GREENHOUSE George test. (Transcript of Press Conference with Norris Bradbury at Los Alamos, New Mexico, September 24, 1954, p. 15.)

⁸³One source states that the SAUSAGE was 244" long; the Moore and Bechanan study (p. 260) says the SAUSAGE measured 243.625" overall.

⁸⁴Most of Mike's weight was in the device casing. A steel cylinder with an outside diameter of 80 inches, foot-thick walls, a length of 20 feet, and made of steel with a density of 4.5 ounces per cubic inch, would weigh over 86 tons.

York.

ACF produced many of the exterior and ballistic casings of nuclear test devices used in later Pacific Proving Ground detonations; late in 1952, ACF established a production facility in Albuquerque, New Mexico to build thermonuclear bomb casings.⁸⁵ The company had manufactured several large "blockbuster" conventional bomb casings for the U.S. Air Force after World War II, and it was the only place in the U.S. that could machine pieces of metal in sizes required for Mike's casing.⁸⁶ ACF had also produced many bomb casings for stockpiled nuclear weapons.

A large cryostat attached to Mike recirculated liquid hydrogen around the secondary stage. The SAUSAGE was only an experimental device -- it was far from being a deliverable military weapon.

Other odd pieces of the SAUSAGE were fabricated by various shops throughout the U.S.; as was done during World War II manufacture of the first atomic bombs, none of the subcontractors was given complete information about how the subassemblies were to be used. Complete coordination of Mike device assembly involved activities in Buffalo, New York; Boston, Massachusetts; Columbus, Ohio; Los Alamos, New Mexico; Los Angeles, California; Eniwetok atoll; and several other widely geographically-dispersed sites.⁸⁷

Figure IV-1 shows the SAUSAGE in its cab at Elugelab; Marshall Holloway is visible in the center of the photo.

At the time of the test, the SAUSAGE was the largest experimental cryogenic device ever constructed. It was an irony of its design that in order to generate the highest temperatures ever achieved on earth, the fuel in the device had to be cooled to one of the lowest temperatures on earth, around -423⁰ Fahrenheit. The liquid hydrogen for the SAUSAGE's cooling system was produced by the cryogenic facility erected at Eniwetok for the GREENHOUSE tests.

The device's fusion stage (the secondary)⁸⁸ contained liquid

⁸⁵Furman, p. 379.

⁸⁶LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, p. 37.

⁸⁷Moore and Bechanan, p. ????.

⁸⁸OPERATION IVY, REPORT OF COMMANDER, Task Group 132.1, states: "...to measure the time between the two fission phases of the device" (p. 11) and "...to allow rough measurement of the time delay between primary and secondary fission reactions" (p.

deuterium within an evacuated chamber, cooled by liquid hydrogen circulating around the chamber. The deuterium in the secondary surrounded the plutonium rod "sparkplug" assembly in the heart of the bomb.⁸⁹ The high-yield TX-5 fission trigger -- whose x-rays would compress the entire secondary thermonuclear fuel assembly -- was located in a separate space atop the secondary. The trigger was not refrigerated.

Even with its load of deuterium fuel, the SAUSAGE was still primarily a fission device:⁹⁰ surrounding the fusion fuel chamber was a very thick layer of uranium-238, tuballoy, which acted both as a "pusher" to compress the secondary, and also as a tamper for the initial thermonuclear explosion within the stage. The uranium also served as an additional fission fuel source when bombarded by neutrons of several energies emanating the fusion and fission reactions within the deuterium fuel.⁹¹ This mass of uranium also contributed

13). A press release dated 18 February 1986 from the Natural Resources Defense Council states (on p. 3) that the first two-stage thermonuclear (Superbomb) test was conducted by the U.S. on 31 October 1952. These statements tend to indicate that Mike was a two-stage device. The unboosted TX-5 primary stage yield was probably between 40 and 70 kilotons.

⁸⁹Letter dated July 23, 1952 from Duncan Curry, Jr., TG 132.1, Subject: Scientific Experimental Programs, Operation IVY (revised). Project 2.6 cites the "temperature of the burn in the region of the spark plug."

⁹⁰"Although MIKE was a thermonuclear, or fusion, device, a significant portion of its energy release resulted from fission processes." (DNA 6036F, p. 25.)

⁹¹Memorandum dated 10 July 1952 for A. C. Graves from W. E. Ogle, Subject: Recent Changes in the IVY Program; "Multiple Neutron Capture in the Mike Fusion Explosion," A. G. W. Cameron, Canadian Journal of Physics, Volume 37, 1959, The National Research Council, Ottawa, Canada, p. 323. The secondary stage in the Mike device did not contain any tritium: tritium that was consumed in it was generated mainly by the fusion of deuterium with deuterium, (see Equation H in "Fusion Physics" under "Weapons Physics" in Volume I) and by the breeding of tritium by collisions of neutrons with deuterium (see Equation M in "Fusion Physics" in Chapter 1). The "sparkplug" neutrons generated more tritium by irradiating the deuterium (which substituted for lithium in the "dry" weapons; see SOURCES OF TRITIUM, D. G. Jacobs, U. S. Atomic Energy Commission, Division of Technical Information, Oak Ridge, Tennessee, 1968, p. 14, and Equation M in "Fusion Physics" under "Weapons Physics" in Volume I). A late 1952 AEC report stated that "the detonation of the SAUSAGE has established the feasibility of achieving a

significantly to the weight of the Mike device.

The SAUSAGE was a very complex device, consisting of a multi-vessel secondary and a tritium-boosted sparkplug:

The thermonuclear secondary, which would be essentially a bottle of liquid deuterium with a "stick" of plutonium mounted inside for a sparkplug, would hang in the center of the big steel casing below the fission primary, behind a heavy blast shield.

The cryogenic system for the SAUSAGE was similar to the system in the storage dewars but simpler. Liquid hydrogen boils at 20° K, liquid deuterium at 23.5° K, which means that liquid hydrogen stores a few degrees colder than liquid deuterium. Taking advantage of that difference, Panda cryogenists designed the double-walled stainless-steel inner dewar in the SAUSAGE to connect through pipes in its upper end to a reflux condenser set in a big tank of liquid hydrogen.

Vaporized deuterium would flow by convection through a pipe to the reflux condenser, where it would cool to a liquid again and flow thorough another pipe back into the inner dewar. The inner dewar, suspended within the big outer casing, was a cylindrical double-walled stainless steel tank with rounded ends with the sparkplug mounted inside it.

thermonuclear reaction in liquid deuterium on a large scale and of using the resultant neutrons to cause extensive fission of natural uranium." (Atomic Energy Commission, Progress Report to the Joint Committee, Part III - Weapons, June 1952 through November 1952, AEC 129/46, December 17, 1952.) As noted earlier, "Chronology of Major Decisions Regarding Thermonuclear Program," USAEC, June 14, 1951, p. 9, states that LASL was researching by June 1951 a thermonuclear device that used a "large mass" of U-235.

The Teller-Ulam principles -- separate stages, radiation coupling, and implosive compression of the fusion fuel capsule before its ignition -- can be applied to both liquid-fueled and solid-fueled thermonuclear weapons. The only significant difference between these two types of devices is the source of tritium produced by neutrons from the fissioning "sparkplug:" deuterium in liquid-fueled weapons and lithium in solid-fueled weapons.

A second evacuated assembly -- the outer dewar -- surrounded the inner dewar. Between the two dewars was located a "floating" copper thermal radiation shield cooled in turn with liquid nitrogen to minimize heat flow between adjacent surfaces.

This shield in turn was surrounded by the inner face of the uranium tamper, which was covered with gold leaf to further reduce heat flow.

The plutonium sparkplug was boosted with gaseous tritium, just a few grams.⁹²

The main nuclear reactions expected to occur were the union of deuterium atoms to form either helium-3 and a 3.265 MeV neutron, or a proton and tritium. Tritium might be formed from either the union of deuterium atoms, or from the union of a neutron and helium-3, if the latter were formed in sufficient concentration. In either case, the tritium would fuse quickly with deuterium, yielding helium-4 and a 14 MeV neutron.

All of the high-energy 14 MeV neutrons would cause fission in the U-235 fusion tamper wrapped around the secondary.⁹³ The ratio of deuterium atoms "burned" to 14 MeV neutrons spawned by the "burning" was expected to vary from 5:1 to 3:1.⁹⁴

The SAUSAGE was erected in a large, hangar-like corrugated aluminum-covered shelter 88 feet wide, 46 feet deep, and 61

⁹²Rhodes, DARK SUN, pp. 486, 490-494. This excerpt is a condensation of a much-longer, more-complete description of the cryogenic aspects of the Mike device presented in the Rhodes book.

⁹³Shelton, p. 6-34. Shelton quotes Alvin Graves of LASL as saying about the IVY Mike test "we used some of those left-over neutrons that were produced to also fission some uranium." Shelton also quotes Graves as saying the fission-to-fusion ratio on the shot was about 50:50, when it was in fact nearly 80% fission.

⁹⁴Operation IVY Pacific Proving Grounds November 1952, Report of the Scientific Director, Measurement of Reaction History, Volume II - Experiments, Results, and Analysis, Mike Shot, Part I, WT-621, E. H. Krause and Staff, Radiation Division, Naval Research Laboratory, Washington, D.C., June 1953, pp. 12, 13.

feet high, on the islet of Elugelab at Eniwetok atoll in the AEC's Pacific Proving Grounds. Elugelab was raised in elevation and built up with sand and coral bulldozed onto it.⁹⁵.

Other structures surrounding the shot cab included an auxiliary building for subassembly and repair work, a tritium storage hut for the "sparkplug" tritium, a telemetry building with special generators and a dehumidified "dry" room, a central dehumidified timing station with firing center and television transmitter, and a 375 foot high radio and television control signal tower and broadcast antenna. **Figure IV-2** shows the IVY Mike shot cab, with its associated radio and TV tower.

A semicircular array of seven enclosed mirrors was positioned from 106 to 201 feet away from the shot cab to reflect early bomb light to "streak" cameras for Project 3.3, which would measure internal temperatures before the Mike device vaporized.⁹⁶ Like the face of a gorgon, the thermonuclear explosion was too dreadful to be viewed directly by the cameras.

Another item of equipment on Elugelab was a 20 ton liquid gas dewar, approximately three-quarters filled with liquid hydrogen, which was to be expended in the test. This dewar was part of the reflux system that circulated liquid hydrogen around the secondary, and it held a ten-hour supply of hydrogen for the device in case of postponements of the test.⁹⁷

The islet was surrounded by diagnostic and test instruments and experiments radiating from the Mike shelter. Most notable of these was an 8-foot square, 8,887-foot long aluminum-sheathed plywood shed or "flume" running from Elugelab across Teiteripucchi island to the adjacent islet of Bogon. The shed enclosed a number of helium-filled plastic ballonets, which

⁹⁵The height-of-burst for Mike was 35'. (OPERATION HARDTACK, April-October 1958, Project 1.4, Physical Characteristics of Craters from Near-Surface Nuclear Detonations, WT-1609 (EX), Headquarters Field Command, Defense Atomic Support Agency, Sandia Base, Albuquerque, New Mexico, May 10, 1960, p. 13.)

⁹⁶List of IVY Test Structures, undated.

⁹⁷Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, pp. A-3, A-6, A-8; "Notes on Pogo Staff Meeting, 29 December 1951, Blast Measurements for IVY and SNAPPER," p. 2. The Moore and Bechanan study (pp. 250, 317) indicates that this dewar was one of the Camco RTDs.

minimized atmospheric absorption and directed gamma rays and neutrons from the exploding test device into various detectors and counters, whose data would be telemetered to other stations. To furnish collimation, lead baffles with small holes in them were interspersed along the helium channel.⁹⁸

The experimental tube -- called the Krause-Ogle box, after its inventors -- linked to the Mike shelter looked like a long train pulling into a station. At Elugelab, the box ended in a set of large and thick concrete shields which channeled radiation into the box.

On Bogon island, the other end of the box ran into a thick sand-and-gravel covered reinforced concrete bunker, containing telemetry equipment to transmit neutron and gamma ray data to a survivable bunker further away from surface zero.

More than 20,000 non-expendable helium bottles (tanks) weighing 200 lbs. each were used to supply gas to fill the ballonets in the box. The tunnel was so long that the Bogon end had to be raised slightly to compensate for the earth's curvature between ends (the earth curves about 6.85 inches per mile).⁹⁹

Diagnostic data would be telemetered to remote recording sites before the box was destroyed by the explosion. The Krauss-Ogle apparatus was to be used for Program 2, which would measure progress of the nuclear reactions in the Mike device.

Figure IV-3 shows the Krause-Ogle box as viewed from Bogon island. **Figure IV-4** shows Elugelab island with the IVY Mike shot cab, mirror array, and Krause-Ogle box running to Bogon. **Figure IV-5** shows the Bogon island end of the diagnostic "flume."

⁹⁸Operation IVY Pacific Proving Grounds November 1952, Report to the Scientific Director, Measurement of Reaction History, Volume II - Experiments, Results, and Analysis, Mike Shot, Part I, WT-621, E. H. Krause and Staff, Radiation Division, Naval Research Laboratory, Washington, D.C., June 1953, p. 14.

⁹⁹Moore and Bechanan, p. 131; OPERATION REDWING, Report of the General Manager Albuquerque Operations, James E. Reeves and K. F. Hertford, USAEC, October 26, 1956, p. 29.

Mike Shot Experimental Programs

A number of other vital diagnostic experiments were carried out during the Mike shot. Two of the most important were Projects 3.3, Hot-Spot Observation, and 3.4, Bomb Case Motion. These experiments were vital to the measurement of Mike's thermonuclear behavior.

Project 3.3 was one of several observations designed to collect information on the sequence of events following the fission phase. Laboratory design of a fusion bomb was strongly dependent upon the extraction of as much data as possible from each field experiment. **Figure IV-6** shows the Project 3.3 experiment schematically.

Since field tests cost millions of dollars, expended many precious kilograms of rare fissile and fusionable materials which had been fabricated and assembled at great expense, and involved several thousand participants, some of them highly-paid scientists and technicians whose time was always at a premium, there was usually a strong imperative to gather as much accurate diagnostic data as possible, in the form of photographs, oscilloscope tracings, seismic readings, radiochemical samples, and activated "tracer" and diagnostic samples.

In some cases, completely independent experiments were designed to observe the same phenomenon, either directly or indirectly, thus increasing the probability of obtaining useful results.

One interesting piece of diagnostic information was the temperature distribution within the Mike device, before its disassembly, as a function of time. The relative efficiency of a fusion reaction is closely associated with its temperature, and several experiments were set up to determine temperatures. Two of these were Projects 2.6 and 3.3.

Project 2.6 was a "time of flight" determination of the exploding device's neutron energy spectrum, which could be interpreted in terms of temperature within the SAUSAGE as it detonated. This was also known as the TENEX, or temperature measurement by neutron energy spectrum experiment.¹⁰⁰

Project 3.3 was designed to measure interior temperature indirectly by determining the time it took for a shock wave to pass through a 1" thick piece of steel. Since the time was directly proportional to the speed of the wave, and the speed

¹⁰⁰Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. D-10.

of sound in steel was known, then the "strength" of the explosive wave could be deduced from hydrodynamic theory. The "strength" of the wave was in turn directly related to the temperature behind the shock wave at that point.¹⁰¹

Three selected areas of the Mike device case were observed by seven cameras; at the top, middle, and bottom of the case, two surfaces were photographed: the outer surface of the case itself and the outer surface of a 1" thick steel plate attached to the case. The time difference between shock arrival at these two surfaces would give the traverse time; shock arrival at each surface would show itself as a sharp rise in light intensity. In the middle of the case, an additional plate was attached and filmed.

Seven "streak" cameras used fixed film trips, which recorded a moving image generated by a mirror rotating at extremely high speed. Each image was accurately timed to a resolution of three "shakes" (0.03 microseconds). The cameras were operated from a fallout-proofed shelter on Bogallua island.

The external areas of the Mike device case selected for observation were near the top, just above the middle, and near the bottom. Seven "hot spot" collimating pipes, helium-filled 35' long tubes, were welded to the Mike device case at these points.¹⁰² The light from these areas was directed through the tubes, out openings in the shot cab enclosure to an arc of mirrors angled toward the lagoon on the Bogallua side of surface zero, and then reflected to the cameras. **Figure IV-7** shows the "hot spot" diagnostic pipes being welded to the Mike device casing.

The pipes served to block all light emanating from other parts of the device casing, and the images photographed by the "streak" cameras were the light shining on the mirrors, and not the explosion of the bomb casing. **Figure IV-8** shows the diagnostic light pipes emanating from the Mike device casing; the pipes were aligned by a latticework of scaffolding supports.

Project 3.4, Bomb Case Motion, photographed the actual disintegration of the Mike casing as the shock waves from the fission and fusion explosions broke through it. The casing would be vaporized before any pieces of it could be physically moved any significant distance.

¹⁰¹Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. D-13.

¹⁰²Memorandum dated 10 July 1952 for A. C. Graves from W. E. Ogle, Subject: Recent Changes in the IVY Program.

A portion of the energy of the exploding primary would be seen as light emitted from the device casing. Photography of the progress of this light would provide information about the nature and behavior of the explosion from its initiation through the sequential steps of case expansion, thermonuclear reaction, and case disassembly. Knowledge of these events and their relative timing would be invaluable to the scientists concerned with the design and theoretical problems involved in thermonuclear devices.

The bomb case was observed directly with two groups of three Edgerton, Germeshausen and Grier (EG&G)-designed cameras on Bogallua island. One group, photographing case disassembly, operated at a speed of 3,500,000 frames per second; the second group took pictures of early fireball growth, at a slower rate of 90,000 frames per second. The case disassembly cameras were expected to take only eight useful frames each in a time of just over two microseconds, so fast would be the vaporization of the device.¹⁰³

These high speeds were necessary to build a photographic record of how the bomb case disintegrated and how the fireball grew. A detailed knowledge of the former was invaluable to those LASL scientists interested in internal temperatures, pressures, reaction rates, etc. Fireball growth rate was an indicator of many performance parameters, including yield.

The early formation of "jets" or "spikes" in the fireball was also of great interest, as they were indicative of non-uniform expansion or compression and might explain the existence of temporary asymmetries in the shock wave.¹⁰⁴

As with the "streak" cameras of Project 3.3, these very-high speed cameras used a fixed-film, rotating mirror assembly with electronic timing devices to close their shutters after the image had been "swept" once, to prevent multiple exposures.¹⁰⁵ Each of the reflected images was focussed onto a section of film strip. The shutter on each of the faster cameras closed after 30 microseconds, when about a hundred 35mm images had been obtained. Since the angular velocity of the mirror face was known, and the distance between lenses was known, then the time between individual frames could be established, thus

¹⁰³Moore and Bechanan, p. 55.

¹⁰⁴FINAL REPORT, JOINT TASK FORCE 132, OPERATION "IVY" 1952, p. C-5.

¹⁰⁵THE TURQUOISE BOOK, OPERATION IVY, pp. 3.3.1-3.4.3.

giving the interval between observed events.

These ultra-high speed cameras were called "frame" cameras and they used spinning mirrors made of beryllium. The mirror assembly, driven by compressed helium, was located within an evacuated chamber to eliminate air friction. The mirror was brought up to speed and maintained at a constant rotation rate before the explosion. Using this technique, not only could a continuous record of time-versus-case movement be recorded, but the time scale would be uniform for the duration of the filming.¹⁰⁶

A similar photographic technique was used to determine the time interval between fission and fusion reactions by measuring the time between appearances of a phenomenon called "Teller light," a very short-duration light produced by the interaction of gamma rays and neutrons with the still air near the test device.¹⁰⁷

Teller light was named after Dr. Edward Teller, who first suggested this experiment; it was first used during Operation SANDSTONE in 1948. The ionization -- and intensity of -- glowing light from the bomb casing was directly proportional to gamma ray intensity, which in turn was proportional to the neutron multiplication rate. "Teller light" was also used to measure the alpha of exploding fission devices.

Another type of camera used during IVY and other U.S. nuclear tests was the so-called "Rapatronic" camera, which was basically a box-type camera equipped with long lenses and electrically-triggered shutters which operated by polarization by a pulsed magnetic field, allowing very fast exposures, on the order of a few microseconds, of the blast fireball which expanded at a rate of more than one hundred times the speed of sound.

Pretest Activities

Preparations for the Mike shot were extensive and meticulous. As was usually done during nuclear tests, activities such as assembly and detonation of the device, necessary evacuation of

¹⁰⁶SPLIT SECOND: THE WORLD OF HIGH SPEED PHOTOGRAPHY, Stephen Dalton, Salem House, Salem, New Hampshire, 1983, p. 18.

¹⁰⁷THE ONO BOOK, OPERATION CASTLE, Appendix I, Annex C, Operation Plan 1-53, Task Group 7.1, Joint Task Force Seven, Headquarters, Task Group 7.1, Los Alamos Scientific Laboratory, 1 October 1953, pp. 65, 68. This document is the CASTLE version of the Turquoise Book for Operation IVY; it lists and schematically illustrates all of the proposed scientific experiments.

personnel, airborne photography and sampling flights, re-entry to the shot area, and recovery of radiological and radiochemical samples and data records were rehearsed at length. Preparations included extensive practices in the continental U.S. that centered on Mike device assembly and air operations.

The first portion of major construction for Operation IVY at Eniwetok was the cryogenics area facility started in late 1951. More than \$10 million was spent for all new construction required at Eniwetok for IVY. Of this amount, approximately \$7 million was for scientific test or test-supporting construction, and the remainder was for base camp facilities.

Scientific test construction included stations to house instruments, recorders, cameras, etc. Test-supporting construction included a 9,000-foot causeway between Elugelab (the shot island) and Bogon, a deepwater pier and ramp, and an airstrip. Base facilities included a 3,000 kilowatt power plant, machine shop, and various buildings for the cryogenics plant.

More than 500 scientific stations were constructed on 30 separate islands in addition to reef locations on Eniwetok atoll. These stations ranged from the simple, such as those on small rafts, to the complex, such as Station 1, the Mike shelter, also known as the shot cab or zero structure. By the end of September 1952, approximately 8,600 cubic yards of concrete had been poured in construction of all facilities. Two temporary tent camps of 250- and 500-man capacity, with roads and utilities, were erected to house scientific and construction personnel.

Fabrication of Cryogenic Equipment

The fabrication of components for the new cryogenics plant began in November 1951 at Columbus, Ohio. A training program for operators started at the Ohio State University Cryogenics Laboratory in Columbus, followed by visits to Air Products, Inc. in Allentown, Pennsylvania, and then to the newly-completed cryogenics laboratory at the National Bureau of Standards Laboratory in Boulder, Colorado, for observation of installations and tests of equipment similar to that planned for the Eniwetok facility.¹⁰⁸

¹⁰⁸Construction of the Boulder facility was announced by the AEC at the end of March 1951; its stated purpose was to design and develop large containers for liquid deuterium. (USAEC Santa Fe Operations Office Press Release dated March 29, 1951; Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 21.)

The first group of TU 2 personnel left Columbus, Ohio, on January 17, 1952 to go to Eniwetok to begin the repair and installation of equipment for the old liquefaction plant in Building 330 on Parry island.

The Cambridge Corporation (Camco, a subsidiary of the Arthur D. Little Company) of Somerville, Massachusetts designed special hydrogen transport dewars during early 1952. Camco used a building at the NBS Boulder laboratory to produce, assemble, and test refrigerated transport dewars and cryogenic parts of thermonuclear weapons.¹⁰⁹

In the fall of 1951, Arthur D. Little had signed a contract with the AEC to design, manufacture, and test zero-loss liquid hydrogen transport vehicles. It quickly became apparent that manufacturing facilities in excess of those then available to Little would be required. Hence, in January 1952, A. D. Little and Carrier Corporation established a jointly-owned affiliate, the Cambridge Corporation, to fabricate this special cryogenic equipment.¹¹⁰

Before 1952, liquid hydrogen had never been successfully stored in large quantities, nor for long periods. Large dewars to hold hydrogen did not exist, either for static storage or for safe transportation. Very little was known about the hazards attendant to moving liquefied explosive gases.

The only work in this field up to now was the liquefaction of hydrogen on a laboratory scale, using heavy fixed equipment at high pressures. To make a mobile apparatus, the liquid would have to be handled at lower pressures with smaller and lighter containers, which would also have to be rugged enough for field use.

These containers had to be capable of operating fully loaded over the highways of the continental U.S. under long-haul conditions and, at the same time, capable of overseas operation, even while being loaded on and off naval landing craft. Loss of gases, especially of deuterium, had to be minimized or eliminated altogether.

As finally constructed, each huge trailer-mounted dewar, or vacuum bottle, held 2,000 liters (about 528 gallons) of

¹⁰⁹Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, p. 47.

¹¹⁰Undated statement from A. D. Little Corporation entitled "Refrigerated Transport Dewars."

liquid gas, weighed 40,000 lbs., and was 35 feet long by 8 feet wide by 12 feet high.

Each unit, called a Refrigerated Transport Dewar (RTD), or "transportainer," carried its own helium refrigerator and power supply (a diesel-driven motor generator). A high vacuum between the walls of the dewar vessel was maintained by a vacuum pump that was a part of the dewar-refrigerator combination. The helium refrigeration system absorbed heat generated inside the dewar by the spontaneous change in state of the liquid hydrogen from orthohydrogen to parahydrogen.

The refrigeration system used in each RTD was based on the Collins expansion cycle. The components designed and developed by Camco included a special refrigerator that used helium gas for a refrigerant; a stainless steel vacuum-insulated dewar with a nitrogen-cooled radiation shield underneath it; and vacuum-jacketed, liquid hydrogen transfer lines with elbows, disconnects, and valves.

The RTDs used suspension systems comprising numerous stainless steel cables arranged to withstand severe shock loads in any direction.

At its inception, the RTD was intended to store liquid normal hydrogen without loss, so the refrigerator was sized to absorb the heat, about four or five watts, produced by the spontaneous exothermic conversion of normal 75% orthohydrogen to parahydrogen. The dewar was sealed so well that when the refrigerator was turned off, the rate of evaporation of liquid parahydrogen was well under one percent -- or 20 liters, when completely filled -- per day.¹¹¹

The Cambridge Corporation designed and manufactured 15 RTDs; the first was delivered within five and a half months after receipt of the contract, and the last was completed just six and a half months later.¹¹² The complex and cumbersome RTD trailers, essentially flatbed platforms with a diesel-driven compressor and a dewar aboard, bore little esthetic resemblance to sleek modern liquid gas transfer vehicles.

These RTDs were fabricated in a Camco plant in Somerville, Massachusetts, a suburb of Boston. The transport dewars were assembled in this plant with certain portions of the refrigerator manufactured by Arthur D. Little, certain portions subcontracted to outside firms, and the remainder

¹¹¹CRYOGENIC ENGINEERING, Russell B. Scott, D. Van Nostrand Company, Inc., New York, 1959, pp. 225-227.

¹¹²Undated statement from A. D. Little Corporation entitled "Refrigerated Transport Dewars."

made in the Camco shops.¹¹³

The successful design, construction, and operation of dewars of this size, within the short time available, were substantial advances in the state-of-the-art of liquefied gas handling technology, and a major contribution to the success of the Mike test.

In February, the cryogenics plant design was approved and construction of the plant at Parry began in March. A shop building, approximately 100' long by 25' wide and equipped with a traveling overhead 20-ton capacity crane, was erected in the cryogenics area on Parry island at Eniwetok to perform the necessary maintenance on the dewar units. In addition, there were seven lean-to buildings, approximately 20' by 40' each, for storage of the dewars between filling and transport.

Between August 24, 1952 and October 29, 1952, these dewars transported 17,500 liters of liquid hydrogen, 1,000 liters of liquid deuterium, and 63,400 liters of liquid nitrogen between Parry island and Elugelab island at Eniwetok.¹¹⁴ The deuterium for the Mike device, as well as the nitrogen and hydrogen, were shipped from the U.S. to Eniwetok in gaseous form in large trailers carrying long pressurized tubes.¹¹⁵

The shot island camp on Elugelab was started and completed in late April. The island was about 1,650 feet long and less than 1,000 feet wide.

Operation TUMBLER-SNAPPER: Prelude to IVY

Operation TUMBLER-SNAPPER at the Nevada Test Site in the spring of 1952 provided valuable training for many of the personnel involved in Operation IVY. Many of the ideas, techniques and methods, and much of the test equipment tried at TUMBLER-SNAPPER determined the diagnostic experiments or techniques to be used at IVY.

For example, five F-84G airborne radiation sampling aircraft participated in TUMBLER-SNAPPER, and the crews gained valuable experience and training for their sampling missions during Operation IVY by actual penetration of heavy radiation fields in the atomic clouds to collect samples of bomb debris.

¹¹³Moore and Bechanan, p. 262.

¹¹⁴Moore and Bechanan, p. 261.

¹¹⁵Rhodes, DARK SUN, p. 490.

In addition, many of the experiments conducted during the SNAPPER phase of the operation were included as feasibility, technique, and theory tests for planned IVY measurements.¹¹⁶ TUMBLER-SNAPPER also included two tests of TX-5 weapons with TOM initiators, the same fission assembly that was to drive the Mike device secondary. (The TX-5 fired during Operation GREENHOUSE had used an URCHIN initiator to reduce its yield for the structures test program.)

One instrument developed before TUMBLER-SNAPPER and successfully tested during the shot series was an integrating radiation dose rate meter called an Integrator, invented at LASL.¹¹⁷ This device was intended to measure the exposure of aircraft and their crews to ionizing radiation.

The Camco transport dewars, which were produced in Somerville and trucked to Boulder, were filled for the first time with liquid hydrogen and put through operational tests in late April. Additional experience in handling long-distance transport of liquid hydrogen was gained when a half-dozen dewar trailers were driven from Boulder to Los Alamos during testing of the Mike device cryogenic system in June. The dewars traveled in a convoy with security escorts.

Assembly testing for the Mike device was first carried out in June at site DP-54 at LASL on an assembly of the inner and outer dewars when the cryogenics system was assembled and connected for filling operations. The entire dewar and reflux system was filled with liquid hydrogen from the dewars driven from Boulder.

Tests at LASL and ACF in Buffalo in June of the first complete tuballoy, dewar, and reflux system for the Mike device offered valuable training to TU 4 personnel. The reflux system had been studied by means of various mockups and scaled-down systems earlier in 1952. The entire cryogenic system was filled with liquid hydrogen in June, using the Camco RTDs.

Some slight modifications to the SAUSAGE cryogenic system were made in July, and the system was filled for a second time. The second test showed that the liquid hydrogen circulation system

¹¹⁶"Tentative Planning for Weapon Development Tests on Operations TUMBLER and SNAPPER to be Conducted at the Nevada Test Site During the Spring of 1952," prepared by William M. Ogle in collaboration with members of the Pogo Staff, J Division, Los Alamos Scientific Laboratory, 18 January 1952, Part I: Program Outline, Weapon Development Tests, Operations TUMBLER and SNAPPER, p. 16; Moore and Bechanan, pp. 233, 240.

¹¹⁷Moore and Bechanan, p. 240.

was ready for shipment to the Pacific.¹¹⁸

A second group of TU 2 technicians arrived at Eniwetok on May 1 to complete the installation for Building 330, the deuterium liquefaction plant, and to begin the assembly and erection of a new Air Products nitrogen liquefaction plant in Building 334. The remainder of the TU 2 personnel arrived before July 15, by which time the installations were complete and ready for test and training.

At the Buffalo plant of American Car & Foundry, the Mike device was completely assembled, without the cryogenic system and with dummy material substituted for the high explosives and nuclear materials in the primary and secondary stages, during the week of July 14-19 in a building that had dimensions similar to the cab then being built on Elugelab.¹¹⁹

The purpose of the construction of the mockup was to familiarize personnel with assembly procedures and to determine if any redesign of the components was necessary.¹²⁰ This assembly gave added experience in the assembly of the secondary fusion stage, which was the most difficult part of the assembly operation. A few days later, all the components of the device except the cryogenics system were assembled for the first time. JTF 132 assumed operational control of the device on July 25. During August, the device was prepared for shipment overseas.

The advance echelon of JTF 132 was established at Eniwetok in early August, and on August 18, the USS CURTISS sailed from San Diego to the Naval Supply Center in Oakland, California to load special equipment and some of the components of the Mike device. The tuballoy and inner dewar assemblies were transported between August 21 and 26 by C-54 aircraft during several trips between Buffalo and the naval air station at Alameda, California, adjacent to the Naval Supply Center.

The reflux dewar assembly was transported from LASL via Santa

¹¹⁸History - Task Group 132.1, 1 May 1952 to 30 June 1952, Maj. Alfred S. Knauf, USAF, HQ Joint Task Force 132, p. 19; Memorandum dated 18 December 1952 to Commander, JTF 132, from Commander, TG 132.1, Subject: Submission of Report of the Commander, Task Group 132.1 on Operation IVY; Moore and Bechanan, pp. 233, 235.

¹¹⁹OPERATION IVY, Report to the Scientific Director, Documentary Photography, WT-612(EX), J. L. Gaylord, Lookout Mountain Laboratory, Los Angeles, California, 1 February 1953, p. 33.

¹²⁰DNA 6036F, pp. 177, 181.

Fe, New Mexico to Alameda between August 22 and 24 by a C-47 aircraft. On August 26, two C-54s left Kirtland AFB in Albuquerque for Alameda; each aircraft carried one complete set of the nuclear components for the Mike device, along with one blast shield and one center rod (sparkplug) assembly.

A special flight carried detonators from Albuquerque to the Naval Weapons Center at Port Chicago, California (not far from Oakland) via the airport at Concord, California. The HE and X-unit assemblies for the TX-5 primary for the Mike device arrived at Port Chicago via rail on August 29.

When loading had been completed at NSC Oakland, the CURTISS sailed to Port Chicago to pick up the remainder of her cargo, and she sailed for Eniwetok on the morning of August 29, soon after the rail delivery of the TX-5 components.

Two armed Coast Guard vessels escorted the CURTISS to the Golden Gate at the mouth of San Francisco Bay, where she was joined by two U.S. Navy destroyer escorts. Patrol aircraft of Navy patrol squadron VP-47, based at NAS Alameda, escorted the flotilla out to distance of 500 miles. Similar aerial coverage was provided when she was within 500 miles of Oahu island in the Hawaiian chain and within 500 miles of Eniwetok, where she arrived early in the afternoon of 12 September (local time).¹²¹

The cryogenics facilities on Parry were substantially completed by August 1. The Camco hydrogen dewars, carefully packaged, were shipped in the holds of Military Sea Transportation Service Victory ships from Oakland, California, to Eniwetok. Four of the dewars left Oakland aboard the aircraft cargo ship USNS SGT. JACK J. PENDLETON (T-AKV-5) on August 2. The remaining four dewars were shipped via the USNS PVT. JOSEPH F. MERRELL (T-AKV-4) on August 21.¹²² These ships arrived at Eniwetok approximately 13 days after sailing. The Camco shop in the cryogenics area on Parry island was fully equipped and operating by August 15.

Operation TEXAN: Rehearsal for IVY

Task Group 132.4, the Air Force support group for Operation IVY, conducted Operation TEXAN, a full-scale rehearsal for IVY, from Bergstrom AFB near Austin, Texas, on August 13. Of the 59 USAF planes scheduled for IVY, 39 were used; a target ship simulating Mike shot ground zero was positioned in the

¹²¹Moore and Bechanan, pp. 178, 180.

¹²²THE SHIPS AND AIRCRAFT OF THE U.S. FLEET, Sixth Edition, James C. Fahey, Ships and Aircraft, Falls Church, Virginia, 1950, 1954, p. 42.

Gulf of Mexico at a distance from Bergstrom equal to the distance from Kwajalein to Eniwetok. Flights were made simulating bomb drop, sampling, drone control, aerial refueling, and cloud tracking missions.¹²³

The purpose of TEXAN was to determine the capabilities of TG 1324.4 men and equipment to fulfill their missions during IVY. Flight operations were staged from Bergstrom AFB, near Austin, Texas; to simulate the crowded conditions that would be encountered at Kwajalein, men and aircraft were limited to a small portion of the airbase. The target area, simulating Runit island at Eniwetok, was a point in the Gulf of Mexico, 369 nautical miles from Bergstrom.

In general, Operation TEXAN was considered a complete success; most of the minor discrepancies and problems which were disclosed could be easily resolved before movement of the task group to Kwajalein.¹²⁴

During TEXAN, a B-50 dropped a modified T-59 training shape (a MK IV bomb shape) that simulated the high-yield MK 18 to be dropped at Eniwetok. The T-59 included battery-driven telemetry equipment on its tailplate, simulating the transit time transmitter that would be installed on the MK 18.¹²⁵

Other shipments of men and materiel were made through Oakland and through Travis Air Force Base in California and Hickam Air Force Base near Honolulu, Hawaii. The bulk of cargo and equipment was shipped by surface vessels during July and August. The movement of personnel and cargo by air gradually increased and reached a peak during September and October.

Special air cargo shipments that had priority over normal air movement included aluminum sheets and plastic ballonets for the helium-filled tunnel in the Krauss-Ogle box. In addition, replacement parts for the cryogenics plant compressors

¹²³SEMIANNUAL HISTORY OF THE AIR FORCE SPECIAL WEAPONS CENTER FOR THE PERIOD 1 APRIL - 31 DECEMBER 1952, Volume I, The Narrative Account, Historical Division, Office of the Adjutant, Air Force Special Weapons Center, Kirtland AFB, New Mexico, 16 December 1952, pp. 309, 310, 311, 345; Moore and Bechanan, pp. 240-243.

¹²⁴Moore and Bechanan, pp. 241-243.

¹²⁵Memorandum dated August 1, 1952 to G. A. Fowler, Organization 5200, Sandia Corporation, from A. P. Gruer, Organization 5224, Sandia Corporation; "Program 2.5, Remote Transit Time Measurement on IVY Operations, Report to the Test (Scientific) Director," Raymond F. White, Division 5222, Sandia Corporation, June 15, 1953, p. 17.

required special airlifts to Eniwetok.

Task Unit 2 activities were closely coordinated with other construction by the Holmes & Narver Company, the AEC's nuclear test site construction contractor, at Eniwetok. Test runs were made to ensure smooth operation of the liquefaction plants and to train operators; the plants went into full production starting August 24 and ending October 28. During this period, several hundred liters of liquid deuterium and liquid hydrogen were produced for transport by the Camco hydrogen dewars.

In addition, purified gases and liquids were provided for Task Unit 4, the Mike assembly group, as needed. Rehearsals for emergency operation in case of shot delay were performed for each of the four liquefaction plants.

To camouflage its design, the Mike device was shipped in sections aboard the USS CURTISS to Eniwetok; the outer case was shipped as a set of thick-walled rings,¹²⁶ and after the

¹²⁶The IVY Classification Officer made a note about the Mike device movement:

Only one anomalous situation occurred during the overall operation. It is generally agreed that the size, weight, and shape of a nuclear device are Restricted Data, but in the case of the Mike device, it was not feasible to produce and transport the case on a Restricted Data basis within the time limit set, if ever.

Furthermore, the Mike case was determined by the scientific authorities not to reveal information on the nature and operation of the Mike device. Consequently, it was arranged that the components of the Mike case, when not identified as nuclear-device components, could be treated as Confidential Security Information, and the Classification Guide reflected this ruling that the size, weight, and shape of the Mike device would be Confidential Security Information.

However, whenever the Mike device was illustrated or photographed as a nuclear device, it was graded as Restricted Data. Later, on a directive from Washington, the Mike device was graded Restricted, not Security Information, in order to avoid the ridiculous situation which results from

possible interpretation of the wording of the Atomic Energy Act of 1946, in defining Restricted Data. After the completion of the operation, at the direction of the AEC, Washington, the grading of size, weight, and shape of the Mike device was raised to Confidential Restricted Data, which conforms to the actual gradings applied to photographs of the assembly.

However, component parts of the case, when not identified as nuclear-device components, are merely treated as accountable property and are given such protection. The component parts are, of course, kept in containers or otherwise separated so that the shape of the complete case is not apparent. (WT-608 (EX), p. 44)

A little further on, this same classification officer commented on some other aspects of the secrecy surrounding IVY; specifically, some results in the field of the rigid security ordered by AEC headquarters in Washington:

The Classification Office maintained a uniform policy that no information relating to the operation would be considered totally unclassified unless it was approved for release by the AEC or DOD authorities. Sometimes this rigid rule worked some unpleasant delays in routine administration, but no alternative seemed apparent in view of the extreme concern expressed by the Washington authorities about any public statement relative to the operation.

The Classification Officer can only comment parenthetically that some of the items treated as classified could not reasonably be considered classified when viewed from the field of operations. Furthermore, it was necessary to use open radio circuits when giving information which was considered highly classified in the Washington headquarters.

In addition, the unique procedures employed for the Mike shot clearly revealed yields of extraordinary magnitude to thousands of personnel cleared only for low-order military information. These are matters of

arrival of TU 4

personnel in mid-September 1952, the Mike assembly program began when the components were unloaded from the CURTISS and transferred for assembly at the shot cab on Elugelab. The CURTISS remained anchored off Elugelab and served as a base of operations during September and October.¹²⁷

A detailed schedule had been established for assembly of the device and for installation of equipment within the cab after assembly started. This schedule took into account the predicted assembly time of the Mike device as well as the practice hydrogen and deuterium fillings and necessary instrument installation.

The only mechanical problems encountered during assembly were difficulties in obtaining tight vacuums in the device itself and in the cryogenics system; some difficulties were experienced in assembling a leak-proof inner dewar. On September 25, vacuum piping was installed and assembly of the outer casing rings began.

The first hydrogen fill was scheduled for October 2, but was delayed three days by a leak in the deuterium filling line. The fill began at 5:30 PM on October 5, and was completed and stabilized in two hours. The first refill occurred at 9:30 AM the following morning, and the second refill at 1:00 AM on October 7. The system was then drained into Camco RTDs, and some minor repairs made to fix leaks in the vapor return line, filling manifold, and refrigerator dewar system.

The outer rings were assembled and in place by October 8. Starting two days later, after the vacuum system was completely assembled, the cryogenic system of the SAUSAGE was filled with hydrogen and later with deuterium.¹²⁸ After each of

security, not of classification, but they are pertinent to the applications of security gradings in an unrealistic manner because of the necessity to enforce directives by individuals not personally acquainted with the situation as it actually occurs. (WT-608 (EX), p. 44)

To the extent that AEC security regulations during IVY superseded and contradicted common sense, the classification officials were quite literally prisoners of their own device.

¹²⁷History - Task Group 132.1, 1 September 1952 to 31 October 1952, Maj. Albert S. Knauf, USAF, HQ JTF 132, p. 22.

¹²⁸History - Task Group 132.1, 1 September 1952 to 31 October

these practice fillings, the system was drained and the liquid gases returned to the dewars for storage.

The final filling operation was started on the evening of October 26, six days before shot date, and proceeded according to schedule, except for difficulty in removing a solid air block in one of the deuterium lines. After the blockage was removed, filling continued normally and the system operated within design limits. The reflux dewar was topped off eight times to replace lost hydrogen before the device was detonated; the last filling occurred just seven hours before firing.¹²⁹

A last-minute change was made in the TX-5 pit design and a new core was flown out to Eniwetok on a C-124; the core arrived on schedule and was quickly installed in the primary aboard the CURTISS.¹³⁰ The new core featured a reduced quantity of plutonium and an increase in its uranium content.¹³¹

At about 8:00 AM (local time) on the morning of October 31, final SAUSAGE assembly started. At 11:00, the TX-5 primary was moved from the USS CURTISS and installed in the SAUSAGE at about 3:00 PM. An hour later, the pit was inserted into the primary.¹³² Final device assembly was completed at 5:00 PM and the evacuation of the majority of JTF 132 personnel from Eniwetok was completed at 6:00.

During the day, the hydrogen reflux system had been periodically topped-off from transport dewars brought by naval landing craft from Parry island. The final dewar was brought to Elugelab at about 9:30 PM.¹³³

1952, Maj. Albert S. Knauf, USAF, HQ JTF 132, p. 23.

¹²⁹History - Task Group 132.1, TG 132.1 Participation in Mike Shot, Maj. Albert S. Knauf, p. 2.

¹³⁰History - Task Group 132.1, 1 September 1952 to 31 October 1952, Maj. Albert S. Knauf, USAF, HQ JTF 132, p. 24; Moore and Bechanan, p. 265.

¹³¹Memorandum for James S. Lay, Executive Secretary, National Security Council dated October 30, 1952 from Gordon Dean, Chairman, USAEC, subject: Approval for Operation IVY.

¹³²History - Task Group 132.1, TG 132.1 Participation in Mike Shot, Maj. Albert S. Knauf, pp. 3, 4.

¹³³Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. A-6.

The firing party check list was completed and the arming team returned from Elugelab to the ESTES at about 1:45 AM on November 1. The ESTES then got underway and sailed from Eniwetok lagoon at about 2:15 AM to a point 30 miles out to sea almost directly south of the shot island, i. e., to about 10 miles south of the southern end of Eniwetok lagoon.

The CURTISS left her anchorage near Elugelab at about 11:10 PM on October 31 and sailed to an anchorage near Parry island at the southeast corner of the atoll lagoon. Early on the following morning, she embarked the last of the remaining task force personnel and then followed the ESTES out to sea at about 3:45 AM and steamed to a point 35 miles due east of Elugelab.¹³⁴

Using television monitoring equipment and radio control circuits, the firing team aboard the ESTES began preparations to fire the SAUSAGE. The television equipment, then in a rather embryonic state, allowed the team to monitor systems and critical equipment within the SAUSAGE and note the position of various timing signals used to activate remotely-controlled cameras and other recording equipment and experiments and Go-No Go interlocks which could stop the firing sequence at any time.

The firing team communicated with the test device by three radio links to activate timing signals and the sequence timer to detonate the device on schedule. The sequence timer activated hundreds of pieces of test equipment according to an established pattern via a series of cams that closed switches at the proper moments.

The device was rigged to be fired after a preprogrammed electromechanically-driven automatic countdown sequence which also activated test and monitoring equipment. At 7:15 AM local time on the morning of November 1, 1952, the Mike device was detonated. At zero hour, ships and test personnel evacuated from Eniwetok lagoon were standing out to sea, about 30 to 40 miles south and east of the shot island.

The Mike Shot

Ten years of theory, research, and engineering culminated at zero hour in a sustained blinding flash of light and a searing wave of heat as the world's first man-made thermonuclear sun rose from the earth.

The explosion of the SAUSAGE was awesome in its appearance and effects. The fireball grew instantly to a diameter of 3 1/2 miles, completely engulfing Elugelab and the surrounding

¹³⁴Moore and Bechanan, p. 270.

atoll, and vaporizing nearly everything within it. Where the islet had been, a huge crater -- 164 feet deep and 6,240 feet in diameter -- was blasted into the sand and coral, and approximately 80 million tons of solid material was thrown into the sky.¹³⁵

The entire Krauss-Ogle box, subjected to pressures of 700 to 1,000 psi at the shot island,¹³⁶ was obliterated and the concrete bunker on Bogon burned and partially destroyed. Waves washed over the atoll and islands surrounding the hole where Elugelab had been.¹³⁷

The blast reached out for miles, crushing and sweeping away everything in its path. Nearby islands were stripped clean of vegetation and animals by the heat and blast waves. The hot mushroom cloud from Mike rose to an altitude of 27 miles and spread out for a distance of 100 miles; according to one account, the cloud soared to 90,000 feet at near-sonic speed.¹³⁸ The cloud stem was roughly eight miles in diameter

¹³⁵"Known U.S. Nuclear Tests, July 1945 to 31 December 1985," Robert S. Norris, Thomas B. Cochran, and William M. Arkin, NUCLEAR WEAPONS DATABOOK Working Paper NWD 86-2, February 1986, Natural Resources Defense Council, Washington, D.C., p. 14; letter dated February 2, 1955 from Herman Hoerlin, Group Leader, J-10, LASL, to Dr. John von Neumann. DNA 6036F states (p. 34) that "The testing process used up much of the limited land space that was available. An extreme case was the IVY Mike shot, which eliminated Elukelab (Elugelab), the island on which the device was detonated, and left a large underwater crater."

¹³⁶"Notes on Pogo Staff Meeting, 29 December 1951, Blast Measurements for IVY and SNAPPER," p. 4.

¹³⁷Before the test, predicted wave heights at various islands were: Engebi, 100 ft.; Rojoa, 50 ft.; Runit, 30 ft.; Japtan, 17 ft.; Parry, 17 ft.; and Eniwetok, 16 ft. (History - Task Group 132.1, 1 May 1952 to 30 June 1952, Maj. Albert S. Knauf, USAF, HQ Joint Task Force 132, p. 51.; Memorandum for Record dated July 3, 1952 from Lt. Col. R. P. Campbell, Jr., DMA/USAEC, Subject: Notes from the Commission's Briefing of July 1, 1952.) Final wave heights were only about 10% of these predictions; water rose two to four feet at Runit and one to three feet at Parry. (Operation IVY, Final Report, Joint Task Force 132, 1952, Headquarters, Joint Task Force Seven, Washington, D.C., 9 January 1953, p. 12.) LASL's J Division had predicted peak wave heights of only three feet at Parry. ("Notes on Pogo Staff Meeting, 29 December 1951, Blast Measurements for IVY and SNAPPER," p. 12.)

¹³⁸A METEOROLOGICAL ANALYSIS OF THE TRANSPORT OF DEBRIS FROM

and the cloud top was 40 miles across. **Figure IV-9** shows the IVY Mike cloud fairly soon after the explosion

The heat pulse from the explosion was felt for miles and lasted for many seconds: a B-36D weapons effects aircraft, at an altitude of 40,000 feet over 15 miles from surface zero, experienced a 93⁰ F temperature rise on its wings. The aircraft also encountered very strong (62% of design limit) horizontal bending moments on its tail surfaces from the air blast wave. These effects led to the conclusion that a B-36D would not be able to deliver a megaton-yield bomb without using either a drone bomber or a drogue parachute on the bomb.¹³⁹

Eyewitnesses to the test offered their own descriptions of the explosion:

The Shot (sic), as witnessed aboard the various vessels at sea, is not easily described. Accompanied by a brilliant light, the heat wave was felt immediately at a distance of thirty to thirty-five miles. The tremendous fireball, appearing on the horizon like the sun when half-risen, quickly expanded after a momentary hover time and appeared to be approximately a mile in diameter before the cloud-chamber effect and scud clouds partially obscured it from view. A very large cloud chamber effect was visible shortly after the detonation and a tremendous conventional mushroom-shaped cloud soon appeared, seemingly balanced on a wide, dirty stem. Apparently, the dirty stem was due to the coral particles, debris, and water which were sucked high into the air. Around the base of the stem, there appeared to be a curtain of water which soon dropped back around the area where the island of Elugelab had been.

The shock wave and sound arrived at the various ships approximately two and a half minutes after the detonation, accompanied

OPERATION IVY, USAEC NYO-4555(Del.), Lester F. Hubert, et. al., Weather Bureau, October 30, 1953, p. 10. Preliminary cloud heights indicated an elevation of 57,000 feet after 1.5 minutes; 108,000 feet after another 1.1 minutes; 110,000 feet within another 0.8 minutes; and 118,000 feet 5.7 minutes after detonation. (Moore and Bechanan, p. 274.)

¹³⁹WT-608 (EX), p. 12.

by a sharp report followed by an extended, broken, rumbling sound. The pressure pulse and the reduced pressure period as perceived by ear were exceptionally long.

Although the upper cloud first appeared unusually white, a reddish-brown color could soon be seen within the shadows of its boiling mass as it ascended to greater height and spread out over the atoll area. At approximately H+30 minutes, the upper cloud was roughly sixty miles in diameter with a stem, or lower cloud, approximately twenty miles in diameter. The juncture of the stem with the upper cloud was at an altitude of about 45,000 feet. Numerous projecting fingers could be observed in the neighborhood of the juncture of the stem with the upper cloud...

The cloud had ascended very rapidly, and soon appeared, in the words of one observer, to have "splashed" against the tropopause. After approximately fifty-six minutes, the entire cloud appeared to have become stabilized at an altitude of over 120,000 feet, although this figure was later questioned also. By this time, deforming effects by winds were becoming apparent though as late as sunset ... distant and high portions of the cloud could still be observed.¹⁴⁰

Another description of the Mike fireball was equally graphic:

The detonation occurred at 0715M, 1

¹⁴⁰DNA 6036F, pp. 187, 188. One of the best photos of Mike's detonation was taken from aboard the U.S.S. ESTES and appears as the frontispiece in the 1962 edition of THE EFFECTS OF NUCLEAR WEAPONS (Samuel Glasstone, ed.; U.S. Atomic Energy Commission, Washington, D.C., April 1962). The predicted height for the Mike cloud was 80,000 feet. (Minutes of the Thirty-Fourth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 23-24, 1953, Washington, D.C., p. 7.) Another account of the explosion places the base of the mushroom cloud between 70,000 and 80,000 feet and the top between 120,000 and 130,000 feet. (A METEOROLOGICAL ANALYSIS OF THE TRANSPORT OF DEBRIS FROM OPERATION IVY, USAEC NYO-4555(Del.), Lester F. Hubert, et. al., Weather Bureau, October 30, 1953, p. 12.)

November 1952 as scheduled. It was observed from the flag bridge of the USS ESTES (AGC-12) at a distance of 31 miles on a line bearing approximately 155⁰ from the shot island.

The initial aspect of the explosion, seen through density goggles, consisted of an immense fireball which appeared on the horizon like the sun when half-risen; however, the angle subtended by the half-disk at maximum was at least twice that of the sun. A rough estimate indicates that its diameter at this time was between 3 and 4 miles. The fireball was not homogeneous but consisted of a bright inner core of approximately two-thirds the total diameter surrounded by a thin relatively dark shell (orange in color as seen through the goggles), the whole being enclosed in an outer, very bright shell which was the limiting region of the fireball.¹⁴¹

Conditions inside the mushroom cloud, as seen from sampling aircraft, were unearthly and not unlike the inside of an active volcano:

Immediately upon entering the cloud, RED LEADER (the flight code) was struck with its intense color. It cast a red glow over the cockpit and his radiological instruments indicated maximum readings. There was no way of knowing how much more radioactive the area actually was than the maximum readings his instruments were indicating. The hand on the Integrator, which showed the rate of at which radioactivity was being accumulated, "went around like the sweep second hand on a watch ... And I had thought it would barely move!" The combination of most instruments indicating maximum readings and the red glow like the inside of a red-

¹⁴¹Extracts from a report by Professor C. E. Palmer to CJTF 132, an eyewitness description recorded shortly after the event, in A STUDY OF CERTAIN OPERATIONAL WEATHER CONSIDERATIONS INFLUENCING THE TEST AND DELIVERY OF HIGH YIELD WEAPONS, by Commander Elbert W. Pate, USN, Staff Weather Officer, Joint Task Force Seven, and Professor Clarence E. Palmer, Professor of Geophysics, University of California at Los Angeles, 30 June 1953, p. 17.

hot furnace was "staggering" and (the pilot) quickly made a ninety degree turn to leave the cloud.¹⁴²

A Scripps Institution of Oceanography survey ship, the RV HORIZON, was anchored at a seamount 72 miles north of Eniwetok to measure long-distance wave action caused by the blast; even this far away, the effects of the shot were impressive:

When Mike finally exploded I was just finishing installing the last instrument. It was dark (tests are always fired just before dawn) and I was standing alone on the raft with seawater over my ankles, gingerly tightening a hot (137 volt) wire with a plastic screwdriver and barely avoiding electrocution.

Suddenly, very suddenly, a new sun dawned -- over Eniwetok to the south -- for (the) Mike (shot cab) and the atoll were well below the horizon, seventy-two miles away. There was no sound at first, just the fantastic first flash of the explosion. Then the sun-like fireball rose slowly above the southern horizon, lighting the sea around it. It rose and dimmed and surged upward into the familiar mushroom cloud.

A minute and a half later the sounds started coming. Not a great boom from the direction of the blast but a series of thunderings from many directions as the sound waves were guided by atmospheric layers and reflected from cloud banks. Then the real dawn broke in the east and in its gray light we could see a mountain of clouds above the blast site.¹⁴³

¹⁴²Moore and Bechanan, p. 277. Radiation intensities in the cloud were in excess of 500 roentgens per hour; at this rate, a lethal dose could be absorbed in less than an hour. (History - Task Group 132.1, TG 132.1 Participation in Mike Shot, Maj. Albert S. Knauf, p. 24.)

¹⁴³"Mike," a reminiscence by Willard Bascom, undated. The estimate of only 90 seconds between the explosion and the sound 72 miles away is probably incorrect; the actual time lapse should have been closer to six minutes. (It took two and a half minutes for sounds of the explosion to reach observers aboard the USS ESTES, 31 miles south of Eniwetok atoll.) During another Pacific test 18 months later at

Colorful accounts of the explosion soon appeared in the press, in spite of the tight security and an almost total news blackout surrounding any public announcement of the test:

The first sign of the explosion came to the men aboard ship in the form of a flash many times brighter than the sun, followed by a wave of heat across their backs (one observer said the heat wave that hit his back was "about 180⁰ F;" another described it as a "momentary touch of a hot iron"). Although the men had their backs turned with their arms across their dark glasses, the blinding light was not kept out.

"It would take at least ten suns to equal the light of the explosion from a distance of thirty-five miles," a navigator wrote.

Ten seconds after zero, the men on ship started turning around to face the direction of the blast.

"I could hardly believe my eyes," one wrote. "A flame about two miles wide was shooting five miles into the air. This lasted for about 7 seconds. Then we saw thousands of tons of earth being thrown straight into the sky. Then a cloud began to form about twenty seconds after the shot." Several observers described the shape of the "mushroom cloud" as about a mile wide at the bottom and from twenty to at least thirty miles wide on top; one said it went fifteen miles into the air.

"You would swear," one sailor wrote, "that the whole world was on fire. It was really something I'll never forget."

At least three eyewitnesses reported that the test island on which the bomb had been exploded disappeared after the blast.

"About fifteen minutes after shot time, the

Bikini, a Japanese fishing boat crew more than 80 miles from that explosion also perceived the omnidirectional nature of the blast noise, noting that it seemed to emanate as much from the sea below the boat as from the air around it.

island on which the bomb had been set off from, started to burn and it turned a brilliant red. It burned for about six hours. During this time it was gradually becoming smaller," one man wrote. "Within six hours a mile-wide island had disappeared."¹⁴⁴

Newspaper reports stated that a hydrogen bomb was detonated, and that it vaporized a half-mile wide and three-mile long atoll into gas and dust; that the blast blew the top off a building on an island 18 miles from ground zero; that heat from the explosion was felt on shipboard 32 miles away; that the cloud rose to 50,000 feet; and that the sound of the blast made a 5,000 lb. bomb explosion sound like a firecracker in comparison.¹⁴⁵

The effects of the Mike shot on the local flora and fauna were devastating:

Rigili is fourteen miles south-southwestward down the lagoon from the Mike shot crater. Yet there the survey team found that the trees and brush facing the test site had been scorched and wilted by the thermonuclear heat. Many of the terns there were sick, some grounded and reluctant to fly and some with singed feathers, particularly the noddy terns and the sooty terns, whose feathers are dark in color. On Runit island on November 6 the story was much the same...

... At Engebi the group went ashore on an island where the sense of desolation was deepened by the presence of a reinforced concrete building, ruptured and shaken but still standing, on the island flat that had been swept by the blast and the succeeding surge of water. The body of a bird was seen, but no living animals and only the stumps of vegetation... Among the specimens

¹⁴⁴BULLETIN OF THE ATOMIC SCIENTISTS, Vol. VIII, 1952, p. 300. These accounts are reprinted from newspaper stories about letters from task force personnel to friends and relatives in the U.S. See following section on IVY security breaches.

¹⁴⁵"Abstract of News Stories Based on Letters," Appendix "H" to Atomic Energy Commission, Security and Public Reporting Aspects of 1952 Pacific Tests, Note by the Secretary, AEC 483/35, November 28, 1952.

collected were fish which seemed to have been burned. On each of these fish the skin was missing from one side as if, the field notes said at the time, the animal "had been dropped in a hot pan."

The work at Engebi finished, the party took the LCM westward through the turbid waters of the shot islands and past the Mike crater, which had the appearance of a new deep channel into the lagoon and the depth of which, after the churning of the sludge had subsided, had been estimated at fifteen fathoms.

When the LCM, turning southwest, stood off the beach of Bogombogo and Bogallua, three miles from the Mike site, the effect of the energy that had been released a week before was startlingly evident. On the survey team's visit to Bogombogo on October 25 the island had been heavily laden by stands of coconut palms and thickly populated by birds. Then it had seemed ideal as a downwind station for sampling after Test Mike (sic), and the group had expected to return there in due time, to make post-test collections.

But when members of the team saw Bogombogo again on November 8 the extent of their miscalculation was terribly clear. The island had been stripped of vegetation by the force and heat of the blast. Palm trees had been burned down to the roots. All animal life, so far as members of the team could tell, had been snuffed out. The same was true of Bogallua.¹⁴⁶

As was expected, damage to the islands was directly proportional to their distance from surface zero. At Engebi, grass was green at its base, but scorched at its tip. The island appeared to have been washed by water that carried burnt debris away, but this may have been due to the blast wind followed by heavy rainout from the mushroom cloud.

¹⁴⁶PROVING GROUND: AN ACCOUNT OF THE RADIOBIOLOGICAL STUDIES IN THE PACIFIC, 1946-1961, Neil O. Hines, University of Washington Press, Seattle, Washington, 1962, pp. 141, 143, 144. Engebi was three miles from Mike's surface zero and Runit was about 13 1/2 miles from Elugelab.

The atoll islands -- from where Elugelab had been down to Bogallua, including Bogon, Bogairikk, and Teiteiripucchi -- were swept clean; only a fine deposit of coral sediment was left over the reef rock. Not a trace remained of the 9,000-foot long Krauss-Ogle box, which suggested that a jet of fire blasted it down, and a violent water wave washed away the debris. Bogon reeked with the smell of decaying dead fish.¹⁴⁷

Figure IV-10 shows the Bogon-Elugelab area before the test; **Figure IV-11** shows the same area after the explosion. Although the entire area was sparsely built-up before the test, nothing was left afterwards.

Figure IV-12 is an aerial view of the same archipelago before the test; **Figure IV-13** is an "after" view, with Elugelab replaced by a crater.

The radioactive cloud from Mike moved slowly to the northwest after the shot and heavy rain showers were observed below it, mostly around the lower cloud stem.¹⁴⁸

Mike's 10.4 megaton explosion was more than 46 times the size of the 225 kiloton GREENHOUSE George shot a year and a half earlier, then the largest fission explosion. Mike was also the largest manmade explosion on earth at its time (this record would stand for only 16 months until surpassed by the CASTLE Bravo shot).¹⁴⁹

One of the reasons for the tremendous yield of Mike was the very-high neutron flux resulting from the compressed liquid deuterium fuel. Because solid lithium-6 deuteride and tritide fuels use many of the newly-spawned neutrons to propagate fusion reactions, the external neutron flux of a solid-fueled device is lower than that of a liquid deuterium-fueled system of equal yield. Neutrons generated by liquid-fueled devices jacketed with uranium cause more complete "burning" of the surrounding uranium tamper.

The neutron flux in the Mike explosion was sufficiently intense to create two new elements from neutron irradiation

¹⁴⁷History - Task Group 132.1, TG 132.1 Participation in Mike Shot, Maj. Albert S. Knauf, p. 20.

¹⁴⁸DNA 6036F, p. 191.

¹⁴⁹As great as Mike's yield was, it was trivial compared with the estimated 500 MT explosive yield of the volcano Krakatao, which blew up in 1883 off the coast of Indonesia, between Sumatra and Java. ("Notes on Pogo Staff Meeting, 29 December 1951, Blast Measurements for IVY and SNAPPER," pp. 12, 13.)

of uranium: einsteinium ($_{99}\text{Es}^{254}$) and fermium ($_{100}\text{Fm}^{257}$); small quantities of these radioactive, short-lived materials were found among radiological samples of post-shot debris.¹⁵⁰

In spite of the tremendous power of the Mike shot, almost 77% of its yield, about eight megatons, came from fission, mostly from the explosion of the uranium-238 tamper around the secondary.¹⁵¹ In this respect, the SAUSAGE was far more a fission device than a fusion device.

The Mike test was as important a milestone for the development of U.S. thermonuclear weapons as the TRINITY test in 1945 had been for the development of fission weapons. Years later, Dr. Herbert York, erstwhile director of the new Livermore, California weapons lab, wrote:

When I reflect back on (the detonation of the IVY Mike shot), as I sometimes do in preparing or giving lectures on the history of the nuclear era, a feeling of awe and foreboding always recurs. Even at that time, I thought of that moment ... marking a real change in history - a moment when the course of the world suddenly shifted, from the path it had been on to a more dangerous one. Fission bombs, destructive as they might have been, were thought of being limited in power. Now, it seemed, we had learned how to brush even these limits aside and to build bombs whose power was boundless.¹⁵²

¹⁵⁰For a while, element 100 was jokingly known as "Pandamonium," after Project Panda. ("An NHP Interview with Theodore Taylor," conducted at The University of Maryland, College Park, MD, 12 April 1989, Nuclear History Program Oral History Transcript #1, p. 28.) Edward Teller suggested the name "eleugelabium" for element 100; the JCAE staff favored the name "tellerium" for element 99. (Memorandum for the File dated January 13, 1953 from John Walker.)

¹⁵¹Table I, attachment to letter dated December 21, 1954 to Dr. Bugher, Division of Biology & Medicine, USAEC, Washington, D.C. from Col. Robert N. Isbell, USAF, Chief, Atomic Operations Division, Air Research & Development Command, USAF.

¹⁵²MAKING WEAPONS, TALKING PEACE, Dr. Herbert York, Basic Books, New York, 1987, p. 69.

The AEC Reports to President Truman

The official AEC preliminary report to President Truman on the Mike test, dated November 1, 1952, stated:

The Atomic Energy Commission wishes to report that at 0715 November 1 (1415 October 31, Washington time), the first full scale thermonuclear experiment was conducted successfully at Eniwetok. You will recall that we discussed certain aspects of this operation with you on June 30, at which time Dr. Norris E. Bradbury, Director of the Los Alamos Scientific Laboratory, described this particular experiment in some detail for you.

From early and incomplete evaluation of results, the yield is estimated roughly to have been more than 6 megatons, possibly as high as 12 megatons; that is, 6 to 12 million tons of TNT equivalent. In comparison with atomic bombs of the Hiroshima type, this is equal to some 50 to 75 in terms of destructive effect.

If the more precise and lengthy methods of measuring yield, which are now in process, confirm the above yield, the detonation exceeds what we had anticipated. We are confident, however, that the principles of radiation ducting and implosion, the recent discoveries that opened the way to accelerated thermonuclear effort, can be applied in full scale devices.

The shot island Elugelab is missing, and where it was there is now an underwater crater of some 1,500 yards in diameter.

No significant fall-out of radioactive contamination occurred. As a precaution against fall-out, Joint Task Force 132 had evacuated Eniwetok Atoll for the shot. They expect to return to the atoll today.

Unfortunately, during the cloud sampling operation one aircraft, an F-84-G (sic), and pilot were lost in Eniwetok lagoon. Otherwise the principal operation missions were conducted without incident and as planned.

Our present plan calls for the conduct of

the second and final detonation of this operation not earlier than November 11, Eniwetok time. Actual conditions encountered on re-entry to Eniwetok may, of course, cause delay beyond this date. This detonation will be a proof-test of a one-half megaton fission bomb dropped from a B-36 aircraft.¹⁵³

Six days later, AEC Chairman Gordon Dean advised President-elect Dwight D. Eisenhower of the success of IVY, that one of Eniwetok's islands was missing, and of the likelihood of Soviet success in determining what had happened:

In a consultation with the President yesterday it was his opinion that I should notify you immediately concerning the weapons program which is being conducted at the present time at Eniwetok.

The significant event to date is that we have detonated the first full-scale thermonuclear device. This took place on October 31st, Washington time. This was an experimental device which demonstrated that a thermonuclear explosion is feasible. It will probably be a year before we will be able to test the first deliverable thermonuclear weapon.

From early and incomplete evaluation of the results, the yield is estimated roughly to have been approximately 10 megatons, that is, ten million tons of TNT equivalent. If the more precise and lengthy methods of measuring yield, which are now in process, confirm the above yield, the detonation exceeds what we had anticipated.

The island of the Atoll (sic) which was used for the shot -- Elugelab -- is missing and where it was there is now an underwater crater of some 1500 yards in diameter.

....

We are attempting to keep the Russians in the dark as to what has been and will be exploded in the test series. In view of the

¹⁵³Memorandum for the President, signed by Gordon Dean, Chairman, U.S. Atomic Energy Commission, Washington, D. C., November 1, 1952.

large number of personnel involved in the operation, and in view of the size of the detonation and the fact that the light could be observed at sea for several hundred miles, it is not likely that we can for long keep from the Russians the fact that there has been a thermonuclear explosion; although to date the weather has played into our hands in that the winds have held the cloud for a period of seven days over the Pacific, making it virtually impossible for the Russians to obtain samples from the cloud and thus determine the nature of the explosion.

Unclassified seismic stations picked up the explosion. They will be able to place the epicenter of what they will probably regard as an underwater earth tremor as being in the neighborhood of the Eniwetok Atoll. We nevertheless do not propose to confirm that the tremor was attributable to our test series.¹⁵⁴

At the University of California's seismic monitoring station in Berkeley, where Drs. Edward Teller, Ernest O. Lawrence, and Luis Alvarez witnessed the IVY Mike shot, the seismographic paper record of the earth's disturbance by the shot was classified and confiscated by representatives of a local AEC office.

Dean's comments to Eisenhower about the usefulness of seismic indication were somewhat disingenuous: an earthquake in the vicinity of the Marshall islands, coupled with public knowledge of AEC experiments at Eniwetok, was bound to raise suspicions of a cataclysmic event such as that described in unofficial and unauthorized newspaper accounts.

The newly-proven hydrogen bomb was as great a leap in explosive power over the atomic bomb as the atomic bomb had been over conventional explosives: the Mike shot was 500 times more powerful than the crude FAT MAN prototype fission bomb detonated in New Mexico during Operation TRINITY in July 1945.

A draft memorandum prepared by AEC Chairman Gordon Dean for President Truman stated: "As Chairman of the Atomic Energy

¹⁵⁴Letter dated November 7, 1952 to Gen. Dwight D. Eisenhower from Gordon Dean, Chairman, USAEC. The letter is notable for the AEC's failure to mention any Soviet seaborne radio-sample collection capability, either surface or subsurface.

Commission, I believe an event has taken place that at least parallels in magnitude, and in my judgment surpasses, the new order of military power that the TRINITY shot, the first atomic blast at Alamogordo in 1945, opened."¹⁵⁵

Nuclear weapon yields could now be measured in megatons of TNT equivalent force, rather than just kilotons. After the Mike shot, the world -- and American foreign policy -- would never be the same. A new threshold of destruction had been crossed, and it would be only a matter of time until other nations followed suit: the Soviets would test a solid-fueled boosted fission device in August 1953, and a true multistage thermonuclear "superbomb" in November 1955.

Another result of the success of the Mike shot was that the sprawling and expensive reactors in South Carolina would not be needed for tritium production, and the reactors were put to use making plutonium and small amounts of tritium for boosted hydrogen bomb cores. The cost of thermonuclear weapons had also been reduced dramatically.

Despite the press "leaks" emanating from test participants, the AEC issued only cryptic public news releases about the results of Operation IVY on November 11 and November 16; the November 16 statement stated that "the test program included experiments contributing to thermonuclear weapons research."¹⁵⁶

The King Shot and Roll-up of IVY

Meanwhile, back in the Pacific, the dummy MK 18 bomb (a T-59 practice unit) was shipped on the USS CURTISS, in storage condition, to be assembled aboard the ship at Eniwetok. The live MK 18 was delayed until October 20, when its parts were completed by Sandia. It was assembled at the Sandia Base in Albuquerque in ready condition, with detonators, fuses, firing system, nuclear components, initiator, and all other special parts and test instrumentation installed, including a transmitter fastened to the tailplate of the bomb to radio

¹⁵⁵Undated draft memo for President's Approval from Gordon Dean.

¹⁵⁶Office Memorandum dated November 17, 1952, to All Field Managers from Richard G. Elliott, Director, Information Division, Santa Fe Operations, USAEC, Subject: Official Commission Announcement of Eniwetok Test Series 1952. The November 11 press release is reprinted in FOREIGN RELATIONS OF THE UNITED STATES 1952-1954, Volume II, National Security Affairs, Part 2, U.S. Government Printing Office, Washington, D.C., 1984, p. 1025.

transit time from the bomb to airborne recording stations carried aboard a pair of B-50s.¹⁵⁷

The live bomb was flown from Albuquerque on November 4, aboard a USAF C-124. It arrived via Travis and Hickam AFBs at Kwajalein on November 7 (local time).

Task Unit 5 personnel left Albuquerque on October 31, and arrived at Kwajalein on November 3. After their arrival, the CURTISS, which served as the TU 5 workshop, was brought alongside a deepwater pier, and the live MK 18 bomb was taken aboard for storage.

On November 8, the dummy MK 18 unit was loaded onto the B-36H strike aircraft, and all final electrical checks were made. The dummy bomb was dropped over the sea on the morning of November 9 near the King surface zero target, northeast of Runit island at Eniwetok atoll. Other ships and aircraft participated in this "dress rehearsal" for King day.¹⁵⁸ The safing and firing systems on the bomb functioned perfectly.

The live MK 18 and all components were checked in preparation for loading on November 12. The weapon was flown from Kwajalein to the test area on November 13, but because of a solid overcast at Eniwetok, the drop was canceled. After a 24-hour delay, it appeared that further delays would be encountered, and the bomb was unloaded from the B-36H and put aboard the CURTISS for check-out and battery replacement on the 15th. At 2:00 PM that day, it was reloaded into the strike aircraft and all final checks made.¹⁵⁹

¹⁵⁷Letter dated June 20, 1952 to Duncan Curry, Jr., Chief of Staff, TG 132.1, LASL, from T. E. Zudick, Sandia Corp., Subject: Monthly Status Reports, Operation IVY, p. 5.

¹⁵⁸Memorandum dated June 26, 1952, to D. F. Worth, Jr., Field Manager, AEC, Sandia, from C. L. Tyler, Manager, SFO, Albuquerque, N.M., Subject: Release of Materials for King Shot, Operation IVY; History - Task Group 132.1, 1 May 1952 to 30 June 1952, Maj. Albert S. Knauf, USAF, HQ Joint Task Force 132, p. 49; "Program 2.5, Remote Transit Time Measurement on IVY Operations, Report to the Test (Scientific) Director," Raymond F. White, Division 5222, Sandia Corporation, June 15, 1953, p. 18. There is no mention of this dummy bomb drop in the Defense Nuclear Agency's official IVY history (DNA 6036F) even though it is well-documented in WT-608 and other IVY reports.

¹⁵⁹"Program 2.5, Remote Transit Time Measurement on IVY Operations, Report to the Test (Scientific) Director," Raymond F. White, Division 5222, Sandia Corporation, June 15, 1953, p. 19.

By 6:30 AM the following day, the King weapon had been loaded aboard the bomber and was enroute to Eniwetok. Inflight insertion of the fissile capsule was performed by the bombardier, following normal procedures. By H-3 hours, the drop aircraft had begun orbiting over the target area, conducting practice runs to get into the allowable drop time window.

The bombing target consisted of a number of brightly painted oil drums mounted on pilings arranged in a cruciform shape at ground zero; small radar reflectors, to improve bombing accuracy, were mounted on the center drums. In conjunction with the target, an offset aiming point consisting of a large radar reflector was positioned 1,900 feet due south of ground zero. The large reflector was detectable by radar from a range of 60 miles; the smaller target was optically visible from a distance of 25 miles. The bomb was aimed visually.¹⁶⁰

The 8,550 lb. MK 18 was dropped from 40,000 feet and exploded 56 seconds later with a 500 kiloton yield¹⁶¹ at 11:30 AM on the morning of the November 16 (November 15 on the eastern side of the dateline) at about 1,480 feet above sea level over the target area 2,000 feet north of Runit island.¹⁶²

The test of Dr. Taylor's high-yield fission design brought its own drama to nearby observers:

KING Shot was perhaps more spectacular in many respects than MIKE Shot -- due, primarily, to the relatively close distance (approximately 11 miles) of observers to ground zero. The heat wave was immediately felt by observers and the growth of the

¹⁶⁰Final Report, Task Group 132.4, Kirtland AFB, New Mexico, January-December 1952, p. 8; Moore and Bechanan, p. 291.

¹⁶¹DNA 6036F, p. 1; ANNOUNCED UNITED STATES NUCLEAR TESTS, July 1945 through December 1990, NVO-209 (Rev. 11), Office of Public Affairs, U.S. Department of Energy, Nevada Operations Office, January 1991, p. 3. Shelton, p. 5-31, identifies the King device as a modified MK 6D weapon; however, an attachment to a letter dated March 24, 1994 from Everet H. Beckner, Deputy Assistant Secretary for Research and Development, Defense Programs, to the author listing the names of many devices tested by the U.S. between 1945 and 1962 includes a MK 18F.

¹⁶²Burst height was set for 1,500 feet; the bomb was falling at 980 feet per second (668 MPH) when it exploded. (Memorandum dated October 11, 1952 to H. E. Lenander, Org. 5230, Sandia Corp., from W. C. Scrivner, Org. 5240, Sandia.)

fireball, the development of the cloud chamber effect, and the formation of the conventional mushroom-shaped cloud were all clearly visible.

In approximately 45 seconds after detonation, the shock wave hit Parry island and the sound was very similar to that experienced when in close proximity to the firing of a 90MM gun. The cloud rose rapidly to a great height, spread out over Parry island and seemed to remain suspended there until it began to disperse. An early report estimated the cloud height to be approximately 67,000 feet with the base of the mushroom at 40,000 feet.

Damage from King was noted as far north of surface zero as Lojwa, where brush was scorched, and as far south as Jinedrol, where brush was burning. Closer in at Billae, palms were broken and burned. Much debris from surface objects was on Runit itself with power poles downed and burning and puddles of water standing a quarter mile from the edge of the lagoon. The airblast over the reef off the northern tip of the island had apparently pushed seawater that far onto the island.¹⁶³

Other accounts of the damage at Runit were more detailed:

At H+75 minutes, a fairly comprehensive report of the damage and contamination on Runit island was received. Although a water reading in the vicinity of the target read zero, a reading of 300 mr (milliroentgens) at 500 feet above the target was registered and the ground reading at the Runit powerhouse was 3 mr. Two telephone poles remained standing; tent-frames were demolished and charred; mud had been thrown almost to the airstrip; there was evidence of a wash-over of the northern end of the

¹⁶³DNA 6036F, p. 206. Lojwa was about three and a half miles from King surface zero; Jinedrol, just over five miles; and Billae, about two miles. Another estimate placed the King cloud top at 66,000 feet with a diameter of 41,000 feet. (Letter dated 17 July 1956 to Lt. Col. Jack James, Program 9, Task Group 7.1, JTF 7, from D. F. Seacord, Jr., Edgerton, Germeshausen, & Grier, Inc.)

island; the radar target had been dished out; Naval Research Laboratory (test structure) revetments were eroded; the blockhouse was left but the sand was gone; discolored yellow water was observed 2,000 yards off the beach, downwind from ground zero; and the reef was unchanged.¹⁶⁴

On the south end of Runit charred 2x4s were noted. Telephone poles were downed and the vegetation had been burned. At the Runit airstrip the field was littered with telephone poles and other debris. The marine pier was demolished and only the pilings were left standing. In the vicinity of the pier the grass had been burned and broken telephone poles and drums were thrown about. The fill about the blockhouse north of the pier on Runit had been pushed away from it.

At the target area on the northern tip of Runit, the radar marker was missing. Water had washed over approximately 1/4 mile of the northern end of the island and pools of water were standing in this area. Vegetation was badly burned. Muddy water was evidenced up to 700 yards of the end of the island, indicating a possibility of water being (rained) out.¹⁶⁵

Within the limitations of the basic design of the pit used for the King test, the shot demonstrated the immediate high-yield capabilities of the 92-point implosion system.¹⁶⁶

Re-entry to the cryogenics area on Parry Island was possible the day after Mike shot (November 2). Test productions showed no damage to the liquefaction plants. After King shot, when the IVY series ended, dehumidification units were installed in the cryogenics buildings and the interiors of the buildings were sealed. Periodic operation of the compressors and other equipment were conducted, and the buildings and machinery checked, cleaned, and lubricated in order to keep them ready for possible future tests of liquid-fueled

¹⁶⁴Final Report, Joint Task Force 132, Operation IVY 1952, p. S-11.

¹⁶⁵History - Task Group 132.1, TG 132.1 Participation in King Shot, Maj. Albert S. Knauf, p. 11.

¹⁶⁶WT-608 (EX), p. 8.

weapons.¹⁶⁷

The major rollup of equipment for TU 3 was completed before the Mike shot. Virtually all equipment, except three dewars and several "transportainers," was ready for evacuation three days before Mike was detonated (the dewar trailers were returned for storage at the NBS lab at Boulder). This left only a very small amount of equipment and materials to be handled after Mike shot.

The overall storage efficiency of the TU 3 Camco dewars, from the time of receiving liquid hydrogen until it was emptied, and including all losses of every kind (from normal heat leaks, accidents, dewar-to-dewar transfers, some experiments, and from malfunction of the equipment) was 95% over a period of approximately 2 1/2 months. Of the most precious commodity handled by TU 3, deuterium, not one liter of 1,000 liters liquefied, stored, and transferred was lost.

Security and the Mike Shot

Another unique and unusual aspect of Operation IVY was the relatively massive breach of security of the tests, compared to previous "closed" U.S. Pacific Ocean nuclear test series, i.e., Operations SANDSTONE and GREENHOUSE; Operation CROSSROADS in 1946 was opened to the world's press.

Before IVY, the AEC and the National Security Council had prepared elaborate justifications for secrecy and for disclosing as little information as possible about the purposes, conduct, and results of the series. The public information plan approved for IVY was to hold to a bare minimum official announcements concerning the tests. An extraordinarily tight curtain of secrecy was to cloak the series.

This plan was the result of an extensive study by an ad-hoc Psychological Strategy Board and the National Security Council prepared in September 1952. The goal of the plan was to acknowledge in advance that a Pacific test series would be conducted in the autumn and acknowledge afterwards that the tests had been completed; that the tests included experiments contributing to thermonuclear weapons research; that AEC executives were pleased with the results; that DOD and AEC commended JTF 132 for its performance during the series; that the United States was conducting weapons development only because of Soviet intransigence in the matter of arms control; and that the U.S. stood ready to enter into effective and enforceable arms control agreements with the Soviets, should they ever wish to do so.

¹⁶⁷Moore and Bechanan, p. 317.

The ad hoc panel of the Psychological Strategy Board included representatives from both the AEC and the State Department; its purpose was to study the psychological impact and possible political and military exploitation of the IVY tests, especially the thermonuclear experiment. It became apparent very soon after the panel was established that there would not be a prepared announcement immediately after Mike shot unless such an announcement was forced as a result of increased press speculation or unless the shot was detected and announced by a foreign power.¹⁶⁸

Two official statements made before and after IVY appeared to have been appropriate for achieving the desired psychological strategy objectives. The first announcement was a joint AEC-DOD press release dated February 18, 1952 which stated only that "preparations for a new series of tests at the AEC's Eniwetok Proving Ground are being carried out."

The first announcement produced a flurry of brief articles in which speculation on the testing of an "H-bomb" gained some currency. By April, more thoughtful articles had begun to appear in leading newspaper columns, predicting that an "H-bomb" would be tested in the fall.

The second press release in September increased this speculation. One Washington, D.C. correspondent for the Associated Press newswire service discussed problems of refrigeration, transportation, and delivery of cryogenic weapons. By the time Mike was detonated, the test was probably among the worst-kept secrets of the AEC.

In fact, some of the pre-test stories about the forthcoming shot were so accurate that the AEC believed that they exceeded the possibility of sheer guesswork. In an effort to determine if someone was leaking restricted data, the AEC requested the Federal Bureau of Investigation to look into several of the stories. No sources of leaks were ever found.¹⁶⁹

The February and September statements conveyed much less information than had heretofore been issued through press releases, press conferences, and pictures after both earlier Pacific weapons development test series, Operations SANDSTONE and GREENHOUSE. Full success of the program of limited public disclosure depended almost entirely on there being no revelations other than through official U.S. government

¹⁶⁸Moore and Bechanan, p. 183.

¹⁶⁹Moore and Bechanan, p. 185.

announcements.

The purpose of this limited public disclosure was to gain a psychologically-strategic advantage for the U.S. The NSC assumed that if there were no breach in the security of test information, the limited public announcements to be made by the AEC and DOD would avoid unwarranted optimism in the U.S. or abroad about U.S. possession of deliverable thermonuclear weapons and consequent let-down when the truth was disclosed; would minimize alarm among U.S. allies and neutral nations; and would leave much room for uncertainty and speculation in Moscow and Peking on the status of fusion and fission weapons development by the U.S.

From a security point of view, items to be protected included any information that revealed the current status of U.S. nuclear weapons development and subsequent production and stockpiling of thermonuclear weapons; the specific times of test detonations; explosion particulars including weapon design details, fireball growth rate, and blast and other nuclear phenomena which could lead to analysis and determination of yield and materials used; and explosion effects.

The reasons for classifying this type of information were numerous and readily defensible. Classification, it was believed, would prevent information revealing the status of the U.S. thermonuclear program from reaching the USSR; receipt of such information by the Soviets might assist them in making decisions on their course of policy and action in foreign relations and military affairs.

Specific test times, particularly that of the Mike shot, were classified because if the USSR had this information, it would help its long-range test detection program. At an AEC meeting on October 16, 1952, Darol Froman of LASL, in response to a question from AEC Chairman Gordon Dean, admitted that if the Soviets had a long-range nuclear test detection system as good as that then possessed by the U.S., they would be able to determine that the IVY Mike device incorporated thermonuclear reactions, but would probably not be able to estimate (at long range) the yields of either the Mike or King shot.¹⁷⁰

Knowledge of shot times might allow the Soviets to determine where and when to send airborne and seaborne collection airplanes, ships, and submarines. Knowing the time of the shot would allow the Russians an exact period within which to examine their seismographic and acoustic detection records to determine the size of the explosion. Knowing when to look and thereby pinning it to a specific part of their data which

¹⁷⁰Minutes of AEC Meeting 767, October 16, 1952.

included the blast, they would then be able to analyze the specific proportions for estimation of the intensity of the burst.

Information about explosion phenomena and effects (such as size of the "mushroom" column, height of the cloud top, fireball size and growth rate, and general destructive effects) was classified because this data would assist in determining yield. A reasonably good estimate of yield might help the USSR design blast- and shock-resistant structures, and would also allow the Soviets an insight into the state of the U.S. thermonuclear weapons program.

Considerations for classification did not take into account the possible political ramifications of Soviet overestimations of the status of the U.S. thermonuclear weapons program and what steps the Russians might take based on such a misjudgment. The AEC, DOD, and NSC all seemed to assume that the Soviets would properly fear any significant U.S. technical advance, and temper their international behavior accordingly.

The Soviets invariably accelerated their own weapons program to offset any perceived U.S. advantage; such perceptions of possible U.S. superiority usually did not restrain Soviet behavior or change its goals. For example, in July 1945, when President Truman told Soviet dictator Josef Stalin that the U.S. had a new and powerful weapon, Stalin ordered his scientists to accelerate their work on a Soviet atomic bomb (at that time, thanks to extensive espionage, Stalin knew more about the status and technical details of the American atomic bomb program than did Truman).

In the draft memorandum prepared by AEC Chairman Dean for President Truman, Dean presented his suggestions for release of information about Mike and his estimations of what impact it might have on the Soviets; Dean apparently favored a strong and detailed statement of the results of the test. He even went so far as to suggest that, now that the U.S. had successfully tested a hydrogen bomb, the Soviets might not respond to a significant escalation of the Korean War.

Dean apparently assumed that the Soviets would suppose that a successful test was an indication of a large stockpile of readily-deliverable bombs, a state of affairs that was nonexistent in the U.S. at this time:

An exploitation of the psychological impact of the thermonuclear detonation of October 31, 1952 is worthy of consideration. Time to do this is on our side, and is still available -- its impact is still of great moment. It should be done as part of a

plan, I suggest, to which I can add comments from the nuclear and personal point of view.

In addition to the announcement itself, with its clear statement of the truth, and its primary objective, to use the information for fulfillment of ethical and moral obligations to our allies and friends, and the presently uncommitted neutrals, there is implied by this, recognizable to those to whom naked power is understandable, an implication of a vast shift in offensive power.

Pravda (Author's note: the official Communist party newspaper) devoted about three lines to the Pacific event -- "We will not be blackmailed." I don't think they will -- but they will pause and consider.

The (U.S.) political and military leaders may decide that, after all other means have failed in bringing the Korean war to a close, that strong offensive action is required -- for example, amphibious landings; bombing of bases, supplies and (troop) concentrations beyond the Yalu River, together with offensives on various parts of the present fronts.

Expansion of the war (by the Soviets) might not take place as a result of such actions, since the power shift (as a result of Mike) has resulted in disequilibrium in our favor. New excursions by the USSR on other parts of her periphery because of the boldness of the moves, partly supported by the new magnitude of available power, may be deferred.

In a recent speech, (USSR Premier Joseph) Stalin said that there was no need for him to fight because with time, the Western allies would fight among themselves. Our technological and power leadership demonstrated recently suggests that with suitable exploitation, the lie may be given to (his) statement by drawing closer the Allies, deciding the wavering neutrals, and even more, by bringing an even greater rift between the USSR and her conquered satellites.

It is proposed that initially the announcement be made thru (sic) the U.N. ... I suggest that the psychological impact and exploitation of the thermonuclear explosion may tend to unite the dissident Western U.N. members.¹⁷¹

To implement the Operation IVY security plan, the NSC suggested and President Truman agreed that all public announcements concerning Operation IVY would be made only by the AEC's Division of Information Services in Washington, D.C. The AEC notified its divisions and offices, and JTF 132, that public statements would come only from Washington, and that no statement or comment should be issued by any other source. Both DOD and the State Department told all their officers in Washington to make no statements and offer no comments on IVY to press or radio.

The commander of JTF 132 personally believed that an announcement should be made immediately after each shot. This would prevent foreknowledge of the scheduled time of each test, and might also serve as a "fire break" to allay more detailed speculation and prevent security leaks.¹⁷²

A detailed classification guide for IVY was issued; among other things, the guide specified that the proposed exact time -- month, day, and year -- of tests, the number of devices to be tested, and weapon types, were all "Secret -- Security Information." The classification guide was based on experience in previous U.S. nuclear test series, and on general AEC and military experience in limiting the spread of such information to only authorized personnel.

A meager public information plan was also prepared; permissible releasable information included only the season of the year in which the tests would be conducted. There would be no press conference after IVY, in contrast to lengthy conferences held after SANDSTONE and GREENHOUSE, and a joint AEC-DOD announcement on the thermonuclear test would be made only if there was widespread press comment.¹⁷³

In response to this plan, the commander of JTF 132 objected strongly, noting that

¹⁷¹Undated draft memo for President's Approval from Gordon Dean.

¹⁷²Moore and Bechanan, p. 184.

¹⁷³Moore and Bechanan, p. 184.

It appears most likely that the press will learn of the shots at an early date after the detonations and will publish stories of varying degrees of accuracy relating to the tests ..

...
It is exceedingly difficult for the Commander JTF-132, his deputies, and high-level observers to avoid press contacts or, when questioned, avoid comments to the press on matters which have appeared in the press ..¹⁷⁴

The task force commander reiterated his recommendation that a joint AEC-DOD press release be made after each shot. His suggestion was overruled.

Classification had proven effective on previous occasions, and it was believed that it would be adequate for IVY, even though the two shots to be detonated during IVY were both bigger than any previous U.S. nuclear explosion and despite the fact that Mike was to be the world's first large-scale thermonuclear test. Neither of these points could be legally communicated to most task force personnel who were not cleared to know them.

However, in what would turn out to be a regrettable oversight (at least from the AEC's viewpoint), the security plan specifically forbade any censorship of personal mail from task force personnel. Peacetime censorship -- the continuing war in Korea was only an undeclared "police action" -- could have been imposed only by the President or the Secretary of Defense; neither chose to do so during Operation IVY.¹⁷⁵

Before IVY, the AEC and DOD considered censorship of personal mail and/or issuance of an official advisory to press and radio stating the nature, in general terms, of the type of nuclear test information that should be kept from hostile foreign intelligence and requesting that such information, however obtained, not be published or broadcast. In this manner, self-censorship by supposedly-cooperative news media would, in effect, turn the media into ad hoc AEC security officers.

Censorship of mail, which was not imposed during either

¹⁷⁴Moore and Bechanan, p. 184.

¹⁷⁵Atomic Energy Commission, Security and Public Reporting Aspects of 1952 Pacific Tests, Note by the Secretary, November 28, 1952. Much of the following discussion of security violations is drawn from this document.

SANDSTONE or GREENHOUSE, would only call attention to the fact that IVY was particularly significant. Asking the news media to police itself would also only turn undue attention toward the operation; in addition, such a request would, in the AEC's words, "bring on a useless renewal of the acrimony over Government security of information measures."

Task Force 132 participants, both military and civilian, were fully briefed on their individual responsibilities for maintaining tight security. Self-censorship was stressed throughout the entire operation; before IVY began, participating JTF 132 personnel were specifically prohibited from discussing preparations for tests, number or types of tests, physical characteristics of the shots, exact date or time of any test, and types or numbers of ships, airplanes, or equipment used for the tests.¹⁷⁶

On October 30, just one day before the Mike shot, AEC Chairman Gordon Dean sent a memorandum to several of his division directors reminding them that "it is of the utmost urgency that there be nothing publicly issued concerning the Mike shot in the IVY series except in cases of life and death emergency or of great pressure in the psychological strategy field."¹⁷⁷

Despite the security briefings and warnings, there was early evidence that IVY security was far from air-tight. Even before the psychological strategy plan was issued, acting JCAE Chairman Carl T. Durham had stated in a newspaper article that the IVY tests "would be extraordinary in nature" and "would open the gravest era in the history of the world."

There were even more severe "leaks" during IVY. On the day of one test, inquiries to the AEC and DOD by TIME-LIFE magazine correspondents indicated knowledge of the exact shot time.¹⁷⁸

¹⁷⁶Memorandum dated 8 August 1952 to Distribution, from Commander, TG 132.1, Subject: Security Indoctrination for IVY; OPERATION IVY, Final Report, Joint Task Force 132, 1952, Headquarters, Joint Task Force Seven, Washington, D.C., 9 January 1953, pp. 26, 27, 28; IVY Information Plan - Letter to all JTF-132 Personnel, Subject: Discussion of Operation IVY Outside Joint Task Force 132.

¹⁷⁷Memorandum to General Manager, Director of Military Application, Director of Information Services, and Director of Biology and Medicine dated October 30, 1952 from Gordon Dean, Chairman, USAEC.

¹⁷⁸Anders, p. 231. The AEC replied "no comment" to the TIME-LIFE inquiry. The Moore-Bechanan history notes (p. 185) that on Mike day, November 1, 1952, a Secret-Security information

In addition, between November 10 and 15, there were 16 separate unauthorized newspaper accounts in the Los Angeles Examiner and in Associated Press and United Press wire service releases from Lima, Ohio describing the Mike shot, all occasioned by personal letters from task force members to relatives and friends in the U.S. that disclosed information that violated security instructions (an "injunction of secrecy" was issued to the recipient of one such letter). On November 17, AEC Chairman Gordon Dean stated that the letters "violated security regulations."¹⁷⁹

The impact of the Mike shot on JTF 132 personnel -- especially those with no previous exposure to nuclear explosions and who knew little of what was happening or what to expect -- was so overpowering and dramatic that "leaks" about it were nearly impossible to suppress.

As a result of these stories, AEC and DOD information officers were deluged with press inquiries and demands for an official press release. Nationwide press coverage described the Mike shot and discussed the AEC security practices. On November 17, an official AEC-DOD statement was issued, announcing an end to IVY and the inclusion of a thermonuclear experiment.¹⁸⁰ Press speculation subsided noticeably for the next few weeks, until speculation began about the next series of Pacific tests.

The AEC estimated that these security breaches might have allowed the USSR to collect fallout samples; disclosed that the Mike shot was the first full-scale test of a thermonuclear device (although no evidence was shown that the test device could be stockpiled); and compromised exact yield (within a reasonably large range).

Still left to protect after these disclosures were the exact yield of the shot; whether the U.S. had detonated a device or a deliverable bomb; specific nuclear composition and

message was transmitted in the clear (unciphered) from the USNS SHANKS giving the time of Mike detonation.

¹⁷⁹Letter dated January 21, 1953 to Barry Biederman from William L. Borden, Executive Director, JCAE. On November 22, the managing editor of the Los Angeles Examiner stated that he had consulted the AEC about his paper's November 10 story; the AEC confirmed this and stated that it had advised against publishing the story. (Moore and Bechanan, p. 185.)

¹⁸⁰Moore and Bechanan, p. 185; memorandum dated November 17, 1952 to All Field Managers from Richard G. Elliott, Director, Information Division, SFO, subject: Official Commission Announcement of Eniwetok Test Series 1952.

technical design details; results of experiments, including blast forces, transit time, alpha, gamma measurements, etc.; specific crater dimensions and damage-versus-distance information; and the exact current state of the U.S. thermonuclear weapons program and whether or not the U.S. had stockpiled thermonuclear weapons.

A nuclear explosion as big as Mike was impossible to conceal: it gave much evidence of its occurrence, in the forms of acoustic, thermal, barographic, hydrographic, visual, seismic, and radiological signals and residues; the rise of blast debris into the stratosphere guaranteed possible worldwide distribution of vital clues as to the composition and operation of the device.

Ironically, local Eniwetok weather at shot time helped maintain security better than task force personnel: the remnants of the Mike shot cloud remained near the atoll for seven days, delaying Soviet sample collection efforts. Much of the cloud "rained out" into the sea. Before the test, the AEC estimated that if the Soviets collected fallout, they would be able to deduce that a thermonuclear device had been detonated, but they would not be able to determine yield, based solely on fallout samples.¹⁸¹

U.S. analysis of Mike fallout samples at LASL revealed the presence of Pu-244 and other heavy elements unique to the shot; the presence of these materials revealed an especially high neutron flux which might have disclosed something of the nature of the device, i.e., that the deuterium was compressed before it was ignited.¹⁸² According to a panel of AEC scientists evaluating possible Soviet intelligence gains, if the idea of radiation implosion had occurred to the Russians, it was possible for trained scientists to calculate the degree of deuterium compression required to insure a

¹⁸¹Hewlett and Holl, p. 4; Minutes of AEC Meeting No. 767, October 16, 1952. The AEC needn't have worried: in his memoirs published in 1990, Soviet physicist Andrei Sakharov claims (p. 158) that U.S. fallout samples concentrated in melted snow collected after the Mike shot were "absent-mindedly poured down a drain." According to Dr. David Holloway, Soviet collection and analysis of radioactive debris from the U.S. Pacific tests were not far enough advanced at this time to permit much useful data to be derived from the Mike shot. ("Moral Leader of A Nation," book review by David Holloway in the BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 47 No. 6, July/August 1991, p. 37.)

¹⁸² Teletype dated December 2, 1952 to James G. Beckerley from Ralph Carlisle Smith.

thermonuclear reaction.¹⁸³

As damaging to national security as these disclosures may have been, they, like the enormous yield of the Mike device, would soon be eclipsed by an even worse breach of secrecy during Operation CASTLE 16 months hence. No prosecutions resulted from the IVY disclosures, possibly because litigation would only have highlighted the revelations, and risked further security breaches.¹⁸⁴

The commander of JTF 132 offered perhaps the best comments about IVY secrecy, and nuclear test secrecy in general:

It is the opinion of the Commander, JTF 132, that no Top Secret, Restricted Data information was leaked to unauthorized person or agencies as a result of violations of secrecy that occurred. It is practically certain that various individuals and press agencies had much more information than they published.

It is a recognized fact that the firing of atomic devices in Nevada cannot be kept secret. It is not quite so evident that such secrecy is impossible at the Pacific Proving Grounds. However, it is impossible to keep information from the public and the press of the large movement from the West Coast (sic) of personnel, supplies, equipment, and materiel by both sea and air -- and the movement of large numbers of aircraft and ships.

Furthermore, seismographic stations, long-range detection, and other scientific means well known to all scientists can definitely determine that a shot has been fired; can determine with a great degree of accuracy

¹⁸³Memorandum dated March 2, 1953 to Mr. Dean, Mr. Murray, Dr. Smyth, and Mr. Zuckert from Drs. H. A. Bethe, N. E. Bradbury, Edward Teller, and J. von Neumann.

¹⁸⁴Letter dated December 15, 1952 to Honorable Everett M. Dirksen, United States Senate, from Gordon Dean, Chairman, USAEC; letter dated May 13, 1954 to Representative Sterling Cole, JCAE, from K. D. Nichols, General Manager, USAEC. By 1979, during the PROGRESSIVE case, the DOE was altogether too willing to highlight improper release of classified information and risk further disclosures, which inevitably occurred.

its time; and can determine with at least a fair degree of accuracy its size and nature.

It is the conclusion of the Commander, JTF 132, that general information that preparations are underway for a test operation and, afterwards, that an atomic device has been fired, cannot be kept secret from the press and the public.

It is the conclusion of Commander, JTF 132, that great effort and considerable expense is expended in attempting to keep secret activities which cannot possibly be kept secret.

It is also the conclusion of Commander, JTF 132, that the significance of test operations at the Pacific Proving Grounds can be considerably de-emphasized by announcing before the tests -- in general terms as was done before IVY -- that a series of tests will be conducted during a coming general period; and, that the importance of innocuous security violations, which are almost sure to occur in one form or another, can be de-emphasized by announcing in general terms shortly after each shot that another atomic device has been fired.

That is generally the method used in Nevada because it cannot possibly be otherwise. Commander Joint Task Force 132 considers it a realistic approach to the problems to conclude that the same principles apply to the Pacific Proving Ground.¹⁸⁵

Despite this sensible suggestion, no such policy would ever be implemented by the AEC or the DOD, and these agencies would attempt -- usually unsuccessfully -- for the next several years to keep details of Pacific tests from the world.

The Wheeler Incident

A far more appalling potential disclosure of technical information about the Mike shot was an extraordinary security violation by Dr. John Wheeler, director of Project

¹⁸⁵Moore and Bechanan, p. 186.

Matterhorn, less than ten weeks after the test. The minor revelations by JTF 132 personnel paled to insignificance alongside it.

As part of a Congressional review of the AEC's performance of the U.S. thermonuclear program, Wheeler had occasion in early January 1953 to receive for his comments a compilation of some of the most sensitive and highly-protected top-secret classified nuclear weapons information in the United States.

Among other things, a document entrusted to his safekeeping revealed the "three concepts" of staging, radiation implosion, and pre-ignition compression of thermonuclear fuels, along with a succinct summary of the American thermonuclear weapons program, including the design and operating principles of the SAUSAGE (specifically the employment of a "sparkplug" to ignite the compressed secondary), important AEC device and code names, and a summary of a written "debate" in mid-1952 between Drs. Edward Teller and Hans Bethe on the pace of development of the U.S. hydrogen bomb program.

Wheeler misplaced a copy of the document on a train ride between Princeton, New Jersey, and Washington, D.C.; although the envelope that held the report was later located in Wheeler's office at Princeton, an exhaustive FBI and AEC search, including interrogation of passengers and crew that had been on the train, as well as a partial disassembly of a railroad car, failed to locate the top-secret paper (it was never found).¹⁸⁶

Wheeler's loss of the paper was a shocking and unforgivable transgression: documents classified this highly were supposed to be carried only by armed guards in locked private railroad car compartments.

There was some suspicion that Wheeler might have been

¹⁸⁶Memorandum dated January 27, 1953 to Bill Borden from Frank Cotter, subject: Professor John A. Wheeler, Loss of Document. A detailed account of this incident, which earned several very-embarrassed AEC commissioners a severe tongue-lashing from President Eisenhower, is described in ATOMS FOR PEACE & WAR, 1953-1961, Richard Hewlett and Jack Holl, University of California Press, Berkeley, California, 1989, pp. 37-41. Wheeler received only oral and written reprimands from AEC Chairman Gordon Dean; for this grievous misdeed, Wheeler's "Q" clearance was neither suspended nor revoked. By comparison, Dr. J. Robert Oppenheimer's "Q" clearance was suspended in December 1953 and later revoked in April 1954 for purely political reasons; Dr. Oppenheimer was never accused of a security breach as serious as Wheeler's.

followed by a Soviet agent intent on stealing the paper:

Mr. Williams stated that the case of the missing paper involving John Wheeler of Princeton is causing the (Atomic Energy) Commission considerable concern. It is claimed that it is not the fact that the paper contained numbers and precise data of classified nature as it is the revelation of nomenclature and phraseology which might be meaningful to foreign agents.

It was interesting that Mr. Williams apparently had deduced or had been informed that the prime deduction from the Wheeler episode was that Wheeler had been tailed by a foreign agent. Apparently, the General Manager's office was unaware of the many possibilities of the paper having been accidentally dropped or lost or that Wheeler's initial recollection of having it aboard the sleeper (car) was proven in error with the finding of the envelope in his office at Princeton which he thought that he had placed the paper in while on the Pullman.¹⁸⁷

Immediately after discovering the apparently irretrievable loss of the paper, the AEC convened a panel of scientists who were intimately involved with the thermonuclear weapons program -- Drs. Edward Teller, John von Neumann, Norris Bradbury, and Hans Bethe -- to evaluate the possible effect of the loss, and the potential usefulness of the paper to the Soviets. The panel concluded:

We have been asked to comment on the value which the document lost by Wheeler might have for the Russians if it has fallen into their hands. We shall not discuss whether it is likely that the document actually has come into their possession.

[deleted]

The lost document also states [deleted] was successfully tested at Eniwetok on November 1, 1952.

¹⁸⁷Memorandum to the Files dated February 12, 1953 from Walter P. Brobeck, subject: Weekly Conference with Walter J. Williams, Deputy General Manager, AEC, February 6, 1953.

[deleted]

Knowing the idea of the radiation implosion, it is possible for trained scientists to calculate the degree of deuterium compression needed to insure a thermonuclear reaction. It is stated in the document (second page, second paragraph) that radiation energy is use for compression, as is also implied by the name "radiation implosion" used throughout the document. How to engineer the flow of radiation from the primary bomb (to the secondary bomb) is a complicated problem but one to which a solution will certainly be found.

Very valuable would also be the information on the time scale of our efforts, [deleted]

[deleted]

If they had the idea before this time, then the lost document will not have added to their knowledge because it contains no detailed technical information.

If they did not have the idea (of radiation implosion) before this time, then the possession of the lost document could change their program from one with an unpredictable time scale to one which can be completed in a definite and limited time.

Apart from [deleted] the document reveals our interest in lithium-6. It makes it clear that we consider the separation of lithium isotopes as technically feasible, (which cannot be very surprising to anyone), and that Li-6 is regarded as useful for some thermonuclear device.

[deleted]

However, knowledge [deleted] revealed by a combination of the lost document and the Russians' own observations on the Mike shot debris, will be sufficient as a basis for a full-scale thermonuclear program.

We have also been asked whether we could recommend any definite steps following the

presumed loss of the document. Clearly, the most obvious step is to make every effort to find out whether the Russians actually got the document.

[deleted]

Concerning positive steps for our own thermonuclear program, there was agreement that the most fruitful step at this time would be an early increase in the production capacity for lithium-6, without prejudice as to the manner and the concentration in which this substance would ultimately be used.¹⁸⁸

In the late winter of 1953, AEC Commissioner Thomas E. Murray suggested that there be a further public elaboration of the results of IVY, partly in response to the published "leaks" by task force personnel:

If it is the case that no agency of the Government (sic) is now specifically charged with planning a program of domestic public information and foreign policy actions called for by the "Mike" shot, I recommend that a letter of the following kind be sent to the President:

Before the "Mike" shot of last November, the Government, upon the recommendation of the Psychological Strategy Board, adopted a policy of "silence" as to the results of the "Mike" shot.

¹⁸⁸Memorandum dated March 2, 1953 for Mr. Dean, Mr. Murray, Dr. Smyth, Mr. Zuckert from H. A. Bethe, N. E. Bradbury, Edward Teller, and J. von Neumann. Even more than forty years later, portions of the lost paper remain so sensitive that references to them are absent from this heavily-censored memo. To date, there has not been any evidence that the Russians actually ever saw the missing paper; however, the "three concepts" believed to have been possibly compromised in 1953 were savagely defended by the U.S. Department of Energy in 1979 during the PROGRESSIVE case (a battle the government ultimately lost). At least one government affiant during the PROGRESSIVE case, Dr. Hans Bethe, was well aware that the "three concepts" may have been irrevocably compromised 26 years earlier, yet he claimed that publication of Howard Morland's technically-incorrect article by the PROGRESSIVE would seriously damage U.S. national security.

However, the value of this silence technique was largely lost owing to security failures which forced a deliberately ambiguous official announcement of successful thermonuclear experiments.

In his final State of the Union speech former President Truman was only dissuaded from going into some details of the weapons implications of the "Mike" shot by the position that this matter was of such importance as to deserve close study by the incoming Administration. (sic)

We believe that in addition to the possible psychological warfare capital to be made out of the successful "Mike" shot, much remains to be done in educating the American public as to the advantages and disadvantages to the Nation (sic) which arose as a result of this test.

If no agency has been specifically charged with planning a program of domestic public information and foreign policy actions called for by the "Mike" shot, we recommend that a panel be established to develop such a program, with representation from the Departments of State and Defense, the Atomic Energy Commission, CIA, and with participation by such other agencies or consultants as may be appropriate.¹⁸⁹

There is no evidence that Murray's suggestion was ever acted upon by the AEC or the Eisenhower administration; however, a certain amount of the "domestic public information" called for by Murray for disclosure to the public was released piecemeal

¹⁸⁹Memorandum dated March 3, 1953 for Chairman Gordon Dean from Thomas E. Murray, Commissioner, USAEC.

within the Oppenheimer hearing transcripts declassified and published by the AEC in the summer of 1954.

Yet another incidence of possible violation of security surrounding Operation IVY apparently occurred in the summer of 1953 when congressional representatives James E. Van Zandt and Carl Hinshaw, both members of the JCAE, made remarks on August 10 regarding the results of IVY. Their comments drew a rejoinder from the AEC's Deputy Director of the Office of Classification who claimed that their remarks compromised secret restricted data.¹⁹⁰

The Spread of the "Three Concepts"

The success of the Mike test set off a race among other world nuclear powers to discover the "three concepts." The presence of einsteinium, fermium, and other high-atomic number transuranic elements, created by the incredibly-dense neutron flux in the highly-compressed secondary of the SAUSAGE could have tipped off the Soviets (and any other interested and technically-proficient party) about the likely operation of the device.

By thoroughly analyzing collected fallout, the Russians could have learned that the thermonuclear isotopes in the Mike device were compressed before their detonation, but the exact mechanism by which the compression was accomplished would not be obvious, except that it was probably not accomplished by means of chemical high explosives.¹⁹¹

At a meeting of the GAC in Washington, D. C. in February 1953, the GAC concluded: "The formation of these very special products (high atomic number isotopes) makes long-range detection of such a shot and inference as to its nature very easy for anyone in the world with requisite radiochemical equipment."¹⁹²

There is some question as to what access the Soviets had to the fallout from the IVY Mike detonation. Local weather kept the fallout from Mike over the Pacific for a week, during

¹⁹⁰Memorandum dated August 11, 1953 to Lewis L. Strauss, Chairman, USAEC, from C. L. Marshall, Deputy Director, Office of Classification, USAEC.

¹⁹¹Hirsch and Mathews, pp. 16-18; letter dated January 15, 1954 to Dr. J. Carson Mark, Director, Theoretical Division, LASL, from Paul C. Fine, Technical Assistant, Division of Military Application, USAEC.

¹⁹²Minutes of the Thirty-Third Meeting of the General Advisory Committee to the USAEC, Washington, D.C., February 5-7, 1953, p. 11.

which much of it fell into the sea.¹⁹³ In addition, as late as January 1954, the majority of the Mike debris remained unlocated¹⁹⁴ (which is not to say that it did not drift into the Soviet Union). A few months later, most of the fallout was believed to have fallen into the sea, with some of it remaining in the upper reaches of the stratosphere.¹⁹⁵

However, there was local fallout in the area of Eniwetok and it is possible that Soviet submarines might have collected samples. Russian seaborne collection capabilities were better at this time than their limited airborne high-altitude collection capabilities; an unidentified snorkel-equipped submarine had been sighted near Eniwetok during early March 1948 just before the SANDSTONE tests began.¹⁹⁶

Before IVY began, security plans assumed that, because of experiences during SANDSTONE and GREENHOUSE, foreign powers would attempt to gain information on JTF 132 activities by clandestine submarine observation.¹⁹⁷

Until the Soviets publish their own comprehensive nuclear weapons program history, this matter will probably remain conjectural. The Soviets collected fallout from Mike in snow, but apparently they did not then possess the requisite

¹⁹³Hewlett and Holl, p. 4. In an article in IZVESTIA on December 8, 1992, Yuli Khariton, Honorary Scientific Director, Federal Nuclear Center of Russia, stated that while it was true that the Soviets used airborne collection equipment to gather atmospheric samples from nuclear weapon tests, the Soviet collection system was not organized well enough by November 1952 to obtain useful results.

¹⁹⁴Minutes of AEC Meeting No. 950, Wednesday, January 6, 1954, p. 3; Minutes of the Thirty-Seventh Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, November 4, 5, and 6, 1953, p. 23.

¹⁹⁵Letter dated April 2, 1954 to L. L. Strauss, Chairman, USAEC, from W. F. Libby.

¹⁹⁶Lilienthal II, p. 301. According to Dr. David Holloway, the Russians began aerial collection of fallout from U.S. Pacific tests in 1954. (Letter from Holloway to author dated 25 January 1991.) It is reasonable to assume that the Soviets eventually developed a technically-proficient airborne radiological sampling capability to support their own nuclear weapons development program.

¹⁹⁷Operations Plan 1-52, Task Group 132.1, Commander, Joint Task Group 132.1, Joint Task Force 132, Operation IVY, 15 August 1952, p. E-3.

technological capability to analyze the material.¹⁹⁸ In addition, some radioactive elements also transmuted during their passage from Eniwetok to the USSR; if submarines had collected fallout samples promptly after the blast and preserved and transported them carefully, a better analysis might have been possible.¹⁹⁹

The Soviets apparently did not concentrate on a search for high atomic number isotopes but instead tested for the presence of uranium-237 and beryllium-7 which resulted from thermonuclear reactions. By the time the fallout reached Russia, these elements had decayed to only natural background levels.²⁰⁰

After the Soviets, the British were the next nuclear power to discover radiation implosion; the source of its discovery by the United Kingdom also remains in question:

Let me tell you one amusing incident. During the war, there was a group of people working at the (Los Alamos) laboratory, known as the British Mission. They were mostly theoreticians, some experimentalists, technical physicists, and so on, and special chemists -- an extraordinary, brilliant group of people. And they were right in the thick of things. Oppenheimer did not believe in compartmentalizing information, thank Heaven.

I copied him, and I didn't believe in it either. So the British knew everything that we knew up to about 1946, when the Mission went home and then the technical contact

¹⁹⁸"From the Interview Given by Yu. B. Khariton to the Journal 'Priroda'," note to "The H-Bomb: Who Really Gave Away the Secret," Daniel Hirsch and William G. Mathews, in Soviet Physics Usp. 34 (5) May 1991 (Uspekhi Fizicheskikh Nauk 161, 153-169), translated from the Russian (after originally appearing in English in the Bulletin of the Atomic Scientists, January/February 1990, pp. 22-30).

¹⁹⁹Hirsch and Mathews, and Andrei Sakharov, claim that the Soviets recovered fallout from Mike in snow collected in Russia.

²⁰⁰"Reminiscences of Different Years" by A. I. Pavlovskii of the All-Union Scientific Research Institute of Experimental Physics, translated to English and reprinted in Sov. Phys. Usp 34 (5), May 1991.

stopped. Sometime after the Mike shot (in November 1952), contact with the British had been re-established on purely fission things.

The British were allies, after all, a small, poor country. And so we had established some degree of technical assistance in this particular field. The discussions went on partly in England and partly here.

It soon became perfectly clear that the British had something else in mind besides the conventional implosion weapon. They hinted around about this and I would have made a small bet that what we were talking about was the same idea that formed the basis of Mike. So I asked if they would be willing to disclose the weight, size and shape of this thing and just what it looked like from the outside. They were, and did so -- just what you could see.

I then hastily asked for a recess because it was perfectly obvious they were discussing something that we had gone through great pains in inventing by Teller and Ulam and others. And they'd invented it too. Whether or not there was technical leakage was never learned.

In due course the Russians discovered it and the Chinese discovered it. So it's not something that if we hadn't done it, it would never have happened. It would have happened anyway, and I'm glad that we did it first.²⁰¹

²⁰¹Norris Bradbury, former Los Alamos Scientific Laboratory director, in REMINISCENCES OF LOS ALAMOS, 1943-1945, Studies in the History of Modern Science, Vol. 5, Lawrence Badash, Joseph O. Hirschfelder, and Herbert P. Broida, editors, D. Reidel Publishing Company, Boston, Massachusetts, 1980, pp. 169-170. This meeting apparently occurred in Albuquerque in 1958. A memorandum for the File dated October 27, 1952 by Gordon Arneson, Special Assistant to the Secretary of State for Atomic Energy Affairs, notes that the United Kingdom would be notified in the near future of the exact time and date of the IVY shots "in order to facilitate British and Canadian interception of the (shot) clouds," as had been done for previous test series. (This memorandum is reprinted in FOREIGN

More recently, further light has been shed on the development of the "three concepts" of thermonuclear weapons in the United Kingdom:

To my amazement, when (in 1955) I reached Aldermaston (the British nuclear weapons development lab, their counterpart to Los Alamos and Livermore), I was assigned the improbable job of uncovering the secret of the ULAM-TELLER invention (more accurately, of course, the ULAM invention), an idea of genius far beyond the talents of the personnel at Aldermaston, a fact well-known to both (William) Cook and (Sir William) Penney.

Under great stress, and with no assistance whatever, I came up with the correct scheme within six months, minor modifications excepted (and also obvious precautions). When presented at a subsequent meeting, a crucial one, judging by the full-dress uniform of the visiting Admiral, my proposal (with working drawings of a primary), the only one offered, was peremptorily rejected by Penney, who declared the matter not to be urgent anyway!

I was supported barely pro-forma, if at all, by Cook. Afterwards Penney demonstrated his complete lack of understanding of the problem in a private talk with Cook and myself. I was not invited to subsequent meetings held to discuss the project.²⁰²

RELATIONS OF THE UNITED STATES 1952-1954, Volume II, National Security Affairs, Part 2, U.S. Government Printing Office, Washington, D.C., 1984, pp. 1036, 1037.) This notification may have been given more for the purpose of allowing British and Canadian scientists to calibrate and improve their own fallout collection and nuclear test detection equipment, rather than to provide technical secrets of the U.S. test devices. The memo notes that "no inkling was given or asked, as to the nature of either shot."

²⁰²Attachment to letter dated 13 May 1985 from Professor Emeritus John C. Ward, theoretical physicist, School of Mathematics and Physics, Macquarie University, North Ryde, New South Wales, Australia, to The Honorable Mrs. Margaret Thatcher, Prime Minister of England. Professor Ward had

The first British thermonuclear device was detonated in May 1957 at Christmas Island in the Pacific Ocean. The first British fission device had been tested in October 1952 at the Monte Bello islands off Australia, about a month before the IVY Mike thermonuclear test by the U.S. at Eniwetok. British aircraft later collected cloud samples from U.S. thermonuclear blasts during Operation CASTLE in 1954 for radiochemical analysis in the United Kingdom.²⁰³

The "Emergency Capability" Program

The IVY Mike detonation, though technically very important, was nevertheless a device shot, just as the attempted "runaway burn" GREENHOUSE George shot of 1951 had been. It was not a test of a deliverable bomb, although materials used in the SAUSAGE cost less in dollars and fissionable materials than materials used in certain currently-stockpiled fission bombs.²⁰⁴

There were still three obstacles to be overcome before the AEC could produce fusion weapons for stockpiling. First, the logistic complications resulting from the use of cryogenic fuels had to be simplified or eliminated. Second, the relatively brief life span of the liquefied deuterium fuel charge would have to be overcome. And third, if the weapons were ever to be employed against anything other than the most unusual targets, their dimensions and weights would have to be reduced from their present gargantuan scale.

One estimate of the time between the successful IVY Mike test and a deliverable weapon was one-and-a-half to two years.²⁰⁵

earlier been led to believe that the discovery of the "three concepts" was "most urgent" at Aldermaston. Ward, a British citizen, resigned in protest of his treatment and returned to the United States.

²⁰³According to Hirsch and Mathews, the British received from the U.S. sometime after 1955 samples of fallout from the 1.6 megaton Soviet thermonuclear test of November 22, 1955. Judging by Penney's behavior, as related by Ward, it is possible that the British might have had more than one group working simultaneously to duplicate the Teller-Ulam concepts, or perhaps analysis of the samples of Soviet and American fallout provided by the U.S. had already allowed the British to deduce the "three concepts."

²⁰⁴"Some H-Bomb Policy Issues as of January 1, 1953," prepared by the Joint Committee on Atomic Energy, United States Congress; "A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, p. 16.

²⁰⁵Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR

Success was nearer than expected: LASL was already interested in the concept of a solid-fueled thermonuclear bomb. The device was given the name ALARM CLOCK, but it had little in common with the previous designs by the same name.²⁰⁶ By the summer of 1952, two candidate thermonuclear warheads (the EC14 and EC16) were in development engineering at Los Alamos.²⁰⁷

Even before the IVY Mike test, the Air Force had believed that, with intensive effort, the U.S. could produce in a few years a relatively inexpensive and easily-deliverable thermonuclear weapon which could be used against any potential target. Ten such weapons, it was thought, could destroy the urban centers of Russia, or if used by the Soviets, the great cities of the U.S. It was therefore imperative for the United States to develop this capability before Russia could wreak the same havoc on America.

An important new requirement for high-yield thermonuclear weapons would be some form of retarding mechanism, to allow the delivery aircraft time to escape the blast following detonation. Development of retarded-fall devices was a high priority with the Department of Defense at the beginning of 1952.²⁰⁸

On March 14, 1952, the Air Force Chief of Staff forwarded a memorandum to the Secretary of the Air Force which included the following statement:

Although the current effort on H-bomb development in the AEC is substantial, it is the belief of the Air Force that this effort is far from the maximum capability of our country. Nothing less than our maximum effort will suffice in this crucially important field. The principal

PROGRAM, p. 14.

²⁰⁶AF ATOMIC ENERGY PROGRAM, p. 201. The latest incarnation of the ALARM CLOCK was eventually dubbed TX-14. See Volume VI. The earlier ALARM CLOCK had proven feasible after it was redesigned to incorporate radiation implosion and solid lithium-6 deuteride. (Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, p. 12; Teller, COMMENTS ON BETHE'S HISTORY OF THE THERMONUCLEAR PROGRAM, p. 2.)

²⁰⁷NUCLEAR WEAPONS DATABOOK, Vol. II, p. 16.

²⁰⁸Report on Increased Military Participation in the Non-Nuclear Portion of the Atomic Weapons Field, Joint Atomic Energy Commission - Department of Defense Study Group, ca. January 15, 1952, p. 7.

requirement for faster progress is a major augmentation of the level of effort on research and experimentation, over and above that presently being carried out at Los Alamos, particularly in terms of arriving at a proven weapon design which is matched to our Strategic Air Weapons System.²⁰⁹

A few weeks later, on April 29, the Air Force stated clearly its thermonuclear weapons objectives:

The objective of the U.S. Air Force is to achieve a capability for delivery of thermonuclear weapons at the earliest practicable date. To attain this objective, development of thermonuclear bombs should be predicated on achieving maximum weapon yield (while) being compatible as to size, weight and shape for delivery by the (B-62) SNARK, the B-47, the B-36, and the B-52.²¹⁰

By this time, the LASL ALARM CLOCK held promise of being a portable thermonuclear weapon. A group at the lab, led by Lothar Nordheim, was striving to meet Air Force size and weight specifications.

In the middle of May, the AEC urged that Sandia be given a bigger role in weaponizing the TX-14:

It is evident, that if we really have meant what we have been saying about LASL and Sandia being a team we require in the weapons field, that we better cut Sandia Corporation in on the thermonuclear business immediately and particularly so if you really expect to produce ALARM CLOCK devices as previously discussed in the quantities mentioned.

Let us look at the situation as it now stands. ACF is supposed to begin design of a droppable weapon this summer. If this weapon functions correctly, we are supposed to be in a position to produce a quantity now supposedly governed by the Li-6 output versus (the) amount required

²⁰⁹AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 225.

²¹⁰AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 211.

by each weapon.

Immediately questions arise as to who is going to produce the fuse, the associated equipment for testing it, who does the ballistics so that we can hit Eniwetok if we drop one? Do they propose to build only one? If they want to drop the first ALARM CLOCK, why I don't know, they better get down to the hard facts of life on how one produces a weapon.

It is strange but it seems to me that this is right where we stood in 1944 in the fission business. The reluctance of LASL to let Sandia do this I do not understand and do not believe we should countenance. If the present setup with LASL free-wheeling with Wright Field and other agencies is how they expect to do the job (then) I have no hope (that) it will be done in the time required.

I believe we should direct Santa Fe Operations office (of the AEC) to have Sandia Corporation take over the ordnance engineering of this job under the technical guidance of LASL and produce a weapon. If we lack confidence in Sandia Corporation to perform this task, we certainly should tell them so and in what respect they are deficient.²¹¹

It appeared at this time that the B-47 would be a far more suitable thermonuclear delivery vehicle than the B-36, and that there was a 50% chance of having a deliverable hydrogen bomb in 1954.²¹²

Soon afterwards, on May 22, an "emergency capability" committee composed of representatives from Los Alamos, Sandia, and the Air Force Special Weapons Center was established to determine the earliest possible date for the deployment of thermonuclear weapons.²¹³ These were to be "quick and dirty"

²¹¹Memorandum dated May 13, 1952 to General Fields from Captain Hayward, subject: Thermonuclear Weapon.

²¹²Minutes of the Thirtieth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, April 27, 28, and 29, 1952, pp. 30, 32.

²¹³"Timetable of Early Thermonuclear Weapons Events," U.S.

TX-14 ALARM CLOCK bombs,²¹⁴ little more than untested weapon prototypes without drogue parachutes and with fuzing and arming systems simplified to the point of being hazardous to their delivery crews. The U.S. nuclear armsmiths' race to beat the Russians was still being run in deadly earnest.

On the same day, Norris Bradbury and Sandia president Donald Quarles wrote to the AEC's Director of Military Application and suggested that the hardware for TX-14 ALARM CLOCK devices be developed as an "emergency capability," i.e., a capability to exist as soon as the necessary lithium-6 were available to insert into the hardware. Dr. Bradbury further proposed that the hardware be designed so that the ALARM CLOCKS could be delivered by existing Air Force bombers.²¹⁵

Technically, the precise meaning of an "emergency capability" device was that if war required use of the weapon, it would have no geometric weight or engineering characteristic which would preclude its being carried on certain manned U.S. aircraft of a type flying operationally at the end of 1953. The device was to be designed so that it could be ground-handled, loaded, and suspended from an aircraft with only slight modification of currently-operational bomb transport, loading, and suspension equipment.

"Emergency Capability" was also understood to preclude standard operational suitability testing, reliability testing, and in-flight insertion capability. The weapon would have to be assembled and readied for drop by technicians before takeoff of the delivery aircraft.²¹⁶

This philosophy paralleled closely that of the preparation for use of the first atomic bombs during World War II. An "emergency capability" weapon could also weigh no more than 50,000 lbs., the maximum bombload a B-36 could carry and still fly to a Soviet target farthest from an overseas SAC base.

Department of Energy, Albuquerque Operations Office,
Albuquerque, New Mexico, September 22, 1986.

²¹⁴Borden-Walker H-bomb chronology, p. 82.

²¹⁵Borden-Walker H-bomb Chronology, p. 82; letter dated May 22, 1952 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL, and D. A. Quarles, President, Sandia Corporation.

²¹⁶Letter dated June 23, 1952 from Gordon Dean, Chairman, USAEC, to Honorable Robert LeBaron, Chairman, MLC. This letter is a slightly-rewritten, almost verbatim version of the Bradbury-Quarles letter of May 22, 1952 to Brig. General K. E. Fields.

In June, the Joint Chiefs of Staff sent to the AEC a formal military requirement for a thermonuclear weapon with a megaton-range yield that would be compatible in size, shape, and weight with available 1954 delivery systems. The JCS stressed the urgency of beating the Russians in the race to produce a deliverable weapon, even if considerable risks had to be taken:

The Joint Chiefs of Staff have directed a review and analysis of the known and prospective targets and target complexes in order to determine the number of instances in which the use of very high yield (500 KT to 40 MT) weapons would be more effective and/or efficient.

It is expected that this review will have application to requirements for very high yield weapons of both fusion and fission types. It will not be possible for the Joint Chiefs of Staff to establish specific production requirements for thermonuclear weapons until the results of this review and analysis have been received and evaluated.

The Joint Chiefs of Staff have requested that you be informed of the following:

a. A military requirement exists for the development of thermonuclear weapons giving yields of one megaton upward; a military requirement exists for the production of such weapons provided that they are compatible in size, shape and weight with delivery systems that will be available in 1954.

b. Upon completion of current target analysis, the Joint Chiefs of Staff will be able to establish more specific requirements in terms of annual rate, estimated totals, and desired yield ranges.

c. Pending completion of the aforementioned studies, the Joint Chiefs of Staff considered that the magnitude of this requirement is sufficient to justify the currently-planned

production of lithium-6. However, based on preliminary studies, it is anticipated that the requirement for these weapons may be considerably greater than indicated by these production facilities. Plans for essential component production facilities should be developed.

d. The Joint Chiefs of Staff urge that the Atomic Energy Commission expand and intensify the exploitation of the thermonuclear reaction wherever its use may be profitable throughout the weapons field.

e. The Joint Chiefs of Staff wish to emphasize that prior production of a deliverable thermonuclear weapon by the USSR would serve to reduce the present U.S. preponderance in atomic weapons, and that such a shift in balance might exert strong influence on Soviet policy decisions. This factor, when taken together with the promise of U.S. thermonuclear programs, is considered to provide adequate justification for an approach which may represent considerable technical risk and large expenditures of funds.

f. The target studies, and hence the resultant requirements, will of necessity be based on assumptions that may be subject to change as additional technological information on these weapons becomes known. Therefore, the Joint Chiefs of Staff wish to be periodically informed concerning Atomic Energy Commission action and progress in the thermonuclear field.²¹⁷

²¹⁷Letter dated June 13, 1952 to Gordon Dean, Chairman, USAEC, from Robert LeBaron, Chairman, MLC; Atomic Energy Commission, The Thermonuclear Weapon Test Program, Report by the Director

At the time, there were two ways to meet this requirement: with either a true multistage thermonuclear device using the as-yet untested Teller-Ulam operating principles, or with a very large semi-thermonuclear fission device (called the LITTLE EDWARD) that would contain a substantial amount of thermonuclear fuel but which was little more than an enlarged version of the BOOSTER tested during Operation GREENHOUSE.

While the latter approach offered a quicker and more certain route to meet the JCS requirement, it was large, unwieldy, of limited yield, and did not offer advantages for future development. Its weight made it unacceptable for use as an ICBM warhead.

A true Teller-Ulam device, however, if feasible, would offer many more opportunities for diversification, such as warheads smaller or more powerful than the LITTLE EDWARD. Both designs could use either "wet" or "dry" fuels.²¹⁸

By the middle of June, the AEC was planning to follow the IVY series with a test of a solid-fueled device that was designed to be used immediately as a weapon. The emergency fabrication and stockpiling of weapons was based on the assumption of success in the 1953 test, the date then set for CASTLE.²¹⁹

Two devices were contemplated for a fall 1953 test (CASTLE), including the current incarnation of the ALARM CLOCK. These devices were viewed as fulfilling the emergency capability requirements. The working assumption at this time was that CASTLE would be successful, with one or more multimegaton devices, and that several "emergency capability" devices would be constructed on a philosophy like that relating to the FAT MAN in 1944, namely, as hand-built improvisational units lacking some modern safety features.²²⁰

The Air Force began logistical planning for thermonuclear weapons in mid-1952. The chairmen of the AEC and the JCAE negotiated a series of agreements which instructed the Air

of Military Application, attachment to Operation CASTLE, Note by the Secretary, December 22, 1952.

²¹⁸Hewlett and Holl, pp. 163, 164; J. Carson Mark interview, LOS ALAMOS SCIENCE, Number 7, Winter/Spring 1983, p. 37.

²¹⁹Progress Report to the Joint Committee on Atomic Energy, December 1951 through May 1952, USAEC, p. 8.

²²⁰Minutes of the Thirty-First Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, June 13 and 14, 1952, p. 5.

Force to work directly with the AEC in the buildup of an early, or "emergency," thermonuclear strike capability.²²¹

The TX-14: The First Emergency Capability Weapon

The first device considered for "emergency capability" weaponization was the TX-14 ALARM CLOCK. Los Alamos would design, develop and fabricate both the internal and external components of the warhead, as well as its handling equipment. Sandia was responsible for the tail assembly and fuzing system, and for bomb ballistics and the design and fabrication of dummy weapons for testing.

The Air Force Special Weapons Center (AFSWC) at Kirtland AFB in Albuquerque undertook modification of ConVAir B-36H and Boeing B-47B aircraft for delivery of the TX-14. The Special Weapons Center was to develop a safe delivery technique and all ground-handling equipment for the bomb. As a longer-range objective, AFSWC was to develop a drogue parachute for production models (the TX-14 without a parachute was dubbed "EC14," for "emergency capability"). Los Alamos, Sandia, and the Special Weapons Center were to collaborate in developing a master operating plan for readying the weapons for strike missions.²²²

A conference of representatives from Boeing, ConVAir, and AFSWC concluded that the weight of the TX-14 should not exceed 50,000 lbs., with a diameter of 62.5" and an overall length of 20 feet. These dimensions were transmitted to Los Alamos, received the concurrence of LASL, and were accepted by the AEC and the Air Force as final maximums.

Two Air Force bombers, the B-36 and B-47, were expected to be able to carry a bomb of these dimensions with relatively slight modifications. The B-36 was by far the easier plane to modify and the bomb bays of two were to be modified and readied for test drops with dummy bombs by the end of 1952. Two B-47s were also to be modified by year's end.

The Air Force expected to have 200 B-47s operational by the end of 1954; however, the B-47 was hard-pressed to lift a 50,000 lb. bomb. A fully-loaded B-47 weighed 185,000 lbs. and very little fuel could be carried if a 50,000 lb. bomb was

²²¹HISTORY OF ATOMIC LOGISTICS, Frederick A. Alling, Historian, Historical Division, Office of Information Services, Air Materiel Command, Wright-Patterson AFB, July 1956, p. 117.

²²²Borden-Walker H-bomb chronology, p. 82; Atomic Energy Commission, Progress Report to the Joint Committee, Part III - Weapons, June 1952 through November 1952, AEC 129/46, December 17, 1952.

aboard. Range improved dramatically with a 20,000 payload.

At this time, when carrying 25-ton bombs, the B-36 was exceedingly vulnerable to fighters and the B-47 was so "short-legged" (i.e., short-ranged) that successful delivery of these enormous bombs was very much in doubt.

On a more favorable subject, bomb loading was not expected to be a problem. The dynamics of a 50,000 lb. bomb were such that it had to be carried in a sling in the aircraft bomb bay. Loading a 25-ton bomb was not an easy matter, but the Air Force had developed during World War II a monstrous hydraulic lift called the B-2 to lift bombs weighing up to 22 tons. **Figure IV-14** shows a B-2 lift hoisting a "blockbuster" conventional bomb shape.

These devices, of which there were only 10 by the fall of 1952, could be modified to lift heavier weights, and two were to be ready for test at Kirtland AFB by the beginning of October. If the B-2 lifts had not existed, it would have been necessary for the Air Force to spend time and money developing an adequate substitute.

Despite orders placed in 1953, there would still be only 10 B-2 lifts in service at USAF bases by January 1955.²²³

Because of logistic problems and the dependence upon the B-2 lifts, "emergency capability" thermonuclear strike missions would have to originate at Kirtland AFB. Presumably, there would be a limited number of advance bases, such as Limestone AFB near Limestone, Maine, which could be used as "stepping stones" to targets overseas. (Limestone was re-named Loring AFB in October 1954.) Facilities to accommodate B-36 operations at Limestone had been completed in December 1950, and the 42nd Bomb Wing of SAC's 8th Air Force was assigned to Limestone in February 1953).²²⁴

By early July, the MLC was depending heavily upon the TX-14 ALARM CLOCK to provide early emergency thermonuclear weapons capability to the Air Force, including possible service as a high-yield weapon for the drone B-47 program of Project BRASS RING:

In reply to your letter of June 23, 1952,

²²³Alling, HISTORY OF ATOMIC LOGISTICS, p. 127.

²²⁴AIR FORCE BASES, Vol. I, ACTIVE AIR FORCE BASES WITHIN THE UNITED STATES, Robert Mueller, United States Air Force Historical Research Center, Office of Air Force History, Washington, D.C., 1989, pp. 327, 328, 329; September 12, 1952 Walker memorandum, pp. 18, 19.

the Department of Defense concurs in the decision to provide an emergency capability on the assumption that the first test of the ALARM CLOCK will be successful. It is clearly recognized that this approach is necessary to fulfill the objectives of the Department of Defense to achieve a capability for delivery of thermonuclear weapons at the earliest practicable date.

In this regard, it is our understanding of your statement of design parameters that the (Atomic Energy) Commission will produce for the Department of Defense a limited number of TX-14 weapons compatible with the B-47 and B-36 as an emergency capability in early 1954. Of course, yield is not a limiting factor in the unmanned B-47, and we would like to be kept informed should yield limitations influence the development time scale appreciably.

The Air Force has delegated to the Director of Research and Development, Headquarters, United States Air Force, the responsibility for insuring that the development of a suitable (bomb) carrier and ancillary equipment proceeds concurrently with the development of the TX-14 by the Atomic Energy Commission.

...

In the early stages of this program, the Department of Defense appreciates that the limiting quantitative factor in obtaining an emergency thermonuclear capability is not the target requirement, but rather is the Atomic Energy Commission development program and production facilities. The statement of quantitative requirements for these weapons is properly a matter for decision by the Joint Chiefs of Staff. The Military Liaison Committee will await your further study and evaluation of the unknown factors involved in the development and production phases of this program.²²⁵

In the meantime, the Wright Air Development Command in Ohio had developed a new ribbon parachute which would prolong the

²²⁵Letter dated July 3, 1952 to Gordon Dean, Chairman, USAEC, from Robert LeBaron, Chairman, MLC.

time-of-fall of a thermonuclear weapon to 200 seconds. After July 1952, several test bomb models were airdropped successfully with the new chutes. Results showed that there was now sufficient time for the delivery aircraft to escape before detonation, and that the B-36 could deliver a parachute-retarded bomb about as accurately as a conventional high explosive bomb.²²⁶ In view of these developments, there was no longer any need for delivery by drone aircraft, and the BRASS RING project was canceled in the spring of 1953.²²⁷

As long as "emergency capability" was limited to the TX-14, the Army and Navy had no capability for thermonuclear weapons delivery, although both services desperately wanted smaller atomic weapons for delivery by short-range missiles and carrier-based aircraft. These services were excluded from the thermonuclear weapons program even after LASL expanded the "emergency capability" program to include the TX-16 and TX-17 weapons (dubbed "EC16" and "EC17," respectively; the EC16 was a liquid-fueled weaponized version of the IVY Mike device), since both of these bombs had the same dimensions and weight as the TX-14.

By September 1952, two months before the IVY Mike shot, the outlook for future thermonuclear weapons appeared to be very favorable; materials availability was a bigger problem than device feasibility:

"Feasibility" of a "thermonuclear weapon" not only is established but undoubtedly has been established for some time. Not only is an H-bomb apparently feasible, but a variety of H-bombs of varying types appear feasible. Quantity production is feasible. A technical point may have already been reached at which (the U.S.) atomic energy program could henceforth rely on hydrogen weapons as the primary instruments of our defense.

At the present time, the thermonuclear program in general has increasing momentum. Los Alamos is probably spending the majority of its time and enthusiasm on thermonuclear developments. The "second laboratory" has been launched. The caveat, of course, is that in a major transition period a number

²²⁶Knaack, POST-WORLD WAR II BOMBERS, p. 128.

²²⁷AF ATOMIC ENERGY PROGRAM, Vol. IV, pp. 213-215. For a summary of BRASS RING, see Knaack, POST-WORLD WAR II BOMBERS, pp. 125-128.

of things can go wrong -- and there are many opportunities for slips.

There is an immediate pressing issue on the production of H-bomb materials which must be met squarely and immediately. First, a decision must be made as to how much lithium production above and beyond the present Oak Ridge plant should be planned. As to tests, lithium-6 (production) is a bottleneck for all H-bomb tests now scheduled; quantity production obviously requires major new facilities.

Secondly, an immediate decision should also be reached on the amount of tritium production over and above all present excess reactivity. Tritium is a clear bottleneck in our quantity production of H-weapons, though not to the test program.

Boosted weapons appear feasible, and in quantity for stockpile. The issues (of) materials mentioned above must also be met for boosted weapons, completely apart from H-bombs.

Decisions on the materials shortage go directly to (1) top-level planning for the H-bomb program in the (Atomic Energy) Commission and (2) military requirements. Requirements for numbers of H-bombs and for quantity production of H-bomb materials are still not established. Furthermore, the prospect of boosted fission weapons must be met on the DOD-AEC level quite apart from H-bombs. If a real H-bomb capability is to be achieved without further delay, requirements and objectives must keep pace with the technical outlook. Delay now will probably come from Washington (rather) than the laboratories.²²⁸

Even with the planned production of a lithium facility at Oak Ridge, the availability of sufficient highly-enriched lithium was also expected to be a stumbling block to weapons production. While this problem was of considerably lesser magnitude than the earlier problem of a Super design

²²⁸September 12, 1952 Walker memorandum, pp. 1, 2. Results of Operation CASTLE substantially reduced tritium requirements for thermonuclear weapons.

requiring three to five years worth of tritium production for each weapon, lithium-6 was still a critical element in the American hydrogen bomb program.

Without further construction of enrichment plants, a year's total output of lithium plus the nation's entire present supply of tritium, starting in the fall of 1953, would still be required for the "emergency capability" which the AEC and DOD jointly wished to achieve by the time of the scheduled Operation CASTLE tests in the Pacific. If the CASTLE tests were successful, and if lithium and tritium production were not expanded, then "emergency capability" would not be at hand until late in 1954.²²⁹

A new, smaller thermonuclear bomb, the TX-15 ZOMBIE, was proposed late in 1952. This device was relatively small and light, and could extend thermonuclear capability to the Army and the Navy. LASL estimated that the TX-15 would weigh 4,000 lbs., with a diameter of about 30 inches, and a length that would not exceed 100 inches. Its yield was estimated at 250 kilotons. Here was a small thermonuclear bomb that could be used for tactical purposes, deliverable by fighter planes.²³⁰ At the same time, it offered the prospect of a family of lightweight, solid-fueled high-yield hydrogen weapons.

This possibility reopened the interservice contentions that had marked the development of lightweight, low-yield fission bombs. The Air Force considered carrying the TX-15 in an F-101A for tactical delivery; the Chief of Naval Operations expressed an immediate interest in the deployment of the TX-15 aboard aircraft carriers. There were even rumors that the Navy would ask the Joint Chiefs of Staff for permission to develop the non-nuclear components of the weapon (thus usurping a function of the Sandia Corporation).²³¹

During a secret briefing to then-President-elect Dwight D. Eisenhower in November 1952, after the IVY Mike shot, AEC Chairman Gordon Dean stated that the AEC planned to test components of a deliverable thermonuclear weapon at the Nevada Proving Grounds during the spring of 1953 and to achieve an "emergency capability" after a full-scale test in the Pacific early in 1954.²³²

²²⁹September 12, 1952 Walker memorandum, pp. 23, 27.

²³⁰September 12, 1952 Walker memorandum, p. 9.

²³¹AF ATOMIC ENERGY PROGRAM, Vol. IV, pp. 218, 219.

²³²Hewlett and Holl, p. 41.

The Air Force and the Hydrogen Bomb: Plans for War

By this time, the hydrogen bomb figured prominently in USAF war plans:

(Headquarters USAF) made clear its position "... that all of us in the Air Force have been a little slow in grasping fully the impact which atomic weapons will have upon the next war, if it comes." Furthermore, should the hydrogen bomb prove successful, the impact would be even more devastating.

A study prepared by SAC's operations analysts demonstrated that the use of hydrogen bombs against disruption targets would increase each bomber sortie effectiveness by a factor between three and ten over the (use of) atomic bombs. Therefore, aircraft capable of delivering the hydrogen bomb could deliver the required crippling blows against an enemy in from 1/3 to 1/10 the number of sorties required by aircraft carrying atomic bombs.

Headquarters USAF firmly believed that the ability to deliver the hydrogen bomb could mean the difference between victory and defeat because of the terrible havoc which could be wrought during the first crucial days of a war.

If development of hydrogen bombs continued unabated at this time, megaton yield weapons would be available for use in late 1953 or early 1954. The B-47 was believed to be incapable of delivering the hydrogen bomb, leaving only the B-36 in the existing USAF weapons inventory as a likely delivery vehicle.²³³

Late in November, the JCAE queried no fewer than 55 persons currently or formerly associated with the U.S. nuclear weapons program as to how much of a lead, if any, the U.S. had over the Soviets in thermonuclear weapons development.²³⁴

²³³HISTORY OF THE STRATEGIC AIR COMMAND, THE B-52: BACKGROUND AND EARLY DEVELOPMENT, 1946 - 1954, Historical Study No. 60, Historical Division, Office of Information, Headquarters Strategic Air Command, pp. 43, 44, 47.

²³⁴Among individuals polled by the JCAE were Robert Bacher, Hans Bethe, Norris Bradbury, Oliver E. Buckley, Vannevar Bush, James B. Conant, Gordon Dean, Lee A. DuBridge, Darol

Many of these people did not have access to classified CIA or AEC intelligence estimates, while several did.

The consensus was summed up on the basis of 22 replies: three estimated a U.S. lead of from three to six years; seven estimated between two to three years; eight estimated one to two years; and four estimated U.S. lead time between no time at all and one year.²³⁵

By the end of 1952, fabrication of the ALARM CLOCK hinged mainly upon the production of enriched lithium-6 by Oak Ridge; overall design of a deliverable TX-14 was underway but would not be firm for several months. Dummy models had been constructed for ballistic and fuzing studies, and drop tests had started in November (see MK 14 history in Volume VI).²³⁶

At the start of 1953, proposals were advanced to use hydrogen bombs as the United States' principal weapon against enemy airfields, naval bases, and cities which might be used to launch attacks on the nation.

Two other specific proposals had also been advanced. First, that an urgent effort be made to have thermonuclear weapons ready in time for tactical use against North Korean and Chinese troops in Korea, assuming that sufficiently-large concentrations of infantry, armor, or artillery would be vulnerably located for attack.

Second, DOD proposed that hydrogen bombs be the central weapon of North Atlantic Treaty Organization (NATO) allies for defense of western Europe against Soviet invasion. These proposals raised the question of whether hydrogen weapons would be used for special purposes or whether they would be a major addition to the NATO arsenal.

The status of national policy on the hydrogen bomb, as of the beginning of January 1953, could be summarized as follows: (1) Testing of prototype bomb models was rescheduled from the fall of 1953 to the spring of 1954; (2) such hydrogen bombs

Froman, Alvin Graves, Marshall G. Holloway, Eric R. Jette, Robert LeBaron, Enrico Fermi, David Griggs, E. O. Lawrence, Willard Libby, Herbert B. Loper, Duncan P. MacDougall, Carson Mark, Thomas Murray, J. Robert Oppenheimer, Kenneth Pitzer, I. I. Rabi, Hartley Rowe, Glenn Seaborg, Cyril Smith, Henry D. Smyth, Lyman Spitzer, Lewis Strauss, Edward Teller, Harold Urey, John von Neumann, John Wheeler, and Herbert York.

²³⁵Borden-Walker H-Bomb chronology, p. 90.

²³⁶Atomic Energy Commission, Progress Report to the Joint Committee, Part III - Weapons, June 1952 through November 1952, AEC 129/46, December 17, 1952.

as could be made from the \$35 million lithium-6 production plant under construction at Oak Ridge were required to be in stockpile by January 1, 1955; (3) no quantity requirements for thermonuclear weapons had been made by either DOD or AEC; and (4) a decision had been reached, and approved by President Eisenhower, that the U.S. program for reducing the time between bomb prototype testing and stockpiling would not include early construction of additional lithium-6 production plants.²³⁷

In February 1953, the TX-16 and TX-17 were added to the "emergency capability" program. Both the TX-16 and TX-17 were designed to have identical external configurations, and production models of the TX-14, TX-16, and TX-17 were expected to have interchangeable nose sections and afterbody assemblies.²³⁸

The "emergency capability" program was showing results by the spring of 1953. Once the new drogue parachute had eliminated the need for delivery by drone B-47s and the yield effects of thermonuclear weapons had been worked out, it was no longer necessary to modify B-47s. They were acceptable thermonuclear weapons carriers as they existed, but they could not carry the first generation of large, heavy, high-yield bombs.

The Mike shot had also produced one very welcome result: an October 1952 Air Force nuclear weapons effects handbook had predicted a 38-second thermal pulse from a 40 megaton bomb; after IVY, this prediction was downgraded to about 15 seconds.²³⁹

The cockpit of an airborne XB-47 weapons effects aircraft exposed to the 81 kiloton GREENHOUSE Dog shot became intensely warm, the metal in the cockpit becoming almost too hot to touch. Another B-47 flown during the IVY Mike shot used a retractable, aluminized heat-reflecting thermal curtain in the cockpit to avoid a similar situation.

Since the new B-52 was not scheduled for production until 1954 (the first B-52s were not deployed to SAC until June 1955), and the B-47 could not carry much more than 22,000

²³⁷"Some H-Bomb Policy Issues as of January 1, 1953," prepared by the Joint Committee on Atomic Energy, United States Congress; Borden-Walker H-Bomb Chronology, p. 91.

²³⁸Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1953 - 30 June 1953, p. 186.

²³⁹King, pp. 43, 44, 53.

lbs. for any distance without in-flight refueling, the Air Force concentrated its modification program on the B-36. The first of these aircraft, readied for thermonuclear weapons delivery, arrived at Kirtland AFB on November 3, 1952; on November 28, Headquarters USAF authorized modification of 36 of the bombers, with 20 of them to be completed by December 1953.²⁴⁰

In a short time, this number was raised to 108; later, USAF decided to modify all B-36s for thermonuclear weapons delivery, except 12 which were set aside for carriage of the RASCAL air-to-ground missile. The modification program was to be completed according to the following schedule: 20 by February 1, 1954; 101 during the remainder of 1954; and 87 between January and June 1955, for a total of 208. Thus the Air Force planned to pass from the "emergency capability" phase to the normal delivery phase simultaneously with the attainment of "emergency capability" weapons.

By May 1953, only four B-36s had been modified. (In March 1954, even the 12 B-36s originally reserved for missile carriage were added to the modification program, making a new total of 220 aircraft.)²⁴¹

In May, LASL sent the first TX-14 Trial Run Model (TRM) to Kirtland. At the time, the weapon had the following physical characteristics:

Diameter	61.4 inches (5'1.4")
Length	222.0 inches (18' 6")
Weight with drogue chute	29,851 lbs.
Weight without chute	28,954 lbs.

The TX-14 used a 64-foot diameter drogue chute. The chute's pack weight with pilot chute and deployment hardware was on the order of 900 lbs.

This TRM, in conjunction with additional practice models and the availability of four modified B-36s, allowed the Air Force's 4925th Test Group (Atomic) at Kirtland to begin

²⁴⁰Minutes of the Thirty-Fifth Meeting of the General Advisory Committee to the USAEC, May 14-16, 1953, p. 25; Alling, HISTORY OF ATOMIC LOGISTICS, p. 118. A later Aircraft Modification Schedule, released in January 1955, stated that 112 B-36s were to be converted to carry H-bombs by January 1955; 212 by July 1955; and 219 by January 1956. The 91 RB-36s in the Air Force inventory were also all to be modified by January 1956. Other USAF bombers, including the B-47 and B-52, were also being modified for H-bomb delivery. (Alling, HISTORY OF ATOMIC LOGISTICS, p. 212.)

practice loading and drop tests.

Although plans for "emergency capability" were expanded to include the EC16 and EC17 in the late weeks of 1952, the TX-14 ALARM CLOCK was still developmentally ahead of the other two in the spring of 1953. Schedules called for an emergency availability of small numbers of TX-14s twelve months hence, and for small numbers of TX-16s and TX-17s by July 1, 1954.²⁴²

By mid-1953, the most desirable time-of-fall for the "emergency capability" weapons had not yet been determined. LASL believed that at least 200 seconds would be required to allow the delivery aircraft time to escape the blast. SAC was equally convinced that manned aircraft could escape safely from free-fall versions of the weapons. Sandia calculations indicated that 120 seconds, two minutes, would provide sufficient escape distance; parachutes contemplated for the "emergency capability" weapons provided 120 seconds time-of-fall if the bombs were released at 36,000 feet and fuzed for airburst at 8,000 feet.

This retardation worked out to a terminal velocity of 250 feet per second, about 170 miles per hour. The TX-14, TX-16, and TX-17 were being fitted with four wedge-shaped stub fins and five spoiler bands, which wind tunnel tests had shown would provide sufficient aerodynamic stability during a free fall. Because of the high density of these weapons, LASL estimate that free-fall terminal velocity would approach Mach 1.4.²⁴³

Drop tests had already started on the TX-14; TX-16 and TX-17 shapes were to be available by July or August, and an accelerated drop test program was to be conducted with these shapes. In addition to the nuclear proof tests of all three weapons -- TX-14, TX-16, and TX-17 -- at the PPG in early 1954, each weapon was scheduled to be drop tested in complete but inert form.²⁴⁴

²⁴²AF ATOMIC ENERGY PROGRAM, Vol. IV, pp. 221-223. Eventually five EC14, five EC16, and five EC17 weapons were added to the stockpile; all were withdrawn within a few months of their deployment. About 200 MK 17 weapons were stockpiled (NUCLEAR WEAPONS DATABOOK, Vol. II, pp. 10, 16). Also see MK 14, 16, and 17 histories in Volume VI.

²⁴³Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1953 - 30 June 1953, p. 189.

²⁴⁴Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1953 - 30 June 1953, p. 189.

The First Soviet Thermonuclear Test

More impetus to the emergency capability program came early in August 1953, when Moscow announced that the Soviets had broken the American thermonuclear monopoly. A few days later, on August 12, the Kremlin was more specific and boasted of a successful thermonuclear test. The Russian claim was confirmed by the Atomic Energy Commission, which had detected the test by acoustic and seismic methods (by seismic vectoring, the location of the test was fixed at a site within six miles of an area used for earlier Soviet tests).²⁴⁵

The announcement by the Soviets that they had their own H-bomb before they had actually tested it caused no small amount of consternation within the AEC. Chairman Lewis Strauss prepared a letter for President Eisenhower to try to explain why the U.S. might not have detected a test of a Soviet fusion device:

Dear Mr. President:

I have been speculating as to the reasons for our failure to detect any test of a Russian atomic weapon subsequent to the third explosion which was announced on _____. You will recall that these Russian explosions were picked up as a result of a monitoring system which I initiated in the fall of 1947 but which did not become fully operational on a 24-hour basis until sometime in late 1948. It is only a speculation, therefore, to assure that the first explosion detected in August 1949 was actually the first Russian test.

Various reasons have been advanced for the fact that apparently no other bombs have

²⁴⁵Statement issued on August 20, 1953 by AEC Chairman Lewis L. Strauss; Zaloga, pp. 100, 101; Minutes of the Thirty-Sixth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, August 17, 18, and 19, 1953, p. 54. Less than two weeks earlier, Strauss had written to President Eisenhower and noted that since the Soviets had not tested a fission device in nearly two years (the last previous Soviet test was on October 18, 1951), they might be getting ready to conduct a thermonuclear test as the next logical step in a weapons development program. ("In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 No. 2 (1989), p. 329.)

been tested since 1951. These reasons range all the way from the extreme, typified by ex-President Truman's statement that the Russians have no bombs and that these were not bomb explosions at all, through the following postulates: (a) that fissionable material, being relatively scarcer for the Russians than for us, they are less prodigal in its expenditure; (b) that they have frozen their (bomb) design and are satisfied with the efficiency of the weapon used in their last shot; and (c) that they have undertaken by some means to mask their tests subsequent to the third shot, such as in a deep underwater explosion in Lake Baikal or in a very deep mine.

Unfortunately, a further alternative has just suggested itself to me and is the occasion for this letter to you. This speculation is based in part upon the fact that Uranium (sic) is relatively scarce in Russia so far as we know, and secondly, upon the fact that as a result of the confession of Klaus Fuchs and the defection of (Bruno) Pontecorvo and others, I have long entertained the belief that the Russians are well ahead of our work in the theoretical aspects of fusion bomb geometries.

I therefore come to the conclusion that since a fission weapon is used only as an igniter of the H-bomb, improvements of its efficiency by the Russians did not need to go beyond the point that they achieved in 1951. There was no further necessity for them to test fission weapons, having perfected the trigger, if their entire effort is and has been in the production of thermonuclear weapons. This theory would account for the fact that there have been no further tests of atomic weapons and, if true, would be very bad news indeed.²⁴⁶

²⁴⁶Draft of a letter August 7, 1953 to President Eisenhower; Allen Dulles, Director of the CIA; W. Sterling Cole, Chairman, JCAE; Charles E. Wilson, Secretary of Defense; Robert LeBaron, Chairman, MLC; John Foster Dulles, Secretary of State; and General Robert Cutler, Special Assistant to the President, from Lewis Strauss. Strauss noted that "this was the draft of a letter which I had intended to write to the

Strauss' concerns about clandestine Soviet testing may have been alleviated a little on August 12, when the Russians fired their thermonuclear test device.

This Soviet test employed a fusion reaction with a boosted configuration involving use of lithium deuteride; rainwater samples contained tritium. The yield was submegaton, on the order of 400 kilotons. It was detonated atop a tower, and was not a deliverable weapon.²⁴⁷ The core of the device consisted of oralloy only; approximately 10%, or 40 KT, of the total yield came from this core. Fifteen to 20 percent of the total yield, or 60 to 80 KT, came from the lithium deuteride, and the remainder of the yield, about 280 to 300 KT, came from fission of tuballoy in the device.²⁴⁸

The Soviets had bypassed a liquid-fueled thermonuclear device in favor of a more militarily-useful solid-fueled weapon.

By late October 1953, the AEC had prepared a detailed analysis of the August 12 Soviet test in terms of yield, materials used, design principles embodied, and deliverability; these points were compared to comparable American progress or results in each area.²⁴⁹

The Soviet device was apparently a large boosted fission device wrapped with lithium deuteride, similar to the LITTLE EDWARD and the earliest ALARM CLOCK designs.²⁵⁰ In a public

President but I covered the substance of it in conversation on the morning of the day that (Soviet Premier) Malenkov announced that they had the hydrogen bomb; four days later, they actually detonated it." The missing date of the latest Soviet test by this time, which had occurred in October 1951 (the Soviets conducted no nuclear tests during 1952) is a shown in the draft document.

²⁴⁷Alexander, p. 19; Hewlett and Holl, pp. 57-59; "Soviets Shown to Have Lagged on H-Bomb in 50's," New York Times, October 7, 1990; NWD 86-3, "Known Soviet Nuclear Explosions, 1949 - 1985, Preliminary List," Jeffrey I. Sands, Robert S. Norris, and Thomas B. Cochran, Natural Resources Defense Council, Washington, D.C., February 1986, p. 11; Zaloga, pp. 95, 96.

²⁴⁸MAKING THE RUSSIAN BOMB FROM STALIN TO YELTSIN, Thomas B. Cochran, Robert S. Norris, and Oleg A. Bukharin, Westview Press, Boulder, Colorado, 1995, p. 18.

²⁴⁹Letter dated October 27, 1953 to W. Sterling Cole, Chairman, JCAE, from Lewis L. Strauss, Chairman, USAEC.

address at the Massachusetts Institute of Technology on November 14, 1990, Dr. Hans Bethe stated:

... Sakharov proposed at an earlier time to have a fission bomb with layers of deuterium and in particular one layer between the active (fissile) material and the uranium reflector; this layer was to consist of lithium. This is the way the 1953 test was made. ... it was a glorified booster of a very high yield.

More detail has been provided recently:

Soon after he joined the (Soviet hydrogen bomb) program in 1948, Sakharov came up with the first idea, the "layer cake" design. He proposed that alternate layers of thermonuclear fuel (deuterium, tritium, and their chemical compounds) and uranium-238 placed in a fission bomb between the fissile material and the conventional high explosive would make a fission-fusion-fission reaction possible, since the high-energy neutrons from the deuterium-tritium reaction would fission the uranium-238.

A uranium casing would compress the deuterium and speed up the thermonuclear reaction. This idea is similar to the "alarm clock" (sic) concept proposed by Edward Teller in late 1946 - a design not pursued in the United States.

The second idea, proposed by V. L. Ginzburg in 1949, was to use lithium deuteride in the bomb. The lithium would provide tritium when bombarded by neutrons, raising the power of the thermonuclear process. Both ideas were used in the first Soviet thermonuclear bomb, detonated in August 1953.

... There is no suggestion that Soviet physicists knew of Teller's "alarm clock" concept. ... At the end of 1953, when the Presidium (now Politburo) approved the

²⁵⁰Similar designs have apparently also been developed recently by Israel. (TRIPLE CROSS, Louis Toscano, Birch Lane Press, New York, 1990, p. 124; "Revealed: The Secrets of Israel's Nuclear Arsenal," The Sunday Times, No. 8,461, London, England, 5 October 1986, pp. 1, 4, 5.)

development of an intercontinental ballistic missile, its warhead was to be a second-generation version of the 1953 Soviet design, not one based on the Teller-Ulam configuration.²⁵¹

A CIA estimate prepared at the time of this first Soviet thermonuclear test concluded that stripped-down Soviet Tu-4 bombers, without aerial refueling and on one-way missions, could reach targets within the continental United States. However, these piston-engine aircraft would be extremely vulnerable to modern American jet-powered interceptors.²⁵² (In the fall of 1952, the CIA had estimated that the Soviets might have stockpiled as many as 50 implosion-type, nominal-yield bombs.)²⁵³

²⁵¹"Moral Leader of A Nation," book review by David Holloway in BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 47 No. 6, July/August 1991, p. 37.) Zaloga (pp. 95, 96) claims that the initial concept of the "layer cake" was to place a layer of liquid deuterium between the fissile material and the surrounding HE; the Soviets realized quickly that the deuterium would not be sufficiently compressed to cause a fusion reaction. A later design included a layer of natural, unenriched uranium around the deuterium to further compress the deuterium and add its own fission-generated neutrons and explosive power to the device. Still later, the deuterium was replaced by lithium-6 deuteride. More lithium deuteride was placed inside the center of the fissile core, further adding a boosting effect to its yield, and earning for the Soviets the historical honor of the being the first country to test a solid-fueled boosted thermonuclear explosive device.

In addition, a photo of a mockup of the "model assembly of the first (Soviet) hydrogen bomb" in an article entitled "Reminiscences of Different Years" by A. I. Pavlovskii of the All-Union Scientific Research Institute of Experimental Physics, translated to English and reprinted in Sov. Phys. Usp 34 (5), May 1991, shows a spherical device resting atop a support stand and bearing a remarkable resemblance to an American naval mine (minus detonator horns.)

²⁵²"Special Estimate, Soviet Capabilities for Attack on the US through Mid-1955," SE-36/1, (supersedes SE-36), Central Intelligence Agency, approved 31 July 1953, published 3 August 1953.

²⁵³"Toward Armageddon: The Foundations of United States Nuclear Strategy," PhD thesis by David A. Rosenberg, University of Chicago, August 1983, p. 157.

The AEC was receiving by this time fairly reliable data on Soviet plutonium production, based mainly on British monitoring of krypton releases from Soviet production reactors. In addition, the longest minimum distance from any point in the Soviet Union to an American-manned test monitoring station was only 3,000 miles, so nuclear explosions as small as 20 kilotons were detectable anywhere in Russia.²⁵⁴

These stations used a combination of seismic, acoustic, electromagnetic, and atmospheric sensors to detect the outputs of nuclear explosions. Fast electromagnetic sensors close to a large multistage nuclear blast could even pick up the individual signals from each of the exploding primary and secondary stages.²⁵⁵

In October 1953, three other early thermonuclear "emergency capability" weapons, the TX-15 ZOMBIE, EC17, and EC24 entered development engineering at Los Alamos.²⁵⁶

At this time, DOD scientists believed that the B-52 could carry a bomb weighing as much as 30,000 to 50,000 lbs., with a yield of 20 megatons. At the same time, lighter and smaller bombs with yields of one or two megatons were projected to weigh only 3,000 lbs. or less, making them candidates for ICBM and IRBM warheads.²⁵⁷

In November, in a memorandum to K. D. Nichols, General Manager of the AEC, Commissioner Thomas E. Murray stated that "the recent evidence of Soviet progress in thermonuclear work places a far greater premium on time in our development and testing of new weapons."²⁵⁸

By the end of 1953, a Joint Emergency Capability Operational Planning Committee (JECOPC) had been formed, comprising

²⁵⁴Minutes of the Thirty-Third Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, February 5-7, 1953, pp. 28, 29.

²⁵⁵Letter dated April 28, 1962 to Glenn T. Seaborg, Chairman, USAEC, from Gerald W. Johnson, Chairman, MLC, attachment to AEC 1077/84, May 11, 1962, Atomic Energy Commission, Test Detection Stations, Note by the Secretary.

²⁵⁶NUCLEAR WEAPONS DATABOOK, Vol. II, p. 16.

²⁵⁷York, MAKING WEAPONS, TALKING PEACE, p. 90; JOHN VON NEUMANN, Norman Macrae, Pantheon Books, New York, 1992, pp. 358, 359.

²⁵⁸Memorandum dated November 24, 1953 to K. D. Nichols, General Manager, USAEC, from Thomas E. Murray, Commissioner, USAEC.

members from LASL, Sandia, Field Command/Armed Forces Special Weapons Project (FC/AFSWP), Air Force Special Weapons Center (AFSWC), and the AEC. This committee was to establish operational procedures for the entire "emergency capability" program.²⁵⁹

Another significant result of the IVY Mike test was a February 1954 Air Force decision to proceed with "crash" development of an intercontinental ballistic missile (which eventually became the ATLAS). This decision was made several weeks before the CASTLE test of the TX-15 ZOMBIE warhead, a descendant of which would be the most likely candidate for a future ICBM warhead.²⁶⁰

The design of the TX-15 looked promising enough to proceed with the development of a relatively-inaccurate delivery vehicle whose error in targeting would be more than offset by the yield of its warhead.

Early in February, the Deputy Secretary of Defense wrote to the Joint Committee on Atomic Energy and advised the JCAE of the factors on which the Joint Chiefs of Staff were basing their requirements for thermonuclear weapons, which were: (1) the capability of the USSR to produce thermonuclear weapons, (2) the capability of thermonuclear weapons to increase the destructive power a stockpile of weapons, (3) the types of targets against which thermonuclear weapons could be profitably employed, (4) possible configurations and yields of thermonuclear weapons, (5) the availability of delivery vehicles for thermonuclear weapons, (6) the military requirement for an expanded thermonuclear program while still maintaining a fission weapon program, and (7) the production capacity of the AEC.

The JCAE was further advised that the stockpile desired by the JCS represented the greatest feasible increase in available stockpile yield without major expansion of AEC facilities, but was not intended to fulfill all military requirements. The JCS recommendations would require a major revision of the present production program of the AEC, but the JCS believed this revision was necessary.

The Joint Chiefs emphasized that as the stockpile of

²⁵⁹Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 July 1953 - 31 December 1953, p. 225.

²⁶⁰York, THE ADVISERS, p. 86; NUCLEAR WEAPONS DATABOOK, Vol. II, p. 17.

thermonuclear weapons grew, stockpiled high-yield fission weapons would eventually be retired completely.²⁶¹

Operation CASTLE

In early 1954, attention of the weaponeers at Los Alamos and Livermore once again turned to the remote Pacific, where the "emergency capability" bombs were to be tested.²⁶²

Objectives of CASTLE

Operation CASTLE was, in many ways, a continuation of IVY: the conceptual design of the weapons tested at CASTLE began about two years before the test series, even before the IVY series.²⁶³ An objective of the CASTLE series was to validate one or more of the "emergency capability" weapons, and produce deliverable H-bombs. Successful CASTLE tests would follow directly from the success of the IVY Mike device: the Teller-Ulam principles had been demonstrated in a liquid-fueled device; the next step in U.S. thermonuclear weapons

²⁶¹"Comments Concerning the H-Bomb Development," Department of Defense, Military Liaison Committee to the Atomic Energy Commission, 21 April 1954; letter dated February 5, 1954 to Honorable W. Sterling Cole, Chairman, JCAE, from Roy M. Kyes, Deputy Secretary of Defense.

²⁶²Much of the following history of Operation CASTLE (except where noted) is drawn from OPERATION CASTLE, Report of Commander, Task Group 7.1, Pacific Proving Grounds, March-May 1954, WT-940 (EX), Los Alamos Scientific Laboratory, Los Alamos, New Mexico, June 1954, pp. 26, 28, 33, 34, 35, 40, 41, 62-69, 72, and 76-78; HISTORY OF OPERATION CASTLE, Pacific Proving Ground, Joint Task Force Seven, Major General P. W. Clarkson, United States Army Commanding, Spring 1954, pp. 1-13, 50-55, 59-62, 75-77, 79, 84, 107, 110-114, 117, 135-150, 152, 153, 221; CASTLE SERIES 1954, DNA 6035F, Defense Nuclear Agency, Washington, D. C., 1 April 1982, pp. 1, 2, 34, 45, 47, 53, 75, 127, 185, 201, 203, 205, 210, 246-250, 255, 267, 271, 283, 296, 310, 339, 368, 372, 405, and 415; OPERATION CASTLE, Pacific Proving Grounds, March-May 1954, Project 9.1, Report to the Scientific Director, Cloud Photography, January 27, 1958, pp. 20-37; and JOINT TASK FORCE 7, COMMANDER TASK GROUP 7.3, Operation Plan No. 1-53, pp. 3, 9, A-1, A-5, J-I-A-1, J-I-1, and J-I-2.

²⁶³Bethe, "Comments on the History of the H-Bomb," p. 49. An Office Memorandum dated July 28, 1952 to Brig. Gen. K. E. Fields, Director, DMA/USAEC, from George F. Kraker, Deputy Manager, Santa Fe Operations Office, USAEC, Subject: Task Force Organization after IVY, proposed "that the Ivy and Castle operations be designated as a single, overall effort to be known as Operation Ivy/Castle."

development was to determine the characteristics of radiation implosion in large-scale solid-fueled devices.

The major objective of Operation IVY had been to verify the Teller-Ulam configuration in a strictly experimental situation. Operation CASTLE was to follow IVY as quickly as scientific and technological developments allowed; one of CASTLE's main objectives was to verify the Teller-Ulam configuration in forms suitable for convenient military use. An early AEC planning document written in mid-June 1952 detailed the relationship between IVY and CASTLE:

Operations IVY and CASTLE are each planned to include the firing of a complete experimental thermonuclear device. IVY is principally directed toward establishing thermonuclear feasibility from a purely experimental device and providing important fundamental information. CASTLE, however, is further directed toward a deliverable emergency weapon in addition to furnishing important information.

In the event IVY is successful, it is anticipated that it will become a matter of great urgency to provide the deliverable version of a thermonuclear weapon with minimum delay. Although CASTLE is tentatively scheduled to occur within one year of IVY, there is no certainty that the lag between an experimental device and a deliverable weapon, which are physically quite different, can be shortened to this extent. The importance of the time lag is such as to suggest it may be a determinant for war or peace.

The short term goal in the thermonuclear field should be to create a deliverable thermonuclear weapon with a minimum lag between experimental device and usable weapon. This means gambling on the success of the [deleted] (the CASTLE device) as we are doing. It seems prudent, however, that this bet be covered by accelerated and comprehensive diagnostic and componentry tests, now evolving in specific detail.

Such information will facilitate exploitation in the shortest time of the possibilities of radiation implosion. It will, on the one hand, provide back-up for an emergency design, and on the other hand, speed the ultimate design(s).

....

The thermonuclear program has as a short term objective the successful demonstration of the first deliverable thermonuclear weapon in order to provide an emergency capability in this field. The long term objective will be the development of optimum thermonuclear weapons, which will proceed along the lines of reducing weapon size and weight, and improving efficiency and utility, as has taken place in the field of fission weapons.²⁶⁴

The weapons test program for CASTLE finally included five Los Alamos and two Livermore lab (UCRL) devices. The most important aspect of the operation, however, was the testing and validation of solid-fueled (lithium deuteride and lithium hydride) weapons.²⁶⁵

Operation CASTLE was the culmination of another phase of the development of the Super that began in 1942, although the

²⁶⁴Thermonuclear Research At The University of California Radiation Laboratory, Director of Military Application, U.S. Atomic Energy Commission, Washington, D. C., June 13, 1952, pp. 1, 2. With respect to the U.S. H-bomb being "a determinant for war or peace," an October 1950 report to the Joint Chiefs of Staff noted that there was an "assumption that the United States must prepare for a change from cold war to total war (before) 1 July 1954." ("Accelerated Missile Program," Enclosure "C" to JCS 160/26, 2 October 1950, "Report by the Guided Missiles Interdepartmental Operational Requirements Group to the Joint Chiefs of Staff on Requirements for Guided Missiles.") In a letter to Dr. Richard Feynman in late March 1951, Dr. John Wheeler had expressed his fear that the U.S. might be involved in a nuclear war with the Soviet Union by the following September. ("In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 No. 2 (1989), p. 312.) Part of the reason for 1954 being considered as the "year of maximum danger" was an April 1950 CIA estimate of Soviet nuclear weapons capabilities which concluded that between mid-1954 and late 1955, the Soviets might possess 200 atomic bombs, a force sufficient to destroy the United States. (Rhodes, DARK SUN, p. 441.) These shared fears of the Soviets were one of the driving forces behind the U.S. "emergency capability" program for thermonuclear bombs.

²⁶⁵AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 297.

devices tested at Bikini in 1954 were light-years removed from designs and principles contemplated in 1942. The test objectives were threefold. The first was to fire six or seven experimental thermonuclear devices, six of whose yields were to be in the megaton range, including proof tests of three "emergency capability" weapons: the EC14, EC16, and EC17.

The test firing of the EC16 was contingent upon the results of the other six shots: if the solid-fueled devices worked satisfactorily, the liquid-fueled EC16 would be canceled.

Second, CASTLE was to gather sufficient diagnostic data to evaluate performance of the test devices; and third, the tests were also to provide effects information from megaton-yield weapons.²⁶⁶ The CASTLE shots were much more heavily-instrumented than even the IVY Mike shot had been: there were 24 separate diagnostic programs, organized into three groups representing the Department of Defense, LASL, and UCRL. The results of CASTLE would largely determine the scope of the continuing "emergency capability" program.²⁶⁷

Planning and Construction

As early as April 1952, the AEC planned to conduct developmental tests of high-yield weapons at the Eniwetok Proving Ground (EPG; later changed to the Pacific Proving Grounds, PPG, when Bikini atoll was incorporated into it) in the fall of 1953 under the code name Operation CASTLE.²⁶⁸ (The tests were subsequently deferred to the spring of 1954.)

At the beginning of May, LASL director Norris Bradbury expected that CASTLE would test four devices in tower or ground shots, with no air drops, around September or October 1953.²⁶⁹

²⁶⁶NWD 86-2 (NRDC), p. 16.

²⁶⁷OPERATION CASTLE, Report of the Manager, Santa Fe Operations, Pacific Proving Ground, Spring 1954, p. 19.

²⁶⁸OPERATION CASTLE, Summary Report of the Commander, Task Unit 13, Military Effects, Programs 1-9, Pacific Proving Grounds, March-May 1954, WT-934 (EX), Headquarters Field Command, Armed Forces Special Weapons Project, Sandia Base, Albuquerque, New Mexico, January 30, 1959, p. 15; letter dated April 25, 1952 from H. D. Smyth, Acting Chairman, USAEC, to Honorable Robert LeBaron, Chairman, MLC; History of Task Group 7.4 Provisional for the Period 15 July through 31 August 1953, Capt. Marvin Polakoff, USAF, et. al., Joint Task Force Seven, Joint Chiefs of Staff, p. 1.

²⁶⁹Letter dated 8 May 1952 to Carroll L. Tyler, Manager, Santa

The first general concept for Operation CASTLE was issued in June 1952. At that time, CASTLE was tentatively scheduled for September and October 1953. Only one high-yield "dry" thermonuclear device was expected to be ready for test at that time, along with three experimental fission devices yielding on the order of 50 kilotons each. At least one, and possibly as many as two, of the fission-type weapons might be detonated in an airdrop.

The statically-fired thermonuclear device was to be tested at Bikini, and was to have "the general configuration of a weapon" and was to be based on an IVY device. The three fission tests were to occur at the northern islands of Eniwetok atoll.²⁷⁰

CASTLE plans noted that the yield from the IVY thermonuclear device could not be accurately predicted at this time. Eniwetok atoll represented an important asset to the U.S. and the AEC, and its future use as a proving ground might be jeopardized if its real estate were vaporized through successive detonations of high-yield thermonuclear devices. Primarily for this reason, Bikini was a more attractive site for large explosions.

The practicability of using one of the several uninhabited islands in the Marshall, Caroline, and Mariana Islands groups was investigated, and a survey of Bikini to determine its suitability as the site of very-high yield detonation was planned.²⁷¹

The testing during 1953 of an experimental thermonuclear device having a "weapon configuration" was indicative of both accelerated AEC research efforts and the imminent development of an effective, transportable thermonuclear weapon. Because

Fe Operations Office, USAEC, from N. E. Bradbury, Director, LASL.

²⁷⁰"Program Plan -- Operation CASTLE," Headquarters Task Group 132.4 Provisional, Kirtland AFB, New Mexico, 3 July 1952, attachment to History of Task Group 7.4 Provisional for the Period 15 July through 31 August 1953, Capt. Marvin Polakoff, USAF, et. al., Joint Task Force Seven, Joint Chiefs of Staff; Report by the Commander, Joint Task Force 132 to the Joint Chiefs of Staff on Armed Forces Participation in the 1953 Experimental Nuclear and Thermonuclear Tests at Eniwetok/Bikini, June 13, 1952, p. 4.

²⁷¹OPERATION CASTLE, PACIFIC PROVING GROUNDS, Spring of 1954, Report of Commander, Task Group 7.5, Joint Task Force 7, p. 4.

of the short time between the end of IVY and the beginning of CASTLE (approximately nine months), preparations for CASTLE could be accomplished most expeditiously by the retention in the Pacific and on the west coast of the U.S. of the personnel, planes, ships, and scientific equipment assembled for IVY.²⁷²

In July, ACF was notified by the AEC to procure steel for the casings of several of the CASTLE devices.²⁷³

Because the high yield of some of the shots might damage base facilities at Eniwetok, and because of the limited dry-land area available at Eniwetok, Bikini atoll was being considered for thermonuclear tests and Eniwetok atoll for the fission shots.²⁷⁴ Holmes & Narver, the AEC's test facilities contractor, made a cursory reconnaissance of Bikini in September 1952 and began an extensive survey the following month.²⁷⁵

The CASTLE program at this time included three LASL tests and one UCRL shot.²⁷⁶ Continuing Air Force dissatisfaction with

²⁷²History - Task Group 132.1, 1 May 1952 to 30 June 1952, Maj. Albert S. Knauf, USAF, HQ Task Force 132, p. 58; letter dated July 2, 1952 to Maj. Gen. P. W. Clarkson, Commander, Joint Task Force 132, from Brig. Gen. K. E. Fields, Director of Military Application, USAEC; memorandum dated 28 August 1952 from Chief of Naval Operations, to Distribution, Subject: Atomic Weapons Effects Program, Operation CASTLE; memorandum dated November 10, 1953 to Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC, from Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Development of Bikini for Operation CASTLE; Report by the Commander, Joint Task Force 132 to the Joint Chiefs of Staff on Armed Forces Participation in the 1953 Experimental Nuclear and Thermonuclear Tests at Eniwetok/Bikini, 4 August 1952, pp. 1, 2, 4, 6.

²⁷³September 12, 1952 Walker memorandum, p. 16.

²⁷⁴Excerpt from Minutes of AEC Meeting 746 on September 11, 1952; letter dated September 20, 1952 to Honorable Robert LeBaron, Chairman, MLC, from Gordon Dean, Chairman, USAEC.

²⁷⁵Memorandum dated August 21, 1952 to Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC, from Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Necessity for Beginning Work at Bikini Atoll; memorandum dated 25 September 1952 to C. L. Tyler, Manager, Santa Fe Operations Office, USAEC, from P. W. Spain, Field Manager, Eniwetok Field Office, USAEC, Subject: Preliminary Reconnaissance of Bikini Atoll.

the pace and emphasis of the thermonuclear weapons program at Los Alamos, along with personal rancor between Dr. Edward Teller and Dr. Norris Bradbury and others at Los Alamos, had led to the creation of the University of California Radiation Laboratory (UCRL) at Livermore, California (now the Lawrence Livermore National Laboratory) in the fall of 1952.

Figure IV-15 shows the Livermore site in early 1954, located at what had been a World War II U.S. Navy airfield. The new lab inherited many of the old Navy buildings and barracks.

The AEC authorized a cryptic press release in case questions arose about the new lab; no mention was made of nuclear weapons development or testing:

The University of California Radiation Laboratory is moving some of its classified work to the Atomic Energy Commission's Livermore site. This work is part of a developmental program of the AEC. About the first week in September, 70 of the Radiation Laboratory personnel will transfer to Livermore and by next summer, approximately 300 will have been shifted there.²⁷⁷

The establishment of a second U.S. nuclear weapons design and development lab, to compete with Los Alamos for Department of Defense nuclear warhead design contracts, may have been good for advancing the state-of-the art of U.S. nuclear weaponry, but it also started an internal arms race between the labs and represented a major escalation of the international arms race with the Soviet Union.

The establishment of the second lab would also not accelerate the U.S. hydrogen weapons program: it would take nearly four years for UCRL to have a successful thermonuclear test.

In November, late in the IVY operation, construction started at Bikini.

By the time Bikini was added to the PPG, plans for its use were firm. Eniwetok would continue to be the main base of

²⁷⁶Memorandum dated November 10, 1953 to Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC, from Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Development of Bikini for Operation CASTLE.

²⁷⁷Letter dated September 10, 1952 to W. B. Reynolds, Business Manager, Radiation Laboratory, University of California, from H. A. Fidler, Area Manager, USAEC Berkeley Area Office, Subject: Press Release.

operations and would be the site of relatively low-yield detonations. Bikini was to be a supplemental site for detonations with yields high enough to damage Eniwetok base facilities or require evacuation of the atoll. All facilities at Bikini would be of temporary construction and kept to a minimum.²⁷⁸

At this time, LASL had initiated a comprehensive program to develop three distinctive two-stage thermonuclear weapons for test in the winter of 1953-1954 and for emergency stockpiling before they were tested, if necessary. LASL was also developing other experimental devices in order to prove new design concepts, and UCRL was doing likewise, having started developing ideas for new thermonuclear bombs in June 1952.²⁷⁹

LASL's plans for the spring 1953 Nevada test series, Operation UPGHOT, included component tests directed toward studies of radiation implosion and of the symmetry of implosion achieved by radiation.

UCRL's plans at this time were to fire two devices during Operation UPGHOT in the spring of 1953 and two large-scale devices named RAMROD and RADIATOR during Operation CASTLE in the fall of 1953. The RAMROD had been under study at UCRL for several months.

The two UCRL UPGHOT tests were expected to yield between 0.5 and three kilotons. These shots would use uranium deuteride, a modification of the wartime LASL uranium hydride formulation, the chief difference being that the UCRL variant was of a higher order of purity than anything ever achieved at LASL. UCRL hoped that the deuteride would be an active source of energy within the core itself.

The uranium deuteride compound itself was not difficult to manufacture and could be done in a small room with simple apparatus. The result of the formulation was a heavy plastic-like cube which was easy to handle and work with.

The new Livermore lab had acquired a UNIVAC computer whose previous claim to fame was that it was the machine used to predict the results of the November 4, 1952 national elections for radio and television audiences (the first time that a computer was used for this purpose).²⁸⁰

²⁷⁸Minutes of AEC Meeting 746, September 11, 1952.

²⁷⁹Report of the Manager, Santa Fe Operations, U.S. Atomic Energy Commission, July 1950 to January 1954, pp. 13, 14.

²⁸⁰Memorandum to the File dated November 10, 1952 from John S. Walker, subject: Project Whitney. An interesting note on the

Most of the basic requirements for the support of CASTLE were determined by early 1953. They included construction of a device assembly area and barge slip on Parry island; refurbishment and reactivation of the IVY-era cryogenic plant on Parry (both Los Alamos and Livermore planned liquid-fueled device tests); and procurement of shot barges, including a deep-water barge and an array of anchors, buoys, chains, and cables to moor it.

Figure IV-16 shows the device assembly area and barge slip on Parry island, with two shot barges moored to the slip.

Other requirements were for the design and construction of superstructures on the barges to support and house the devices and accompanying personnel; construction of shot cabs on three islands to shelter devices and adjacent equipment; and construction of a massive shot control station bunker on Enyu island at Bikini and numerous instrument stations throughout both Bikini and Eniwetok atolls.

At Bikini, construction of a 4,500-foot airstrip on Eninman island, a causeway, a pier, and a dredged channel and various camps had already started. Other requirements were a task group voice and teletype radio link between atolls; three or four large voice radio networks at each atoll; sampling planes with a capability of obtaining radioactive cloud samples at altitudes up to 55,000 feet; an improved weather service capable of forecasting winds at high altitudes and making wind runs at altitudes up to about 100,000 feet in the vicinity of zero point up to shot time; and a large number of four-wheel drive vehicles for moving about the atoll islands.

With regard to sampling aircraft, the capabilities of the B-47, the Martin B-57, "featherweight" B-36, the ConVAir B-60, and English Electric Canberra were all considered. Of these aircraft, the projected capabilities of the B-57 looked most impressive; however, the plane would not be available for many months, until the latter part of CASTLE.

The B-57 was still undergoing teething problems. The airplane buffeted at Mach 0.72, and Martin was unable to locate the source of this vibration. The company had already lost one B-57 through mid-air disintegration.

The plane also wallowed considerably at its ceiling altitude, which was limited to only 48,600 feet. In addition, the Wright J-65 jet engines used on the plane were having severe production and operation problems, due in part to substitution of Studebaker and Buick engines for Wright

use of the UNIVAC to predict the outcome of the election was that many commentators refused to believe its projections of an Eisenhower victory over Adlai Stevenson.

engines; the former required modification before they could be used on the B-57. As a result of these and other difficulties, only 16 B-57s would be produced by the end of 1953.

The bomber's range was also so limited that direct flights from the west coast of the U.S. to Hawaii were not possible. Bomb-bay fuel tanks to extend the plane's range were not available.

Partially for these reasons, the Air Force chose to use F-84G and two "featherweight" B-36s for sampling.²⁸¹ However, it appeared that even these aircraft would not be able to sample at altitudes above 50,000 feet.²⁸²

At Eniwetok atoll, the device assembly area on the southern end of Parry island provided a specially secured, central location for working on nuclear devices. It included support and shelter of the assembly teams, machine tool facilities, and high explosive magazines. This area was completed in late February 1954.

Several areas at both Eniwetok and Bikini had to be repaired or reconstructed in the aftermath of a furious typhoon (Hester) which swept over both atolls at the end of December 1952, causing much wind and water damage.²⁸³ Winds up to 70 knots were accompanied by ocean waves 100 to 300 feet beyond the normal tide-line on Eniwetok island.

²⁸¹History of Task Group 7.4 Provisional for the Period 15 July through 31 August 1953, Capt. Marvin Polakoff, USAF, et. al., Joint Task Force Seven, Joint Chiefs of Staff, pp. 11, 12; memorandum for General Mills dated 16 October 1953 from Brig. Gen. Howell M. Estes, Jr., Commander, TG 7.4 Provisional, subject: B-57 Aircraft for CASTLE, attachment to history previously cited. "Featherweight" B-36s were high-altitude reconnaissance B-36D PEACEMAKERS stripped of all non-essential flying and crew comfort equipment and all armament except the tail turret. (Knaack, POST-WORLD WAR II BOMBERS, pp. 35, 36.) Of the three "featherweight" B-36s in the Air Force in 1953, Task Group 7.4 would use two.

²⁸²Memo for the Record dated 13 August 1953 by Col. D. W. Kesling, USAF, Deputy Commander, TG 7.4 Provisional, subject: Visit to Los Alamos Scientific Laboratory, attachment to History of Task Group 7.4 Provisional for the Period 15 July through 31 August 1953, Capt. Marvin Polakoff, USAF, et. al., Joint Task Force Seven, Joint Chiefs of Staff.

²⁸³Letter dated 13 January 1953 to Honorable Gordon E. Dean, Chairman, USAEC, from Maj. Gen. P. W. Clarkson, Commander, JTF 132.

At Eniwetok, Parry island was not badly damaged, but there was extensive damage to installations at Eniwetok island. Roads were washed out, tents destroyed, three buildings partly collapsed, a freighter pier was badly damaged, and a weather station was knocked out of commission. Debris was scattered widely, and much of the shoreline around Eniwetok island was washed away.²⁸⁴

Navy requirements for the new test series included a large carrier-based helicopter group and a helicopter landing barge; the use of a ship to transport devices and major components to the forward area and to furnish close support to shot barges; and the use of an LSD (Landing Ship Dock) to transport shot barges between atolls. The barges and test devices could not be risked on the open sea between Eniwetok and Bikini; the enclosure of the barges by the hull of the LSD would also foil any possible surveillance by snooping Soviet submarines near the atolls.

Other naval requirements included frequent inter-atoll sealift; seaplanes for inter-atoll airlift when the airstrip on Eninman was not available; and ships with a capability of supporting prolonged operations, including actual device firing, from afloat. The need for many additional landing craft and DUKW's, four-wheeled amphibious trucks, was immediately evident.

Development of the Department of Defense weapons effects experiments entailed additional support and participation by the Navy, including two drone Liberty ships, five tugs, two salvage vessels, one destroyer mine sweeper (DMS), two patrol planes, and modification of several landing craft.

In addition to the manned and drone sampling planes and control planes for the drones, the following support was required from the Air Force: planes for weather reconnaissance, for DOD effects projects and for documentary photography; L-13 liaison aircraft and helicopter airlift at Eniwetok and Bikini until arrival of the carrier-based helicopter group; and several C-47s for trips between Eniwetok and Bikini.

Many special flights were required from the Military Air Transport Service (MATs) for transportation to the forward area of devices and components not ready in time for surface shipment and for the return to the U.S. of radiochemical and radiobiological samples after each shot.

²⁸⁴Report of the Field Manager, Eniwetok Field Office, to the Manager, Santa Fe Operations, AEC, July 1, 1950 to July 1, 1953, pp. 13, 14.

MATS was also required to provide regularly-scheduled airlift between Travis and Hickam Air Force bases (in California and Hawaii, respectively) and Eniwetok for passengers and urgent freight; and the Military Sea Transportation Service (MSTS) was required to carry many tons of test and construction materiel to the forward area.

The task force assigned to CASTLE was Joint Task Force Seven, carrying on the designation of the flotilla that had conducted Operations CROSSROADS and SANDSTONE in 1946 and 1948, respectively. JTF 7 was a successor to JTF 132 that had conducted Operation IVY; JTF 132 was renamed JTF 7 on February 1, 1953. As such, JTF 7 carried over many of the task units from JTF 132. More than 10,000 men were in JTF 7.

CASTLE Task Units

Task Unit 2 (TU 2), Production, operated the cryogenic plants at Parry island and manufactured liquid deuterium for use in nuclear devices for both LASL and UCRL. TU 2 personnel were from an AEC contractor, Herrick L. Johnston, Inc., of Columbus, Ohio.²⁸⁵ The use of cryogenic fuels for two of the test devices, LASL's JUGHEAD and UCRL's RAMROD, was given much consideration. Plans were made to guarantee that production and transport of liquid hydrogen isotopes would be concurrent with shot schedules.

TU 3, Special Materials and Facilities, handled and delivered liquefied gases produced by TU 2 to users in TU 4 (LASL Assembly) and TU 14 (UCRL Assembly). Like TU 2, this unit was composed of employees of an AEC contractor, the Cambridge Corporation (Camco) with field offices in Boulder and Denver, Colorado. Only five refrigerated transport dewars (RTDs), instead of the eight used during IVY, were used during CASTLE to move liquefied gases around Eniwetok for TU 4 and TU 14.

TU 4, LASL Assembly, was composed of LASL personnel and its contractor organization, American Car & Foundry (ACF), Inc., of Albuquerque, New Mexico (ACF began weapon manufacturing activities in Albuquerque in 1953 following construction of a new plant in the fall of 1952).²⁸⁶ In early 1954, the AEC signed a "definitive contract" with ACF to extend from February 1954 to June 1959. This contract replaced a previous

²⁸⁵CASTLE SERIES 1954, DNA 6035F, Defense Nuclear Agency, Washington, D. C., April 1982, p. 56.

²⁸⁶"AEC Plans Further Consolidation of Weapons Production Complex," USAEC press release No. J-280 dated Friday, December 9, 1966, USAEC, Washington, D. C.; September 12, 1952 Walker memorandum, p. 18.

letter contract under which "certain engineering, design, fabrication, testing, and operating work" had been carried out for more than a year by ACF in Buffalo.²⁸⁷ (By January 1956, a \$2.5 million expansion project at ACF Albuquerque was nearing completion.)²⁸⁸

The LASL-designed devices were actually assembled by ACF except for the Nectar shot (ZOMBIE) device, which was fabricated at LASL. TU 4 worked primarily at Parry island, preparing devices for installation on shot barges.

The devices were assembled within the secure area on Parry and transported to shot sites by barges carried aboard the USS BELLE GROVE (LSD-2). A ramp was available within the parry secure area so that devices to be detonated on islands at Eniwetok or Bikini could be taken aboard an LST (Landing Ship Tank) by truck.

The barge-detonated devices were assembled in a shelter (called a cab) aboard the barge, which was moored in a specially constructed slip at water's edge in the assembly area. This slip was equipped with a large overhead crane to handle heavy loads.

TU 6, the Firing Party, armed (and disarmed) and fired the test devices. Members of TU 6 checked the various electromechanical links that prevented premature device detonation and removed them just before activating the firing mechanism. All the Bikini devices were detonated by signals sent over a wire system from the control bunker at Station 70 on Enyu, which was manned by TU 6 only for shot Bravo, the first Bikini detonation in Operation CASTLE.

In general, two teams were used for arming and firing: the arming team made tests of the X-units (the firing sets on the devices) and firing racks before the final assembly of the devices, and the inspection and test checks of the firing circuitry, including the proof-testing of Go-No Go interlock circuits. The firing team operated the shot control equipment.

The work of TU 6 was conducted in close liaison with TU 4 (assembly team for LASL devices), TU 14 (assembly team for UCRL devices), and TU 15 (signal and firing circuits team). The arming and firing teams used detailed countdown checklists enumerating each step in shot preparation and measurements associated with arming and firing of each device.

²⁸⁷Major Activities in the Atomic Energy Programs, January-June 1955, USAEC, Washington, D.C., July 1955, p. 32.

²⁸⁸Major Activities in the Atomic Energy Programs, July-December 1955, USAEC, Washington, D.C., January 1956, p. 39.

The arming team for "dry" solid-fueled devices would be augmented by eight to ten cryogenic engineers for the two "wet" liquid-fueled shots. A firing bunker with a 300-foot tall radio control tower was built and partially buried at Station 70 on Enyu island at Bikini atoll for use during all the Bikini shots.

An existing control bunker, Building 311, on Parry island would be used for the Eniwetok shots. Once again, detailed contingency plans were made by the firing party for the event of weapon/device misfire, HE-only or low-order nuclear detonation, or for delay, stop, or postponement of shots.

The original operations plans called for firing of the various devices from land-based control stations based on Enyu island at Bikini atoll and Parry island at Eniwetok, using cable-wired electrical circuits. These circuits terminated at the various zero points; each device would fire fifteen minutes after the last button was pushed by a firing team member. The bomb was monitored and recording instruments started in the last few minutes by a precise electromechanical timer.

After contamination from the Bravo shot made Station 70 at Enyu radiologically unsafe, the remaining Bikini tests were initiated by a radio signal from the USS ESTES relayed to the firing system through equipment in the evacuated bunker. Three 250-watt Motorola radio links were used to start the timing signals and the sequence timer and to actuate the emergency stop at Station 70 if necessary. A voice time and countdown broadcast was transmitted in the same manner except that the last 15 minutes were tape-recorded and automatically transmitted from Station 70 on Enyu.

The test at Eniwetok, Nectar, was wire-detonated from Parry island.

During Operation CASTLE, no misfires or other delays were caused by arming or firing operations. Problems with the weather delayed many shots, requiring the repeated arming and disarming of all but two devices.

Another new wrinkle in this test series was the inclusion of UCRL personnel in larger numbers than ever before. In the summer of 1952, UCRL's thermonuclear role had been extended beyond merely developing and setting up diagnostic instrumentation and recording equipment. UCRL had also been empowered to conduct research and development leading to new or extended weapon design concepts, and by the end of 1952, UCRL was preparing to test thermonuclear weapon prototypes during Operation UPSHOT in the spring of 1953 and during

CASTLE in late 1953 or early 1954.²⁸⁹

For CASTLE, UCRL was to design, fabricate, and assemble two devices and perform certain diagnostic experiments associated with these shots and LASL shots. TU 14, UCRL Assembly, composed entirely of UCRL employees, assembled the solid-fueled Koon shot device (MORGENSTERN)²⁹⁰ in the shot cab on Eninman at Bikini and the liquid-fueled RAMROD device that was to be detonated during the Echo shot. TU 12, another UCRL task unit, and TU 14 combined totaled 272 men.

Most of the work on MORGENSTERN was completed before shot Bravo; the RAMROD device was nearly finished when the Echo shot was canceled on April 13, 1954.

During the stateside phase of CASTLE, before moving to the PPG, TU 14 designed, fabricated and mocked-up the assembly of the two devices. This work was coordinated with GMX, W, and J-Divisions at LASL, Camco and the NBS at Boulder (for the liquid-fueled RAMROD device), and Consolidated Western Steel, who constructed and assembled the RAMROD case and supporting structure.

TU 15, Timing, provided timing signals transmitted by wire and radio that were primarily intended to coordinate the device firing systems; however, this unit also serviced all experiments requiring pre-shot instrument starting and reference timing signals. About 70 employees of Edgerton, Germeshausen and Grier (EG&G), an AEC contractor, manned this task unit.²⁹¹ No difficulties were experienced with the hard-wire or radio signals.

Once again, as at SANDSTONE and IVY, the USS CURTISS was the principal shipboard facility for project laboratory and device assembly team basing and office space, as well as for

²⁸⁹Atomic Energy Commission, Progress Report to the Joint Committee, Part III - Weapons, June 1952 through November 1952, AEC 129/46, December 17, 1952; memo dated 18 December 1952 to Carroll L. Tyler, Test Manager, from J. C. Clark, Deputy Test Director, J-Division, LASL, Subject: Operation Plan, Continental Atomic Tests, Spring 1953.

²⁹⁰The MORGENSTERN which in German means "Morning Star," was named after a 16th century European weapon which was either a single-piece or articulated club with a spiked ball on either its end (if a single piece) or at the end of a short chain (if articulated). (WEAPONS: AN INTERNATIONAL ENCYCLOPEDIA FROM 5000 BC TO 2000 AD, The Diagram Group, St. Martin's Press, New York, 1990, pp. 17, 18.)

²⁹¹DNA 6035F, pp. 56, 57.

transportation of weapons and devices to Parry island from the U.S. The USS ESTES served as the main joint task force command facility, and as a "backup" firing station in event of unplanned evacuation from Bikini or Eniwetok. The BELLE GROVE was used to transport shot barges from Parry island to Bikini.²⁹²

The BELLE GROVE had to be modified slightly to carry the barges between Eniwetok and Bikini. Because the barges with their shot cabs were significantly higher from keel to cab top than the landing craft normally carried in the well deck of the LSD, a portion of the superstructure deck over the well had to be removed. A section of the deck was dismantled and stored in San Diego, and two more sections were removed in the forward area at Eniwetok. Following movement of the barges, these two sections were replaced so the ship could continue her secondary role as a tender for landing craft at Bikini.

Selecting the CASTLE Test Devices

Uncertainties about device designs, materials availability, and military requirements were reflected in many changes to CASTLE plans. Between October 1952 and May 1953, devices or weapons and experiments were added to or deleted from the schedule with a frequency that produced a constant revision of the test facilities required by the scientific programs.

Before February 1953, definitive planning for CASTLE was considerably curtailed by the lack of a firm definition of the scope of the tests and associated experiments and diagnostics. Some of the limitations affecting planning at that time were:

- a. Initial device designs and engineering could not be pursued until a general agreement was reached on the types and/or systems of thermonuclear devices desired, which was in turn partially dependent upon certain diagnostic information from the Nevada Operation UPHOT-KNOTHOLE series in the spring of 1953.
- b. Having determined the design or model to be followed, production capabilities on certain critical materials, including lithium-6 and boron-10, appeared to be another limiting factor affecting specific test dates.

²⁹²OPERATION CASTLE, Operation Plan No. 1-53, Task Group 7.1, 8 December 1953, pp. 8, 9, 12, A-1, A-2, A-3, A-4, A-5, D-1, D-3, K-IV-1, P-1.

- c. Bomb site locations: With the advent of such high-yield shots as GREENHOUSE George and particularly IVY Mike, LASL realized that careful consideration had to be given to bomb site locations in the PPG. The primary concern was the residual radiological effects which might affect post-CASTLE operations, although loss of dry-land real estate was another serious concern. The use of barges as shot platforms appeared to be one way to alleviate some of the radiological problems, but the effectiveness of this method would not be known until after CASTLE ended.
- d. The detonation of the device: Only after the location and probable yield of each shot were settled could it be determined whether or not the device should be fired by radio command or direct wire circuit, and whether or not television observation of control equipment, as was done during IVY, would be essential to support the firing system.
- e. Diagnostic measurements desired: Theoreticians could not determine, beyond standard experiments, the desired diagnostic experiments until the device designs became firm. The lack of knowledge as to the type and extent of diagnostics delayed critical early construction in the PPG. The experiments dictated such things as camp locations, blockhouse design and location, tower placements, etc.²⁹³

At the beginning of October 1952, the basic requirements for CASTLE included two shots of the IVY Mike variety, i.e., on the order of 10 MT yield, and two smaller shots of GREENHOUSE George size, i.e., on the order of 200-300 KT. It appeared, in a significant departure from earlier planning to conduct all megaton-yield tests at Bikini, that one Mike-type shot would have to be fired at Bikini and the other Mike-type plus the two George-type shots would have to be fired at Eniwetok to meet reasonable time schedules.

The only way that two thermonuclear devices could be fired at Bikini would be to allow approximately 90 days between shots to refurbish facilities. An alternative plan was to fire one thermonuclear shot at Bikini and one at Eniwetok.²⁹⁴

²⁹³CASTLE, TG 7.1 History, Installment 1, November 20, 1953.

²⁹⁴Memorandum for Record dated 7 October 1952, signed by Brig. Gen. K. E. Fields, Director of Military Application, USAEC, and Maj. Gen. P. W. Clarkson, Commander, Joint Task Force 132, Subject: Conference on CASTLE Planning at 1030 hours, 7 October 1952, War Room, JTF 132, Parry Island, Eniwetok atoll.

Later in October, LASL was proceeding with plans for as many as six CASTLE shots, including one in the megaton range, two in the high-kiloton range, and one drop test. LASL assumed that UCRL's Project Whitney would also conduct one megaton-range and one kiloton-range experiment. Tests would still be held at both Bikini and Eniwetok.²⁹⁵

In early November this schedule of shots was also extended to Operation REDWING, then planned for the Pacific in 1955. Both series would comprise four tests; during both CASTLE and REDWING, UCRL would detonate one multimegaton-yield device. If additional shots were required, airdrops might be added to the original four ground-level tests.²⁹⁶

By November 15, no fewer than five devices were under consideration for firing during CASTLE, including a 100 KT LASL mockup casing test on a tower at Runit at Eniwetok; a 250 KT LASL tower shot on Eberiru; a one- to five-megaton UCRL shot at ground level on Bikini; a one- to four-megaton LASL shot on a barge anchored in the IVY Mike crater at Eniwetok; and a one-megaton LASL test on a platform on the reef at the western side of Eniwetok atoll.

The two solid-fueled, megaton-yield LASL devices were slated for test during the latter part of the series to allow more time to accumulate sufficient lithium-6.²⁹⁷ At this time, a pilot plant to produce enriched lithium at Oak Ridge was to be in operation by the middle of March or early April 1953; even if this startup date were met, six weeks would be required to "shake down" the plant and another six weeks to build to full production. At this rate, there was no way that CASTLE could be conducted before 1 October 1953.²⁹⁸

²⁹⁵Letter dated October 23, 1952 to Brig. Gen. K. E. Fields, Director, Division of Military Application, USAEC, from Darol Froman, LASL.

²⁹⁶Letter dated 3 November 1952 to Norris E. Bradbury, Director, LASL, from Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC; letter dated 3 November 1952 to Honorable Robert LeBaron, Chairman, MLC, from Gordon Dean, Chairman, USAEC.

²⁹⁷Conference Minutes dated 15 November 1952, Headquarters, Joint Task Force 132, San Francisco, California; Memorandum for Record dated November 19, 1952 from Lt. Col. William R. Sturges, Jr., Subject: Trip to Eniwetok Atoll, November 7, 1952 through November 18, 1952.

²⁹⁸September 12, 1952 Walker memorandum, pp. 22, 23.

An alternative and more efficient and economical process of enriching lithium might be found; just how much lithium-6 might be required and which substitute materials for it, if any, could be developed were also still unanswered questions.²⁹⁹

Current requirements for tritium were a different problem. The amount required per weapon was relatively uniform, but there was a major policy question as to whether or not a substantial number of future stockpiled weapons should also use tritium, either for boosting or for significant thermonuclear yield. At this time, there was so little tritium on hand that this question was largely moot.

The excess reactivity at the AEC's Hanford works was on the order of only 5% and a few lithium slugs were being used to produce tritium there, but only as a byproduct of their major use, which was to even out the distribution of neutrons in the piles.

Excess reactivity at the Savannah River reactors was approximately 20%; most or all of this could be employed to produce tritium. Even with this capacity, not enough tritium could be generated in time to start immediate stockpiling of thermonuclear bombs and/or boosted weapons in any significant quantity. It would take one to two years to build more tritium-production reactors, even if construction were assigned a high priority and the high cost of the piles could be afforded by the AEC's budget.³⁰⁰

Also in November, planning had begun for Operation DOMINO, a series of nuclear tests originally scheduled for Nevada in the fall of 1953 following Operation UPSHOT; DOMINO was to further test candidate primaries for CASTLE thermonuclear devices, including a tower shot of the still-viable but ever-elusive gun-type primary.³⁰¹

Preliminary DOMINO plans included tower shots of six LASL devices, three LASL device airdrops, and two UCRL tower

²⁹⁹September 12, 1952 Walker memorandum, pp. 24, 25.

³⁰⁰September 12, 1952 Walker memorandum, pp. 26, 27.

³⁰¹Letter dated November 17, 1952 to Honorable Robert A. Lovett, Secretary of Defense, from Gordon Dean, Chairman, USAEC; letter dated January 14, 1953 to William L. Borden, Executive Director, JCAE, from Col. Jack L. Armstrong, USAF, Executive Officer, Division of Military Application, USAEC. DOMINO was later rescheduled for spring 1954 and canceled in July 1953 after UPSHOT-KNOTHOLE ended successfully.

shots.³⁰²

On November 18, LASL had in mind four thermonuclear experiments for the next Pacific Proving Ground tests. These included the TX-14 ALARM CLOCK; the ZOMBIE, or TX-15; the TX-16 JUGHEAD, a deliverable version of the IVY Mike (SAUSAGE) device, and the TX-17 RUNT. At a meeting of AEC commissioners, Dr. Norris Bradbury pointed out that the TX-16 and the ZOMBIE were additions to the experiments already planned for CASTLE.

Now that it was known that the SAUSAGE worked, DOD was expected to request a "weaponized" version of the device. The major technical hurdles to be overcome in weaponizing the Mike device would be reduction of casing size and weight and incorporation of a cryogenic system into a bomb.

The TX-14 test might use an extra SAUSAGE casing, left over from IVY. A yield of one megaton might result from the ZOMBIE shot, which would be detonated on a barge and instrumented sparsely. Another version of the TX-15 would make use of a smaller and lighter ballistic case appropriate for a deliverable weapon. The success of the device could have important consequences.

To carry out a Pacific series, including all four thermonuclear shots, it might be necessary to postpone CASTLE from the fall of 1953 to the spring of 1954. Bradbury listed several reasons for delay. First, it was highly unlikely that ACF and the Cambridge Corporation could have the TX-16 ready before the spring of 1954.

Second, because of adverse weather, it was virtually impossible to conduct a four-shot series in the Pacific in the fall. There would be much better weather during the spring; JTF 132 had been lucky to get in two shots during IVY late in 1952.

Third, both the ALARM CLOCK and the TX-16 would make use of information from casing mockup tests during Operation UPSHOT; because of the short interval between UPSHOT in the spring of 1953 and CASTLE in the fall, there would be precious little time to apply data from the UPSHOT tests to useful casing designs for the TX-14 and TX-16.

The simultaneous fall 1953 conduct of Operation CASTLE in the Pacific and Operation DOMINO in Nevada would preclude DOMINO from being anything more than airdrops and collection of simple yield data.

³⁰²Diary of Harry H. Haight, undated, received by DNA 2 January 1980.

Bradbury also noted that tests he had described did not include experiments planned for Project Whitney, the nascent UCRL thermonuclear weapons program.³⁰³ At this time, Whitney personnel were visiting LASL for conferences and library work. UCRL had requested one shot each during Operations UPSHOT and CASTLE.³⁰⁴

In response to a question as to what effect a successful liquid-fueled TX-16 test might have on the AEC's plans for lithium-6 production, Bradbury replied that enriched lithium would be required under any circumstances in order to produce more deliverable and manageable solid-fueled thermonuclear weapons.

Bradbury pointed out that available real estate at Eniwetok for the conduct of high-yield tests "was being gradually destroyed" and that Bikini would have to be used for the conduct of future thermonuclear experiments.

At this meeting, AEC Commissioner Eugene M. Zuckert (later to become Secretary of the Air Force) stated that he was concerned about postponing CASTLE until 1954. Having initiated the thermonuclear program and having successfully detonated a high-yield thermonuclear device, the nation might not be able to delay the test of a deliverable hydrogen weapon.

Zuckert also pointed out that most of the American public believed that the U.S. already had a deliverable hydrogen bomb, and that current and future foreign policy might be based on this mistaken assumption.

AEC Commissioner Henry DeWolf Smyth asked if LASL could accelerate development of a deliverable liquid-fueled thermonuclear weapon, if only a clumsy prototype.

In reply to these comments, Bradbury stated that LASL was proceeding as quickly as possible and that the limiting factor was the availability of scientific manpower. He believed that a deliverable liquid-fueled thermonuclear weapon could be made, but he did not see how it could possibly be ready for test before the spring of 1954.³⁰⁵

Implementation of the results of IVY into weapons then under

³⁰³Letter dated February 3, 1954 to Lewis L. Strauss, Chairman, USAEC, from Sterling Cole, Chairman, JCAE.

³⁰⁴September 12, 1952 Walker memorandum, p. 11.

³⁰⁵Minutes of AEC Meeting No. 776, November 18, 1952.

development prompted LASL to recommend at the end of November that CASTLE be expanded to include six tests and that the operational period be delayed from fall 1953 to early spring 1954, due in part to a shortage of lithium-6, as well as all the other reasons for delay cited earlier by Bradbury. (In late November, most of the available lithium-6 in the U.S. was expected to be consumed by the CASTLE tests.) This proposal was accepted, with the proviso that one thermonuclear model might still be tested, "on an emergency basis," in late 1953.³⁰⁶

Following the mid-November meeting with the AEC, Bradbury wrote to Brig. Gen. K. E. Fields, the AEC's Director of Military Application, and summarized current laboratory plans for CASTLE:

The extremely successful behavior of the recent MIKE shot has emphasized, not only the important future of radiation implosion, but the practical fact that a cryogenic thermonuclear system works. Subject, of course, to important additional information to be obtained regarding thermonuclear systems and radiation implosion engineering in Operation UPSHOT, we believe that the main outlines of Operation CASTLE can now be forecast for planning purposes along the following lines.

The Los Alamos Scientific Laboratory may presently be expected to propose four shots for CASTLE having the following general characteristics:

- (a) A test of the TX-14 ALARM CLOCK generally following the description of this device as given in the program for the current calendar year. This device is being designed with "emergency weapon capability" as a criterion. It is expected to have a yield in the megaton range, and fairly elaborate instrumentation will probably be appropriate in its test at Eniwetok.
- (b) A test of the TX-17 RUNT device; this is a device falling in a size and weight package similar to that of the

³⁰⁶Borden-Walker H-bomb chronology, p. 89.

TX-14. It is hoped that "emergency weapon capability" can also be ascribed to the tested form of this device although its actual characteristics cannot be given any precision until after the UPSHOT test. Fairly elaborate instrumentation will probably be appropriate to the test of this device at Eniwetok.

- (c) A device named JUGHEAD having the same basic design as the SAUSAGE device of Operation IVY but reduced in size and weight so as to have the same "emergency weapon capability" as the ALARM CLOCK but retaining the cryogenic character of the recent MIKE shot. Such a device would utilize less expensive materials than the ALARM CLOCK but would have much greater technical and logistic complications.

A direct comparison of these systems appears to the Los Alamos Scientific Laboratory to be of extreme importance. The yield of such a device should be expected to be in the megaton range, but elaborate instrumentation should not be required in connection with its test at Eniwetok and an adequate observation of yield only would be satisfactory.

- (d) A device named ZOMBIE [deleted] In either event, no elaborate instrumentation would be required and a measurement of yield alone would be sufficient. The yield of this device is thoroughly uncertain at this time, but is regarded as unlikely to be of weapon interest. However, in the event that this expectation is incorrect, it should be regarded as requiring test under circumstances in which megaton yields would be acceptable.

Bradbury went on to declare that CASTLE should now be firmly scheduled for the spring of 1954. A modified UPSHOT program would contain two and possibly three shots which would bear directly upon the design of the ALARM CLOCK, RUNT, and JUGHEAD devices.

By September 1952, over half of the UPSHOT series was

allocated to thermonuclear device components tests.³⁰⁷ The stated purposes of UPSHOT at the end of 1952 were "to conduct component studies in connection with the thermonuclear program and to develop further experimental data on external initiators and new configurations of fissionable material in fission weapons."

Operation DOMINO, then slated for the spring of 1954 or the fall of 1953, was to be conducted in Nevada for similar purposes; Operations CASTLE and REDWING were then to be Pacific Proving Ground test series of full-scale deliverable thermonuclear devices.³⁰⁸

UPSHOT was now scheduled well into May 1953 and might be subject to further delays. LASL believed that several months would be required to evaluate and apply UPSHOT results to CASTLE device designs, and that an early CASTLE date just a few months after UPSHOT would be detrimental to full use of UPSHOT-derived design data.

Operation UPSHOT, and its coincident DOD effects tests series, Operation KNOTHOLE, would also require the attention and attendance of many personnel needed for CASTLE, and much of the necessary construction and instrumentation for CASTLE could not be completed until after the Nevada tests, due in part to the requirement for better CASTLE shot yield estimates based on results of some of the UPSHOT tests.³⁰⁹

In addition, delays in lithium-6 production were getting to be such that it was a real question as to whether or not enough of it would be available in the quantities required for CASTLE any earlier than the spring of 1954. In August 1951, the AEC had approved construction of a small lithium isotope separation plant at Oak Ridge; the plant was to be in operation by fall 1952.

However, late in September 1951, the AEC requested Oak Ridge to produce lithium deuteride by September 1953 in a new plant that was to have twice the capacity of the first.³¹⁰

³⁰⁷September 12, 1952 Walker memorandum, p. 10.

³⁰⁸Progress Report to the Joint Committee on Atomic Energy, June through November 1952, Part III, Weapons, December 29, 1952, p. 8.

³⁰⁹History of Task Group 7.4 Provisional for the Period 15 July through 31 August 1953, Capt. Marvin Polakoff, USAF, et. al., Joint Task Force Seven, Joint Chiefs of Staff, p. 8.

³¹⁰Hewlett and Holl, p. 163.

The proposed UCRL tests of Project Whitney, when added to LASL experiments, made it mandatory that CASTLE be scheduled so that the total length of the operation, including the inevitable weather and technical delays, would not lead to times of increasingly bad weather (as would be the case for a fall Pacific series) but rather to even better weather (as would be the situation in the spring).

Finally, the time required to prepare the TX-16 for test was, in LASL's opinion, already at an absolute minimum on the schedule proposed and no earlier test of a cryogenic system was possible under any circumstances. In addition, it did not appear that appropriate military logistic support for the TX-16 would be ready much sooner. (The Air Force began building cryogenic facilities at several air bases in the spring of 1953.)³¹¹

Logistics of "Wet" Bombs

The logistics of the stockpile-to-target sequence for "wet" bombs were quite complex, compared to those of "dry" bombs. To begin with, if liquid-fueled weapons were stored unarmed and unloaded, then they would have to be filled with liquid deuterium, in the same way that early liquid-fueled rockets were prepared for launch. However, this fueling-up would take quite some time, time that might not be available in the event of an unexpected enemy attack.

On the other hand, if liquid-fueled weapons were maintained fully-loaded, ready for deployment to waiting bombers, then the equipment required to constantly refrigerate, replenish, and circulate liquid hydrogen through the bombs would necessitate many pumps, compressors, insulated piping, and power stations to generate electricity for all this equipment.

To protect them from attack or sabotage, liquid deuterium and liquid hydrogen or nitrogen dewars might have to be located underground. If the bombs were stored empty, then a high-pressure liquid gas transfer system would be required to fuel the bombs quickly, probably before they were loaded aboard bombers. All of these fueling and handling operations would be very dangerous, involving very cold and explosively-flammable hydrogen gases.

Once loaded aboard their delivery aircraft, "wet" bombs would still require a small hydrogen recirculation system on the

³¹¹Minutes of the Thirty-Fifth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 14, 15, 16, 1953, p. 25.

aircraft to keep the deuterium in a liquid state and to replenish lost deuterium (the TX-16 vented deuterium at a rate of 50 liters per hour). This heavy equipment could be lifted only at expense of delivery aircraft speed, range, bombload, and ceiling.

"Dry" bombs would require only storage in bunkers with controlled temperature and humidity to prevent the possibility of any moisture infiltrating the hygroscopic lithium deuteride in the secondaries. Practically no other complications would be encountered; "dry" weapons could be even be kept loaded aboard quick-reaction alert bombers "cocked" and poised for immediate takeoff in event of attack.

LASL was also quite aware that although a test of a very high-yield liquid-fueled system had been successfully conducted, actual weapon capability existed only for the SAUSAGE, which was far from deliverable. The "emergency capability" for the lithiated solid-fueled ALARM CLOCK was primarily dependent upon the availability of large quantities of lithium-6; the amount required per weapon was not yet completely known.

TX-14 "emergency capability" would not be fulfilled until the weapon was tested, but the device was likely to be ready for test before sufficient Li-6 for the test had been produced.

Even after a successful test, the TX-14 would not be deployed in quantity, and a true "emergency capability" for thermonuclear warfare achieved, until lithium was also available in adequate quantity. If the TX-14 did not work, the liquid-fueled TX-16 would have to become an "emergency capability" weapon, with all its cumbersome and unwieldy cryogenic handling equipment and special technicians to nurse it to its target.

By this time, a suggestion had been made to divide CASTLE into two phases, one in the fall of 1953 and a second in the spring of 1954. LASL opposed such a proposal because it would require almost continual overseas residence of task force and technical personnel for six months during some of the worst Pacific winter weather, including the typhoon season between November and January.

The second phase of CASTLE would conflict with Operation DOMINO, and it would not make "emergency capability" weapons ready any sooner, even if solid-fueled weapons were shown to be achievable.

Operation DOMINO, already postponed to the spring of 1954, might either be delayed further or conducted on a very limited basis in Nevada in the fall of 1953. In the latter event, only the simplest experiments on air-dropped fission devices would be performed. LASL hoped to test deuterium-

based initiators during DOMINO if the external initiator program was ready by then for further testing. A fall 1953 DOMINO series would closely resemble the small-scale tests conducted during Operation RANGER in early 1951.³¹²

UCRL's Project Whitney, which was to test two thermonuclear devices, one of which was cryogenic, required an early determination of the levels of support by the NBS at Boulder, Colorado; ACF; and Camco. UCRL needed ordnance engineering support, as well as cryogenic support.³¹³

UCRL planned to test two moderated uranium deuteride-fueled devices during UPSHOT (these became the Ruth and Ray shots), and two large-scale thermonuclear devices (the RAMROD and MORGENSTERN) during CASTLE. The latter two systems would not be ready until the late winter or early spring of 1954 so UCRL agreed with LASL's recommendation for a spring 1954 series in the Pacific.³¹⁴

In early December, CASTLE scope was changed again, to embrace at least five detonations, three of which would be megaton-yield thermonuclear shots.³¹⁵ In the middle of the month, six thermonuclear shots were planned.

The first would be a test of the cryogenic liquid-fueled UCRL RAMROD device to be detonated on either a 200-foot high tower or in a ground-level cab on Eberiru island at Eniwetok. Yield was expected to be on the order of 250 KT. This would be followed by a test of a "dry" LASL device to be detonated on a tower or ground-level cab located on the northwest tip of Teiteiripucchi island at Eniwetok, or possibly on a barge or jetty located on or near the IVY Mike crater. Yield was expected to be two to five megatons.

The third shot would be a test of the UCRL megaton-yield solid-fueled MORGENSTERN device on a tower located on Eninman island at Bikini. The shot would be fired from an AGC-type (command) ship. The fourth shot would be a LASL "dry"

³¹²Letter dated 28 November 1952 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

³¹³Teletype dated November 28, 1952 to Carroll Tyler from K. E. Fields.

³¹⁴Letter dated November 24, 1952 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from Herbert F. York, Director, UCRL.

³¹⁵Letter dated December 5, 1952 to Gordon Dean, Chairman, USAEC, from Maj. Gen. P. W. Clarkson, USA, Commander, JTF 132.

thermonuclear-fission device to be detonated on a barge in Bikini atoll lagoon; yield was expected to be about 250 KT. Pending instrumentation requirement changes, the device might be located on Runit island at Eniwetok and fired first.

LASL would also detonate the fifth and sixth shots; the fifth was the liquid-fueled JUGHEAD with an expected yield in the megaton range. This test would also be a barge shot in Bikini lagoon. The final shot would be a "dry" device also detonated on a barge in Bikini lagoon and fired from an AGC-type ship.³¹⁶

At a joint AEC-MLC conference on December 18, Gordon Dean and Kenneth Fields pointed out that for technical reasons, it would probably be necessary to postpone Operation CASTLE from the fall of 1953 until early 1954. Postponing the test of an ALARM CLOCK until early 1954 would make it possible to test three deliverable thermonuclear weapons, each yielding a megaton or more.

If the postponement were approved, LASL would develop three different thermonuclear weapons which would make it more likely that a deliverable bomb could be produced than if one weapon were tested in the fall of 1953 and the other two were excluded from test.

A deliverable weapon would most likely be developed if three different types were developed independently and tested simultaneously in the spring of 1954. The alternate types of devices to be tested and the current problems in their evolution were discussed at the meeting.

In answer to a question, AEC Director of Military Application Capt. John Hayward stated that the only device that would possibly be ready in anything approaching deliverable form by the fall of 1953 would be a design which LASL currently believed to have the least probability of success of all the candidate devices.

If this design were supported to the exclusion of others, it would mean a delay of six months before other types could be tested. An additional reason for not testing a deliverable form of the ALARM CLOCK and other solid-fueled weapons in late 1953 was the desirability of using results of Operation UPSHOT in their development.

MLC Chairman Robert LeBaron noted that a requirement for an

³¹⁶Memorandum dated 16 December 1952 to Brig. Gen. K. E. Fields, Division of Military Application, USAEC, from Alvin C. Graves, J Division Leader, LASL, Subject: Operational Concept of CASTLE.

emergency capability ALARM CLOCK already existed, and that aircraft were being modified to carry it. Gordon Dean replied that although the AEC was working on development of an emergency capability for the ALARM CLOCK, it was also simultaneously working on other approaches toward deliverable thermonuclear weapons.³¹⁷

At an AEC meeting four days later, the commissioners were notified by Norris Bradbury that a decision on the start date for CASTLE was required immediately: the lab was preparing to take preliminary steps in thermonuclear weapons development that would make it extremely difficult, if not impossible, to conduct the test series in the fall of 1953.

In response to a question by Chairman Dean whether the U.S. would have as many ALARM CLOCKS ready for deployment in June 1954 whether the TX-14 was tested in late 1953 or early 1954, General Kenneth Fields stated that as many bombs could be stockpiled by then; it was just a matter of having a tested weapon at an earlier date.

The AEC also noted that while DOD was asking for an emergency capability weapon as soon as possible, the Department of Defense was at the same time unwilling to keep JTF 132 together in standby status until early 1954 (during December 1952, the Korean War was still straining U.S. military resources).

Dean also wondered if DOD's requirement for a tested weapon in the fall of 1953 was so great that the Department was willing to risk a test failure and the eventual postponement of development of more certain weapons as a result of holding an early test?³¹⁸

LASL director Norris Bradbury continued to argue for a postponement for three major reasons. First, information to be derived from shots during Operation UPSHOT could be used to assist in designing and selecting the best of several competing weapon models now being considered.

Second, the exact date by which sufficient quantities of lithium-6 would be available for a test of the ALARM CLOCK was still problematical. It now appeared that tests of devices requiring lithium-6 could not be conducted earlier than the spring of 1954.

Third, it was highly unlikely that an operational ALARM CLOCK

³¹⁷Minutes of Seventy-Fifth Joint AEC-MLC Conference, December 18, 1952.

³¹⁸Minutes of AEC Meeting No. 793, December 22, 1952.

-- the TX-14 -- could be ready before the spring of 1954. Development of adequate military logistic support for the TX-14 even by then was questionable.

LASL proposed development of three possible thermonuclear weapon models (the TX-14, TX-16, and TX-17) which would all be ready for test by the spring of 1954. A successful full-scale test would be necessary before any of the models could be accepted as an emergency weapon, and final selection and design of CASTLE test devices would be critically influenced by the results of UPSHOT in the spring of 1953.³¹⁹

The AEC and MLC met jointly again on December 23. Reviewing the proposed CASTLE Program, Gordon Dean addressed the postponement of the ALARM CLOCK test until early 1954. He stated that fabrication of the TX-14 would continue and that the weapon would be stockpiled at the proposed rate, unless it was superseded by a more technically- or economically-attractive bomb.

Dean outlined the developmental work underway at LASL and reported on the possible advantages of delaying CASTLE in order to test two additional thermonuclear weapons at the same time as the ALARM CLOCK. This option would not substantially delay the availability of emergency capability weapons.

With this understanding, the MLC agreed to a postponement of CASTLE but asked that every effort be made to start the test series by the end of February 1954. As far as JTF 132 was concerned, the Air Force component was ready to maintain a 50% level of current personnel; the Navy would reduce its task groups considerably until after the start of 1954; and the Army would maintain adequate groups in standby status.³²⁰

The LASL and UCRL Technical Programs for 1953-1954

The LASL and UCRL technical programs for calendar year 1953 and fiscal year 1954 were submitted by the respective lab directors (Norris Bradbury and Dr. Ernest O. Lawrence) late in December 1953. Neither lab had any concrete plans for DOMINO tests, but both were planning major activities during UPSHOT and CASTLE.

³¹⁹Atomic Energy Commission, The Thermonuclear Weapon Test Program, Report by the Director of Military Application, attachment to Operation CASTLE, Note by the Secretary, December 22, 1952.

³²⁰Minutes of the Seventy-Sixth Joint Conference of the AEC and MLC, December 23, 1952.

To the extent permitted by other activities, LASL planned to continue theoretical computation of the burning process of non-equilibrium thermonuclear reactions in liquid deuterium. In the field of equilibrium reactions, LASL would give extensive attention to "dry" systems using Li-6, deuterium, and U-238. Depending upon the results of component tests during UPSHOT, LASL hoped to prepare an "emergency capability" weapon, the TX-14, for test during CASTLE, which was planned to be a sequence of possibly as many as four experimental detonations to study radiation implosion systems.

LASL also planned to return to a 40-hour, five-day work week at the beginning of January 1953, after nearly three years of 48-hour, six-day work weeks. Lab staffing had grown sufficiently to allow a normal schedule; LASL estimated that by mid-June 1954, there would be 3,050 people on "The Hill."³²¹ (At the beginning of February 1953, there were 2,900 persons at LASL.)³²²

The new University of California Radiation Laboratory also planned both equilibrium and non-equilibrium thermonuclear experiments, many that were ideas of Edward Teller, including various new geometrical arrangements and fuel combinations for thermonuclear weapons. UCRL also planned two low-yield detonations during UPSHOT, and possibly as many as three full-scale device tests during CASTLE, although specific fuels had not yet been chosen for them.³²³

The UCRL UPSHOT tests were to be conducted to provide information required for the design of thermonuclear devices planned for Operation CASTLE.³²⁴

LASL's UPSHOT plans included a series of six shots. The first was an experiment called the "XR3," a tower shot using a MK 5 warhead with a Type D pit to continue the experimental study first started during Operation SNAPPER of the yield of

³²¹LASL Technical Program for CY 1953 and FY 1954.

³²²Minutes of the Thirty-Third Meeting of the General Advisory Committee to the USAEC, Washington, D.C., February 5-7, 1953, p. 13.

³²³Letter dated December 24, 1952 to J. J. Flaherty, Manager, San Francisco Operations Office, USAEC, from Donald Cooksey, Associate Director, UCRL, for E. O. Lawrence, Director, Subject: Program of Weapons Project, UCRL.

³²⁴Letter dated January 12, 1953 to Gen. Kenneth E. Fields, Director, Division of Military Application, USAEC, from Donald Cooksey, Associate Director, UCRL.

implosion bombs versus initiation times; "XR3" yield was expected to be about 15 KT.

A second test, named "Hamlet," was to be a test of a "radical implosion system;" its objective was to reduce the amount of fissionable material required to provide a given moderate yield or to improve the performance of a small implosion system.³²⁵

"Hamlet" was expected to yield 37 KT and was to use a TX-13 HE assembly with a Type D pit. The laboratory also planned to test a MK 6 bomb with a TOM initiator and a Type D pit.

Other LASL UPSHOT experiments would, like their UCRL counterparts, be oriented toward both guiding and confirming design considerations of CASTLE thermonuclear devices. Because of the extreme complexity of the computations and the highly uncertain physical constants involved in the design of "dry" test devices, experimental information would be the only reliable guide to design. (CASTLE would soon reveal, in a truly spectacular fashion, the limitations of LASL's knowledge in at least one critical area.)

LASL noted that although the UPSHOT tests were basic to the understanding and control of the use of radiation implosion for thermonuclear weapons, there were other types of information which might ultimately be necessary for weapons design but which could not be included in UPSHOT plans. For example, because of yield limitations on Nevada tests, no information would be obtained on the various factors affecting progress of a thermonuclear explosion.

Nor could there be obtained any clear data on the general subject of the progress of the mixing of heavy and light materials. An experiment (called "Mixmaster") was designed and considered for a while; the test was to have been an observation and measurement of the effects of mixing of light and heavy materials, as would happen during the explosion of

³²⁵McPhee, in THE CURVE OF BINDING ENERGY (pp. 113, 114), described "Hamlet" as follows:

One time, for example, Ted was asked to see how well he could do "in a certain yield range" in terms of "high efficiency, high compressions, high criticality" -- no fancy innovations, just the best implosion bomb he could make within the parameters given. The result was Hamlet, the most efficient pure-fission bomb ever exploded in the kiloton range.

thermonuclear devices.

However, LASL concluded that although the experiment would provide useful data on mixing, the information would not be in a form that could be readily applied to weapons then under design, so the experiment was withdrawn from the UPSHOT schedule.

Nonetheless, LASL proposed several radiation implosion tests, including two tests of a primary called the RACER for use in TX-14, -16, and -17 bombs. The RACER would use a TOM internal initiator.³²⁶ (By early February 1953, UPSHOT comprised six LASL and two UCRL experiments.)

At the end of December, the AEC was proposing a two-month long, six-shot CASTLE test series starting in February or March 1954. The six detonations would all be thermonuclear shots; four would have yields in the megaton range and four shots would be fired at Bikini atoll.³²⁷

At this time, the UCRL testing program comprised a number of objectives; among them were some new and radical proposals:

The fundamental contributions to the Weapons Program resulting from the success of the tests proposed by UCRL in CASTLE would be:

a. the design of radiation implosion systems of small dimensions, [deleted] Such designs, if successful, could lead to significantly smaller thermonuclear weapons of diameters as low as 36 inches with yields in the range of [deleted]

b. [deleted]

c. [deleted] Because of the peculiar design [deleted] it may be possible to initiate such a reaction in [deleted] deuterium using the large flux of 14 MeV

³²⁶Letter dated January 6, 1953 to Brig. Gen. Kenneth E. Fields, Director, Division of Military Application, USAEC, from N. E. Bradbury, Director, LASL, via Carroll L. Tyler, Manager, Santa Fe Operations, USAEC.

³²⁷Atomic Energy Commission, Operation CASTLE, Note by the Secretary, December 30, 1952.

neutrons issuing from the
thermonuclear reaction.

d. [deleted] These systems have never
been studied experimentally and little
is known concerning the cross sections
involved.

It is clear from these objectives that
UCRL's immediate test program is not
directed toward early design of a
deliverable weapon, but rather toward long
range studies of radiation channel design
and hydrodynamics of fuel assembly.

UCRL planned to participate in only two UPSHOT shots and
would probably participate on four CASTLE shots, including
two originated by LASL. UCRL planned to record the reaction
history on the latter: measuring the alpha of the primary,
time between primary and secondary explosions, and progress
of the secondary explosion.

UCRL was striving for a CASTLE test program that could be
well-rounded while still capable of being supported by the
new lab's small staff. UCRL did not plan to test any
"emergency capability" weapons; LASL planned to fulfill the
need for an early "emergency capability" by testing the ALARM
CLOCK, and by developing a practical liquid deuterium-fueled
weapon (the JUGHEAD) based on the SAUSAGE that had been
tested during the IVY Mike shot.

UCRL's CASTLE shots would test new compression geometries and
fuel assembly methods; the soonest the lab would be ready
might be January 1, 1954, a year hence. Since at least one
test would be liquid-fueled, the lab had to establish a
cryogenics engineering staff and laboratory.³²⁸

At the beginning of January 1953, LASL hoped to postpone
indefinitely any firm decision regarding both the scope and
execution date for Operation DOMINO.³²⁹

On January 7, General Fields wrote to Norris Bradbury and

³²⁸Atomic Energy Commission, UCRL Testing Program, AEC 605,
December 30, 1952, with attachment, "Weapons Program
Planning, University of California Radiation Laboratory,
Livermore, California," Herbert F. York, December 8, 1952.

³²⁹Letter dated January 5, 1953 to Carroll L. Tyler, Manager,
Santa Fe Operations Offices, USAEC, from N. E. Bradbury,
Director, LASL, Subject: Committee to Study Operational
Future -- Nevada Proving Grounds.

confirmed the change of date for the start of CASTLE and tied "emergency capability" weapon deployment directly to successful testing in 1954:

- a. The CASTLE operation will take place in early 1954.
- b. TX-14s will be fabricated at the maximum rate allowable by the availability of materials to the end that an emergency capability in that number will exist in the stockpile when the first weapon model is available for proof test.
- c. LASL should be prepared to support scientifically a short overseas test operation in the Fall of 1953 on an emergency basis if scientific developments indicate the necessity therefor.
- d. In the absence of an emergency overseas operation, DOMINO may be scheduled for the Fall (sic) of 1953. If it is conducted, every attempt will be made to hold it to a RANGER-type operation.

The Department of Defense has been notified of our intention to postpone CASTLE to early 1954, although the possibility of an overseas test in the Fall of 1953 on an emergency basis was noted.³³⁰

In the middle of the month, the AEC hoped that at least three of the six CASTLE shots would result in megaton-yield weapon models.³³¹ A week later, DOMINO plans had progressed to the point that LASL was requesting a series of three to five airdrops to test new initiators, among other things.³³²

The AEC was going to have to construct several new facilities

³³⁰Letter dated January 7, 1953 to Norris Bradbury, Director, LASL, from Brig. Gen. K. E. Fields, Director of Military Application, USAEC.

³³¹Letter dated January 15, 1953 to Honorable Carl T. Durham, Acting Chairman, Joint Committee on Atomic Energy, from Gordon Dean, Chairman, USAEC.

³³²Letter dated 23 January 1953 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

to provide materials for hydrogen weapons, including a lithium-6 separation plant, a manufacturing and fabrication facility for deuteride and hydride weapon components, and another plant for the isotopic separation of boron-10.³³³

The availability of thermonuclear weapon fuels became increasingly important as CASTLE planning continued. At the beginning of February, there was enough tritium on hand in the U.S. to see the labs through the end of 1953; excess reactivity in two of the Savannah River reactors would be used to produce more for the CASTLE tests in 1954.³³⁴

Another important hydrogen bomb component was boron-10, used to direct the flow of neutrons within the weapon casing and as a source of lithium-7 when bombarded by neutrons. There was currently a shortage of B¹⁰; the AEC had plans to set up a new plant to produce this vital material. A planned \$21 million emergency "postum" reactor had been canceled by DOD in mid-October 1949.³³⁵

Boron has a high cross-section for the absorption of slow neutrons which fission U-235 and Pu-239, but a low cross-section for the absorption of fast neutrons which fission U-238.³³⁶ Because of this characteristic, boron-10, when loaded into the secondary stage of a two-stage weapon, would prevent predetonation of the "sparkplug" by stray neutrons from the primary without interfering with the subsequent fissioning of the U-238 fusion tamper around the secondary.

³³³Memorandum to the Files dated January 16, 1953 from Wayne P. Brobeck, subject: Meetings with Walter Williams on January 9 and 16, p. 2.

³³⁴According to September 12, 1952 Walker memorandum, p. 22, there was enough tritium on hand in September 1952 for device shots in both Operations UPSHOT and CASTLE, as well as for research programs at all AEC laboratories.

³³⁵Letter dated November 23, 1949 to Senator Brien McMahon from David E. Lilienthal, Chairman, USAEC.

³³⁶Memorandum dated February 2, 1948 to Harry O. Whipple from Thomas N. White, Subject: Vault Design; letter dated July 15, 1969 to Dr. H. Peter Metzger, Chairman, Colorado Committee for Environmental Information from David F. Miller, Information Officer, USAEC Nevada Operations Office. The latter notes that the Rulison device, used in an application of "peaceful" nuclear explosives to stimulate the release of trapped underground natural gas, was a "fission device using U-235" and that a "large quantity of boron would be used to 'isolate' the nuclear explosive from earth-reflected neutrons."

Any lithium-7 -- a fusion fuel -- that was generated during the process would be an extra bonus (although the utility of lithium-7 to the early U.S. thermonuclear weapons program would not be fully appreciated until after the CASTLE Bravo shot).

Boron also played a role in increasing the compressive plasma pressure around the secondary, leading to higher thermonuclear efficiency.³³⁷ By "doping" with boron-10 the structural foam holding the secondary in place within the casing, the secondary might be compressed more highly, at a cost of some radiated neutrons.

Lithium-6 production quotas would be met by the construction of a new plant at Oak Ridge, called an "Alloy Development Plant" (ADP) to mask its true purpose. The plant was expected to be producing Li-6 by the middle of March or early April 1953, although not necessarily at full capacity.³³⁸

In order to squeeze the last gram of U-235 out of uranium ore sources, the AEC agreed with a DOD request to use depleted uranium -- consisting almost solely of the U-238 isotope -- for weapon parts which had in the past been made from natural uranium containing both U-235 and U-238 isotopes.³³⁹

By February, the AEC and DOD had decided that February 15, 1954, was the earliest possible date for the first CASTLE detonation. Although CASTLE had been planned for the fall of 1953, perhaps as early as September, it was evident by January 1953 that the operation would be delayed by as much as six months.

Some of the reasons for this delay were changes in design of CASTLE test weapons and devices; changes in the scheduled availability dates of certain materials needed for use in the weapons and devices (primarily delays in lithium-6 deuteride production: large-scale production of lithium-6 began at Oak Ridge in August 1953);³⁴⁰ and the potential conflict for test

³³⁷Winterberg, THE PHYSICAL PRINCIPLES OF THERMONUCLEAR EXPLOSIVE DEVICES, p. 77.

³³⁸September 12, 1952 Walker memorandum, p. 22.

³³⁹Minutes of the Thirty-Third Meeting of the General Advisory Committee to the USAEC, Washington, D.C., February 5-7, 1953, pp. 14, 16, 31. "Depletalloy," or depleted uranium, contained from 99.85% to 99.97% uranium-238. Natural uranium consists of about 99.3% U-238 and about 0.7% U-235.

³⁴⁰Hewlett and Holl, p. 163.

personnel with the DOMINO test series then scheduled for Nevada in the fall of 1953.

A tentative schedule based on the new start date was drawn up, including tests of four Los Alamos weapons and devices and two UCRL devices. One of the UCRL weapons used a "new principle" then under development at Livermore; UCRL was simultaneously pursuing two thermonuclear schemes.³⁴¹

Of these shots, four were to be detonated at Bikini and two at Eniwetok; three were to be island shots, two barge shots, and one barge or island. Except for the RAMROD UCRL device, all shots were predicted to be in the megaton range. The tests would be conducted between mid-February and the end of March 1954 at one-week to ten-day intervals.³⁴² (This was an extremely optimistic schedule, given delays in past Pacific Proving Ground series caused by weather and technical problems.)

It looked now like the TX-16 JUGHEAD would probably be weaponized by the time that CASTLE was completed; a few units could probably be made within a year. LASL was responsible for providing an "emergency" thermonuclear capability, although a specific description of what this capacity should be had not yet been furnished to the lab. The outlook for the ALARM CLOCK was less favorable.³⁴³

The AEC's Savannah River plant had begun producing deuterium oxide in January. The Dana plant was operating at 75% capacity; one problem holding up production was corrosion of the mild steel used in the separation columns (the Savannah River columns used stainless steel).³⁴⁴

The idea of using barges as CASTLE shot sites was proposed at

³⁴¹Minutes of the Thirty-Third Meeting of the General Advisory Committee to the USAEC, Washington, D.C., February 5-7, 1953, pp. 2, 11.

³⁴²"History of Task Group 7.4 Participation in Operation CASTLE, 1 January 1953 - 26 June 1954," Historical Division, Office of Information Services, Air Force Special Weapons Center, Kirtland AFB, New Mexico, 8 November 1954, p. 88.

³⁴³Minutes of the Thirty-Third Meeting of the General Advisory Committee to the USAEC, Washington, D.C., February 5-7, 1953, p. 1.

³⁴⁴Minutes of the Thirty-Third Meeting of the General Advisory Committee to the USAEC, Washington, D.C., February 5-7, 1953, p. 18.

this time to reduce radioactive contamination, to make better use of the limited amount of real estate available (some of which was bound to be vaporized), and to speed up the tests by scheduling assembly operations at the barge slip at Eniwetok and moving each barge into position (by Navy LSD) about five days before the scheduled detonation.

This made it possible to plan on using the same concrete test equipment bunkers for several shots, and the same zero point, if necessary or desirable, regardless of the high radiation levels to be expected on the islands in the vicinity of preceding shots.

Use of barges anchored in the lagoon also left the reef intact, preserving the lagoon, and protecting seaborne operations in the lagoon from strong ocean currents and high ocean waves.

All of the CASTLE shot schedules were based on having acceptable firing weather on the day each shot was scheduled. No such good fortune was expected, however, and in all operational planning, weather was a problem of great concern. Completion dates anywhere from two weeks to two months later than originally scheduled seemed reasonable. (The sixth and last shot in the CASTLE series was actually fired 29 days after its originally-scheduled date.)

At this time, the AEC expected to conduct two continental atomic test series in 1953, Operations UPSHOT in the spring and DOMINO in the fall, and both an overseas (CASTLE) and a continental (TEAPOT) test series in the fall of 1954.³⁴⁵ DOMINO was essentially reserved for "quick and dirty" drops of stockpile bomb models and was to emulate the 1951 Operation RANGER as closely as possible.³⁴⁶

The only other forthcoming continental test series planned at this time was JULEP, a weapons development and weapons effects test series in the fall of 1955. Future overseas test series included REDWING, at Eniwetok in the spring of 1955; WIGWAM, off the U.S. west coast in the spring of 1955; and HARDTACK, at the PPG a year later.³⁴⁷

³⁴⁵Letter dated February 8, 1953 to Honorable Robert LeBaron, Chairman, MLC, from Gordon Dean, Chairman, USAEC.

³⁴⁶Minutes of the Thirty-Third Meeting of the General Advisory Committee to the USAEC, Washington, D.C., February 5-7, 1953, p. 8.

³⁴⁷Memorandum dated May 18, 1953 to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, from Capt. John T. Hayward, USN, Acting Director of Military Application, USAEC; Operation TEAPOT, Report of the Test Manager, Joint Test

To avoid the widespread disclosure of classified information that occurred during IVY, the AEC proposed on February 9 that all mail, both civilian and military, be censored during CASTLE by designated civilian and military representatives.³⁴⁸ Censorship of civilian mail during peacetime was unprecedented; however, the AEC was still smarting over the many "leaks" during IVY. (As events would have it, one of the biggest technical secrets of CASTLE evaded censorship and burst forth in a blaze of international headlines.)

Unfortunately for the AEC, peacetime censorship of civilian mail would require the enactment of federal legislation by Congress; the AEC noted that the imposition of censorship on military personnel while allowing civilians unbridled letter-writing would undermine morale and would probably be ineffective.

Censorship of both civilian and military mail would only make hiring of construction personnel (always a problem at remote Pacific sites) much more difficult. Legal complications surrounding official censorship would only call more undue attention to CASTLE; the receipt of censored mail by relatives and friends of task force personnel would probably result in an even greater effort on their part to learn what was happening. In addition, mail censorship was not in effect even in the Korean combat zone.

With these considerations in mind, the task force commander urged that voluntary censorship be continued, and that the AEC, within security guidelines, make as much information as possible public before the test series began.³⁴⁹ (In an abrupt departure from IVY practice of maximum secrecy and virtually no public information release, the AEC concurred with this recommendation at the end of November 1953.)³⁵⁰

The GAC recommended at this time that "since the results of

Organization, Nevada Test Site, Spring 1955, p. 7.

³⁴⁸Memorandum dated February 9, 1953 to George P. Kraker, Deputy Manager, Santa Fe Operations Office, USAEC, from Capt. John T. Hayward, USN, Deputy Director of Military Application, USAEC.

³⁴⁹Letter dated 24 March 1953 to Chief of Staff, U.S. Army, from Gen. P. W. Clarkson, USA, Commander, Joint Task Force Seven; memorandum dated May 18, 1953 to Brig. Gen. K. E. Fields, USA, from Capt. William L. Guthroe, USN, Subject: DMA Position Regarding Mail Censorship for CASTLE.

³⁵⁰Letter dated November 30, 1953 to Honorable Sterling W. Cole, Chairman, JCAE, from Lewis L. Strauss, Chairman, USAEC.

the test programs are certain to affect the optimum composition of the stockpile with respect to weapon types, and since the information will be most useful before the stockpile increases to the point that weapon refabrication becomes an unmanageable task, we are led to favor an increase in the weapon testing capabilities in the near future."

The GAC also noted that both lithium-6 and tritium were essential materials in the weapons program, and, if available in quantity, would have an important impact on the spectrum of types of weapons maintained the stockpile.³⁵¹

LASL Plans for Operation UPSHOT

By late February, LASL's plans for UPSHOT had become quite definite; they were to advance both fission and fusion weapon technology. Lab director Norris Bradbury prepared a lengthy briefing on LASL's UPSHOT plans in which he presented the forthcoming tests in the larger contexts of contemporary U.S. nuclear weapons development and testing philosophies:

The primary responsibility for the development of new atomic weapons as well as the improvement of existing devices is carried out for the Atomic Energy Commission by the Los Alamos Scientific Laboratory. Supplementary work in this field is also being carried out by the Radiation Laboratory of the University of California at Livermore.

Research in the field of atomic weapons is always directed at getting the most for one's money (i.e., fissionable material) under the particular conditions required by the use or delivery system of the weapon. Accordingly, the LASL directs its efforts both towards increasing the efficiency of use of active material in a given weapon system, and to increasing the scope and flexibility of atomic weapons so that they may be available for the maximum number of applications to problems of warfare.

Research on atomic weapons means, of course, not only research on the best ways under various circumstances to use fissionable material but research on the use of cheaper materials for atomic

³⁵¹Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, pp. 35, 36.

weapons. Thermonuclear reactions, sometimes publicly described as the H-bomb, fall into this field.

Laboratory research and study can investigate nuclear phenomena or the other phenomena of potential nuclear weapons only up to the point before a nuclear reaction starts. There are two reasons for this: one, a chain reaction requires the use of expensive fissionable material; and two, after the chain reaction starts, one has an atomic bomb and it would be impractical to build and staff a new Los Alamos after every experiment. (Author's note: a rare flash of Bradbury humor.)

With stand-in materials, the LASL can investigate in considerable detail most of the behavior of an atomic bomb up to the time when, with real materials, it would actually start to explode. However, the explosive phase of the reaction is not only the most interesting and the most important phase, but it is the phase which is least susceptible to calculation. It involves temperatures as hot as the interior of stars and the properties of materials at temperatures, pressures, and densities obtained nowhere else on earth.

Temperatures of ten million degrees, densities that would make a cubic foot of water weigh a ton, pressures of millions of atmospheres, and times of hundred-millionths of a second (the "shake") are commonplace in these circumstances.

As an example of how difficult the exploding phase of an atomic bomb is to calculate, one may go back to the TRINITY bomb of 1945. After all the theorists had made their best calculations (really, their best guesses), they nodded wisely and said it will probably be about five to ten kilotons. But it came out nearer 20.

A few years ago, the LASL made the most extensive calculation that could be devised using the most modern electronic computers of this same system. This time, however, the answer came out nearly twice as big as the bomb actually went. Clearly, only

experiment can really say exactly what is going to happen and can point the way to real improvements in atomic weapons. This our empirical formulae cannot do.

The details of the Los Alamos and Livermore (UPSHOT) experiments are technically so complicated that little use would be served by describing their intimate details. None of these particular experiments are models of actual weapons although this is, of course, not necessarily true of the shots being conducted for other agencies. Moreover, all of the LASL experiments have as primary objects additional information (generally of many sorts) related to the processes which go on when an atomic bomb explodes under one or another type of nuclear system.

Thus, an extremely detailed and complicated program of experimental observations is both fundamental and essential and the actual "bang" or yield of the explosion is, in many cases, of secondary interest or of diagnostic interest in furnishing information as to what went on, and how, inside the chain reacting system.

The accompanying slide shows the list of bombs and the rather affectionate names assigned to them, the agency responsible for the particular test, and the probable yield. This latter figure, as indicated above, is not of significance in itself in experiments of this sort, but is mentioned here only to give some relative idea of how the explosion size which will be seen compares with the explosions from conventional weapons in the stockpile.

To give some idea of what sorts of fields are being explored by the LASL, the first and eighth shots, Annie and Harry, are primarily explorations of the hydrodynamics of the assembly of fissionable material and what goes on in the very late stages of this assembly, particularly if the initiation of the chain reaction is delayed.

Annie is actually the third (test) of a series (of tests), two of which have been conducted in earlier test operations. The

result of (this series of tests) will give us much information which can be applied to increasing the efficiency of use of fissionable material in new weapons. (Author's note: Annie, a test of the XR3 device, was to determine the third data point on an empirical curve of yield versus initiation time; Harry was the "Hamlet" device test.)

The second and seventh shots, Nancy and Simon, are explorations of scientific phenomena urgently needed by the LASL in connection with the research program on thermonuclear and related systems. These are not thermonuclear weapons; indeed, they have little relationship to any actual weapon, real or imaginary. They are actually pure nuclear experiments in the realm of temperatures and pressures which can only be obtained by setting off an atomic bomb.

The fourth shot, Dixie, is an experimental observation of a new and cheap method for initiating a chain reaction as well as furnishing more light on the question of when and how such reactions start.

The fifth shot, Badger, is an experiment to explore a potential new technique of not only increasing the efficiency of burn-up of fissionable materials in atomic bombs, but an exploratory experiment in the further use of cheap materials in nuclear explosions. Again, the device is not a weapon, but if it works, the ideas found effective therein will speedily find their way into weapon application.

The two (University of California) Radiation Laboratory experiments (shots Ruth and Ray) are explorations of the nuclear properties of certain systems, a knowledge of which may prove useful for both conventional and thermonuclear research programs.

It will be noticed that most of the LASL shots are on towers rather than dropped as airburst bombs. This is primarily because of the detailed instrumentation which requires that not only the precise time of detonation be known, but that the

device be exactly placed and that complicated instruments with electronic recording be in its immediate vicinity.

With yields of the order of magnitude indicated, such tower shots can be conducted with complete safety. Only when the necessary information can be obtained equally well from a free air experiment are these employed.³⁵²

Early in March, a meeting was held between representatives of LASL, the AEC, the Armed Forces Special Weapons Project (AFSWP) and DOD to review CASTLE plans.³⁵³ At this meeting, AFSWP representatives presented a requirement for a barge shot in deep water outside Bikini lagoon.

Although LASL agreed to relocation of one of the barge shots to a position outside the lagoon, with certain restrictions, the estimated yields of the devices then scheduled were too high to make a satisfactory test probable. In view of this and the additional support involved, the proposal was dropped.

The Air Force Proposes an Air Drop

Another consideration in the late winter of 1953 during the early planning for Operation CASTLE was an airdrop of a test weapon; the suggestion was first made in February by Dr. Hans Bethe. The airburst of a high-yield weapon at an operational height would test the whole delivery system, obtain scientific information, and demonstrate dramatically to the Soviet Union that the U.S. had an "emergency capability" for the deployment of thermonuclear weapons. The Air Force had already successfully dropped unarmed TX-14 ballistic shapes over

³⁵²Memorandum dated 26 February 1953 to C. L. Tyler, Manager, Santa Fe Operations Office, USAEC, from N. E. Bradbury, Subject: Nevada Briefings. This document is rather unusual in that it was originally classified only "Confidential," the lowest level of AEC-DOD classification; however, the information in the briefing paper is sufficiently vague and general to warrant this low classification. The "rather affectionate names" assigned to the tests probably refers to the feminine names assigned to the majority of the shots. As noted below, these unclassified shot names were mnemonics for classified device names.

³⁵³The AFSWP was a successor to the AFSWC and a predecessor to the Defense Atomic Support Agency (DASA), which is now the Defense Nuclear Agency (DNA). The acronym AFSWC was later used to designate the Air Force Special Weapons Center.

Nevada and New Mexico.³⁵⁴

One interest from a military point of view would be an opportunity to compare thermal radiation from the air-dropped IVY King shot with a larger weapon. However, Norris Bradbury was strongly opposed to an airdrop for a number of reasons:

In spite of the fact that the devices to be tested in Operation CASTLE are being designed with "emergency capability" in mind, the operation is primarily an experimental test program in the field of weapon development. Accordingly, the diagnostic experiments are still of the highest importance for it is not expected that the design of radiation implosion weapons or their further improvement will stop with this test program.

Therefore, we strongly believe that no step should be taken which diminishes the amount of appropriate and relevant experimental information which can be obtained. Every effort is being made to simplify the experimental program and to include only those experiments which are essential and have a proper balance between their cost and the information which they give.

However, all shots are expected to carry as a minimum some experimental observations including the behavior of the primary bomb (alpha), the radiation transit time to the secondary bomb, and the photographic behavior of the case.

The ALARM CLOCK is presently planned not as a barge shot but as an island shot and it is likely that the most extensive instrumental observations will be carried out on this experiment.

The remainder of the Los Alamos shots are planned as barge shots located near atoll islands so that photographic and other observations can be made on them. We would regard it as absolutely impossible, were it otherwise desirable, to drop the JUGHEAD at

³⁵⁴Teletype dated February 17, 1953 to Carroll Tyler, Santa Fe Operations Office, USAEC, from V. G. Huston, USAEC, Washington, D.C.

the time of the presently planned test shot inasmuch as the associated cryogenic logistics and test drops are actually coming to completion at or shortly after the test date.

The dropping of the ZOMBIE would, presumably, be a task within Air Force capabilities. However, the TX-15 is the last of the devices scheduled for final design freezing and will be the last to be constructed. This is necessarily so in view of the state of knowledge of systems of this nature.

Accordingly, only the ALARM CLOCK is actually available as a device for consideration for a drop test. We would regard the sacrifice of the experimental observation of the (test) as too high a price to pay for an extremely dubious demonstration of "emergency capability." Actually, emergency capability is far more easily, cheaply, and effectively demonstrated by methods other than using a live bomb.

Moreover, it should be recognized that the yield of the ALARM CLOCK is probably more unknown than any of the devices to be tested and may vary from [deleted] to [deleted]. Such a priori ignorance would seem to be extremely unfortunate for any effects instrumentation and might seriously jeopardize its effectiveness.

It should be recalled that the development status of the (thermonuclear bombs) in emergency capability closely parallel the status of the Nagasaki and Hiroshima bombs in the sense that the actual reliability of any of the(ir) components will be far from exactly known.

Thus, there will exist the possibility of failure or malfunction due to completely minor and extraneous reasons. These reasons will not be known, but an unnecessary stigma will be attached to the device which may be impossible to remove and which would not have occurred had a proper experiment been made.

It must be recalled that the definition of

"emergency capability" is that only (unknown) characteristics of the device will prevent its delivery by existing aircraft in time of war. Models of these devices that have the same reliability now expected of conventional weapons will require long and arduous further study, development, and field testing of the Sandia Corporation type.

The certainty of any air drop cannot be guaranteed at this time. Thus it would seem necessary for J-Division to plan on both an air drop and a barge capability for a given test if this philosophy were agreed upon. Accordingly, for this reason alone, the introduction of an air drop does not simplify the test program but rather complicates it.

Even were it certain that an air drop were possible, the resulting operation is not necessarily less complicated except for the fact that few observations are made. Not to do a test at all, of course, is a simplification of the same sort. That air drops themselves present complications is apparent when the question is raised as to whether the (air)strip at Eniwetok (and by the same yardstick, Kwajalein) is long enough to permit the take-off of a B-36 loaded with a (50,000 lb. bomb).

If it is not, then one must go back to Hickam (AFB) and the question of taking off in the vicinity of Honolulu under these circumstances is a very dubious one. Finally, the accuracy of a drop under parachute conditions is such as to complicate enormously the photographic problem.

Furthermore, if the parachute should stream (for which there is always a probability), all such observations would be lost. Accordingly, we cannot agree that an air drop under these circumstances and in the present stage of development really simplifies the Eniwetok test operation.

... In the particular instance, we are less inclined to regard specific effects of weapons of this class as exceedingly important. Our reason for this belief

stems primarily from the fact that this is the biggest weapon that we know how to build.

It will give the largest effects of any weapon that we know how to build, but a knowledge of precisely what these effects are seems not to be of crucial importance in advance of actual use.³⁵⁵

The airdrop proposal was later shelved because it would have placed an additional burden on the hard-pressed staff of J-Division at LASL, and because there was nothing ready to airdrop in early 1954. The bombs proof-tested at CASTLE -- which ranged in weight from 6,520 to 39,600 lbs. -- were built so they could have been dropped from an aircraft; however, ballistic casings, fins, and aerial fuzing systems would have to be attached.

All the test weapons were finally fired on the ground or on barges by electronic command, and there was very little ballistic data available on any of the casings at the time. Airdrop tests on the TX-16 were not scheduled for completion until May 1954. (TX-14 drop tests began in November 1952; TX-16 airdrop tests started in December 1953 and ended in April 1954.)

LASL scientists were very reluctant to put a test device which they knew very little about on board an aircraft, carry it around, then drop it without knowing exactly what condition it was in for the last half hour or so before shot time. In-flight monitoring equipment for thermonuclear weapons was virtually nonexistent at this time.

LASL stated that if Washington decreed it, the lab would airdrop something, but would not guarantee that it would go

³⁵⁵Letter dated 20 February 1953 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL. MK 15 Mod 0 bomb design was not released until October 1954, well after CASTLE had ended. The airstrip at Eniwetok was 6,800 feet long; Kwajalein's main runway was originally only 6,700 feet; it was later extended to about 10,000 feet. A major bombing error during the Cherokee airburst shot of Operation REDWING, two years later, would underscore Bradbury's concerns about gathering useful data from high-yield air drops. Note that in this document at least, the terms "primary bomb" and "secondary bomb" have replaced the earlier terminology of "auxiliary" and "principal" bombs as stages of U.S. thermonuclear weapons. "Streaming" refers to failure of a parachute to properly deploy and inflate.

off, and that it would probably prevent or delay subsequent shots. LASL also believed that stockpiling of an "emergency capability" weapon might well be delayed by such a test.³⁵⁶ (The airdrop test was finally postponed until Operation REDWING in 1956).

The CASTLE shot schedule was changing almost weekly by now. At the beginning of March, the plan was to fire the first shot, a test of the liquid-fueled LASL JUGHEAD device, at Eniwetok, in the vicinity of the Mike crater or Rigili; the second shot on Eninman island at Bikini; the third, UCRL's first thermonuclear test, at Bikini, on a barge in the lagoon or at a site on the Yurochi-Aomoen island chain; the fourth and fifth shots on barges at Bikini; and the last on Eberiru island at Eniwetok.

LASL and UCRL noted that "none of the bomb yields is guaranteed within a factor of three." LASL also expected that as soon as the first shot was fired at Bikini (the second test in the series), the base camp and airstrip on Bikini would be destroyed or contaminated, and all subsequent operations at Bikini would have to be conducted from shipboard (this was a very prescient observation, as things turned out). Shots on Eniwetok would be fired from a control building on Parry island.

Just two weeks later, in mid-March, the first CASTLE test was to be fired on a barge over deep water off the southwest coast of Bikini; the second on a barge in Bikini lagoon near Namu island; the third on a barge near Yurochi; the fourth on Eninman; the fifth on Eberiru at Eniwetok; and the final shot on Engebi at Bikini.³⁵⁷ There was still typhoon damage to be repaired at Eniwetok.³⁵⁸

The GAC Meets to Review CASTLE, March 1953

At the 34th meeting of the GAC in Washington, D.C. on March 23, Gen. K. E. Fields addressed the issue of lithium-6 requirements. Final goals for production could be set in the

³⁵⁶"Summary of Notes on a Meeting Held at Los Alamos, N.M., 3-4 March 1953, To Discuss CASTLE Planning with Particular Reference to TU-13 (DOD Program)," Los Alamos Scientific Laboratory, Los Alamos, New Mexico, March 1953, pp. 14, 15.

³⁵⁷Teletype dated March 16, 1953 from USAEC, Santa Fe, to USAEC, Washington, D.C.

³⁵⁸Memorandum dated 24 March 1953 to Director of Military Application, USAEC, from Maj. Gen. P. W. Clarkson, USA, Commander, Joint Task Force Seven, Subject: Use of Kwajalein as Base for Air Task Group.

fall; however, he listed possible yearly production quantities for lithium enriched to both 30% and 95% Li-6 concentration for Fiscal Years 1954 through 1956.

A budget of \$150 million had been requested for a second lithium separation plant that would have three times the capacity of the first plant now nearing completion at Oak Ridge. Estimates of the cost of Li-6 had risen recently due to technical problems with the electrolytic separation equipment. Current boron-10 requirements were also reviewed during this meeting.

Dr. John von Neumann commented on bomb requirements for Li-6; he noted that it was not yet clear what concentration of lithium-6 vs. lithium-7 was necessary for optimum tritium production during a thermonuclear explosion, or if even unenriched lithium might suffice.

Dr. Willard F. Libby pointed out that lithium-7 could also be used in a bomb to generate tritium (see equation F in "Fusion Physics" under "Weapons Physics" in Volume I) and that it might not be wise to proceed with a \$150 million lithium isotope production facility until the cross-sections of lithium-7 for tritium generation had been measured in the relevant energy range, which might not be obtainable in a laboratory and would have to await a full-scale thermonuclear test using Li-7. (Results of CASTLE would soon prove that Libby was more right about this point than he probably imagined.)

Dr. von Neumann said that the concentration of lithium-6 required for various solid-fueled modifications of the SAUSAGE design (the IVY Mike device) might depend upon weapon geometry, and that Li-6 would be required sooner or later under any circumstances.

AEC Chairman Gordon Dean noted that the expansive plans for UPSHOT-KNOTHOLE and CASTLE did not enjoy the support of everyone in the Eisenhower administration. In the executive branch, the test program was being questioned more and more as a "waste of fissionable material." Clearly there was a need for the AEC to do a better 'sales job' on the values and lessons of nuclear testing, and of its importance to national security.

The GAC also noted that if a new uranium enrichment facility at Portsmouth, Ohio were completed as planned, its existence might allow more flexibility in weapons design and use. An abundance of fissionable material would lead to the possibility of using smaller, less-efficient, but more easily-deliverable weapons. Since it was likely that small warheads might be required for new air-defense, naval, and tactical weapons, much more plutonium and enriched uranium should be

made available.³⁵⁹

A week and a half after this GAC meeting, in response to possible cuts in the U.S. nuclear weapons testing budget for 1954, MLC Chairman Robert LeBaron wrote to Gordon Dean and asked him to prepare a review of future test requirements and in particular, to furnish justifications as to:

- a. Whether the presently-contemplated scale of tests is essential to the development of weapons in accordance with DOD needs.
- b. Whether the scale of military support planned or contemplated for such tests is the minimum required.
- c. Extent of test(ing) after CASTLE which is essential to the (acquisition of) knowledge of the military effects of atomic weapons required for the proper employment of and defense against atomic weapons.
- d. Any methods or measures which can be taken to reduce the costs of such tests without serious loss of such information.³⁶⁰

Operation UPSHOT began in Nevada on March 25. By April 1, design details of CASTLE devices depended upon examination of UPSHOT results, and not upon Operation DOMINO, unless a specific determination of the yield of a CASTLE device primary was required. The elapsed time from the end of UPSHOT to the start of CASTLE was the minimum period required to integrate experimental information from UPSHOT into CASTLE designs.

Results of CASTLE, on the other hand, would not be analyzed in time to have any great bearing upon the next Nevada test series, Operation TEAPOT, now slated for the fall of 1954 or the spring of 1955. It was very likely that TEAPOT would support the next Pacific series scheduled for either fall 1955 or spring 1956 (Operation REDWING) in the same manner

³⁵⁹Minutes of the Thirty-Fourth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 23-24, 1953, Washington, D.C., pp. 4, 8, 9, 13; letter dated March 25, 1953 to Gordon Dean, Chairman, USAEC, from I. I. Rabi, Chairman, GAC.

³⁶⁰Letter dated 3 April 1953 to Honorable Gordon Dean, Chairman, Atomic Energy Commission, from Robert LeBaron, Chairman, MLC.

that UPSHOT supported CASTLE. LASL believed that TEAPOT might well be postponed until the spring of 1955 to allow full analysis of CASTLE results and preparation of suitable and well-planned experiments during TEAPOT and REDWING.

The lab thought that a minimum of a year was required between a preliminary experiment and the time a large and complicated device based upon it was tested. LASL noted that much of the work on forthcoming CASTLE devices was based on assumptions regarding UPSHOT results, and if these assumptions were incorrect, then CASTLE might be delayed and much money wasted. If TEAPOT were held in the spring of 1955, then REDWING should not begin before the spring of 1956.

LASL fully recognized the possibility of failure for at least one CASTLE device; no failure foreseen for CASTLE could be fixed with a quick test. Even if all the CASTLE devices were successful, there would still be a period in which the test systems would have to be engineered into more tractable and militarily-usable weapon systems.

Improvements in efficiency and reductions in weight might be incorporated. The lab believed that for future testing, the combination of both proof-testing of fully-engineered systems and development testing of ideas arising from TEAPOT results should be included in REDWING and that the next Pacific series should not be conducted before the spring of 1956.

The lab also recognized that while CASTLE was being conducted as a rather "quick and dirty" operation to test and ready for deployment megaton-yield "emergency capability" bombs, future improvements in thermonuclear weaponry would be a difficult and unspectacular process and would probably be largely directed toward decreasing warhead weight while losing only a minimum of yield.

Present USAF and USN bomber aircraft would gain only slightly in performance from modest decreases in bomb-load weight. New aircraft that might be more adaptable to carriage of lighter devices, and which would have correspondingly better combat performance, were still a number of years in the future; weapon and aircraft development might reasonably proceed together.

A lighter-weight hydrogen weapon such as the ZOMBIE merely did better what was already done and did not open up any new capabilities.³⁶¹

³⁶¹Office Memorandum dated 1 April 1953 to C. L. Tyler, Manager, Santa Fe Operations Office, USAEC, from N. E. Bradbury, Director, LASL, Subject: Weapon Test Program.

By April 10, the CASTLE schedule had changed again; it included three barge shots and two island shots at Bikini and one island shot at Eniwetok. Now the first shot, named Bravo and using LASL's SHRIMP device, would be fired on February 15, 1954 on a barge anchored over deep water off the southwest coast of Bikini. Ten days later, the second shot, a test of LASL's ZOMBIE, would be conducted as the Nectar shot at Bikini, on the reef 2,500 yards southwest of Namu island.

Shots three and four, named Union and Yankee, would be detonated on March 8 and 17, respectively, from barges in the vicinity of Yurochi and Aomoen islands at Bikini, firing LASL's solid-fueled ALARM CLOCK and cryogenic JUGHEAD devices. Shot five, named Koon, was now slated for March 24 at Eninman, using UCRL's MORGENSTERN device. The last test, named Echo, at Eberiru island at Eniwetok on April 3, 1954 would be the detonation of UCRL's liquid-fueled RAMROD device.³⁶²

Under this plan, Eniwetok would be the main base of operations and all cryogenic work would be done on Parry island. Five of the six devices would be assembled on Parry and Eberiru; those to be fired at Bikini would be transported by barges, naval ships, and trucks to their respective zero points. A firing station would be set up at Enyu island at Bikini, with a six-person or smaller firing party. All new construction was to be completed at Eniwetok and Bikini by January 1, 1954.³⁶³

Operation UPSHOT-KNOTHOLE: Prelude to CASTLE

In the same way that Operation RANGER had assisted in the definition of weapon designs for Operation GREENHOUSE, Operation UPSHOT-KNOTHOLE at the Nevada Proving Ground between mid-March and early June 1953 was conducted in part to develop test devices for CASTLE.³⁶⁴

³⁶²"History of Task Group 7.4 Participation in Operation CASTLE, 1 January 1953 - 26 June 1954," Historical Division, Office of Information Services, Air Force Special Weapons Center, Kirtland AFB, New Mexico, 8 November 1954, p. 88.

³⁶³"Summary of Notes on a Meeting Held at Los Alamos, N.M., 3-4 March 1953, To Discuss CASTLE Planning with Particular Reference to TU-13 (DOD Program)," Los Alamos Scientific Laboratory, Los Alamos, New Mexico, March 1953, pp. 3, 4, 5, 7, 9, 11; teletype dated March 17, 1953 from USAEC, Santa Fe, to USAEC, Washington, D.C.; memorandum dated 7 April 1953 to Distribution from Commander, Task Group 7.1, Subject: General Concept of Operation CASTLE.

³⁶⁴Operation UPSHOT, scheduled to be the next Nevada test

As exemplified by earlier Nevada test series, including Operations BUSTER-JANGLE in 1951 and TUMBLER-SNAPPER in 1952, UPSHOT-KNOTHOLE was also a joint series of both weapons development and military effects tests. UPSHOT was the development test phase, while KNOTHOLE was mainly oriented toward DOD interests.

By the end of 1952, LASL had succeeded in producing an impressive variety of fission weapons, was well along the way toward realizing very small and very large yield nuclear bombs, and was wrestling with the problem of creating deliverable fusion bombs. UPSHOT-KNOTHOLE provided an opportunity to run mockup tests leading to the development of weapons to be tested during CASTLE: the shot schedule for CASTLE would largely depend upon the results of these mockup tests; one CASTLE test would be of the ALARM CLOCK.³⁶⁵ In addition, the new weapons lab at Livermore needed field test information to design its thermonuclear devices.

The devices to be fired in CASTLE depended largely upon the results of these mockup tests in UPSHOT-KNOTHOLE. UPSHOT, the weapons development phase, included tests for both LASL and UCRL, and KNOTHOLE was largely effects tests for DOD and the Federal Civil Defense Agency. In some cases, it was possible to conduct experiments for both labs by setting instruments on opposite sides of a single shot tower.

The results of UPSHOT-KNOTHOLE revealed that hollow-core nuclear systems could greatly increase the efficiency of implosion weapons. The series included tests of radiation implosion mockups, containing small amounts of enriched Li-6, and primaries as design prototypes of TX-14, TX-16, and TX-17/24 thermonuclear systems to be tested during Operation CASTLE.³⁶⁶

series after SNAPPER, was originally slated for the fall of 1952. (Atomic Energy Commission, New Code Name, Report by the Director of Military Application, November 16, 1951.)

³⁶⁵Excerpt from Minutes of AEC Meeting No. 746 on September 11, 1952.

³⁶⁶Letter dated 11 June 1952 from N. E. Bradbury, Director, LASL, to Dr. J. Robert Oppenheimer; Progress Report to the Joint Committee on Atomic Energy, December 1952 through May 1953, USAEC, June 22, 1953, p. 7; letter dated August 11, 1953 from R. W. Henderson, Sandia Corporation, to Marshall G. Holloway, Director, X Division, LASL. The latter letter promises fabrication by ACF and shipment by Sandia to LASL of dummy forward assemblies of TX-14, -16, and -17 weapons for use during UPSHOT-KNOTHOLE.

UPSHOT-KNOTHOLE also produced further information regarding boosting with deuterium gas, external initiators, and initiation via deuterium.³⁶⁷ Operation UPSHOT-KNOTHOLE was characterized as "the most important continental tests held thus far;" successful test results provided increased assurance that a deliverable thermonuclear device would emerge from CASTLE.³⁶⁸

The AEC noted that five of the detonations -- Nancy, Ruth, Badger, Simon, and Climax -- advanced the U.S. thermonuclear weapons program and that the remaining six shots were all important to the advancement of standard fission weapons. For example, one shot (probably Harry, a test of Dr. Theodore Taylor's "Hamlet" device) confirmed a theory which was later applied several designs.³⁶⁹

³⁶⁷Progress Report to the Joint Committee on Atomic Energy, December 1952 through May 1953, Part III, Weapons, June 22, 1953, p. 7.

³⁶⁸Atomic Energy Commission, Semiannual Report to the President, AEC 21/12, Report by the General Manager, July 27, 1953.

³⁶⁹Memorandum dated July 29, 1953 to Distribution from Morse Salisbury, Director, Division of Information Services, USAEC, Subject: Statement That Might Be Issued if Authoritative Sources Criticize NPG Operations. See Appendix 1 for other UPSHOT-KNOTHOLE details and MK 13 history in Volume VI for further "Hamlet" details. According to Buford C. (Carl) Lyon, Plans and Operations Officer for the UPSHOT-KNOTHOLE tests, the tests were named after randomly-chosen names which, in some cases (e.g., "Encore" for "effects"), had some connection to the purpose of the test. For example, Grable (for "gun," and named after the actress Betty Grable) led to Harry (as in Harry James, Betty Grable's bandleader husband between 1943 and 1965). (Telephone conversation with Carl Lyon on September 12, 1991.) In reality, several of the unclassified LASL test names were matching first-letter mnemonics for the then-classified test device names or test purposes, as shown below.

<u>Unclassified Test Name</u>	<u>Test Purpose/Device Name</u>
Nancy	Nevada Zombie
Dixie	Deuterium (boost test)
Badger	Buzzard
Simon	Simultaneity
Encore	Effects
Harry	Hamlet
Grable	Gun (280MM AFAP)
Climax	Cobra

Of the thermonuclear-related UPSHOT tests, Nancy, Badger, and Simon were tests of the RACER primary for application to the "emergency capability" weapons; Ruth and Ray were UCRL tests of uranium hydride-fueled primaries; and Climax tested a COBRA primary for use in the TX-15.

In particular, shot Climax, the eleventh and final UPSHOT-KNOTHOLE event, was "essential to the thermonuclear program" since "the energy release of the device must be known precisely for the CASTLE series."³⁷⁰

LASL had expected to use the newly-developed RACER primary for its TX-14, -16, and -17 tests in CASTLE; however, the behavior of the core was inconsistent: the test of the core in a TX-14 mockup in the Nancy shot was expected to yield 35 to 40 KT; it finally gave only 24 KT. Likewise, tests of the RACER in TX-16 and -17 mockups in the Badger and Simon shots, respectively, yielded 23 and 43 KT; the shots were also expected to yield 35 to 40 KT.

In addition to yield predictability and reproducibility, it was vital that the primary furnish the proper quantity and strength of x-radiation required to adequately compress the secondary. This was an important consideration when fission devices were used as lithium and uranium compressors, as opposed to other applications of fissile explosive devices where x-ray output might not matter as much.

In light of the problem with the RACER, LASL decided to add an additional UPSHOT experiment to test an alternate core, the COBRA. A MK 7 bomb was withdrawn from the U.S. war-reserve stockpile, and delivered to Sandia on May 20 for final checkout and assembly with the test core.³⁷¹

The UCRL HYDRIDE I and II devices, and their respective tests (Ruth and Ray) were exceptions to this naming scheme, as was the LASL Annie shot, which was a test of the XR3 device. UPSHOT-KNOTHOLE device names are from a memorandum to Brig. Gen. Kenneth E. Fields dated December 7, 1954 from James E. Reeves and Alvin C. Graves, Subject: Operation TEAPOT, pp. 5 and 6; association of devices to specific shots is from an undated page of handwritten notes that lists UPSHOT-KNOTHOLE device names and the yields and tower heights of their affiliated tests (shot names do not appear in these notes).

³⁷⁰Memorandum for Mr. Lewis L. Strauss from Gordon Dean, Chairman, USAEC, May 26, 1953, Subject: Proposed Eleventh Shot.

³⁷¹Teletype dated May 11, 1953 from George P. Kraker, USAEC, Albuquerque, to Brig. Gen. K. E. Fields, DMA, USAEC,

The composite uranium-plutonium COBRA core was the result of design work that had just been completed at Los Alamos on some very important "new principles" that would be used during CASTLE; this work was completed in time for a COBRA test to be included in UPSHOT-KNOTHOLE rather than in a fall 1953 Nevada test series (Operation DOMINO).³⁷²

The precise energy release of the COBRA had to be known before CASTLE, and an additional UPSHOT-KNOTHOLE test would save money, compared to the cost of conducting DOMINO, and give greater assurance of adequate emergency capability.³⁷³

Use of the COBRA in thermonuclear device primaries had been planned for CASTLE; at least one DOMINO shot was to have tested the COBRA before CASTLE. Firing of the COBRA during UPSHOT-KNOTHOLE would obviate need for DOMINO.³⁷⁴ Since development and fabrication of the COBRA had been completed sooner than expected, it was available for immediate test. If the COBRA test resulted in unexpected behavior or if the core required redesign and further testing, then DOMINO could still be conducted on a contingency basis.

The COBRA would be fired within a Type D pit (the type of pit also used in shots Annie, Dixie, Encore, and Harry); following a successful test of the Type D pit in MK 7 HE, the pit would have been effectively proven by nuclear detonations in bombs of the MK 5, 6, and 7 classes (including the TX-13 as a MK 6-class test).

Although cancellation of DOMINO would delay certain tests of developmental interest to the weapons labs, it would jeopardize nothing of immediate urgency to the stockpiling of weapons. In addition, a successful test of the COBRA at this time would also accelerate the "emergency capability" program

Washington.

³⁷²Hewlett and Holl, pp. 156; Hewlett and Holl mislocate DOMINO to the Pacific.

³⁷³Memorandum dated May 26, 1953 for Mr. Lewis L. Strauss from Gordon Dean, Chairman, USAEC; letter dated May 27, 1953 to Honorable Sterling W. Cole, Chairman, Joint Committee on Atomic Energy, from Gordon Dean, Chairman, USAEC; teletype dated 7 May 1953 to Gen. K. E. Fields, Director of Military Application, USAEC, from C. L. Tyler, Test Manager.

³⁷⁴Memorandum dated September 7, 1954 to Brig. Gen. K. E. Fields, Director, Military Application, USAEC, from W. B. McCool, Assistant Secretary, USAEC; Subject: Questions Raised in August 30, 1954 memorandum from Commissioner Murray to the Chairman.

of megaton-range bomb development and make success of CASTLE more likely. If trouble occurred, there was still time before CASTLE to redesign the COBRA.³⁷⁵

Near the end of May, Strauss sent two memoranda to President Eisenhower concerning the additional Nevada test and to request the expenditure of additional oralloy and plutonium. In a memorandum dated May 25, Strauss expressed concern about conducting the added shot, because of fallout in Utah from earlier tests. As an AEC commissioner, Strauss had been "dubious of tests of atomic weapons in the United States because of unfavorable public reaction," such as that now occurring.

Strauss first recommended to Eisenhower that the shot not be fired:

The (Atomic Energy) Commission has informed me that, if the test must be conducted at Eniwetok atoll, it will involve a delay of three months. This will not matter if the test is successful. If the test should be a failure, it would involve a corresponding delay in the 1954 tests.

There is no question of the greater convenience of the Nevada testing ground, but I have the feeling that [deleted] that should be tested on the continent and I recommend against it at this time.³⁷⁶

The following day, Strauss had changed his mind:

Following our discussion yesterday, I have again reviewed that matter with the AEC and the Chairman of the MLC. The consensus

³⁷⁵Office Memorandum dated May 11, 1953 to General K. E. Fields from Captain John T. Hayward, USN, Acting Director of Military Application, USAEC, Subject: General Manager's Meeting, May 11, 1953; Memorandum dated May 19, 1953 for Mr. James S. Lay, Executive Secretary, National Security Council, from Gordon Dean, Chairman, USAEC, Subject: Increase in Allowance of Fissionable Material for Expenditure in UPSHOT-KNOTHOLE; letter dated May 19, 1953 to Honorable Robert LeBaron, Chairman, MLC, from Gordon Dean, Chairman, USAEC.

³⁷⁶Memorandum for the President dated May 25, 1953 from Lewis Strauss, subject: Request of Atomic Energy Commission for Your Approval to Increase Amounts of Oralloy and Plutonium for One Further Shot in Current Series.

is that the cost of transferring the test to Eniwetok atoll and the delay incident thereto outweigh possible further public criticism directed at the atomic tests in Nevada.

While some of the protests are based on the shock (Author's note: blast) effect, most are the result of fallout of radioactive particles and this has usually occurred when the explosions have been set near ground level. The AEC has assured me that this shot, instead of being on a tower, will be an air drop and will be detonated at an elevation of at least [deleted] in order to minimize the possibility of radioactive particles settling in adjacent country.

I could not shake (the) AEC estimate that it will entail (a) three to four month delay to set up the test in the Pacific and since I regard it as important that the operational H-weapon should be proof-tested in the spring (of 1954) as scheduled, (I) withdraw (my) objection voiced yesterday and recommend that you permit expenditure of the necessary [deleted] and the conduct of the test as planned.

Eisenhower was not pleased with this last-minute test; in a handwritten and initialed note appended to Strauss' May 26 memo, he stated:

For Adm. Strauss: I don't like it -- and I don't like to have the time element argued in such a case. The proper officials should have brought the query up three months ago. However, in the circumstances, it's approved.³⁷⁷

At the beginning of June, AEC Chairman Gordon Dean wrote to Robert Cutler, national security adviser to President Eisenhower, and explained the need for an additional UPSHOT test. Dean had met with Eisenhower on May 27 to discuss this issue:

³⁷⁷ Memorandum for the President dated May 26, 1953 from Lewis Strauss, subject: Request of Atomic Energy Commission for Your Approval to Increase Amounts of Orallo and Plutonium for One Further Test Shot in Current Series.

Last week we discussed briefly the reasons for our recent request for authorization to expend fissionable material for an additional shot at the Nevada Proving Ground. I think it would be useful to you to state these reasons more specifically, and this is the purpose of this memorandum.

As you know, following confirmation of certain principles of design in the Ivy operation last November, the Atomic Energy Commission proposed and the Department of Defense accepted several proposals for development and testing of thermonuclear weapons on an emergency basis.

It was in January of this year that several alternate concepts were approved, in addition to a design then in progress, and design of these alternates began at once. At the same time, the date (early 1954) and preliminary scope of Operation CASTLE were set.

In all the designs proposed (for detonation during CASTLE, there is required a primary fission bomb or device. [deleted]

During UPSHOT, similar but smaller devices (Author's note: the RACER primary) were used on the Nevada Zombie and Simultaneity tests on March 24 and April 25, 1953. The first gave a lower yield than expected (Author's note: 24 KT instead of a predicted 35 to 40 KT) and the second (test gave a yield) higher (than expected).

This experience and the fact (that) the core to be used has never been tested in any high explosive system, let alone the TX-15 ZOMBIE system, emphasized the need of full scale test of the proposed primary device. These same facts pointed up the emergency of test in that should the performance of the device not be satisfactory, a different design would have to be completed and tested prior to CASTLE.

For the above reasons -- the emergence of a specific design after the start of UPSHOT, and the need to test it immediately -- we were compelled to submit our recent request for an additional

UPSHOT test.³⁷⁸

The eleventh and final UPSHOT-KNOTHOLE test, Climax, was a test of a 900 lb., 30" diameter W-7 weapon with a COBRA core in a Type D pit dropped by a B-36 over the NPG on June 4; the shot yielded 61 KT.³⁷⁹ A bomb rack to handle the TX-7 aboard the bomber was removed from an F-84 and put on the B-36 before the drop.³⁸⁰

At the time, Climax was the highest-yield test ever fired at the NPG: estimated yield was 70 KT $\pm$ 10%; since the shot was fired as an airburst, there was little off-site fallout.³⁸¹ The COBRA core was subsequently used as a primary in the ZOMBIE, and RACER IV cores were used as primaries for the SHRIMP, RUNT I, RUNT II, and ALARM CLOCK thermonuclear test devices during CASTLE.³⁸² **Figure IV-17** shows the Climax shot fireball.

After the Climax shot, the AEC decided that no additional test of the COBRA was necessary and that the device, modified slightly, would be satisfactory for CASTLE and that its adoption would probably be recommended. It now appeared unnecessary to plan for additional tests of the COBRA or of other weapons or devices in the fall of 1953. DOD in particular had supported the addition of the Climax shot to UPSHOT-KNOTHOLE because it would obviate the need for DOMINO.³⁸³

³⁷⁸Memorandum for Mr. Robert Cutler dated June 1, 1953 from Gordon Dean, Chairman, USAEC. A very fragmentary and obscure discussion of the decision to test the COBRA is on pp. 256 - 258 of Anders, FORGING THE ATOMIC SHIELD.

³⁷⁹Atomic Energy Commission, COBRA Shot Nevada Proving Ground, Note by the Secretary, May 29, 1953.

³⁸⁰"Climax - Shot 11, Upshot-Knothole-[deleted]," undated.

³⁸¹Atomic Energy Commission, Proposed Additional Shot for UPSHOT-KNOTHOLE Series, Report by the Director of Military Application, May 13, 1953; Minutes of AEC Meeting No. 863, May 18, 1953.

³⁸²"A Handbook for Operation CASTLE," The Pogo Staff of J Division, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, 1 January 1954, p. vii; "History of Task Group 7.4 Participation in Operation CASTLE, 1 January 1953 - 26 June 1954," Historical Division, Office of Information Services, Air Force Special Weapon Center, Kirtland AFB, New Mexico, 8 November 1954, pp. 45, 61, 70, 79, 81, 84, 119, 121.

A formal announcement of the cancellation of DOMINO was made by the AEC in early July; the test series was identified only as "an Autumn 1953 series:" the code name DOMINO was still classified at this time.³⁸⁴

One of the less-savory incidents during UPSHOT-KNOTHOLE occurred late in May. Some of the earlier shots had dumped considerable fallout over parts of southern Utah, especially around St. George, about 135 miles east of the NTS, and there was pressure on the President to halt testing in Nevada.

On May 27, when Gordon Dean met with President Eisenhower to ask authorization for the COBRA test, Eisenhower

expressed some concern, not too serious, but made the suggestion that (the AEC) leave "thermonuclear" out of press releases and speeches, (and) also (the terms) "fusion" and "hydrogen". The President says "Keep them (the American public and the Soviets) confused as to 'fission' and 'fusion.'"³⁸⁵

Construed in the best possible light, Eisenhower may have been concerned about undue public attention to the possibility that thermonuclear tests were being conducted in Nevada (these types of tests were not being done in 1953 but would be during later years.)

After UPSHOT-KNOTHOLE ended, the AEC prepared a secret "statement that might be issued if authoritative sources criticize NPG operations;" the document summarized results of

³⁸³Minutes of 877th AEC Meeting, June 17, 1953; memorandum for the Chairman, U.S. Atomic Energy Commission from James S. Lay, Executive Secretary, National Security Council, Subject: Increase in Allowance of Fissionable Material for Expenditure in Operation UPSHOT-KNOTHOLE.

³⁸⁴Draft Press Release dated July 6, 1953 attached to Office Memorandum dated July 7, 1953 to Distribution, from Rodney L. Southwick, Acting Chief, Public Information Service, USAEC, Subject: Proposed Release on NPG Test Schedule.

³⁸⁵"Ike Wanted Public Confused on Nuclear Tests," unidentified newspaper dated April 20, 1979; "Eisenhower Wanted the Public 'Confused'," Palo Alto Times, Palo Alto, California, April 20, 1979; "Ike Put Lid on Fallout Warning," Las Vegas Sun, Las Vegas, Nevada, April 20, 1979; "Ike Wanted Confusion at NTS," unidentified newspaper, April 21, 1979; "Ike: 'Confuse' The Public," Nevada Appeal, Carson City, Nevada, April 20, 1979.

the recent test series, possibly in anticipation of further criticism such as that leveled against the AEC in March 1953:

The Atomic Energy Commission believes that the test operations at the Nevada Proving Ground, including those of the 1953 spring series, have well served the national interest.

The four continental test series of the past 2 1/2 years have speeded the weapons development program of this nation by at least one year and possibly two as compared with what would have been possible had those tests been (conducted) at the Pacific Proving Ground. The cost in dollars has been less than ____ that of carrying on these operations in the Pacific, and the savings in scientific manpower have been invaluable.

The most recent series yielded results of the highest importance. A number of the detonations advanced the development of newer weapons. Others were important to the improvement of standard weapons. For example, one confirmed a technique which is now being applied to increasing the striking power of the fissionable material in our weapons. (Author's note: a reference to the Harry shot, which was a test of a "radical implosion system;" its objective was to reduce the amount of fissionable material required to provide a given moderate yield or to improve the performance of a small implosion system.)

Another proved-in the atomic warhead for large caliber artillery projectiles. A third produced knowledge of the effects of weapons which will lead to more efficient use of both tactical and strategic atomic weapons should the need arise.³⁸⁶

UCRL and Operation UPSHOT-KNOTHOLE

Another less-promising line of experimentation during UPSHOT-KNOTHOLE was pursued during shots Ruth and Ray, which were

³⁸⁶"Statement That Might Be Issued if Authoritative Sources Criticize NPG Operations," August 3, 1953. There is no evidence that this paper was ever used. The blank space for comparative costs is in the original and was not filled in.

tests by the new University of California Radiation Laboratory at Livermore to demonstrate the feasibility of uranium deuteride as a fuel for the primary of a radically different type of low-yield tactical thermonuclear explosive proposed by Dr. Edward Teller.

Neither shot was as successful as UCRL had hoped it would be; Ruth's meager 200 ton yield did not even completely destroy its shot tower.³⁸⁷ (See **Figure IV-18.**)

Before the UH₃ test, the shot was characterized as "a different type of shot from those which had been fired prior to this time in Nevada;" the shot " was expected to be of a very low-order detonation ... a possibility existed that no fission would take place."³⁸⁸

The results of these shots provided a forum for Dr. Bradbury to criticize UCRL plans for CASTLE. Relations between the labs had been strained since the inception of the Livermore group; most of this ill feeling was directed from Los Alamos, even to the point of LASL occasionally failing to circulate communications of mutual interest to UCRL (to its credit, UCRL did not reciprocate these slights and usually updated LASL about subjects of consequence to both labs).

There was still a strong antipathy between Bradbury and Edward Teller, which was soon to be exacerbated by Teller's self-serving and damaging testimony against Dr. J. Robert Oppenheimer during the latter's security clearance/loyalty hearings in mid-1954.

In addition, the well-established and historic New Mexico

³⁸⁷"The Development of Nuclear Explosives," Lowell Wood and John Nuckolls, in ENERGY IN PHYSICS, WAR AND PEACE, Hans Mark and Lowell Wood, eds., Kluwer Academic Publishers, Norwell, Massachusetts, 1988, pp. 315, 316. After Ruth, LASL technicians noted derisively that UCRL should use either a bigger bomb or a smaller tower for its next shot. In a letter to the author dated 11 October 1991, Dr. Herbert York defended the Ruth and Ray shots, noting that the tests were "an idea that several of us (probably including Teller) adopted as a good way of doing an early experiment to get our feet wet .. They were not thought of as any sort of weapons prototype, for (a weapon) primary or other .." York claims that the tests were not failures (as the CASTLE Koon shot would be), even though the calculated most-probable yields of the shots were possibly twice as high as the tests actually yielded.

³⁸⁸4925th Test Group (Atomic), Operational Summary Report, 1953 Continental Nevada Spring Test, Operation UPSHOT-KNOTHOLE, p. 1D.

lab, to which the entire United States owed a debt of gratitude for ending World War II, did not welcome the upstart California interlopers who might one day be serious competitors for weapon designs that had heretofore been a LASL monopoly. UCRL's attitude also tended to reflect -- especially in its communications -- the flamboyant hyperbole and eternally-optimistic outlook of Dr. Teller.

The FORTUNE Article, May 1953

Bradbury was also quite upset by an anonymously-authored article that had just appeared in the May 1953 issue of FORTUNE magazine. The article, entitled "The Hidden Struggle for the H-Bomb: The Story of Dr. Oppenheimer's Persistent Campaign to Reverse U.S. Military Strategy," included, among other things, a thinly-veiled attack on LASL and Dr. J. Robert Oppenheimer, and the dissertation implicitly credited all recent U.S. thermonuclear weapon advancements, including the successful IVY Mike shot, to Dr. Edward Teller and his new Livermore laboratory.

The article was particularly critical of Oppenheimer, painting him as a conspirator intent on weakening the U.S. militarily. Among other things, Oppenheimer was accused of trying to block the establishment of UCRL; of opposing accelerated hydrogen bomb development; of using his chairmanship on the GAC to delay thermonuclear research; and of trying to influence scientists not to work for LASL.

Most of the article was untrue, inaccurate, and full of uncredited quotes. That it was published anonymously showed that FORTUNE lacked sufficient integrity to stand behind its falsehoods. Unfortunately, the article could be neither refuted in detail nor adjudicated for libel without recourse to, and publication of, highly classified documents and information.³⁸⁹

The flash point for Bradbury's ire was an AEC document about

³⁸⁹THE OPPENHEIMER CASE: SECURITY ON TRIAL, Philip M. Stern and Harold P. Green, Harper & Row Publishers, New York, 1969, pp. 201-204; "The Hidden Struggle for the H-Bomb: The Story of Dr. Oppenheimer's Persistent Campaign to Reverse U.S. Military Strategy," anonymous, FORTUNE, Vol. 47 No. 5, May 1953, pp. 109, 110, 230; Hewlett and Holl, pp. 47, 48. One of the most persistent abuses of the classification system set up by the Atomic Energy Act has been its use to anonymously slander or discredit individuals with whom the author of an article or document might disagree. It is nearly impossible to publicly refute these types of charges -- which are based wholly or in part on classified data -- without revealing even more classified information.

future full-scale weapons tests; the paper said of present UCRL test plans:

It is recognized that UCRL may be interested in a more intensive test program than is required by LASL due to (UCRL's) more basic and fundamental research and testing and the possibilities for more observable advancement in their particular weapon area.

Bradbury claimed in response that the UCRL program was not more basic and fundamental than LASL's, and he did not believe any current UCRL programs seemed to offer opportunities for technological advancement beyond those of current LASL efforts.

Bradbury noted that although it was widely believed that UCRL was credited with pioneering thermonuclear weapons development, every hydrogen device in which Livermore was currently interested -- including the solid-fueled MORGENSTERN and the liquid-fueled RAMROD -- drew its explosive energy primarily from ordinary fissionable material.³⁹⁰ The same was essentially true for all three of LASL's "emergency capability" designs, the JUGHEAD, RUNT, and ALARM CLOCK, which depended for most of their power on a U-238 tamper around their secondary stages.

As far as LASL knew, UCRL was interested in developing a certain type of radiation implosion geometry which might have

³⁹⁰Minutes of the Thirty-Fifth Meeting of the GAC on May 13, 1953, state (p. 11):

Mr. (Representative Carl) Hinshaw asked about a shortcut to the H-bomb and whether the Russians might have hit on it. It developed that what he had in mind was "some of these various projects that have been going on at Livermore" involving fission and fusion. Dr. (Isidor) Rabi said that radiation implosion will make a great difference in weapon development, that both Livermore and Los Alamos were pursuing this line, and that in the Nevada tests those by Los Alamos were very encouraging and those by Livermore very instructive.

Rabi chose his words carefully to characterize the two UCRL UPSHOT failures; they were instructive, in the sense that they showed what Livermore was trying would not work in its present form.

some application in weapons of rather high yield in reasonably small size (according to the anonymously-written May 1953 FORTUNE article, Teller was at this time attempting to encourage the design of tactically-deliverable thermonuclear weapons).³⁹¹ Such weapons would use primarily fissionable materials with some boosting.

This approach paralleled and duplicated to some extent LASL's own boosted weapons research, although the New Mexico lab used a different form of implosion geometry which was much more amenable to calculation and with which LASL had gained considerable experience (UCRL may have been experimenting with conical or cylindrical critical masses at this time).

Bradbury pointed out that the recent UCRL experiments in Nevada (the UPSHOT-KNOTHOLE Ruth and Ray shots) did not constitute "fundamental research," but were instead attempts to determine nuclear cross sections of a material that was of only problematic weapon interest; in LASL's opinion, the experiments might have been carried out more easily and much more cheaply, in terms of expenditure of fissionable materials and time, in a laboratory, rather than in full-scale field tests.

LASL had explored and discarded uranium hydride cores several years earlier without recourse to full-scale testing (see "Advances in Pit Design" in Volume II); such systems, if made workable, offered only questionable possibilities for weapons application. The results of Ruth and Ray showed that such prospects were even less than UCRL had originally estimated.

Bradbury also did not think very highly of UCRL's current plans for tests of the MORGENSTERN and RAMROD devices during CASTLE. The LASL director noted that the two weapons labs had completely different philosophies with regard to full-scale weapons testing, and that Los Alamos had warned the AEC in the past that the UCRL program of "basic and fundamental research in their particular weapon area" in the CASTLE program was very likely going to interfere directly with LASL's program to provide maximum national "emergency capability" in high-yield weapons in the shortest possible time.

LASL was not counting on any useful information coming out of the UCRL shots (which was a wise decision, considering what would finally happen a year later during the UCRL CASTLE tests; UCRL would not have a successful thermonuclear test until the spring of 1956).³⁹² Relations between the two

³⁹¹FORTUNE, May 1953, p. 110.

³⁹²Letter dated 6 May 1953 to C. L. Tyler, Manger, Santa Fe

competing national weapons labs would not improve for many years.

Lithium requirements for CASTLE and "emergency capability" weapons were reviewed in mid-May at the 35th meeting of the GAC. Dr. Hans Bethe stated that lithium-6 was useful mainly for large weapons, and for all thermonuclear devices because it could be a source of tritium. Dr. Edward Teller stated that from half-as-much to as much lithium-6 should be produced as oralloy at Oak Ridge. (By February 1953, the Paducah - Oak Ridge oralloy production complex was turning out 17 kilograms of oralloy per day.)³⁹³

Making what would turn out to be a stunningly erroneous assumption, Dr. Bethe claimed that ordinary lithium-7 "won't work because it doesn't give tritium." Lithium-7 would generate tritium (see Equation F in "Fusion Physics" in "Weapons Physics" in Volume I) if it were struck by neutrons in the three to four million electron volt range (3-4 MeV); however, even though neutrons with energies between three and 17 MeV might be spawned by a thermonuclear reaction, these neutrons were not expected to have a high cross-section for tritium generation.

Dr. Bradbury noted that one of the objectives of CASTLE was to determine how normal lithium, unenriched beyond about 7.5% in lithium-6, would behave during a thermonuclear explosion. Results of CASTLE would show quite convincingly how useful lithium-7 could be.

Also unknown at this time was whether a thermonuclear reaction would propagate in LiD. According to results of computations by Project Matterhorn, propagation seemed reasonably well assured in lithium reasonably enriched in Li-6; secondary stage diameters considered by Matterhorn showed that little was gained by greater enrichment.

Preliminary calculations by Project Matterhorn had shown that, up to a point, there was a vital linkage between the degree of lithium enrichment and thermonuclear weapon

Operations Office, USAEC, from N. E. Bradbury, Director, LASL. Among small devices considered by UCRL in the spring of 1953 were the so-called "water boiler" two-stage gadgets, named after an early wartime LASL test reactor. These devices appeared to be prohibitively expensive in their use of U-235 and were of dubious utility. (Minutes of the Thirty-Fifth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 14, 15, 16, 1953, pp. 15, 21.)

³⁹³Minutes of the Thirty-Third Meeting of the General Advisory Committee to the USAEC, February 5-7, 1953, pp. 16, 17.

diameter and weight. The more highly-enriched in the lithium-6 isotope, the less fuel was required for a given yield and the smaller in diameter and lighter the resulting secondary stage could be.³⁹⁴ By this time in CASTLE planning, the AEC's lithium enrichment facilities (the "Alloy Development Plants") were still not up to full production capacity, so lithium enriched to varying percentages of Li-6 was expected to be available to be used in CASTLE test devices.

Lithium-6: The CASTLE Bottleneck

At this time, there were two different and promising methods of separating lithium-6 from lithium-7, called "elex" and "orex".

"Elex," or "electrical exchange," was a process wherein mechanical agitators in large shallow trays mixed an amalgam of lithium and mercury with an aqueous solution of lithium hydroxide. After flowing through a number of stages, lithium-6 tended to concentrate in the amalgam. Lithium-7 could also be extracted by electrolysis from the hydroxide solution. An electrical current was passed through the amalgam-hydroxide mixture to prevent chemical reaction of water and lithium.³⁹⁵

"Orex," short for "organic exchange," was a process that used lithium chloride or bromide dissolved in an organic solvent, dimethyl formamide.³⁹⁶ "Orex" allowed for a simpler plant design in which pulsed columns replaced the multitudinous trays, stirrers, and baffles used in the "elex" process.³⁹⁷

The \$150 million cost of the new lithium separation plant at Oak Ridge had dropped to \$20-25 million for an "orex" facility that would produce 10 lb. of lithium-6 per day. Two "elex" facilities and three "orex" plants were planned to be in operation by November 1954; one "elex" and an "orex" were to be on line in August and September 1953, respectively. With "orex," the cost of lithium enriched to 95% in Li-6 would not be much greater than lithium enriched to just 30% in Li-6.

³⁹⁴Minutes of the Thirty-Fifth Meeting of the General Advisory Committee to the USAEC, May 14-16, 1953, p. 13.

³⁹⁵Hewlett and Holl, p. 163.

³⁹⁶NUCLEAR CHEMICAL ENGINEERING, Second Edition, Manson Benedict, et. al., McGraw-Hill Book Company, New York, 1981, p. 800.

³⁹⁷Minutes of the Thirty-Third Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, February 5-7, 1953, p. 31.

It looked now like enough Li-6 would be available on schedule for CASTLE, but the schedule was tight. Lithium-6 was expected to cost, pound for pound, about 25% as much as an identical amount of uranium-235.³⁹⁸ About 230 pounds of raw lithium-containing ore had to be processed to obtain a single pound of lithium, and of this pound, only 1.2 ounces were lithium-6, the most-desirable end product (although ordinary Li-7 would soon be discovered to be very useful in thermonuclear weapons).³⁹⁹

Late in May, in one of its semiannual reports to the JCAE, the AEC noted that progress in the design of deliverable thermonuclear weapons was being furthered by advances already made in fission weapon design, especially those related to size and weight reductions. The AEC described current fusion weapon development plans:

The successful test of the thermonuclear device, designated the SAUSAGE, at Operation IVY on November 1, 1952 has permitted the Commission to redefine the scope and direction of the thermonuclear development program for the next year or more.

Broadly viewed, development of thermonuclear weapons is proceeding along two main, but interrelated lines, one being a highly expedited program for emergency capability under the primary direction of Los Alamos Scientific Laboratory to secure a deliverable weapon from among three potential designs.

The production of prototypes will parallel the development program in order to insure the earliest possible capability. In a parallel effort, the University of California Radiation Laboratory at its Livermore, California site is investigating thermonuclear weapon design concepts involving [deleted]

Shortly after Operation IVY, it was decided to attempt to secure a deliverable

³⁹⁸Minutes of the Thirty-Fifth Meeting of the General Advisory Committee to the USAEC, May 14-16, 1953, pp. 35, 36.

³⁹⁹"Meet Lithium: The Stuff of H-Bombs -- and Milady's Powder," Clifford B. Hicks, POPULAR MECHANICS, July 1958, pp. 57-61, 232.

thermonuclear weapon through simultaneous development of three possible models as the most probable way to assure early success. The plan established provides for concurrent production of non-nuclear components for each prototype in sufficient numbers to constitute an emergency capability in the event one or more of the prototypes is proven by full-scale test to be successful.

The three proposed weapons are designated the TX-14 ALARM CLOCK, TX-16 JUGHEAD, and TX-17 RUNT. [deleted]

Each of these weapons is scheduled to be tested at Operation CASTLE, to be held in the spring of 1954.

The thermonuclear program of the University of California Radiation Laboratory is concerned with developing the design of two weapons, MORGENSTERN and RAMROD, which will involve novel features [deleted] These proposed weapons are also scheduled for testing at Operation CASTLE, but such engineering as may be required to permit weapon production will be done after the tests have proven the feasibility of the new designs.

Designs of the TX-14, TX-16, and TX-17 had been frozen in April; the first stockpile units were due in mid-1954 and would be stockpiled immediately.

The report to Congress also noted that LASL was investigating the design of a combination fission-fusion weapon which would incorporate the mechanics of radiation implosion used in thermonuclear systems in an attempt to achieve high yields and efficiencies in bombs weighing less than 10,000 pounds.⁴⁰⁰ (This became the TX-15, which incorporated Dr. Stan Ulam's December 1950 fission weapon-design ideas; see MK 15 history in Volume VI.)

The report also outlined plans for future tests and described the current status of the U.S. nuclear weapons stockpile. A

⁴⁰⁰Atomic Energy Commission, Progress Report to the Joint Committee, December 1952 through May 1953, Part III - Weapons, May 29, 1953, pp. 1-4. This document shows clearly that the two experimental UCRL designs were not part of the "emergency capability" program.

total of six forthcoming nuclear test series were now planned.

In Nevada, Operations TEAPOT was slated for the fall of 1954, after CASTLE, or the spring of 1955. Operation JULEP would follow in either the fall of 1955 or the spring of 1956. Each of these series comprised five LASL tower shots and one LASL airburst test; UCRL was scheduled for one tower shot and one airburst in each operation.

In the Pacific, Operation CASTLE was still slated for the spring of 1954; LASL would have one ground and three barge shots, with two UCRL ground shots. Operation REDWING, in the spring of 1955, would include at least three LASL and one UCRL ground shots. Operation WIGWAM in the spring of 1955 was to be a deep underwater effects test of modified stockpiled warhead (finally conducting the canceled "Charlie" shot of Operation CROSSROADS in 1946). Operation HARDTACK in the spring of 1956 would include three LASL and one UCRL ground shots.⁴⁰¹

The current U.S. nuclear weapons stockpile consisted mainly of MK 5, 6, and 7 strategic and tactical aerial bombs, with a few high-yield MK 18 strategic bombs and some gun-type MK 8 bomb and MK 9 artillery shell assemblies. The number of nuclear assemblies being added to the stockpile was influenced by the current output of fissionable materials, weapon fabrication capacity, the types of assemblies being produced, and changes in the level of material in the production "pipeline."

There had been a slight drop in the rate of nuclear assembly production during the first quarter of 1953, as compared to the 1952 annual rate. This was caused mainly by a change in military requirements, and because fissionable material was diverted to the UPSHOT-KNOTHOLE tests and to reactor development projects.

To increase warhead production, a new core fabrication-assembly-inspection plant at Rocky Flats, Colorado was nearing completion, and was expected to be fully operational shortly after July 1, 1953. In addition, a government-owned tract of land on the Spoon River, near Macomb, Illinois, had been secured as the future site of a new high explosives processing and assembly plant, to supplement present HE fabrication facilities at the Iowa Ordnance Plant near Burlington, Iowa and the Pantex Ordnance Plant near Amarillo, Texas, and to meet future DOD requirements for implosion

⁴⁰¹Progress Report to the Joint Committee on Atomic Energy, December 1952 through May 1953, Part III, Weapons, June 22, 1953, p. 9.

weapons.

Detailed design of the Spoon River facility had begun and construction was scheduled to begin in July 1953 and be completed by January 1955. The new facility was to be operated by the Thompson Products Company to take over some of the production engineering and prototype production work then being done by the Salt Wells Pilot Plant of Naval Ordnance Test Station in Inyokern, California. (The Spoon River plant was canceled in early November 1953 when the AEC decided it did not require additional HE plant capacity due to a drastic cut in implosion weapon production in anticipation of thermonuclear weapons production.)⁴⁰²

The AEC also had by this time eight widely-dispersed continental U.S. secure weapons storage sites in operation, all on military bases; the latest had become operational earlier in May. Four more were scheduled to be completed between July 1953 and January 1955.

With these facilities, there would be adequate storage capacity until the fall of 1955; after that time, a new scheme, using satellite sites near existing national storage sites, would be employed to provide even greater capacity by storing weapons in warehouses as "standby," rather than operational, warheads.⁴⁰³

CASTLE Preparations Continue

In the meantime, preparations for CASTLE continued. By the end of May 1953, 95% of the scientific structures had been designed, including items such as the Parry island device assembly area, a Bikini island underwater cable system to detonate devices far from the shot control bunker, modifications to the Parry island cryogenic plants, and the basic major scientific stations. Final surface zero positions for the six shots then in the test program were selected in June, and the location of all scientific stations with respect to these zero points were established.

⁴⁰²Atomic Energy Commission, Progress Report to the Joint Committee, June 1953 through November 1953, Part III - Weapons, AEC 129/55, November 27, 1953, p. 12; Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, pp. 7, 14, 36, 52; USAEC press release dated November 10, 1953.

⁴⁰³Atomic Energy Commission, Progress Report to the Joint Committee, December 1952 through May 1953, Part III - Weapons, May 29, 1953, pp. 11, 12, 14, 19; Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, p. 52.

Early in June, plans for CASTLE included six shots: one tower or cab shot and one barge shot at Eniwetok atoll, and two barge and two tower or cab shots at Bikini atoll. The interval between shots was expected to be a week or ten days. The question as to whether the TX-15 was to be fired had not yet been decided.

It was unlikely that a decision would be made on this matter before September 1953 since the results of one particular UPSHOT-KNOTHOLE test had to be analyzed, the design of a primary finalized, and fabrication possibilities studied before this decision could be made. Also, the TX-15 proposal was in a very preliminary stage. Tentatively, it was expected to be a thermonuclear bomb reduced in size and weight well below the TX-14, TX-16, and TX-17.⁴⁰⁴

On June 10, the Division of Military Application received an internally-generated request to review the advisability of conducting the UCRL MORGENSTERN test during CASTLE, in light of a number of factors, including the abysmal performance of the two uranium deuteride-fueled devices during Operation UPSHOT-KNOTHOLE. The DMA believed that technical gains likely to be realized in a test of the MORGENSTERN device as then proposed did not warrant the money that would be spent on it for CASTLE.⁴⁰⁵

In the middle of the month, the scope of the CASTLE operation was expected to be firmed up some time in September when the lithium-6 manufacturing capability of the "Alloy Development Plant" at Oak Ridge was better known and when final LASL and UCRL test proposals were presented to the AEC and DOD.⁴⁰⁶

Operational Order No. 1-53 for Operation CASTLE was issued on June 20 by the Commander of Joint Task Force Seven. The order listed the shots shown in Table 3-1.

Table 3-1 CASTLE Shot Schedule, June 1953

<u>Shot & Code Name</u>		<u>Date</u>	<u>Site</u>
1	Nectar	15 Feb 1954	Bikini, on reef 2500 yds

⁴⁰⁴Letter dated June 3, 1953 to Honorable Robert LeBaron, Chairman, MLC, from Gordon Dean, Chairman, USAEC.

⁴⁰⁵Memorandum dated June 10, 1953 to Col. Huston from Donald F. Mastick, Technical Assistant, DMA, USAEC, Subject: MORGENSTERN Test.

⁴⁰⁶Memorandum dated June 16, 1953 for Mr. (Marion V.) Boyer from Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC.

			SW of Namu island
2	Union	25 Feb 1954	Bikini, 6900' s. of Yurochi and 3 nautical mi. from Aomoen (barge)
3	Bravo	8 Mar 1954	Bikini, on barge NE of Union test site
4	Koon	17 Mar 1954	Bikini, in cab on Eninman island
5	Echo	24 Mar 1954	Eniwetok, tower on Eberiru island
6	Yankee	3 Apr 1954	Eniwetok, on barge over IVY Mike crater

This schedule and shot sequence were changed soon afterwards. The first shot would now be Bravo on 1 March 1954, followed by Union on 11 March. The remaining tests would be Nectar, March 22; Koon, March 29; Echo, April 7; and finally Yankee, on April 14.⁴⁰⁷

Following the failure of UCRL test primaries during UPSHOT-KNOTHOLE, Dr. Teller proposed that an immediate test of an alternate design be conducted in Nevada before CASTLE. This primary was between 23" and 28" in diameter; an airdrop test was suggested.⁴⁰⁸ (This test was never conducted.)

Late in June, another meeting was held at Los Alamos to discuss CASTLE project plans, problems, and support requirements. Evacuation plans for task force personnel were considered, and it was decided that at Bikini, for the first shot and quite possibly for all subsequent shots, it would be necessary to evacuate everyone to ships, except for a very small firing party which would remain in the reinforced concrete control bunker on Enyu island.

At Eniwetok, only emergency evacuation in case of heavy fallout would be required. The possibility that any Bikini shot might make living ashore radiologically unsafe was emphasized (this was in fact what finally happened after the

⁴⁰⁷"Commander's Concept of Operations," Annex B to CJTF Seven Operational Order No. 1-53, 20 June 1953, p. B-1; "General Concept of Operation CASTLE (Revision #1)", 17 August 1953.

⁴⁰⁸Memorandum dated June 24, 1953 to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, from R. E. Cole, Director, Office of Engineering and Construction, USAEC, Subject: Proposed Test.

Bravo shot). The need for adequate shipboard facilities to finish the Bikini operation from afloat was presented at this time and was reaffirmed later when more definite housing, office, laboratory, shop, and work-space requirements became available.

A training program was started on July 1 at the facilities of Herrick L. Johnston, Inc., in Columbus, Ohio, to give basic instruction in cryogenics during a two-week period. The training program was later transferred to the liquefaction plants at Parry island for practical experience. This program continued until December 15, and included plant modifications to make operation easier. These modifications included a spare Joule-Thompson heat exchanger and an additional deuterium oxide distillation column, together with a freon refrigeration unit.

On July 10, Operational Order 1-53 was revised. Predicted yield for Nectar was changed to 2-4 MT; for Union, 4-6 MT; for Bravo, 2-8 MT; for Koon, 4 MT; for Echo, 50-400 KT; and for Yankee, 8-9 MT.

Construction at Bikini and Eniwetok was very advanced by mid-July. In the course of preparations for CASTLE, approximately 700 scientific data collection stations were built at Eniwetok and Bikini. These stations housed many types of still and motion picture cameras, as well as a wide variety of remote recording and telemetry instruments.

A 2,900-foot causeway westward over the reef near Namu island in the northwestern portion of Bikini was built, terminating at an artificial island that became the detonation point for shot Bravo, the first test of the CASTLE series. **Figure IV-19** shows the Bravo shot cab, also called Station 20, under construction.

Experience during Operation IVY with the 9,000-foot long plywood tunnel, which contained a controlled atmosphere for diagnostic measurements of the Mike shot, led to a decision to use pipes between zero points and recording stations for the same general purpose during Operation CASTLE. Pipe sections 8 5/8" in diameter by 40 feet long were butt-welded end-to-end to form arrays suspended on supports a few feet above the ground.

For the Bravo shot, a twelve-pipe, 7,500 foot array was erected running from the shot cab to a station on Namu; for Koon, a two-pipe array, 5,600 feet long, was built running from the shot cab on Eninman island to Reere island; and for shot Echo at Eniwetok, a two-pipe array, 2,700 feet long, was constructed from the shot cab on Eberiru island to recording stations at Aomoen island. Precise survey work was required to align these arrays perfectly: the pipes had to be aligned within one inch from one end to the other.

An array of twelve mirror towers was also built in an arc on the artificial one-acre shot island created for the Bravo test. These mirrors would reflect early bomb explosion light from selected points on the bomb casing to a series of remote high-speed cameras several miles away.

Figure IV-20 shows the Bravo shot cab after it was completed, with the diagnostic pipe array running from its back side, and part of the mirror array in the background.

Figure IV-21 shows the Bravo pipe array terminating in a recording bunker on Namu island. **Figure IV-22** is an aerial view of Station 20 and its mirror array, with Namu in the background.

At this time, in conjunction with SAC, the Sandia Corporation had been reviewing the problem of optimum utilization of a combination of the projected high-yield fusion weapons and current and projected fission weapons. This review resulted in systems studies on the relative efficiencies of fission and fusion weapons used against selected idealized targets; atomic weapon stockpile composition versus SAC capabilities; and maximum yield delivery capabilities.

These studies were expected to have considerable influence on the directions of development of future fusion weapons and the quantity and yield of the future U.S. nuclear weapons stockpile.⁴⁰⁹

In mid-July, LASL reviewed the status of the Ostrich program, the lab's thermonuclear program, for the MLC. Four devices were under consideration for test in CASTLE, as shown in Table 3-2:

Table 3-2 Ostrich Test Devices

<u>Name</u>	<u>Length (inches)</u>	<u>Diameter (inches)</u>	<u>Weight (lbs.)</u>
TX-14	223.5	61.4	30,000
TX-16	296.7	61.4	42,000
TX-17	296.7	61.4	42,000
SHRIMP	171.5	52.9	23,000

These were all two-stage devices.⁴¹⁰

⁴⁰⁹Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1953 - 30 June 1953, p. 215.

⁴¹⁰Report of the Manager, Santa Fe Operations, USAEC, July

The length and weight of the SHRIMP were for the warhead alone, without tail structure which would add some length and about 2,000 lbs. in weight. Post-UPSHOT component tests related to studies of or improvements in these devices might still be necessary before CASTLE.⁴¹¹ The SHRIMP was a scaled-down TX-17 for carriage by B-47s.

The TX-17, also known as the RUNT device, would use normal unenriched lithium, mostly Li-7, for its thermonuclear fuel. The TX-14, or ALARM CLOCK, was also a "dry" device that would use lithium enriched to 95% in the Li-6 isotope. The liquid deuterium-fueled JUGHEAD was similar to, but smaller than, the IVY Mike device, the SAUSAGE.

The SHRIMP, which had not yet been assigned an AEC "TX" number, was an experimental device with geometry similar to that of the RUNT but of considerably smaller weight and size and containing partially enriched lithium (60% Li-7 and 40% Li-6).

UCRL's Project Whitney would field two test devices for CASTLE: the solid-fueled MORGENSTERN and the liquid-fueled RAMROD. The main effort at Livermore was being directed toward CASTLE; the ultimate purpose of the two test devices was the production of easily-deliverable thermonuclear weapons.

Edward Teller summed up the Livermore program at this time in a memo to Lewis Strauss. He discussed the proposed UCRL CASTLE test devices, and -- perhaps in light of UPSHOT-KNOTHOLE -- what might happen if the tests were not successful (which Teller considered unlikely; subsequent events were to show that he was wrong again). He also mentioned UCRL's small weapons program and requested Strauss' guidance on how deeply Livermore should be involved with this line of research:

In case (the MORGENSTERN device) in the CASTLE test will give a yield of a megaton, I believe that within the following year, easy modifications will produce a bomb which in yield, expense in fissionable material, and deliverability will compare favorably with the biggest fission weapons.

Furthermore, a slightly longer development will lead to considerably higher yields

1950 to January 1954, 1 January 1954, pp. 13, 63.

⁴¹¹Memorandum to All Concerned dated July 16, 1953 from Paul C. Fine, Division of Military Application, USAEC, Subject: Trip Report, MLC Tour, July 5-11, 1953.

[deleted]

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While development seems to progress rapidly, efficiently, and satisfactorily, everybody is worried here about the question what would happen if the MORGENSTERN did not function as well as expected. I hope very much that the role of this Laboratory (sic) as a weapons research group justifies us taking risks and that the Laboratory will be encouraged to proceed even in case our first gamble is not fully successful.

In the meantime, I should like to tell you that my personal opinion is that the MORGENSTERN has a good chance of success.

[deleted]

In addition to these high priority projects, a small fraction of the laboratory's effort has been devoted to two other enterprises which are small weapons and controlled thermonuclear reactions.

In the last weeks, I have been particularly interested in the small weapons. This interest comes partly from possible applications both for artillery and for anti-aircraft purposes. But one of the main reasons for my interest is that the field of really small weapons (between 8" and 12" diameters) has received little attention, the problem is difficult, and there seems to me to be quite a bit of room for invention in this field.

If we are to be successful in this field, it seems clear that an additional effort will be needed since the thermonuclear work occupies fully and satisfactorily the present capabilities of the Laboratory.

Furthermore, this work will require quite a number of tests. I hope that ways and means can be found to perform such tests without undue fanfare, with little organizational expense and with no unreasonable contamination.

My personal feeling is quite strong that

the small weapons program is an important one, but perhaps we are trying to undertake too many things, and I would be most grateful to you if you would let me know in case you think that we are going too far.⁴¹²

Strauss replied to Teller a week later, and asked if certain test devices were collaborative LASL-UCRL efforts, or only UCRL projects. In reply to Teller's request for his opinion on the current UCRL program, Strauss said

I note that I have not answered your question as to whether or not you are "going too far" with a small weapons program. I feel that I do not know enough yet to have a positive opinion but I have, for a long time, felt that the "family" of weapons is growing too large.⁴¹³

Teller replied to Strauss on July 28 and explained that tests of the MORGENSTERN and RAMROD devices were entirely Livermore activities. He added that, "apart from exchange of information, friendly cooperation in numerous small matters, and the important sharing of the test facilities at Eniwetok, Los Alamos (was) not participating in (the Livermore) tests."

This was not entirely true, since the use of LASL-designed primaries in the MORGENSTERN and RAMROD would have a decisive, and ultimately fatal, effect on the UCRL shots.

Teller also noted for Strauss that UCRL would be conducting an extensive diagnostic program for one of the LASL tests at Eniwetok, for which the final device to be employed had not yet been decided. This diagnostic experiment amounted to approximately one-half of UCRL's planned diagnostic work during CASTLE.⁴¹⁴

Also at this time, the JCAE requested that the AEC provide it with information concerning the very low-yield weapons

⁴¹²Memorandum dated July 13, 1953 entitled "Present Status at Livermore," signed by Edward Teller. Next to the last paragraph quoted is a handwritten notation, presumably by Strauss, that says "I do not know - but I fear so."

⁴¹³Letter dated July 20, 1953 to Dr. Edward Teller from Lewis L. Strauss, Chairman, USAEC.

⁴¹⁴Letter dated July 28, 1953 to Lewis L. Strauss, Chairman, USAEC, from Edward Teller.

program of UCRL's Project Whitney, including the following:

1. The type of devices now being studied.
2. The timetable associated with these weapons -- that is, are they regarded as possible weapons of the very distant future, or may they be brought to fruition during the next few years, given a sufficiently vigorous developmental program?
3. The resources now being devoted to this effort terms of numbers of personnel, dollars, and laboratory facilities.
4. A statement of any plans the AEC may have for accelerating the tempo of this program.⁴¹⁵

At the same time, the Chairman of the JCAE wrote to the Chairman of the MLC, enclosing a copy of the letter to the AEC, and asked the MLC to furnish a statement on the probable military utility of low-yield nuclear devices. Were they now regarded as potential special-purpose weapons of great interest to the U.S. armed forces, but with only limited utility? Was it possible that these devices might have a profound and far-reaching effect on the U.S. military posture?⁴¹⁶

At an AEC meeting late in July, in response to a question on the relative importance of the various devices to be tested, LASL test director Alvin Graves said that there were compelling technical reasons to test each of the six devices during CASTLE.⁴¹⁷ The AEC had come under pressure from President Eisenhower to reduce the scope of American nuclear testing following unexpectedly-heavy fallout during UPSHOT-KNOTHOLE and because of budget restrictions.

By this time, forthcoming Nevada test series included Operation TEAPOT in the fall of 1954, and Operation JULEP in the spring of 1956.⁴¹⁸

⁴¹⁵Letter dated July 29, 1953 to Lewis L. Strauss, Chairman, USAEC, from Chairman, JCAE.

⁴¹⁶Letter dated July 29, 1953 to Robert LeBaron, Chairman, MLC, from Chairman, JCAE.

⁴¹⁷Minutes, Atomic Energy Commission Meeting No. 893, July 23, 1953, p. 2.

The expansive Nevada and Pacific U.S. test series did not enjoy the wholehearted support of the Eisenhower administration: just a few months earlier, in March 1953, the U.S. nuclear weapons test program had been characterized by some government officials as a "waste of fissionable material."⁴¹⁹

However, Norris Bradbury had pointed out earlier in July that the fissionable material expended in testing to date had resulted in a 400 to 500% increase in the utilization of available material. Bradbury assumed that one six-shot Nevada series of tests might be held every year and one Pacific series every 18 to 24 months.⁴²⁰

Noting that the IVY Mike yield had been almost twice as great as predicted, Graves stated that the behavior of thermonuclear systems was not well understood and was virtually unknown for lithium deuteride-fueled systems. The JUGHEAD test could be deleted, if necessary, since it was almost certain to work. Other devices involved many unknowns which could only be solved by full-scale test.

With regard to "emergency capability," Dr. Graves stated that additional casings and components were being made for the TX-14 and that ballistic drop tests had started. If these tests were successful, the U.S. would have a deliverable thermonuclear weapon soon after CASTLE.

On the other hand, if the CASTLE tests showed that only a device using highly-enriched lithium was a practical weapon, then the military requirement for a short-term, "emergency capability" thermonuclear weapon could not be immediately fulfilled, due to a shortage of Li-6. (There was no shortage of deuterium, in either liquid or gaseous forms.)

The SHRIMP now appeared to offer the best long-term possibility but it would not be feasible to develop it into a deliverable weapon for about one to two years after CASTLE. He added that if the TX-17 (RUNT) worked, it might be the most desirable of the three proposed "emergency capability" weapons (TX-14, -16, and -17) as it was the cheapest of the

⁴¹⁸Operation TEAPOT, Report of the Test Manager, Joint Test Organization, Nevada Test Site, Spring 1955, p. 7.

⁴¹⁹Minutes of the Thirty-Fourth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 23 and 24, 1953, Washington, D. C.

⁴²⁰Memorandum to All Concerned dated July 16, 1953 from Paul C. Fine, Division of Military Application, USAEC, Subject: Trip Report, MLC Tour, July 5-11, 1953.

three since it used normal, unenriched lithium and involved no cryogenic systems.

In reply to a question from a commissioner about when LASL would know enough about weapon statuses to allow the AEC to conduct CASTLE in February 1954, Graves replied that by the middle to the end of September 1953, developmental work on the four LASL and two UCRL test devices should be sufficiently advanced to allow the AEC to determine the start date for CASTLE.

Other factors affecting this date included progress of construction at the Bikini and Eniwetok test sites and the availability of lithium-6. Optimistically, the AEC expected to have enough Li-6 on hand by January 1954 to conduct the test series. Here again, the Commission hoped to know for certain by September 1953 after the first large-scale "Alloy Development Plant" had been in operation for a short while.⁴²¹

Graves then briefly described the ZOMBIE system, a boosted two-stage fission weapon with a yield comparable to the expected yield of the SHRIMP, but with several advantages over the latter weapon. The ZOMBIE would be smaller and more compact than the SHRIMP, and would be considerably safer, permitting conventional in-flight insertion of the primary nuclear core.

The ZOMBIE had not been recommended for test during CASTLE as its characteristics were highly calculable and reliable performance data could be obtained without full-scale test. (See MK 15 history in Volume VI.)

Reviewing proposed locations for the Bikini atoll shots, Graves said that there was a strong possibility that the airstrip and base camp on Bikini island might be destroyed by one of the shots.

CASTLE was currently scheduled to begin on February 15, 1954 and end soon after April 3, weather permitting; at worst, the series was not expected to go beyond May 15. A period of bad weather was expected to begin in the Marshall islands around June 1. The second shot, a test of the ALARM CLOCK, might be delayed or rescheduled, depending on the rate of Li-6 production.⁴²²

The first three CASTLE tests were to be high-yield shots required to assure an "emergency capability" with

⁴²¹Minutes of the Thirty-Sixth Meeting of the General Advisory Committee, August 17, 18, and 19, 1953, p. 11.

⁴²²Minutes of AEC Meeting No. 863, July 23, 1953.

thermonuclear weapons and to test large bombs compatible with the B-36. All three required tritium to boost their primaries and some also required boron-10.

The ALARM CLOCK, JUGHEAD, and the RUNT were essentially military requirements; they would be useful against three or four "point" targets located within a few miles of each other. According to SAC studies, use of these bombs might reduce by 50% the total number of sorties required by strategic bombers against Soviet targets. Opinion was divided as to whether or not these weapons had any tactical value.⁴²³

The total fission yield from CASTLE was now expected to be more than twice the total of the fission yield from all previous U.S. tests.⁴²⁴

The fourth shot, a test of the SHRIMP, was also high-yield but of a device that was smaller and lighter than the first three, was to develop a bomb compatible with the B-47, which was then in service with SAC in slightly larger numbers than the B-36.⁴²⁵

However, the SHRIMP was not included in the "emergency capability" program because it would not be available as a weapon for 12 to 18 months. (The SHRIMP eventually evolved into the MK 21 thermonuclear bomb.) In the meantime, the B-47 had successfully carried and dropped a TX-14 mockup.⁴²⁶ (The B-47 originally featured a 25-foot long bomb bay, which was later shortened to just 15 feet following thermonuclear bomb developments after CASTLE.)

The SHRIMP might also be deferred because it appeared to be quite expensive in comparison to 60-inch diameter, high-yield fission weapons. If the diameter of the ZOMBIE could be

⁴²³Minutes of the Thirty-Sixth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, August 17, 18, and 19, 1953, pp. 11, 12.

⁴²⁴Minutes of the Thirty-Sixth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, August 17, 18, and 19, 1953, p. 11.

⁴²⁵At the end of 1953, SAC had 185 B-36s, 137 RB-36s, and 329 B-47s; a year earlier, there were 154 B-36s, 114 RB-36s, and only 62 B-47s in the SAC inventory. (THE DEVELOPMENT OF STRATEGIC AIR COMMAND, 1946-1986, J. C. Hopkins and Sheldon A. Goldberg, Office of the Historian, Headquarters Strategic Air Command, Offutt AFB, Nebraska, 1 September 1986, pp. 37, 42.)

⁴²⁶Minutes of the Thirty-Sixth Meeting of the General Advisory Committee, August 17, 18, and 19, 1953, p. 11.

reduced to 30", this weapon might be useful as a missile warhead; it also might furnish a limited capability for low-altitude, pull-up "toss" delivery from tactical aircraft.⁴²⁷

Dr. Herbert York, of the University of California Radiation Laboratory in Livermore, defended his lab's two proposed shots by claiming they would show the way to exceptionally small, low-yield (less than a kiloton) thermonuclear weapons. He also noted that, despite their apparent failure, the two UCRL shots during UPSHOT provided significant data and information for development purposes at Project Whitney.

UCRL was also seeking to develop other small weapons and primaries with yields between one and ten kilotons and larger primaries and secondary systems yielding between 10 and 40 kilotons.⁴²⁸

The MORGENSTERN device was to be the prototype of a class of thermonuclear weapons with a wide range of yields, weights, and diameters. The RAMROD was similar to the MORGENSTERN, except that the former was liquid-fueled. The RAMROD was to provide information which was unattainable from the MORGENSTERN alone, and which was essential in order to design weaponized versions of the MORGENSTERN, especially if it were to be done over a wide range of parameters, as planned.⁴²⁹

At the end of July, the AEC reported to President Eisenhower that three potential "emergency capability" designs were under investigation and that production of prototypes would parallel development so that stockpiling could begin as soon as one or more of the test devices proved out.⁴³⁰ In preparation for CASTLE, LASL had requested from ACF (via Sandia) dummy forward assemblies of TX-14, TX-16, and TX-17 weapons, with copies of blueprints showing weights, sizes, shapes, and center of gravity locations of these components.⁴³¹

⁴²⁷Minutes of the Thirty-Sixth Meeting of the General Advisory Committee, August 17, 18, and 19, 1953, p. 12.

⁴²⁸Hewlett and Holl, p. 165; "The Necessity For and Value of Continental Tests," paper prepared by Darol Froman at LASL with the aid of information from Livermore, August 28, 1953.

⁴²⁹Letter dated September 21, 1953 to Gen. Kenneth E. Fields, Director, DMA/USAEC, from Herbert F. York, UCRL-Livermore Laboratories.

⁴³⁰Atomic Energy Commission, Semiannual Report to the President, AEC 21/12, Report by the General Manager, July 27, 1953.

By this time, the composition of the U.S. nuclear stockpile was changing rapidly. All weapons stockpiled before mid-1950 had been retired. Seven new weapon types were being delivered to nine separate storage locations, including a full range of implosion weapons, two gun-assembly type weapons, and one missile warhead. Rate of deliveries of weapons to stockpile during fiscal 1953 (July 1952 - June 1953) were triple the fiscal year 1950 delivery rate.⁴³²

To support new thermonuclear weapon development, LASL and UCRL were designing two-stage nuclear-explosive systems; Sandia was designing and testing fuses and bomb casings; ACF was manufacturing heavy bomb casings; the NBS Cryogenics Laboratory was supporting liquid-fueled weapon system research; and Camco was engineering and producing dewars for both field test and military use.⁴³³

At the beginning of August, the AEC's Dana facility was producing 20 tons of heavy water per month; this water could be separated electrolytically into oxygen and deuterium gas. The Savannah River plant was producing 24 tons of heavy water per month, giving the AEC a total monthly production rate of 44 tons.⁴³⁴

The Test Program Comes Under Fire

Late in August, in response to earlier criticism of the entire U.S. nuclear weapons test effort, LASL and UCRL offered a defense of testing in general, and of testing untried thermonuclear devices in particular. This statement also offered insight into the state-of-the-art of weapon operation knowledge at the two laboratories.

The labs noted that whereas the behavior of fission weapons was relatively predictable by now, the behavior of large-scale thermonuclear systems was much more unknown and unpredictable without further testing which was likely to be protracted and expensive:

⁴³¹Letter dated August 11, 1953 to Marshall G. Holloway, LASL, from R. W. Henderson, Sandia Corporation, regarding Dummies for CASTLE.

⁴³²Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, p. 13.

⁴³³Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, p. 15.

⁴³⁴Memorandum to File dated August 10, 1953 from Ed Heller, subject: Weekly Williams Conference.

A new factor has recently entered the general problem of determining the amount of full-scale testing so as to match appropriately the progress in other facets of development. Whereas a combination of results from basic experimental physics, theoretical calculations, and component experiments which can be performed at Los Alamos can give reliable estimates of all pertinent physical conditions at the beginning of the explosion process in almost any fission weapon configuration, this is not true for devices dependent upon radiation for (fuel) assembly and compression.

Not only are the calculations much more difficult and uncertain for the assembly phases of these newer devices, but the basic data are often less reliable (if known at all) and, still worse, simple experimental checks of predicted behavior during assembly cannot be made without a nuclear detonation.

Thus, where full-scale nuclear detonations for fission weapon development purposes have been made with the primary objective of obtaining information about the explosive and disassembly phases of the process, similar tests are now required for thermonuclear development to obtain information upon both assembly and disassembly phases.

This uncertainty upon these two phases of function of a proposed type of device can easily lead to more than twice as much testing as might be required if only one phase were relatively uncertain.

Another factor influencing the choice of the optimum amount of testing of thermonuclear devices as compared with pure fission devices involves the great difficulty of measuring the desired quantities affecting radiation implosion during its progress. This means that, in the new field (of radiation implosion-driven devices), test experimentation has become much more complicated and costly in manpower and dollars.

This factor tends to hold down the number

of such tests because the diversion of effort required for a high testing rate would so handicap other necessary activities as to impede over-all progress. Nevertheless, it is clear that relatively more tests are needed in thermonuclear weapon development than in fission weapon development, and their demand upon budget, and particularly upon technical manpower, renders it most important to carry out such tests as cheaply time-wise and dollar-wise as possible. ...

....

As explained above, components testing involving nuclear detonations as well as tests of large-yield devices are even more necessary in thermonuclear weapons development than for fission weapons. [deleted] The engineering of these (fusion) devices into manageable, practical (and convenient) bombs requires much development and many "component tests" involving nuclear detonations. Operation CASTLE, planned for Eniwetok in the Spring of 1954, consists of a very heavy schedule of tests of devices whose yields are expected to be so great as to (preclude their) testing (in the continental U.S.).⁴³⁵

In direct response to the earlier (anonymous) charge that the U.S. nuclear weapons testing program was "a waste of fissionable material)," the labs noted the value of previous weapons tests:

In the more distant past, testing activity was not well balanced with other activities. The need for test information at the time of TRINITY was so urgent and so obvious that a large fraction of the national stockpile of fissionable material was used up during a hot war in which it might have been put to direct military use.

The CROSSROADS tests were essentially valueless to weapons development and the

⁴³⁵"The Necessity For and Value of Continental Tests," paper by Darol Froman prepared at LASL with the aid of information from Livermore, August 28, 1953. Tests during Operation CASTLE would soon show both LASL and UCRL how much more they had yet to learn about thermonuclear explosives.

growing demand for test-type information again became (urgent) in 1947 (which led) to SANDSTONE. (Author's note: at the time of CROSSROADS, the entire U.S. nuclear weapons stockpile consisted of only nine implosion weapon cores and HE assemblies; two were expended solely for effects testing during CROSSROADS.)

Another high surge in the demand for (test) information arose before RANGER. The very great and sudden improvements in the national stockpile capability resulting immediately after SANDSTONE and after RANGER are proof, not only of the value of full-scale testing, but also of the fact that testing activity had been at too low a level compared with other activities.

We should never again allow one of our major activities to fall so far behind progress in all other major lines for, if we do, these activities will soon reach the point of diminishing returns. The Los Alamos Scientific Laboratory does not yet feel that the rate of testing is as rapid as the generation of new ideas would really warrant.

To support requirements for more-frequent testing in Nevada, LASL provided a long list of areas for further weapons development, including very small weapons economical in their use of fissionable material; weapons that could withstand high impact decelerations; lightweight high-yield weapons; radiation implosion systems in conventional sizes and yields; special-purpose weapons; more versatile, safer, and interchangeable weapons; replacements for especially costly, complicated, or potentially unreliable weapon components; and thermonuclear weapons of reduced weight and dimensions that would use less critical material.

Tests were specifically required to create efficient, small, low-yield (1/4 to 4 kiloton) bombs; to further "boosting" studies; to help design a particular small implosion weapon, if there were military interest in its development; to measure the "fizzle" yield while simultaneously determining another point on a yield-vs-initiation time curve (the "XR" series; see initiator history in Volume II); to proof-test the behavior of uranium-233 (see Volume II) and highly-irradiated Pu-240-bearing "dirty" plutonium that would be produced in commercial power reactors; to determine basic data for the thermonuclear program; and to study the containment time and radiation ducting afforded by various

thermonuclear bomb casings.⁴³⁶

UCRL was also interested in Nevada tests of exceptionally-small complete weapons and two-stage devices with mocked-up secondaries; test yields were expected to range from subkiloton up to 40 KT.

Early in September, AEC Commissioner Eugene Zuckert, possibly envisioning imminent nuclear combat with the Soviet Union, suggested a "quick and dirty" test of one of the "emergency capability" weapons; he said that "for Moscow purposes, I would rather have a capability of a couple in March 1954 (when the nights are longer) than a few more in July."⁴³⁷

By the middle of September, as a result of several readiness review meetings, the AEC decided that completion of preparations for the first CASTLE shot by February 15, 1954 was improbable, and that March 1, 1954, was a more realistic date for the first shot. The AEC still hoped to achieve "emergency delivery capability" shortly after the first CASTLE test. The new lithium-6 plant at Oak Ridge was now in operation, and it reached full production at the end of the month.⁴³⁸

Another CASTLE planning issue that came to the fore in September was the availability of sufficient quantities of enriched lithium and lithium deuteride for "emergency capability" weapons. The more the CASTLE devices relied on large amounts of these materials, the less likely that Los Alamos and Livermore could meet scheduled stockpile dates. Even beyond this, there was no assurance that a "dry" weapon would work; if the lithium-fueled devices failed, Los Alamos would have to fall back on the cumbersome cryogenic liquid-fueled JUGHEAD bomb.

To alleviate the lithium production problem, a second large-scale lithium enrichment plant was to be established at Oak Ridge (the first high-volume facility had been authorized in

⁴³⁶"The Necessity For and Value of Continental Tests," paper prepared by Darol Froman at LASL with the aid of information from Livermore, August 28, 1953; Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, p. 24.

⁴³⁷Memorandum for Chairman Strauss dated September 3, 1953 from Eugene M. Zuckert, Commissioner, USAEC.

⁴³⁸"Outline of Oral Report to NSC," September 17, 1953 (paper is dated September 15, 1953), no author; U.S. Atomic Energy Commission, "The Atomic Energy Program," Semiannual Report to the President as of December 31, 1953, February 1, 1954.

the summer of 1951 and had started producing in August 1953). Quantity requirements for lithium-6, tritium, and boron-10, all critical components of the solid-fueled "emergency capability" weapons, were still uncertain.⁴³⁹

There was a definite requirement for fully-enriched lithium (95% Li-6, 5% Li-7); weapons using enriched lithium would give high yields at relatively lower cost per unit yield.

At an AEC meeting on September 22, the CASTLE test series and the "emergency capability" program were reviewed. Lithium-6 production was still a determining factor in weapon availability, assuming that CASTLE had even limited success. A thermonuclear version of the ZOMBIE had replaced a test of the RUNT (TX-17) on the CASTLE shot schedule; LASL believed that the ZOMBIE would be a useful and promising weapon because of its relatively low weight per unit yield. ZOMBIE weight was expected to be around 6,000 to 7,000 lbs.; yield was predicted at one-and-a-half to two megatons.

The six-shot test series at this time is shown in Table 3-3.

Table 3-3 CASTLE Shot Schedule, September 1953

<u>Device/ Shot Name</u>	<u>Proposed Date</u>	<u>Location</u>	<u>Predicted Yield</u>
SHRIMP (Bravo)	March 1, 1954	Bikini	4 MT
ALARM CLOCK (TX-14) (Union)	March 11	"	4 MT
ZOMBIE (Nectar)	March 22	"	1.5 - 2 MT
MORGENSTERN (Koon)	March 29	"	2 MT
RAMROD (Echo)	April 7	Eniwetok	0.2 MT
JUGHEAD (Yankee)	April 14	"	6 - 8 MT

Four LASL shots of the SHRIMP, ALARM CLOCK, ZOMBIE, and

⁴³⁹Letter dated September 15, 1953 to Honorable Robert LeBaron, Chairman, MLC, from Lewis L. Strauss, Chairman, USAEC. Strauss succeeded Gordon Dean as AEC Chairman on July 2, 1953.

JUGHEAD devices, and two UCRL tests of the MORGENSTERN and RAMROD devices, were planned. Although the test of the JUGHEAD was somewhat contingent upon the results of the preceding LASL tests, there was a high likelihood at this time that it would be fired.

The RUNT (TX-17) had been pulled from the test schedule because LASL was beginning to doubt its feasibility. Because the TX-17 RUNT used normal unenriched lithium, if successful, it would be a comparatively cheap weapon and could presumably be produced sooner and in greater numbers than weapons such as the ALARM CLOCK requiring lithium enriched to 95% in the Li-6 isotope. In addition, the RUNT had many components in common with the ALARM CLOCK and the JUGHEAD.

However, LASL was becoming increasingly dubious about the satisfactory nuclear performance of the RUNT; it was very doubtful from small-scale laboratory experiments that normal lithium would be a suitable thermonuclear fuel, and accurate cross section measurements still needed to be completed.

The probable yield of the RUNT was very uncertain. (Operation CASTLE would soon show LASL just how uncertain.) AEC commissioner Lewis Strauss noted that for the "emergency capability" program, a timely deployment of a usable, high-yield weapon was a more important consideration than cost.

Components and cases for the TX-14 and TX-16 were being fabricated at ACF in Albuquerque and unretarded drop tests of the TX-14 were underway. It was not definite that the TX-14 would work; although the JUGHEAD would probably work, its weight (42,000 lbs.) and cryogenic requirements made aerial delivery more difficult than for the ALARM CLOCK.

If the logistic and cryogenic problems could be solved in a practical manner, then the TX-16 would provide an early "emergency capability" sooner and could be produced in larger numbers than the TX-14 because liquid deuterium was currently more plentiful than lithium-6. However, a temporary shortage of tritium would limit the number of TX-16s that could be made in the next nine months.

The AEC noted that both the TX-14 and TX-16 would be obsolete in two years when new, lighter solid-fueled bombs such as the SHRIMP would be available.

As far as lithium production was concerned, the AEC planned to produce lithium enriched to 40% Li-6 until the requirement for the SHRIMP was met and then to produce 95% Li-6 enriched lithium for the ALARM CLOCK and other weapons. The AEC examined the possibility of testing the TX-17 first, and if it were successful, delaying the remainder of the CASTLE series until enough Li-6 to provide an "emergency capability" had been produced.

If the enriched lithium that would be consumed in the remaining CASTLE shots were devoted to "emergency capability" weapons, then more weapons could be produced and stockpiled. If, however, the purpose of CASTLE was to explore several approaches to an eventual high-yield, low-weight weapon, then the entire series should be conducted.⁴⁴⁰ How many "emergency capability" weapons might be stockpiled in the near future depended directly upon which types proved successful during CASTLE.

Limited production of highly-enriched lithium restricted acceleration of a test of the ALARM CLOCK. Enough lithium enriched to 95% Li-6 for the ALARM CLOCK would be available from the ADP on about December 1, 1953, if an immediate decision were made to halt production of 40% enriched lithium. It would take about 40 to 45 days after the last shipment of lithium-6 deuteride from Oak Ridge for the completion of fabrication of ALARM CLOCK components at Los Alamos, air shipment to the Pacific Proving Ground, and assembly for test.

Therefore, the earliest date for the firing of an ALARM CLOCK would be around January 15 to February 1, 1954, but at the expense of delaying the rest of the CASTLE program. The present CASTLE schedule called for the first shot on March 1, and a test of the ALARM CLOCK on March 11.

An earlier test of the ALARM CLOCK would not advance the "emergency capability" schedule, unless other CASTLE tests requiring 95% enriched lithium were canceled or proposed; this would make enough Li-6 available for one or two additional ALARM CLOCKS. The extra expense and the difficulties of conducting a later test operation to fire the remaining CASTLE devices would be enormous.

Various aspects of the TX-16 were reviewed, including its availability for flight and drop testing. The Cambridge Corporation and the National Bureau of Standards' Boulder, Colorado cryogenics laboratory were both working on the TX-16, in conjunction with ACF, Sandia, and LASL. As soon as an assembled JUGHEAD was available, flight testing would begin (drop tests of the TX-16 did not begin until December 1953).

The AEC believed that the TX-16 offered the greatest assurance of both a successful and a high-yield weapon, despite its complicated cryogenic support requirements, and directed that every effort be made to start stockpiling TX-16s no later than January 1954.

⁴⁴⁰Minutes of AEC Meeting No. 917, September 22, 1953.

The Commission also reviewed the possible addition of a RUNT test to the six CASTLE tests proposed by LASL and Project Whitney (the MLC had been pressing for its inclusion in the series). Firing of the RUNT was contingent upon the results of the SHRIMP test. If the SHRIMP worked well, there was a chance that the RUNT would also work.

If successful, the RUNT would offer a means, independent of lithium-6 production, to augment the "emergency capability" program. Its test would be a hedge against total failure of the test series, and against limited availability of the ALARM CLOCK. In addition, testing both the RUNT and the SHRIMP would give very valuable information that should lead to lighter and smaller thermonuclear weapons in the future.⁴⁴¹

At another AEC meeting a week later, Gen. Kenneth Fields, Director of Military Application, reiterated that the JUGHEAD was the most promising "emergency capability" candidate because it was similar to the SAUSAGE tested nearly a year earlier. Quantity production of the ALARM CLOCK was still subject to the availability of lithium-6, and the RUNT, while using normal lithium unenriched in Li-6, was likewise subject to great uncertainty about whether or not it would work.

Therefore, the best course to attain an early "emergency capability" would be to stockpile the JUGHEAD at the earliest possible date, carry out the CASTLE program as now planned, and test the RUNT if results from the SHRIMP shot were sufficiently encouraging.

For this reason, CASTLE was again expanded with the addition of a seventh shot (a test of the RUNT).⁴⁴² A seven shot program was not much more difficult than a six shot series; the additional test did not require additional facilities beyond the barge from which it was to be fired.

Fields also believed that results of CASTLE might make it possible to reduce the weight of the ALARM CLOCK and future hydrogen bomb designs.

Norris Bradbury noted that LASL was in agreement with the proposal to stockpile the JUGHEAD as the most promising "emergency capability" weapon. From the standpoint of which weapons should be tested, he said that the ZOMBIE was an interesting possibility but not exciting. For planning purposes, LASL needed a final decision on whether six or

⁴⁴¹Draft Minutes of Executive Session of Commission (Meeting) on September 22, 1953, undated.

⁴⁴²Letter dated November 4, 1953 to Honorable Robert LeBaron, Chairman, MLC, from Lewis L. Strauss, Chairman, USAEC.

seven devices would be tested; if the series were limited to six, the desirability of one shot versus another had to be debated and decided.

Dr. Marshall Holloway discussed the status of work on the cryogenic elements of the JUGHEAD and the status of testing of various components, and Air Force plans for handling logistical and bomb delivery problems associated with liquid deuterium. New dewars had been designed, and refrigerating units developed, to permit continuous cooling of the dewars in transit where necessary.

Dr. Bradbury believed that experience gained during the past year in handling liquid hydrogen had offered valuable lessons that would simplify cryogenic logistics.⁴⁴³

Two days later, at a joint AEC-MLC conference at the beginning of October, Air Force General James E. Briggs, noting that the SHRIMP appeared very desirable from a reduced weight standpoint, as compared to the three "emergency capability" TX-14, TX-16, and TX-17 weapons, asked how soon a "weaponized" version might be available. General Fields replied that a deliverable SHRIMP would probably be ready in 18 to 24 months.

Funds in the amount of \$75 million had been authorized by the AEC for construction of a second "Alloy Development Plant" lithium enrichment facility at Oak Ridge; it would have a production capacity twice that originally planned. The new plant was to be ready for operation in about 18 to 24 months.⁴⁴⁴

The facility would use the "colex" column exchange process rather than a mixer-settler mercury amalgam process used in the first ADP. ADP number 2 was to be erected at Oak Ridge.

When a question arose as to whether the use of highly enriched lithium-6 in thermonuclear weapons was based on absolute need or upon scientific need, LASL noted that use of concentrated Li-6 would permit higher yields, thus reducing the overall size of a weapon designed for a specific yield, or providing higher yield for the larger size.

The AEC might face a difficult decision if the solid-fueled CASTLE devices proved to be outstandingly successful, leading to a big program to stockpile them. This program might require every available gram of U-235, thus depriving the

⁴⁴³Minutes of AEC Meeting No. 920, September 29, 1953.

⁴⁴⁴Minutes of the Eighty-Fifth Conference of the Atomic Energy Commission - Military Liaison Committee, October 1, 1953.

plutonium production reactors at Savannah River and Hanford of the fuel they required.⁴⁴⁵

October 1953: The AEC Reviews CASTLE Plans

An AEC paper circulated at this time went into great detail about the objectives and history of the CASTLE program and the evolution and interrelationships of test devices for the series; it also stressed the many unknowns still surrounding the operation:

Operation CASTLE embraces both short-term and long-term goals for the thermonuclear program. The short-term goal is to prove in an emergency capability with one or more thermonuclear weapons currently being engineered for production and delivery. The long-term goal is to test new designs which should lead to thermonuclear weapons that are smaller, lighter, more deliverable, and perhaps of higher yield in the future.

The tests of the TX-14, TX-16, and TX-17 are directed toward determining the content of the emergency capability program. The test of the SHRIMP may enable the weight of the emergency capability weapons to be reduced and, together with the tests of the MORGENSTERN, RAMROD, and ZOMBIE should point the way to the next generation of thermonuclear weapons.

The earliest feasible date for the start of CASTLE tests is March 1, 1954. The CASTLE program recommended is believed to be the maximum practicable program.

....

In a letter of June 23, 1952, the AEC requested MLC agreement to proposals for an urgent program to (1) fabricate the first deliverable version of a thermonuclear weapon, the ALARM CLOCK, for test toward the end of 1953 and (2) establish an "emergency capability" for use of this device shortly after its first test if (the test) turned out to be successful. ... The possibility was envisaged at that time that

⁴⁴⁵Memorandum for File dated October 6, 1953 from Wayne P. Brobeck, JCAE, subject: Weekly Conference with Walter Williams, Friday, October 2, 1953, p. 2.

other radiation-implosion (sic) devices might be included in the test operation.

The successful MIKE shot in Operation IVY on November 1, 1952 stimulated a vigorous program for developing a deliverable cryogenic weapon patterned after the MIKE device and called the JUGHEAD. Subsequently, calculations on the feasibility of combining the virtues of a dry fuel, as in the ALARM CLOCK, with the favorable implosion characteristics of the uranium tamper as in the SAUSAGE, led to a dry, two-stage device called the RUNT, which would utilize normal lithium.

Finally, the possibility of applying radiation implosion on a smaller scale to boosted fission devices led to the idea of the ZOMBIE, which would employ U-235, lithium-6, and tritium to attain a yield of a megaton or more, but would have a weight and size in the range of ordinary fission bombs.

In order to explore (several) designs for emergency capability and thereby improve the odds of achieving at least one successful weapon, the Los Alamos Scientific Laboratory proposed on November 28, 1952 that CASTLE be expanded to include the (ZOMBIE). In order to accomplish this program, Los Alamos recommended that CASTLE be held in the spring of 1954.

UCRL-Livermore proposed on November 24, 1952 that two thermonuclear devices, the MORGENSTERN and RAMROD, be tested in CASTLE to investigate the potentialities of small-scale radiation implosion and stated that these devices probably could not be ready for test until the late Winter or early Spring of 1954. ... By letter of January 2, 1953, the (AEC) proposed to the MLC that CASTLE be held as early in 1954 as technical progress permitted ...

In the course of further study of various thermonuclear weapon designs at Los Alamos in 1953, consideration was given both to a RUNT containing normal (unenriched) lithium and to a RUNT containing lithium partially enriched in lithium-6. The latter would have a better chance of success and give a

higher yield, but would be dependent upon lithium-6 production.

Meanwhile, in the Spring and Summer of 1953, the Air Force indicated the importance of reducing the weight of thermonuclear weapons for delivery by the B-47, a medium bomber being produced in large numbers, and for delivery by the (B-58) Hustler, a new medium bomber under development. Los Alamos responded by proposing that the enriched RUNT be scaled down to a 3/4 scale version called the SHRIMP with a reduction in weight from 42,000 to 25,000 pounds.

Since the SHRIMP has been under design for only a few months, engineering for production and delivery is not as far advanced as in the case of the ALARM CLOCK, RUNT and JUGHEAD, which are the emergency capability weapons.

The programs for thermonuclear development, testing, and emergency capability have been going forward with great emphasis and speed. The importance of these programs is increased by the fact that the Soviet Union conducted an atomic test on August 12, 1953 that involved both fission and thermonuclear reaction.

....

The purpose of the JUGHEAD and ALARM CLOCK tests is to prove in emergency capability weapons. The third emergency capability weapon, the (unenriched) RUNT, was not included in the (September 1953) Los Alamos proposals (for CASTLE testing) because of considerable doubt as to its nuclear performance and because of the previous assumption that Los Alamos would be limited to four shots in the CASTLE operation.

However, if the RUNT were successful it would have the advantage of being a dry weapon independent of lithium-6 production; it would be a hedge against failure or limited availability of the ALARM CLOCK and it would also be a hedge against logistic and delivery problems that may arise from the cryogenic requirements attendant to the JUGHEAD.

Because the JUGHEAD offers the greatest assurance of good nuclear performance, it must be heavily depended on in the early emergency capability program. However, better decisions regarding the relative merits of JUGHEAD and ALARM CLOCK and future production can be made after CASTLE on the basis of actual results and experience gained from testing each type.

This is particularly important since the present situation indicates a large military requirement for emergency capability weapons soon after CASTLE.

A preview of the performance of the RUNT will be afforded by the test of the SHRIMP. The SHRIMP is essentially a RUNT design reduced in scale but enriched in lithium-6. If the yield of the SHRIMP were to be low because of inadequate propagation of the thermonuclear reaction, then the RUNT would not merit testing. On the other hand, if the yield of the SHRIMP were high, the chances for the RUNT would be increased.

Since the test of the SHRIMP is to be the most thoroughly instrumented one in CASTLE, detailed information should be available on its performance. Because of the potential advantages of the RUNT in the emergency capability program, it therefore appears advisable to include a test of the RUNT in CASTLE, contingent on the performance of the SHRIMP.

The tests which are intended primarily to provide information for the future development of thermonuclear weapons are the ZOMBIE, MORGENSTERN, and RAMROD tests. The goal is to obtain data which will enable the next generation of thermonuclear weapons to be of smaller size, lighter weight, higher efficiency, or greater yield.

Results of the test of the ZOMBIE which has a [deleted] will allow current emergency capability weapons to be redesigned and their weight reduced by as much as 10,000 pounds, with a smaller reduction in weight possible for the ALARM CLOCK. The tests of the RAMROD and MORGENSTERN are aimed at the development of weapons having yields of a

megaton or more for weights and sizes in the range of present fission weapons.

In regard to the timing of the CASTLE operation, Los Alamos proposes that March 1, 1954 be set as the target date for the first test. This is based on the rate of supply of lithium-6, the time required for fabrication, shipment, and assembly of weapon components, and the schedules for construction of test sites and installation of equipment at Bikini and Eniwetok.

Although there is no one phase of the preparations which prohibits the start of CASTLE a little earlier, every phase is (scheduled) so tightly that March 1 seems the best target date to set for the initial detonation. ...

Scheduling the date of the first CASTLE test for March 1 does not adversely affect the time when the first emergency capability is to be achieved. The plan is to initiate emergency capability with JUGHEADs, and these are the thermonuclear weapons that can be stockpiled with greatest assurance of their nuclear performance prior to the CASTLE tests. In accordance with the directives of the Commission on September 5, 1953, every effort will be made to have JUGHEADs available for emergency capability in January 1954.

Consideration was given to limiting the CASTLE program to a maximum of four shots and deferring the remaining tests to a later operation. While this would be favorable from the point of view of simplifying and shortening the CASTLE operation, it would not be consistent with the maximum effort toward thermonuclear progress. It would also involve greater expense inherent in two separate operations.

As to the relative difficulty of seven versus six shots, it is now apparent that a seventh shot would be feasible, at (a) reasonable cost in time, dollars, and efforts, if the barge technique used for the ALARM CLOCK, JUGHEAD, and RUNT shots proves favorable enough for extension to an additional barge shot at essentially the

same site.

Radiation hazard of these over-water shots appears sufficiently low to permit firing at the close time intervals anticipated. The instrumentation mounted on the islands could, if not damaged, service three shots probably as well as two.

The CASTLE program recommended in this paper is believed to be the maximum practicable program. It covers all available possibilities for providing an emergency capability and should furnish an excellent basis for the future development of thermonuclear weapons.

A seven shot CASTLE program provides the greatest assurance of achieving an "emergency capability" by minimizing the possible consequences of unanticipated problems with JUGHEAD and assures earliest development of better weapons to exploit AEC's programmed capacity to produce lithium-6.

Any alternative plan such as postponing a portion of the shots until the Fall of 1954 would be wasteful, costly, and harmful to either the immediate "emergency capability" program or future progress in the field of thermonuclear weapons.

The magnitude and complexity of a seven-shot program is not to be underestimated, but must be reckoned with in view of the importance of thermonuclear weapon progress. It therefore appears ill-advised to consider anything short of a maximum effort for the CASTLE program.⁴⁴⁶

In a companion report, Dr. Norris Bradbury of LASL noted that if the lab were to be limited to only four experiments during CASTLE, he would rather fire the partially-enriched SHRIMP than the unenriched RUNT.

Bradbury also appended a long table that listed general characteristics and nuclear material requirements of the

⁴⁴⁶Atomic Energy Commission, CASTLE Program, Report by the Director of Military Application, AEC 597/23, October 1, 1953.

JUGHEAD, SHRIMP, ALARM CLOCK, and RUNT, as well as some additional general information on other thermonuclear weapon systems which might be of comparative interest. The tabulated material requirements included the amounts of or alloy, plutonium, tritium, boron-10, and lithium (at various degrees of enrichment) for the devices; characteristics included warhead weights, diameters, lengths, and anticipated yield ranges.

Bradbury also attached two tables listing probable monthly production rates of "emergency capability" bombs, including flight and drop test units and stockpile units. He noted that there remained the possibility that the ALARM CLOCK might show unsatisfactory behavior and that it might not be desirable to put lithium-6 into this particular "emergency capability" system.

If that were to happen, it was recommended that the 95% enriched lithium produced up to that time be stockpiled for future use in another more promising system, and that the ADP facility immediately revert to the production of lithium enriched to 40% Li-6 which could be used in enriched RUNTs whose behavior was certain to be satisfactory if the SHRIMP test were satisfactory, and which in any case would give higher yields than similar systems that did not contain enriched lithium.

He also warned the AEC that if "case difficulties" (radiation ducting problems) prevented both the ALARM CLOCK and JUGHEAD from being successful, then the RUNT would certainly not work and the Commission would have to face the fact that there would be no thermonuclear "emergency capability" because of the 50,000 lb. weight limitation imposed by current Air Force bomber payload-versus-range limits.

In an appended letter, Dr. Herbert York outlined UCRL CASTLE plans, which were basically to explode two thermonuclear devices; one was hoped to be the prototype of a class of weapons having a wide range of yields, weights, and diameters. The other was similar to the first, but with a different composition of its fuel; this design was to produce information which was unattainable from the first design, but which was also essential to the development of weaponized models over a wide range of parameters.⁴⁴⁷

About a week later, AEC Chairman Lewis Strauss reported the AEC's CASTLE recommendations to the chairman of the MLC; Strauss noted that testing all three candidate devices (the

⁴⁴⁷Letter dated 22 September 1953 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

TX-14, TX-16, and TX-17) "provides the greatest assurance of achieving an 'emergency capability' by minimizing the possible consequences of any unanticipated delay in solving the difficult engineering problems of the JUGHEAD," and that tests of the ZOMBIE, RAMROD, and MORGENSTERN would "assure earliest development of better weapons" and that the AEC's lithium-6 production program was being phased in concurrently with thermonuclear weapons development.⁴⁴⁸

The CASTLE shot schedule drawn up in April 1953 remained relatively unchanged, except for the addition of another shot in late September, until late October when the final schedule firmed up. This revised schedule called for seven shots to be detonated beginning on March 1, 1954, as shown in Table 3-4.

The most likely yields are shown below the shot name, followed by a predicted yield range. Actual shot dates (local time) are shown in parentheses below originally-scheduled shot dates. The dates of all shots after March 1 changed several times during the operation, as well as the locations for a few of the tests. The Yankee shot was fired in the Union crater; the Nectar shot was moved to the IVY Mike crater at Eniwetok; and the Romeo shot was detonated in the Bravo crater. The Echo shot was eventually canceled.

Table 3-4 Original "Final" CASTLE Shot Schedule⁴⁴⁹

<u># / Name</u>	<u>Laboratory</u>	<u>Device Name</u>	<u>Location</u>	<u>Date</u>
1 / Bravo 6 MT (4-8 MT)	LASL	SHRIMP (TX-21 pro- totype)	Bikini-on reef 2,950 from SW tip of Namu	1 March (1 March)
2 / Union 3-4 MT (1-6 MT)	LASL	ALARM CLOCK (EC14)	Bikini-barge in lagoon near Yurochi	11 March (26 April)

⁴⁴⁸Letter dated October 7, 1953 to Honorable Robert LeBaron, Chairman, MLC, from Lewis Strauss, Chairman, USAEC.

⁴⁴⁹Letter dated 26 January 1954 from Lewis L. Strauss, Chairman, U.S. Atomic Energy Commission, Washington, D.C., to President Dwight D. Eisenhower; memorandum dated October 30, 1953 to Dr. Gordon M. Dunning, Division of Biology and Medicine, USAEC, from Capt. W. L. Guthrie, USN, Chief, Test Branch, DMA/USAEC, Subject: Changed Concept of CASTLE; Semiannual Historical Report, Headquarters, Field Command, The Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1954 - 30 June 1954, John Wendell Bailey, Lt. Col., QMC, Field Command Historian, 30 June 1954, p. 333.

3 / Yankee	LASL	JUGHEAD	Bikini-barge	22 March
8 MT (6-10 MT)		(EC16)	at Union	(5 May)
			shot site	
4 / Echo	UCRL	RAMROD	Eniwetok-	29 March
125 KT (65-275 KT)	(cryogenic)		Eberiru	(canceled)
			island	
5 / Nectar	LASL	ZOMBIE	Bikini-barge	5 April
1.8 MT (1-2.5 MT)		(TX-15)	at Union	(14 May)
			shot site	
6 / Romeo	LASL	RUNT	Bikini-barge	15 April
4 MT (1.5-7 MT)		(EC17/24)	at Union	(27 March)
			shot site	
7 / Koon	UCRL	MORGENSTERN	Bikini-	22 April
1 MT (1/3-2.5 MT)		(solid-fueled)	Eninman	(7 April)
			island	

There were a number of very logical reasons for this shot sequence. Firing the 40%-enriched lithium-fueled SHRIMP first allowed analysis of its results to be applied to the subsequent 95% enriched ALARM CLOCK and unenriched RUNT shots, for better yield prediction and for any required last-minute, minor on-site redesign, as had been done with the IVY Mike device.

Whether or not the RUNT would be fired depended directly and almost entirely on results of the SHRIMP test: if the SHRIMP performed well, then an unenriched system would probably work at least reasonably well. There was no question that if the highly-enriched secondary in the ALARM CLOCK could be ignited, it would burn even better than the partially-enriched secondary in the SHRIMP.

If the SHRIMP "fizzled," then the RUNT test would very probably be canceled. Since the SHRIMP, RUNT, and ALARM CLOCK all used the LASL RACER IV primary, problems with one device might carry over into the others. If the SHRIMP "fizzled," LASL would have determine the cause almost immediately, and either cancel both the RUNT and ALARM CLOCK shots, or, if the primary were the problem, use another primary, possibly the tested COBRA.

The probable high-yield test of the liquid-fueled JUGHEAD was slated to follow the first two tests of solid-fueled devices; after JUGHEAD was fired, unless all of the first three shots "fizzled," the U.S. would have successfully proof-fired at least one deployable "emergency capability" weapon: either the "dry" TX-14 or the "wet" TX-16, or both.

This order of priority of shots was in keeping with the

"emergency capability" program which dictated that all possible efforts must be directed toward early stockpiling of a deliverable thermonuclear weapon.⁴⁵⁰

All shots after Yankee would be exploratory and/or developmental tests of possible future weapons; all devices except the JUGHEAD and RAMROD contained lithium enriched to varying degrees in the lithium-6 isotope.

All of the Bikini shots would be fired from the control bunker firmly emplaced on Enyu island. The Eniwetok shot would be fired from a control station on Parry island.⁴⁵¹

One weapon readied for test at Bikini was the EC16 JUGHEAD. This was a weaponized version of the LASL SAUSAGE device; the design had been modified to allow the liquid-fueled weapon to be airdropped. It was essentially a large vacuum bottle of liquid deuterium cooled by recirculated liquid hydrogen. At least one B-36 bomber had been fitted experimentally with pipes and liquid hydrogen dewars for carriage and delivery of the EC16.

The EC16 was only intended to be tested as a backup in the event that all the solid-fueled "emergency capability" devices failed.⁴⁵²

Late in October, experimental quantities of lithium-6 were being prepared for CASTLE. Lithium enriched to 40% in the Li-6 isotope was now available to Los Alamos and lithium enriched to 95% Li-6 was being produced slightly ahead of schedule.

In addition, since early August, the AEC's output of heavy water had quadrupled, to a rate of 44 tons per week from the Dana and Savannah River facilities.⁴⁵³

At the start of November, the first shipment of fissionable material to Eniwetok was scheduled to leave California on or about December 20.⁴⁵⁴

⁴⁵⁰CASTLE, TG 7.1 History, Installment 1, November 20, 1953.

⁴⁵¹Table 1, Final Approved Shot Schedule, OPERATION CASTLE, Report of the Manager, Santa Fe Operations, Pacific Proving Ground, Spring 1954, p. 4.

⁴⁵²York, THE ADVISERS, p. 85.

⁴⁵³Memorandum for the File dated October 26, 1953 from Edward L. Heller, subject: Weekly Conference with Walter Williams, October 23, p. 2.

The GAC met again between November 4 and 6; topics of discussion included both the CASTLE program and future thermonuclear weapon development and stockpiling.

The committee noted that the JUGHEAD had long-term undesirable problems with regard to logistics and readiness, especially if large numbers were to be maintained in the stockpile. Special handling equipment would have to be kept ready, as well as large quantities of highly flammable liquid hydrogen and deuterium.

Both the ALARM CLOCK and the RUNT were to be ready in deliverable form before CASTLE, in expectation of the successful detonation of at least one of them. After the tests, the AEC might be faced with the problem of redesigning these 40,000 lb. weapons for greater efficiency. In a smaller 25,000 lb. class, a weaponized version of the SHRIMP might be available in about a year.

In a 10,000 lb. class, a weaponized ZOMBIE might be in hand by the end of 1954. Livermore was trying to fit one of its designs into an existing bomb casing.

For the immediate future, the Air Force was most interested in very heavy weapons; in five to seven years, more emphasis on the intermediate and lighter weights was anticipated. Norris Bradbury suggested that weight classes of interest between 1954 and 1959 or 1961 would be 40,000 lbs.; 20,000 lbs.; 10,000 lbs.; and below 10,000 lbs. (These categories would later be formalized as Class "A," Class "B," Class "C," and Class "D" thermonuclear weapons.)

General Kenneth Fields indicated that recent trends in fission weapons had emphasized 30" and 60" bombs; while it was possible that the ZOMBIE and MORGENSTERN -- if successful -- would certainly have an impact on the 60" diameter weapon field, by virtue of their yields either overlapping or surpassing those of stockpiled 60" diameter fission bombs, the effect was hard to foresee at the moment.

Fields also noted that the Li-6 production program was coming along quite well and that 35% of current U-235 production and 18% of current plutonium production capacity was earmarked for thermonuclear weapons. The plutonium-making capacity was being diverted to tritium production as well as direct plutonium production for thermonuclear bomb primaries.

The ADP program was in full swing, with substantially greater production than expected from the "elex" process. No

⁴⁵⁴Letter dated November 4, 1953 to Honorable Robert LeBaron, Chairman, MLC, from Lewis L. Strauss, Chairman, USAEC.

difficulty was anticipated in meeting Li-6 requirements on time for CASTLE. The second ADP was to use a "colex" or "column exchange" separation process, which involved the separation of lithium isotopes by exchange between a lithium amalgam and an aqueous solution of lithium hydroxide. Lithium-7 was concentrated in the amalgam phase; the process required large quantities of mercury.

The second ADP was to start small-scale operation in April 1955 and come up to full capacity by the following October. It would cost \$70 million, including \$13 million for mercury (the first ADP, that used the "elex" separation process, cost \$45 million). The second plant was expected to produce enriched lithium at a cost \$3 per gram; it was likely that a third separation facility would be needed some time in the future, depending on the isotopic concentration.

This possible future requirement would be considered after CASTLE; although "orex" might be of some promise for the third ADP, the "colex" process appeared to be more economical.

Lithium would be produced in three degrees of enrichment of Li-6, including 7.5% (i.e., unenriched), 40%, and 95%.⁴⁵⁵

A new boron-10 production plant was being constructed at a cost of \$1.5 million to generate 250 kilograms of boron-10 per year. It was expected to be in operation by the first part of 1955.

Dr. Herbert York stated that a new directive for small fission weapons development at Livermore would probably not interfere seriously with UCRL's thermonuclear program, although it would divert some of the available skilled talent (shades of LASL ca. 1950!). York indicated that Project Whitney was studying two possible types of thermonuclear gadgets; one was a 45" diameter system weighing 15,000 lbs. in one version or 11,000 lbs. in another. This particular system should fit into a MK 6 bomb ballistic casing.

The other system would be either a 30" diameter, 5,000 lb. device or a 20" to 24" diameter, 3,000 lb. gadget. If both types of devices worked, they would lead to two lines of weapon design which could be interpolated to smaller, lower-yield weapons or extrapolated to larger, higher-yield designs.

At the other extreme, even larger thermonuclear weapons were considered. Dr. John von Neumann argued that SAC was

⁴⁵⁵Letter dated November 23, 1953 to Honorable W. Sterling Cole, Chairman, JCAE, from Lewis L. Strauss, Chairman, USAEC.

extremely confident of its ability to deliver big bombs and wanted the highest possible yield ("the largest possible bang," as quoted in the meeting minutes), so the power of the largest bomb should be maximized.

The SHRIMP, in the 20,000 lb. weight class, had the best current yield-to-weight ratio; nothing of high enough yield was presently available in the weights that SAC preferred. The JUGHEAD had logistic disadvantages, and the ALARM CLOCK was an unknown quantity. Dr. von Neumann felt that a weapon should be developed with the largest possible yield in a 50,000 lb. weight class.

There was an extended discussion of the pros and cons of such weapons. GAC Chairman Dr. Isidor Rabi believed that the logistical problems of the JUGHEAD were overrated and that sufficient quantities of liquid deuterium would be available; also, the unenriched RUNT might work, and it could be "souped up" with Li-6 for a higher yield. It would be unwise to develop, produce, and stockpile a new device before it could be tested.

Dr. Rabi was inclined to urge that the SHRIMP be expedited for "emergency capability" rather than suggest a new weapon development. Dr. von Neumann finally agreed to withhold a recommendation on the issue until after CASTLE. The GAC agreed with the AEC's Division of Military Application recommendation to include a test of the RUNT in CASTLE.⁴⁵⁶

Another subject of discussion at the GAC meeting were the present U.S. capabilities to monitor Soviet nuclear testing. A network of seismic and acoustic stations was already operational; two other stations, one in Maine and one in Washington state, used electromagnetic detection methods. The latter two sites had not been operational during the latest series of Soviet tests, but would be "proved in" (tested) during CASTLE and the next Soviet series.⁴⁵⁷ (The next Soviet series after CASTLE began in mid-September 1954.)

⁴⁵⁶Minutes of the Thirty-Seventh Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, November 4-6, 1953; letter dated November 7, 1953 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC. The meeting minutes note that the third session on November 5 was interrupted for twenty minutes by a "practice air raid alarm;" such drills were a common occurrence in major U.S. cities during the early to mid-1950s.

⁴⁵⁷Minutes of the Thirty-Seventh Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, November 4-6, 1953, p. 8.

Results of monitoring to date had pinpointed the Soviet nuclear weapons test site to within 12 miles latitudinally, and zero miles longitudinally. As a result of recent Soviet shots, specifically, Joe-4, -5, -6, and -7, the GAC feared that the Russians might have made technical advances "of a sort not known to us," especially in the development of more-energetic high explosives or in the discovery of "some new physical principle."

At this time, the AEC had no idea what the latter three Soviet shots had tested.⁴⁵⁸

In the middle of the month, Dr. Edward Teller prepared an estimate of U.S. H-bomb production capability for the next five years, based primarily on lithium-6 availability:

In preparing an estimate of potential H-bomb capacity in the next five years, the controlling factor has been taken to be the production of lithium-6. The lithium-6 capacity, used in this estimate, is based on the following assumptions:

(1) That ADP Plant No. 1, now in operation at Oak Ridge, will continue to produce at a rate of four to five lbs. (of Li-6) per day, with an isotopic concentration of 95%, until April 1954, and after April 1, 1954, production will be increased to nine to ten lbs. per day.

This increase is expected to be brought about by the use of additional electrical equipment, which is scheduled for installation on or before that date. This plant costs approximately \$40,000,000 and uses the elex process for concentrating the isotope lithium-6.

(2) That ADP Plant No. 2, which has just been authorized, will be in full-scale operation on October 1, 1955, and will produce at the rate of 55 lbs. per day of lithium-6 with an isotopic concentration of 95%. This plant will cost \$125,000,000 and will

⁴⁵⁸Minutes of the Thirty-Seventh Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, November 4-6, 1953, pp. 5, 12, 15, 33, 34, 37.

use the colex process.

(3) That ADP Plant No. 3 will be authorized in the spring of 1954 and will come into operation on July 1, 1956. This plant is assumed to cost \$150,000,000 and to have a capacity of 80 lbs. per day of lithium-6 with an isotopic concentration of 95%.

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On the basis of these assumptions, the production of lithium-6 and the weapons capacity will be as follows:

ESTIMATE OF LITHIUM 6 PRODUCTION

<u>Year</u>	<u>ADP #1 Elex (lbs)</u>	<u>ADP #2 Coxex (lbs)</u>	<u>ADP #3 (lbs)</u>	<u>Total (lbs)</u>
1953	260	0	0	260
1954	3065	0	0	3065
1955	3600	4900	0	8500
1956	3600	20000	15000	38600
1957	4000	23000	30000	57000
1958	4500	25000	35000	64500
Totals	19025	72900	80000	171925

Teller closed by noting that while it would take two or three years to make a major change in the lithium-6 production figures, an all-out research effort to develop thermonuclear weapons requiring less lithium-6, or lithium enriched to less than 95% in the Li-6 isotope, or best of all, no Li-6 at all, might result in much more rapid stockpiling of large numbers of weapons.⁴⁵⁹ Unknown to Teller, and to all the other LASL and UCRL physicists, such a program of lithium-6 production would soon be reduced significantly as a result of Operation CASTLE.

⁴⁵⁹Memorandum for File dated November 12, 1953 from Edward Teller, subject: Estimate on H-Bomb Capacity in the Next Five Years. The memo contains another deleted column to the right of the total Li-6 column shown; this column presumably contains the numbers of weapons that could be made from the amounts of Li-6 available. These estimated weapon numbers would become purely academic after CASTLE when the utility of Li-7 became apparent.

November 1953: The AEC Reports on CASTLE to Congress

At the end of November, in its second semiannual 1953 progress report to the congressional Joint Committee on Atomic Energy, the AEC reiterated the purposes of CASTLE and provided a few more details on the test devices and projected stockpile dates:

The two-fold objectives of the thermonuclear weapons program as stated (earlier) continue to be: First, achieving as quickly as possible an emergency capability to deliver early-design thermonuclear weapons, and, second, creating over a longer term improved-design thermonuclear weapons which are smaller, lighter, and more easily delivered, or possibly of higher yield, than the emergency capability prototypes.

The immediate focal point of the thermonuclear program has been the preparation for Operation CASTLE, scheduled to begin in the Pacific in March 1954, at which seven thermonuclear shots are to be fired, five at Bikini and two at Eniwetok. Three of the seven shots are of "emergency capability" weapon prototypes and four are to test new design possibilities.

The results of Operation CASTLE should furnish guidance in selecting the most effective emergency capability weapons for stockpiling, in judging the value of lithium-6 as a thermonuclear fuel, in comparing the relative advantages of "wet" and "dry" thermonuclear systems, and in evaluating other data needed to improve thermonuclear designs.

Of the five models containing lithium, the RUNT is the only one employing normal lithium exclusively (i.e., in four of the designs, the isotopic concentration of lithium-6 will be greater than normal). Energy yields of up to eight megatons TNT equivalent are predicted. Plans for quantity production and for future development of thermonuclear weapons remain flexible, and will be determined by the results of the CASTLE tests.

The thermonuclear weapons program is dependent upon the work of both Los Alamos

Scientific Laboratory and University of
California Radiation Laboratory
(Livermore); Los Alamos Scientific
Laboratory will be responsible for five
shots, and University of California
Radiation Laboratory for two.

.....

(Fusionable materials to be expended during
CASTLE include) small quantities of tritium
used for boosting the primary bomb fission
reactions involved in these devices.

... The design of each of the three
emergency capability weapon prototypes was
released for procurement during the third
quarter 1953, and components for each of
the models are now being fabricated. The
first production unit of each of two
prototypes, the TX-14 and TX-17, are
scheduled to enter stockpile in the first
and third quarters of 1954, respectively.

The stockpiling of the TX-17 depends on
CASTLE results; however, if needed, TX-17s
will begin stockpiling at the beginning of
1954.

To meet minimum interim requirements for
thermonuclear weapons, facilities completed
during the summer have been placed in
operation to produce lithium-6 and
deuterium gas. Sufficient lithium-6 has
already been produced in the new Alloy
Development Plant (ADP) at Oak Ridge to
satisfy the CASTLE requirements, and
current output is providing amounts for
limited production of weapons.

A second plant (Alpha 5) is now being
designed and equipped to produce larger
quantities of lithium-6. At Hanford,
tritium is being produced from a special
reactor loading similar to that carried out
three years ago. Substantially greater
amounts of tritium will be produced at
Savannah River, beginning in the second
half of 1954. Boron-10 will be separated in
a facility now under construction at the
Lake Ontario Ordnance Works.⁴⁶⁰

⁴⁶⁰Atomic Energy Commission, Progress Report to the Joint

Sufficient lithium-6 for the CASTLE devices was also ready by the end of the month; subsequent production was devoted to use in a limited number of "emergency capability" weapons. (Deuterium gas shipments from the AEC's Savannah River heavy-water separation plant had started in July; the gas was combined chemically with lithium at Oak Ridge.)⁴⁶¹

To build up U.S. thermonuclear "muscle" as rapidly as possible, the Joint Chiefs of Staff proposed in December new stockpile-composition requirements that allocated available production capacity mostly to high-yield thermonuclear weapons and low-yield tactical fission weapons.⁴⁶²

The Los Alamos Technical Program for CY 1954 and FY 1955

The annual Los Alamos Scientific Laboratory program for calendar year 1954 and fiscal year 1955, submitted to the AEC in early December, while concerned foremost with thermonuclear weapons, nonetheless placed continuing emphasis on fission weapons development:

...

- IV. Research on fission weapons, including their theory, the hydrodynamics of implosion, high explosive systems, nuclear components, boosting techniques, and initiator systems will be continued at essentially the present level (approximately 40% of the laboratory total effort). Outstanding among the specific studies and objectives proposed in this field are:

...

Development of a specific, low-yield
[deleted] economical [deleted]
weapon in both the MK 7 and MK 12
geometry, as air-defense and
tactical use weapons.

Committee, June 1953 through November 1953, Part III - Weapons, AEC 129/55, November 27, 1953, pp. 1-4; Atomic Energy Commission, Draft Foreword to the Progress Report to the Joint Committee, June through November 1953, AEC 129/56, December 9, 1953, pp. 1, 2.

⁴⁶¹U.S. Atomic Energy Commission, "The Atomic Energy Program," Semiannual Report to the President as of December 31, 1953, February 1, 1954.

⁴⁶²Hewlett and Holl, pp. 166, 167.

Development of a small implosion bomb system (approximately 15" overall diameter) [deleted] for air-defense purposes.

Development of optimized primary bombs for two-stage weapons in the 40,000 - 50,000 lb., the 17,000 - 20,000 lb., the 8,500 lb., and the 3,000 - 4,000 lb. classes.

...

- V. Research on two-stage or "thermonuclear" weapons will occupy an even greater fraction of the Laboratory effort than heretofore. Approximately 15% of the total laboratory effort is expected to be involved in this field, excluding the primary bomb problems.

General exploration of the problem of propagation of burning in thermonuclear fuel will continue, including, it is hoped, a final analysis of the now somewhat uninteresting problem of the classical "Super." Specific systems which will occupy the attention of the Laboratory during this period are:

The preparation of a high-yield weapon system [deleted] based on the ZOMBIE device for CASTLE having a weight of approximately 8,000 lbs., a diameter of 34.5," an anticipated yield of approximately [deleted] at an approximate cost of [deleted] not yet determined.

This development (upon which the results of the whole CASTLE test programs bear) has the highest immediate priority of two-stage weapon development in the Laboratory, and a usable weapon is expected to exist at the end of calendar year 1954.

The preparation of a high-yield weapon system based on the SHRIMP to be tested at CASTLE and having a weight on the order of 17,000 - 20,000 lbs., which the LASL is informed is close to a weight which

should not be exceeded for optimal B-47 use in this type of weapon.

This weapon, again dependent in detail upon the results of the CASTLE test program, is expected to have a yield of [deleted] at an approximate expenditure of [deleted]

[deleted] It is expected that this weapon will be in usable form before the end of this program period. This development cannot be seriously undertaken until after CASTLE, but at that time it will have a priority second only to that of [deleted] above.

The preparation of a high-yield weapon system of the cryogenic type but at a weapon weight not to exceed 20,000 lbs. will be considered as a competitor to [deleted] after CASTLE. More fundamental difficulties connected with propagation may exist in this type of a system at these dimensions, and definitive studies on this point can be attempted profitably only after the CASTLE test program. The associated cryogenic engineering, however, should be easier than that of the IVY SAUSAGE device.

After CASTLE, it is proposed to examine systems smaller than [deleted] with respect to their relative cost-weight-yield characteristics compared to conventional and heavily-boosted implosion weapon systems.

It is not unlikely that an interesting [deleted] system may exist at the 3,000 - 4,000 lb. level. However, the specific possibilities of such systems, as well as the size-weight limitations within which they must be designed for most effective delivery vehicle use, cannot be described until well after CASTLE.

After CASTLE, the [deleted] system will be examined for appropriate

future modifications. Such modifications within the general framework of the emergency capability weapons may include changes in the nuclear engineering to increase the efficiency, as well as changes in the ordnance engineering characteristics for easier assembly and handling.

More extensive modifications which would be appropriate to wholly new weapon systems (such as a 45,000 lb. [deleted] will be considered after more immediate questions relative to existing models have been considered.

The LASL program plan closed by noting that the lab expected to test five devices each during both Operation CASTLE in the Pacific in 1954 and Operation TEAPOT in Nevada in 1955.⁴⁶³ (By March 1954, plans were afoot to break TEAPOT into two separate series -- TEAPOT I and II -- and stage the former during the fall of 1954 and the latter in the spring of 1955.)⁴⁶⁴

UCRL's Plans for 1954-1955

In mid-December 1953, UCRL issued its technical development program for calendar year 1954 and FY 1955. The lab proposed work on both large thermonuclear systems and small fission systems; some specific weapons were also planned.

UCRL noted that design and fabrication of the RAMROD and MORGENSTERN devices was complete, and that final assembly and firing of these systems would occur during CASTLE. The RAMROD was mainly a test of implosion geometry while the MORGENSTERN was intended to be the prototype of a family of weapons using the same implosion system.

In the future, UCRL intended to design and test lithium-6 solid-fueled devices of less than a megaton yield, and to design a special primary specifically for use with two-stage thermonuclear weapons. Such a primary was expected to lead to a considerable weight reduction and several ideas had been suggested and would be investigated theoretically. It might be

⁴⁶³Los Alamos Scientific Laboratory Program for the Period Calendar Year 1954 and Fiscal 1955, attachment to letter dated 11 December 1953 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury.

⁴⁶⁴Operation TEAPOT, Report of the Test Manager, Joint Test Organization, Nevada Test Site, Spring 1955, p. 7.

possible to test-fire the new primary during Operation TEAPOT in 1955.

A small spherical bomb and a gun-type device might also be ready for test during TEAPOT; the lab hoped to fire two spherical implosion devices and either a gun-type device or a linear-implosion system, in addition to the new primary for thermonuclear weapons. Although test devices for TEAPOT would depend heavily on results of CASTLE and laboratory developments in the next year, it appeared in December 1953 that TEAPOT would include possibly as many as three airdrops.⁴⁶⁵

Uranium hydride (deuteride) research would continue at UCRL, with anticipation of its use alone and with metals in various geometries including six- to eleven-inch diameter gun-assembly systems, and 12 to 18 inch diameter spheres. Both critical mass measurement and HE assembly tests would be conducted on hydrides.

Two extensions of the MORGENSTERN were planned. Both were to be weapon prototypes, with ballistic cases and timing and firing systems provided by Sandia. One version would weigh about 19,000 lbs. in its air-drop configuration; further refinement of the ballistic casing and use of a special primary might bring the weight down to 15,000 lbs. (This became the BASSOON device, a prototype of what would eventually become the MK 41 bomb.) This device was tentatively scheduled for completion at about the end of 1954.

The second version would be considerably smaller, weighing between 6,000 and 7,000 lbs.; refinement of this design might reduce its weight to 3,000 lbs. (This eventually evolved into the MK/W-27 bomb and warhead.) This system was tentatively slated for completion in the middle of 1955.⁴⁶⁶

By this time, the weapons development program at Project Whitney was divided into two broad categories: the hectoton (100 tons) low-yield devices group, and the megaton (one million tons) high-yield devices group. The design ideas of the hectoton group embraced several projects, covering a very wide range of efficiencies, yields, sizes, and principles of operation. The megaton group was concerned mainly with detailed work on the CASTLE test devices.⁴⁶⁷

⁴⁶⁵Letter dated December 23, 1953 to Honorable Robert LeBaron, Chairman, MLC, from Lewis L. Strauss, Chairman, USAEC.

⁴⁶⁶Letter dated December 16, 1953 to J. J. Flaherty, Manager, San Francisco Operations Office, USAEC, from Herbert F. York and E. O. Lawrence.

The initial operational concept of Operation CASTLE had been proposed at the end of November 1952. During the intervening year between the initial and final concepts of CASTLE, numerous changes had been made in the operation's scope, shot sequence, and shot locations.⁴⁶⁸ By December 1953, CASTLE was to consist of seven thermonuclear shots, six of which would have yields in the megaton range. Major weapon assembly work would be done at Eniwetok atoll and the majority of the shots would be conducted at Bikini, the first tests at that atoll since the CROSSROADS operation in the summer of 1946.

All major RTD (refrigerated transport dewar) repair and maintenance would be done on Parry island at Eniwetok. Six of the seven devices and weapons would be assembled at Eniwetok: five on Parry island, and one on Eberiru island. At Bikini, one device would be assembled on Eninman island. Preliminary assembly, testing of experimental equipment, and shot rehearsals would be done on Parry for all LASL shots.

CASTLE test device sizes were considerably smaller than the gargantuan IVY Mike device of 1952 (the SAUSAGE). The largest CASTLE device, the RUNT, weighed 39,600 lbs. and measured 225" long by 61" in diameter. The smallest device, the ZOMBIE, weighed only 6,250 lbs. and was 110" long and 34.5" in diameter. All of the CASTLE devices could have been made aircraft-transportable, and all but two of the test devices were later weaponized.

By the end of 1953, only implosion fission weapons were being stockpiled; production of gun-assembly weapons had ended earlier when stockpile requirements were fulfilled. Although there were no programs underway for major modifications to stockpiled weapons, there were 42 formal weapon programs in progress, more than half of which were missile warheads.⁴⁶⁹

As the year ended, plans were being made for the transition from thermonuclear development to production and stockpiling. LASL and UCRL were to develop full explosive systems; Sandia was responsible for the casings and fuzing subsystems; ACF, for heavy case components; the National Bureau of Standards Cryogenics Laboratory was to research and develop means of

⁴⁶⁷Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 July 1953 - 31 December 1953, p. 246.

⁴⁶⁸OPERATION CASTLE, Report of the Manager, Santa Fe Operations, Pacific Proving Ground, Spring 1954, p. 22.

⁴⁶⁹Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, p. 19.

producing liquid hydrogen, deuterium, and nitrogen; and the Cambridge Corporation was to engineer and produce dewars.⁴⁷⁰

To Share or Not to Share: The U.S. and the U.K.

At this time, some thought was given to sharing highly restricted American nuclear weapons data with England. Although the British had been full partners with the U.S. during the World War II Manhattan Project, cooperation and sharing of data had declined considerably, although not ended completely, since the end of the war and in the aftermath of several widely-publicized and highly-damaging spy scandals in England, starting with the Fuchs affair in January 1950.

In 1951, Britain had proposed to test her first atomic bomb at Nevada; however, this proposal was later scrapped when it became clear that there was no way to allow this without divulging classified U.S. information to the U.K., and after Britain decided to set off an overwater blast to measure the effects of an atomic bomb on a harbor.⁴⁷¹

The AEC pondered means by which data might be shared without undue risk of compromise (the Atomic Energy Act forbade release of this type of data to foreign governments) and the benefits that might mutually accrue to the U.S. and the U.K. from sharing information:

Proposed amendments to the Atomic Energy Act, if enacted, would permit broader and freer cooperation and interchange of information with other nations. To assist in consideration of the relationship with the United Kingdom which would be desirable, this memorandum outlines the advantages which might be sought from

⁴⁷⁰Report of the Manager, Santa Fe Operations, U.S. Atomic Energy Commission, July 1950 to January 1954, p. 15.

⁴⁷¹Memorandum of Conversation by the Special Assistant (Arneson) to Secretary of State dated August 2, 1951, subject: Request for a Meeting of the Combined Policy Committee, reprinted in FOREIGN RELATIONS OF THE UNITED STATES 1951, Volume I, National Security Affairs & Foreign Economic Policy, U.S. Government Printing Office, Washington, D.C., 1979, pp. 748-750; letter dated August 31, 1951 to Gordon Dean, Chairman, USAEC from Chairman, JCAE; letter dated September 16, 1951 to Senator Brien McMahon from Gordon Dean, Chairman, USAEC; letter dated 19 December 1951 to Gordon Dean, Chairman, USAEC, from William C. Foster, Acting Secretary of Defense; letter dated January 8, 1952 to Brien McMahon, Chairman, JCAE, from Gordon Dean, Chairman, USAEC.

broader cooperation, the prospective advantages, the principal areas within which cooperation might be undertaken and the available indications as to the likelihood of benefit, and the kinds of cooperative activities which might be undertaken.

ADVANTAGES

Cooperation with the United Kingdom would seek the following advantages:

- a. It could strengthen the community of feeling between the two countries and thereby serve as a support to our alliance.
- b. It could increase the military readiness of the two countries to meet and repel attack.
- c. It could bring about greater efficiency within the U.K. atomic energy program, thereby conserving scarce materials and other resources.
- d. It could contribute to the progress and efficiency of (the U.S.) atomic energy program.

Decisions to seek these advantages in a general program of broader cooperation or in particular areas must take into account both the factual basis for anticipating benefits, and the risks to security which would be incurred.

The degree to which these objectives are achieved will depend, among other things, on the spirit in which cooperation is conducted, the areas affected, and the extent of the cooperative effort.

The first of the above objectives ("a" through "d") is the primary concern of the Department of State and the second the Department of Defense. The last two are the primary concern of the Atomic Energy Commission.

In considering the advantages which might be gained for the U.S. program from further

cooperation with the United Kingdom, several basic factors should be recognized:

a. In addition to direct gains to the technical and military programs of the U.S., the U.S. can gain through the strengthening of the military and technical programs of an ally with the resultant increase in combined strength.

b. The important question is whether the U.S. will derive enough benefit to make cooperation worthwhile, rather than whether the U.S. will "get more out of it" than the U.K.

c. A small amount of novel information may be more significant to the U.S. than a large amount would be to the U.K. Our program is clearly in a better position to exploit novel ideas than is that of the British. Furthermore, because of the scale of our program, any improvements in operations might lead to substantial savings.

d. Evaluation of advantages of cooperation should not be based on specific exchanges of information; from any single exchange, one party may benefit but little. Benefits from the whole program may be much clearer than the benefit resulting from separate parts of it. Cooperation in one field may, furthermore, be necessary to secure cooperation in another.

e. Identification of specific benefits, particularly in advance of an exchange, may be very difficult. As a minimum, however, gains should result from criticism and comparison of the two programs, and from a broadening of the scientific base on which the programs rest.

f. There is reason to believe that specific benefits would result from further cooperation: a number of outstanding scientists are participating in the British program, various phases of the program have been differently oriented than our own, and different handicaps have been overcome.

Furthermore, past cooperation has proved beneficial.

DISADVANTAGES OF COOPERATION

Cooperation with the U.K. would involve wider dissemination of sensitive information than at present, and would thus be attended by increased risk of compromise, to the extent that sensitive information is in the hands of a larger number of people or is afforded less adequate security protection than in the U.S.

In assessing the seriousness of risk in particular cases, a balancing factor may prove to be that, under proper security safeguards, information should leak to the USSR only after some time if at all, and the lag before information would come into the hands of the USSR and then be used by them might permit cashing in immediate and substantial gains to the U.K. and U.S. in the meanwhile.

As a matter of prudence, the principle of "need to know" should be strictly applied in considering dissemination of sensitive information to the U.K., as (well as) to additional groups within the U.S. project. Furthermore, it must be recognized that information transmitted beyond the U.S. is henceforth outside U.S. control, even while our alliance remains firm and U.K. policies and leaders remain unchanged.

As industrial applications become imminent, problems of commercial position and patent rights can be expected to arise.

The proposed amendment of Section 10 of the Atomic Energy Act will require a determination that transmission of restricted data will not adversely affect the common defense and security. Progress of the U.K. in attaining comparability of security standards and procedure, and in particular in completing the backlog of investigations of personnel employed on the U.K. project, will affect the ability of the Commission to make such a determination.

Since at the present rate it will require several years to complete clearances of presently-employed personnel, extension of cooperation may require an effort to secure the ordering of the priority of investigations to correspond as closely as possible with the areas in which cooperative activities are desired. Such activities can then be compartmentalized to include only cleared personnel.

It might also, of course, be desirable to seek intensification of the U.K. investigatory program. Acceleration of the investigatory program may be facilitated by the recently announced plans for setting up the U.K. project as a corporation, thus freeing it from the Ministry of Supply.

AREAS OF COOPERATION

Exploration of the various parts of the respective (weapons) programs will be necessary to ascertain proper areas of cooperation. However, a brief staff review indicates that throughout the scope of the atomic energy activity there are areas where cooperation or exchange of information could prove fruitful. Available examples of possibly beneficial cooperation are included in the following list:

a. Background Scientific Information

Includes instrumentation; biological and medical research on radiation effects; research with radioactive isotopes; nuclear physics and chemistry; related research in physics, chemistry, and metallurgy.

Most basic scientific work is now unclassified. Of what remains classified, some topics (e.g., basic metallurgy of plutonium, some nuclear cross-sections) are at present excluded from the Technical Cooperation Program. Cooperation in these fields would be mutually advantageous, and might be achieved by opening up Area Four of the Technical Cooperation Program, on "Nuclear and Extra-Nuclear Properties of

the Elements."

There would also be advantage in a freer cooperation which would release basic scientific work from the irritating and arbitrary rules of the present program.

b. Production Technology

Information in this area is generally excluded from the present cooperation programs, and offers considerable promise of benefit. The U.K. has apparently successfully built and operated (uranium) ore processing facilities, production reactors, separation facilities, diffusion plants, and weapons manufacturing facilities, and is in the process of building two power reactors. It would therefore bring a background of experience to any interchange, despite the smaller scale of their program.

(1) Weapons design and Production. One route to achieving greater yield is improvement of chemical explosives. Success in this direction could effect gains both in the reduction of size of large fission weapons and in the possibility of making smaller fission weapons of simple design and greater economy of fissionable materials.

It is known, for example, that the British are expert in the field of chemical explosives, and may have made significant advances in this field. The GAC has recently recommended exploration of this possible area of cooperation.

(2) Technology of production of feed and fissionable materials. Further exchange of information on the fundamental metallurgical properties of uranium, plutonium, and their alloys would be of considerable benefit to the U.S. production program, since the U.K. has undertaken extensive work in this

field.

Much of their work has been of a fundamental nature, while the work in AEC production facilities has been confined to the development of specific processes; consequently, a free exchange of information would be of mutual value. An exchange of information on methods of preparing and fabricating uranium also would be useful, including methods of reduction and powder metallurgy.

The U.K. scientists have also conducted an extensive program in basic chemical engineering as related to solvent extraction processes and have developed specialized extraction equipment.

The British made many contributions to gaseous diffusion technology during the war years and have been proceeding with an independent development since that time, directed toward the construction and operation of a gaseous diffusion plant in the U.K.

Since the U.S. has a large capital investment in gaseous diffusion plants, and even minor improvements in operation result in large annual savings, an exchange of information in this area would be desirable.

Studies conducted in the U.K. of stored energy in graphite and radiation damage to graphite (Author's note: a reference to "Wigner's disease") are of considerable interest to us because of the large graphite reactors at Hanford.

Some cooperation in this field has been initiated, resulting in the recent controlled release of stored energy in the Brookhaven reactor. Continued cooperation in this area, including an exchange of experiences with production-scale reactors, would be mutually advantageous. Cooperation

in the technology of heavy-water reactors is also of interest to us.

....

c. Rate and Scale

(Includes number of weapons in stockpile, present and future rates of production of weapons and fissionable materials, capacity of plants, explosive equivalent of individual weapons and total stockpile of those weapons, and rate of ore procurement.)

Rate and scale information may on occasion be usefully exchanged for specified purposes. Information on rate of ore procurement, for example, is now regularly exchanged as a basis for planning and allocations.

Information concerning availability of weapons, yields, etc. has importance for the tactics and strategy of combined operations. In other cases, information on plant capacities or on stockpile (size) may be inferred from information made available for other purposes, such as combined intelligence activities.

Useful exchanges on production processes, plant, or equipment should not be barred because information on plant capacities or output could be incidentally inferred.

d. Information on Utilization and Effects of Weapons

Operational information about atomic weapons (such as information on size, weight and shape, methods of delivery; fuzing and nuclear safing; storage and maintenance; and yields) and weapons effects data are of importance in maximizing the capabilities of the countries for active and passive defense against atomic attack.

Considerable pre-war coordination is necessary to achieve these goals. Since the British have to a considerable extent been more oriented toward defense

than the U.S., they may have developed novel ideas and approaches which would be valuable.

At the present time there are severe limitations on the communication of operational information. It is not yet clear to what extent the need for communicating weapons effects data can be satisfied under the Technical Cooperation Program.

While these matters are substantively of primary concern to such agencies as the Department of Defense and the Federal Civil Defense Administration, the Commission should be aware of advantages to be gained in these areas, since restricted data would be involved in the exchanges.

It does not appear that there are any major areas in the atomic energy field which should a priori be excluded from exploration for cooperation or exchange of information in some degree. For information of the highest degree of sensitivity -- that relating to advanced technical accomplishments, production and military capabilities, deployment of stockpiles and forces, and military and AEC planning -- the justification for cooperation and exchange of information will need to be specific and assured, and the measures to minimize risk (of compromise) most thorough; nevertheless, it is in those areas that the importance and direct advantages can on occasion be greatest.

The risk incurred will depend in large part on the value of the information to a rival, and should be assessed with due weight to the independent progress of rival projects.

KINDS OF COOPERATIVE ACTIVITIES

The extent of cooperation undertaken can be defined in terms of the kind of activities involved, as well as the areas of information. These activities may vary within wide limits. The present technological cooperation program places off-limits information directly and

primarily related to weapons and to plants for the production of weapon parts and materials.

The remaining areas are further circumscribed by confining them largely to research and development information. Cooperative activities are in the form of limited exchanges of information or services for specified purposes.

An expanded program of cooperation with the United Kingdom might involve merely the addition of new areas and topics within the limits of the existing cooperation program, or it might involve relaxation of present limitations to permit to some extent the inclusion of presently off-limits areas and the undertaking of joint activities.

The range of cooperative activities that could be undertaken is indicated in the following list; the degree of cooperation might vary from area to area:

- a. No cooperation.
- b. Cooperation limited to specific exchanges of visits or documents on an ad hoc basis.
- c. Exchanges of papers, visits, services, materials, and equipment within specified areas.
- d. Joint planning of work. (This would involve combined planning of efforts and division of labor, but separate execution of agreed responsibilities.)
- e. Joint conduct of some research, development, testing, and production activities. (This would include combined participation of personnel of both countries in specified projects as well as coordinated planning of the work.)
- f. Integration of the two (U.S and U.K.) programs.

OTHER CONSIDERATIONS

Extension of cooperation with the U.K.

would require consideration of the interest of other nations in the British Commonwealth, particularly Canada, and of other allies and ore-supplying nations. General arrangements between the U.S. and U.K., supplementing or replacing the present modus vivendi would be desirable if there is significant change in the scope or character of cooperation. Changes in present administrative arrangements would also have to be considered.⁴⁷²

A month later, the sharing of very specific weapons-related information was considered:

1. Can information on size, shape, weight, center of gravity, moments of inertia, and data for loading, carrying, and releasing stockpile fission weapons be transmitted to the United Kingdom?

2. Three ways to be explored:

- (a) declassification

- (b) Modus Vivendi

- (c) Legislation.

3. Declassification:

AEC in 1950 agreed to declassify size, shape, weight, center of gravity and moments of inertia for fission weapons only (and revealing nothing on internal characteristics including fuzing and firing mechanisms) in order that non-Q cleared personnel in contractor's plants could be used (so) production schedules might be met.

...

4. If declassified today the reasons for declassification would be different and transmission of detailed (unclassified) blueprints would be required. Certain details on internal characteristics of weapons would be revealed or would be quickly inferred. The reasons for

⁴⁷²Atomic Energy Commission, Considerations Relating to Cooperation with the United Kingdom, Memorandum by Policy Analysis Staff, attachment to AEC 190/101, November 25, 1953.

declassification would be to supply certain weapons characteristics to foreign nationals.

5. Transmission of blueprints would reveal (directly, or by inference) for stockpile fission weapons or fission weapons in development for early delivery to stockpile, the following:

- (a) That we have implosion and gun-type weapons, some with special purposes, such as BETTY and AFAP.
- (b) Our range of weapons capabilities for SAC, TAC, and special purposes.
- (c) By inference, the kinds of planes we use, what ranges, missions, speeds, and the type of (bomb) mountings (internal or external).
- (d) Sufficient ballistic characteristics to be used with a United Kingdom bombsight.
- (e) Loading mechanisms, carrying details, and release mechanisms.
- (f) Sufficient data on fuzing and firing mechanisms to make settings for detonation by delay, contact, time, and radar. Also implications of baro(metric) devices. Included will be wiring, circuitry in aircraft for checking out the weapon in flight, prior to release.
- (g) In-flight insertion systems, manual and automatic.
- (h) Nuclear safing system of the MK 18 and therefore strong implications of nuclear characteristics of weapons.
- (i) Characteristics of latest shapes of both planes and bombs, i.e., F-101 system.
- (j) Implications of tactics for toss and lob (bombing) systems because special fuzing and release mechanisms will be required.

- (k) Certain revelations of nuclear characteristics through check-out system for external initiators.

Most of the foregoing are specific weapons design and fabrication details of varying degrees of sensitivity. Thus, size, shape, weight, and moments of inertia is not the sole issue in transmitting detailed blueprints to the United Kingdom.

Declassification requiring blueprint transmission under these circumstances might open the Executive (branch of the U.S. government) to serious charges.

6. Exchanges under Existing Law

There are permissible exchanges of Restricted Data under present law, as in recent addition to the Modus on weapons effects. The value of any permissible additional exchange would probably be negligible because it would have to exclude all information on design and fabrication of atomic weapons.

7. Legislation

A mode to accomplish is included in the Commission's current legislation.

Comment:

Of three possibilities, there seems, if time is of the essence, one method - declassification, by which the Commission can effect transmission of these data. It involves a serious risk.

Declassification and transmission of data to the United Kingdom would mean that we have no security objections to publication of this information.

Care, of course, could be taken to see that the information is not published openly.

The safe way is to wait for favorable action on our legislative proposals.⁴⁷³

Not until the summer of 1958 would the U.S. have a regular exchange of nuclear weapons data with England; when it finally happened, the scope and extent of the data would be such that the British nuclear weapons program would be taken over almost completely by U.S. designs and weapons.⁴⁷⁴

January 1954: The AEC Publicly Announces CASTLE

On January 8, 1954, the AEC publicly announced plans for another series of "weapons tests of all categories" at the Pacific Proving Grounds.⁴⁷⁵ No date was given, although the Commission did state that men and materiel were soon to be enroute to the Marshalls:

During this month, men and materiel will begin moving to the Pacific Proving Grounds of the Atomic Energy Commission to carry out a further phase of a continuing series of weapons tests of all categories. These tests are part of the Atomic Energy Commission research program for the improvement of weapons, pursuant to its responsibility under the AEC Act. These tests will be conducted by Joint Task Force Seven (JTF 7)... There will be no observers other than U.S. officials.

"Leaks" about the new test series had appeared as early as the first week in November 1953; one United Press (UP) wire service account "rather accurately describe(d) the activities in the next spring," including what was then "highly classified information on the nature of the tests."⁴⁷⁶

⁴⁷³Draft memo by McKay Donkin dated December 17, 1953. BETTY was the MK 7 depth bomb; AFAP refers to nuclear artillery projectiles. TAC was the USAF's Tactical Air Command. The "F-101" system referred to may have been the W-5/MK 96 external shape carried by the F-101A (see MK 5 bomb history in Volume VI).

⁴⁷⁴Hewlett and Holl, pp. 537, 538.

⁴⁷⁵On January 26, AEC Chairman Lewis Strauss wrote to President Eisenhower and invited him to visit the Pacific to witness a test; the President, in a hand-written note appended to a copy of the AEC invitation, replied: "Don't believe I can do it. I'd like to talk to you about these sometime."

⁴⁷⁶Letter dated November 6, 1953 to Dr. James O. Beckerley, Director, Office of Classification, USAEC, from Ralph Carlisle Smith, Assistant Director of Classification, LASL.

A week before the January AEC announcement, an "informed source" told UP that "the Armed Forces will test a plane-dropped hydrogen bomb at the Eniwetok-Bikini proving grounds;" that "the weapon may be less powerful than that (device) tested in 1952" and that "it is so large that it must be fitted into the B-36 with a shoehorn."⁴⁷⁷ While there was no air-drop planned for CASTLE, as might have been inferred from this statement, it was nonetheless a relatively accurate characterization of the TX-16 JUGHEAD.

Early in January, AEC Chairman Lewis Strauss wrote to President Eisenhower and requested final approval for CASTLE and permission to expend the required fissionable and fusionable materials, including oralloy, tuballoy, plutonium, tritium, lithium-6, and deuterium:

This letter is written to obtain approval for the full-scale atomic test program planned to begin on March 1, 1954 and scheduled to extend through April 1954 at the Pacific Proving Grounds. CASTLE, the code name assigned to the operation, has been planned in coordination with the Department of Defense ... It is planned to detonate not more than seven shots in the CASTLE series.

The short term aspects of the series, tests of devices named ALARM CLOCK, JUGHEAD, and RUNT, are directed toward determining the content of the emergency capability program. Tests of devices named ZOMBIE, RAMROD, and MORGENSTERN concern longer-term aspects of the thermonuclear program and should provide answers to problems indicating the direction of development in the next generation of thermonuclear weapons.

The firing of the RUNT device scheduled as the sixth shot is contingent upon

Smith was extremely bitter in this letter, describing what he perceived as a double standard of prosecution concerning minor security violations in the field versus major "leaks" to the press in Washington, D.C. A double standard of enforcement of classification regulations has been a hallmark of the U.S. atomic energy program since its inception. The Progressive case in 1979 was probably one of the best examples of this abuse.

⁴⁷⁷BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 10, 1954, p. 61.

evaluation of diagnostic information obtained from detonating the SHRIMP, scheduled as the first shot in the series.⁴⁷⁸

At the end of the month, the AEC informed President Eisenhower:

Of the seven thermonuclear shots to be fired (during CASTLE), three will be prototypes of deliverable weapons and four will be thermonuclear devices to test new design possibilities. The results of CASTLE should furnish guidance in selecting the most efficient emergency capability weapons for stockpiling, in judging the value of lithium-6 as a thermonuclear fuel, and in evaluating other data needed to improve thermonuclear designs.

Components of two of the prototypes to be tested are now in emergency production, and these two models (Author's note: the TX-14 and -16) are scheduled to enter stockpile on a "hand built" basis in February 1954. The third (Author's note: the TX-17) may be stockpiled beginning in the summer of 1954, depending on CASTLE results. Production schedules now being formulated for the period of the next two or three years are expected to emphasize the fabrication of thermonuclear weapons.⁴⁷⁹

CASTLE Test Operations

As was the case in earlier Pacific island test series, the logistics for CASTLE were staggering. Among the thousands of tons of materiel flown or shipped to Eniwetok were many tons of radioactive and hydrogenous substances, including polonium, curium, tritium, uranium-235, uranium-238, plutonium, and lithium-6 deuteride and lithium-7 hydride.

Several tons of weapon components, including one device measuring 137" x 67" x 89," weighing four tons, were flown by four U.S. Air Force special air missions aboard military

⁴⁷⁸Letter dated January 11, 1954 to The President, The White House, from Lewis L. Strauss, Chairman, USAEC.

⁴⁷⁹U.S. Atomic Energy Commission, "The Atomic Energy Program," Semiannual Report to the President as of December 31, 1953, February 1, 1954.

transport aircraft from California and New Mexico to the Pacific. Sealed pressurized containers of lithium hydride were kept under constant hourly inspection during these flights.⁴⁸⁰

Other special nuclear materials were transported by MSTs (Military Sea Transportation Service) beginning in November 1953 when the USNS PVT. JOSEPH E. MANN (TAK-253) sailed from Oakland Army Depot in California with 27 steel containers of lithium hydride aboard.

Also kept under constant surveillance was the USS CURTISS. Carrying highly classified AEC test materials, including weapons components, she sailed under full wartime conditions from the naval weapons depot at Port Chicago, California (now the Concord Naval Weapons Station), on January 10, 1954. She was escorted by a flotilla of destroyers and watched over by land-based aircraft when she was within 500 miles of the California coast, Hawaii, and Eniwetok. While the Pacific tests series GREENHOUSE and IVY were conducted during the Korean War, wartime security precautions were maintained even after the Korean "police action."

Soviet Eavesdropping on the Pacific Proving Grounds

There was little doubt in the 1950s that Soviet surface ships and submarines observed U.S. nuclear tests in the Pacific; the two entrances to Eniwetok lagoon in the deep entrance between Japtan and Parry islands and in the wide passage between Eniwetok and Igurin islands were seeded with an array of 17 acoustic hydrophones, spaced between 250 and 350 yards apart, installed before Operation IVY in 1952 and monitored by an underwater detection detachment listening around the clock for intruding submarines and surface craft.⁴⁸¹

⁴⁸⁰"Movement of CASTLE Test Materials to the Pacific Proving Grounds," memo dated 4 December 1953 from Major General P. W. Clarkson, U.S. Army, to Commander, Military Air Transport Service, Andrews Air Force Base, Washington, D.C. The test device described in this memo was probably part of the RAMROD device: all of the test devices, with the exception of ZOMBIE, RUNT II, and part of the RAMROD, were shipped to Eniwetok by surface vessel in the first week of January 1954. Part of the RAMROD device and the entire ZOMBIE device were flown out to the Pacific in early March (this December 1953 memo lists the scheduled arrival date at Eniwetok for the "device" as 8 March 1954).

⁴⁸¹Annex J to Joint Task Force 7, Commander Task Group 7.3, Operation Plan 1-53, 31 December 1953, p. J-I-1; Moore and Bechanan, p. 174.

During GREENHOUSE, the IVY equipment was preceded by an array of ten sonobuoys, at intervals of 1,200 yards. Two buoys were laid in the deep entrance and eight in the wide passage; they were monitored remotely by radio from Eniwetok island. The usefulness of these buoys left much to be desired: small craft were detectable only within ranges of 300 to 400 yards, so it was possible that boats could have slipped into the lagoon by passing through the 400 to 600 foot "silent zone" between buoys.

In addition, during GREENHOUSE, IVY, and CASTLE, had a submarine been detected trying to enter the lagoon, patrolling destroyers were usually so far away as to be useless for defense or attack.⁴⁸²

In addition to the sonobuoy detection system, both Navy surface vessels and long-range patrol aircraft watched the seas around the PPG during test periods; however, these resources were usually meager in number and stretched very thin over large empty ocean areas. On several occasions, small Japanese fishing boats were overlooked.

These precautions were typical of the security measures taken during U.S. nuclear testing. The area within several miles around nuclear test devices in Nevada and the PPG was deserted for the last few hours before a shot and it was theoretically possible, although highly unlikely and extremely difficult and dangerous, for a group of commandos or saboteurs to slip in unnoticed and disarm, damage, or steal the device. Any tampering with firing circuits would have been immediately apparent unless elaborate steps were taken to maintain the appearance of electronic continuity.

For this reason, elevators on shot towers were disassembled and removed after a device was armed, which meant that disarming teams had to climb a ladder on the outside of the tower, making emergency disarming very difficult.⁴⁸³ Shot cabs on the ground, towers, or aboard barges were usually locked

⁴⁸²Report dated 16 June 1951 to Commander Joint Task Force Three, from Commander Task Group 3.3, Subject: Report of Participation in Operation GREENHOUSE, pp. A-1, A-2, A-3.; Joint Task Force Seven, Commander Task Group 7.3, History of Operation CASTLE, Installment Number 4, Period of 8 April through 15 May 1954, p. 19.

⁴⁸³"Background Information on Nevada Nuclear Tests," Office of Test Information, Nevada Test Organization, Nevada Test Site, Mercury, Nevada, May 1, 1957, p. 33. Bernard O'Keefe, in NUCLEAR HOSTAGES, pp. 152-156, describes a particularly harrowing disarming experience during Operation UPSHOT-KNOTHOLE in Nevada in 1953.

and sealed after the last member of the arming party had departed.

Contingency plans were made to repulse any type of armed or surreptitious entry during test operations, all the way up to a full-scale war. During predawn tower shots in Nevada and the Pacific, the tower was illuminated by floodlights at its base and kept under observation until a short time before zero hour. Any intruders would be seen as they scaled the tower legs.

The remoteness of the Marshalls made them especially exposed. As early as Operation SANDSTONE in 1948, the AEC realized that both Bikini and Eniwetok were vulnerable to a sea-launched attack:

In a recent letter to the Commission, you evidenced your interest in exploring the hazards of "conventional" attack. In this connection it might occur to you to inquire whether the Joint Chiefs of Staff have considered the remote possibility that the Russians might seize upon the opportunity presented by the forthcoming tests at Eniwetok to deliver an attack there.

This would be dictated (with) the obvious aim of neutralizing our atom bomb superiority for a considerable period of time by canceling out a large number of our specialists, most particularly assembly groups if, for example, the presumed enemy has any information as to the critical importance of those technically-equipped individuals. They will be concentrated in a very limited area, and their presence in the area at a given time will be fairly convincingly indicated by the volume of radio traffic.⁴⁸⁴

These concerns were well founded: during SANDSTONE, a total of 11 submarine contacts were reported within a restricted area around Eniwetok. One was categorized as an actual submarine, two were probable contacts, seven were "doubtful," and one was a U.S. sub. The Navy asserted that it would first fire on any submerged or surfaced sub, before attempting to identify it as friendly or hostile. The Navy concluded that "it (was) certain that submarines did (approach Eniwetok)

⁴⁸⁴Memorandum to Secretary of Defense dated 17 March 1948 from AEC Commissioner Lewis L. Strauss. "Canceling out" is a cute euphemism for "killing."

during SANDSTONE".⁴⁸⁵

Partly for this reason, the PPG was protected during tests by air, ground, and naval forces; between test series, a permanent garrison force remained at Eniwetok on what must have been one of the most isolated duty stations in the world. During Operation SANDSTONE, Eniwetok was protected by an armed contingent equipped with small arms, including M-1 rifles, carbines, and .30-caliber machine guns.⁴⁸⁶ After 1952, Eniwetok was permanently garrisoned, but Bikini was usually totally abandoned and unguarded.

Even more extreme measures were taken to prevent Russian intelligence gathering: after every Pacific test series, temporary structures and towers were torn down; remnants of tower footings dynamited; radioactive craters bulldozed and filled or flooded with seawater; and all useful evidence destroyed to avert possible sample-gathering by unfriendly forces.⁴⁸⁷ Some of these security measures also reduced radiological hazards to personnel during subsequent test series.

The AEC believed these precautions to be wise: after Operation SANDSTONE, one paper concluded that by on-site study of Eniwetok and its environs within a month of the last shot in May 1948, skilled nuclear scientists from a foreign country might be able to deduce or calculate the following restricted data:

- The efficiency of each of the three test devices, accurate to a factor of two.
- That sufficient plutonium was present to allow collection

⁴⁸⁵"Operation SANDSTONE, Nuclear Explosions, 1948/Atomic Weapon Tests, Operation SANDSTONE, 1948, Volume I, Report to the Joint Chiefs of Staff," Headquarters Joint Task Force Seven, San Francisco, California, June 1948, pp. 121-123.

⁴⁸⁶OPERATION SANDSTONE, NUCLEAR EXPLOSIONS 1948, Atomic Weapons Tests, Operation SANDSTONE, 1948, Annex I Part II, Volume I, Report to the Joint Chiefs of Staff, Section VIb, Chapter XIII, p. 79.

⁴⁸⁷"Information from Post-Shot Study of Test Sites," H. Mayer and F. Reines, Appendix D to SANDSTONE Report, Vol. III, Scientific Director's Report of Atomic Weapon Tests at Eniwetok, 1948, Sandstone Handbook of Nuclear Explosions, Frederick Reines, ed. 1 August 1949, pp. 307-326; OPERATION SANDSTONE, NUCLEAR EXPLOSIONS 1948, Atomic Weapons Tests, Operation SANDSTONE, 1948, Annex I Part II, Volume I, Report to the Joint Chiefs of Staff, Section IX, p. 119.

of samples that would permit study of its chemistry and nuclear physics.

- That evidence could be found indicating that one bomb contained plutonium and enriched uranium while the other two contained only uranium.
- That the presence of a large amount of polonium would probably suggest a possible initiator.
- That the test devices were not gun-assembly type weapons.⁴⁸⁸

Landing parties or intelligence-gathering could be easily supported by submarines. At the time of CASTLE, U.S. naval intelligence estimated that the USSR had 19 ocean patrol and 37 medium-range type submarines in the Pacific area (the same numbers had been used for GREENHOUSE security planning three years earlier).

Of these vessels, the U.S. believed that at least four to six ocean patrol submarines could be maintained in the Eniwetok-Bikini operational area indefinitely and that this number could be increased for short periods.⁴⁸⁹ Unobserved submarine approach within one mile of major islands of the two atolls was possible; surreptitious approaches could be made within 1,500 yards of other islands of the atolls.⁴⁹⁰

⁴⁸⁸"Restricted Data Obtainable from a Post-Shot Examination of Eniwetok Atoll," undated, but probably ca. mid-1948.

⁴⁸⁹Draft Operation Order No. 1-53, Headquarters, Joint Task Force Seven, Washington, D.C., 1 May 1953, p. C-I-1. The Soviet submarine fleet in the Pacific was sizable: by 1956, there were 102 boats in the area of the Sea of Japan, based primarily at Vladivostok; this number had increased from 92 in 1951. (TRENDS AND FORCE LEVELS AND DISPOSITION OF MAJOR NAVIES SINCE WORLD WAR II, Floyd D. Kennedy, Jr., et. al., CNR 145, Center for Naval Analyses, Alexandria, Virginia, June 1989, pp. E-2, E-3.) GREENHOUSE security plans estimated that there were 81 Soviet subs in the Pacific fleet in the spring of 1951.

⁴⁹⁰Operation Order CJTF 132 No. 1-51, Headquarters, Joint Task Force 132, Washington, D.C., 26 November 1951, p. B-1; Operation Order No. 1-53, Headquarters, Joint Task Force Seven, 20 June 1953, p. C-1. The distance between the huge Soviet Pacific submarine base at Vladivostok and the Marshall Islands is on the order of 4,200 statute miles, or an 8,400-mile round trip, well within the 10,000-mile range of the diesel-electric-powered "Zulu" class subs built between 1952 and 1957. (SOVIET SUBMARINES: DESIGN, DEVELOPMENT, AND TACTICS, Jan Breemer, Jane's Information Group, Coulsdon,

Landing parties from Soviet submarines might, if they wished, either sabotage communications and technical installations on the atolls, or plant instrumentation and sample collection equipment of their own that could be retrieved later. Enemy forces might also, by means of electronic countermeasures emanating from ships, planes, or submarines, interfere with task force communications and signals.⁴⁹¹

Suspicious submarines were a feature of American Pacific Proving Ground nuclear tests starting with the second series, Operation SANDSTONE, in 1948. As early as February 20, 1948, two months before the first shot, the destroyer escort USS RABY (DE-698) made sonar contact with a submarine about 26 miles from Eniwetok. In company with the destroyer USS PERKINS (DD-877), the RABY tracked the contact for about six hours before losing it. The incident was categorized as a probable submarine at deep submergence; the submarine was traveling at an estimated seven knots.

Just four days later, the RABY and her sister destroyer escort USS SPANGLER (DE-696) picked up another possible submarine contact 17 miles from Eniwetok and tracked it for more than 13 hours.

On March 3, a Navy PBM patrol bomber visually sighted a submarine 20 miles from the atoll; when spotted, the sub was running at about five knots with just its snorkel above the surface of the sea. The outline of the sub's hull was clearly visible when the plane passed just 200 feet above the sea. The submarine dove deeply when it realized it had been seen.⁴⁹²

During Operation GREENHOUSE in 1951, a number of unidentified subsurface contacts were made by patrolling U.S. Navy ships and aircraft. With only two destroyers available at that time (the Korean War made high demands on naval warships), the surface patrol could provide little more than token security and this only in the immediate vicinity of the island in the northern part of Eniwetok atoll on which the shots were to be detonated (tall shot towers would have been easily visible through a submarine periscope).

To make best use of the available patrol ships and aircraft,

Surrey, England, 1989, p. 87.)

⁴⁹¹Draft Operation Order No. 1-53, Headquarters, Joint Task Force Seven, Washington, D.C., 1 May 1953, p. C-4.

⁴⁹²OPERATION SANDSTONE, NUCLEAR EXPLOSIONS 1948, Atomic Weapons Tests, Operation SANDSTONE, 1948, Annex I Part II, Volume I, Report to the Joint Chiefs of Staff, Section VII, pp. 23, 24, 58-64.

searches were designed to give "practical" coverage during periods between shots, and "saturation coverage" during critical periods between H-48 hours and H+24 hours, that is, two days before and one day after each shot. Between shots, most long-range patrols by radar-equipped P2Vs were flown at night when there was the greatest probability of catching a submarine running on the surface or recharging its batteries.⁴⁹³

In addition, only nine P2V-type long-range patrol aircraft (four P2V-3Ws and five P2V-2s) were available in 1951 to cover 30,000 square miles of ocean. In one GREENHOUSE report, the Navy noted that "had unfriendly submarines made any determined effort to observe the operation, the patrol forces, both air and surface, would have been inadequate ... Fortunately, no such act was made and there is no evidence that unfriendly submarines were in the area at any time. Although several contacts were reported, final evaluation in every case was non-submarine."

These contacts included "disappearing radar contacts," all of which were classified as weather phenomena. Other contacts were classified as sightings that were finally evaluated as "other than submarines." However, on April 12, a surfaced submarine was spotted and reported by a MATS aircraft; a patrol airplane sent to investigate reported a "disappearing radar contact."

A second patrol plane later dropped sonobuoys (underwater listening devices connected to floating surface radio transmitters) after sighting an oil slick; the sonobuoys picked up the sound of cavitations (propeller disturbances) at a rate of 100 RPM on an easterly course southeast of Bikini atoll. The contact was evaluated as a probable unidentified submarine and the area was kept under surveillance for 12 more hours with no further contact.⁴⁹⁴

At the time of GREENHOUSE, the latest U.S. intelligence estimates indicated that the USSR had 77 submarines of various type in the Pacific, and that 10 other long-range subs for this fleet were under construction. The deployed boats were based at Port Arthur, Sovetskaya Gavan, Vladivostok, and Petropavlovsk.

⁴⁹³Joint Task Force Three, AG File, ca. May 8, 1952, pp. A-3, A-4.

⁴⁹⁴Report dated 16 June 1951 to Commander Joint Task Force Three, from Commander Task Group 3.3, Subject: Report of Participation in Operation GREENHOUSE, pp. 7, A-9 - A-13; Joint Task Force Three, AG File, ca. May 8, 1952, p. A-6.

Of the 77 Soviet subs then in the Pacific Fleet, 12 were long-range ocean patrolling type; 37 were medium range types; and the remaining 28 were limited to coastal patrols. There were no high-speed Type XII (a wartime German design) boats in the Pacific. The Navy believed that at least 42 ocean patrol and medium range subs were capable of cruising to and patrolling near Eniwetok atoll, which was about 2,500 miles from the nearest sub base.

The Navy thought that the Soviets could maintain a continuous patrol off Eniwetok to observe GREENHOUSE. By using 10 long-range subs, a continuous patrol of two boats could be maintained offshore. With maximum effort, the Soviets might be able to increase this to a total of six boats.

With direct and photographic observation, the Soviets might be able to learn the time of all shots and make fairly accurate yield estimates, based on the rate of fireball expansion and changes in its state as it expanded. Relatively simple cameras, with shutter speeds as low as 100 frames per second, could be used to gather this data, presumably through periscopes.

These cameras might be triggered by photocells or by monitoring U.S. time signals just before the shot. Since the islands of Eniwetok were almost flat with no natural elevation over 13 feet, and since most were also no wider than three-quarters of a mile, photography of task force activities would not be difficult.

Landing parties might be able to closely examine and photograph blast damage to large structures, take soil samples, and make radiographic and radiological measurements.

The Navy concluded that in order to frustrate Soviet intelligence-gathering efforts, surface vessels would have to be kept at least 100 miles away, and submarines would have to be kept clear of the atoll and forced to submerge below periscope depth.⁴⁹⁵

These counter-submarine efforts, however, could be obviated by regular U.S. patrol patterns and schedules; in addition, there was practically no way to keep submerged submarines away from shot sites as zero hour approached, since all aircraft and surface vessels were severely restricted in how closely they could approach these sites until after the test devices were detonated.

⁴⁹⁵Operation Plan CTG 3.3 No. 1-51, Appendix II to Annex E, The Soviet Submarine, RAdm. R. H. Cruzen, Commander Task Group 3.3, Joint Task Force Three, Task Group 3.3, Pearl Harbor, Territory of Hawaii, 31 January 1951.

Prowling unidentified submarines were also detected during Operation IVY. On October 1, 1952, the destroyer escort USS O'BANNON (DDE-540) reported a sonar contact approximately four miles north of Eniwetok atoll. Although contact was sharp, no Doppler or hydrophone sounds were detected. Contact was lost after about seven minutes.

The next day, a P2V patrol plane reported a radar submarine contact approximately seven miles northeast of Parry island, outside Eniwetok atoll. Sonobuoys were dropped and machinery noises were detected. Two destroyer escorts and another P2V were dispatched to the contact area but a further combined air and sea search turned up nothing.

On November 9, about a week after the Mike shot, an Air Force B-29 reported, within an hour and a quarter, three separate sightings of a submerged submarine proceeding in a northerly direction at four to six knots between 11 and 14 miles north of Parry island. A P2V, an F4U fighter, and two destroyer escorts were dispatched, and sonobuoys were dropped at the site of the contact. Although the search was continued for four hours, no further contact was made.⁴⁹⁶

During the late evening hours of the same day, the USS RENDOVA (CVE-114), while at anchor in Eniwetok lagoon, made a brief radar surface contact just off Eniwetok island. The contact disappeared but a search plane sent to investigate dropped sonobuoys and received brief contacts of unidentified noises on two buoys dropped in the first pattern. A pair of destroyer escorts, assisted by a patrol plane, continued the search with no further results.⁴⁹⁷

The AEC also noted that the USSR could conceivably conduct airborne photo and radiological reconnaissance of the Bikini-Eniwetok area from Far East bases, specifically, the Sakhalin or Kurile islands. With prior knowledge of exact CASTLE shot dates and times, reconnaissance flights could be properly timed to permit aircraft to remain in the target area for brief periods.

To the extent that CASTLE shot dates and times eventually

⁴⁹⁶Moore and Bechanan, pp. 177, 178, 246; Commander Task Group 132.3, History of Operation IVY, prepared by LCdr. D. K. De Jarnatt, USN, Staff Historian, undated (but subsequent to November 1952), p. 128.

⁴⁹⁷Commander Task Group 132.3, History of Operation IVY, prepared by LCdr. D. K. De Jarnatt, USN, Staff Historian, undated (but subsequent to November 1952), pp. 128, 129.

slipped drastically from all previously-announced schedules, these changes would have obviated the threat of any on-site Soviet air sampling or photography; even had they known the original shot schedule, the Russians would have been baffled by the unpredictable and seemingly capricious shot schedule.

Only ships and submarines that could remain in the Bikini-Eniwetok-Kwajalein area for long periods would have been useful, and their usefulness would have depended largely upon their ability to correlate the radio traffic volume with impending shots, or to actually eavesdrop on the shot countdowns.

The Soviet Tu-4, a "Chinese copy" of the wartime U.S. Boeing B-29, was the only airplane that might be able to overfly the PPG; the bomber would have to be equipped with bomb-bay fuel tanks to give it greater range. Maximum unrefueled combat-load radius of a standard Tu-4 was estimated at this time at only about 2,000 statute miles, about 400 miles short of the aerial distance from the Sakhalin Island to Bikini and about 350 miles short of the distance from Vladivostok.⁴⁹⁸

With a single in-flight refueling,⁴⁹⁹ the Tu-4 might extend

⁴⁹⁸Moore and Bechanan, p. 167. Soviet sources later stated that an Tu-4 had a 3,170 statute mile range with a six-ton bombload. (Zaloga, pp. 84, 263.) If this is true, then a Tu-4 carrying extra fuel and reconnaissance equipment and operators would have been even harder-pressed to reach the Marshalls and return safely to Soviet territory. Zaloga also states that by 1953, the CIA was estimating Tu-4 with a combat radius of 1,700 miles and a combat range of 3,100 miles with a 10,000 lb. bomb load. The aircraft was estimated to take 16 hours to fly 3,100 miles. (Zaloga, p. 263.) In this context, "range" is the distance that an aircraft can fly unrefueled under given operating conditions from takeoff until its fuel supply is exhausted. Combat radius is the distance for any given aircraft on a combat mission with a specified fuel-weapons-crew-equipment load and flight plan; it represents the maximum distance at which an aircraft can operate, under given conditions, and return to its point of origin without midair refueling. Because combat flights typically entailed taking off with heavy loads, climbing to high altitudes, flying a circuitous route to the target, and usually either climbing or descending over a target, and increasing speed in the vicinity of the target, the combat radius was considerably less than half the distance the same aircraft could fly under combat conditions. (Knaack, POST-WORLD WAR II BOMBERS, p. 6.)

⁴⁹⁹The Soviets had developed a primitive hose-and-drogue refueling system for the Tu-4 by late 1949 or early 1950. (Zaloga, pp. 76, 77, 263.)

its "loiter" time over the Marshalls to over four hours; however, there was no evidence by 1954 that the Soviets had developed any significant in-flight refueling capability. The Soviet Third Long-Range Air Force, stationed in the Far East, was estimated by early 1954 to be equipped with approximately 190 Tu-4s. None of these was known to be modified for long-range reconnaissance.

The only other Soviet aircraft known to have greater range potential than the Tu-4 was the Type 31 prototype (the Tu-85) heavy bomber which had been observed in flight for the first and only time during the July 1951 Aviation Day airshow in Moscow. The Type 31 appeared to have an unrefueled combat radius of about 3,000 miles and considerably in excess of this if combat loads of bombs and ammunition were replaced by fuel. (The Tu-85 was later canceled when a new generation of Soviet jet bombers reached the drawing boards).⁵⁰⁰

Although there was no evidence by early 1954 that the Type 31 was in series production, it was possible that the Soviets might have equipped an early model with photo or radiological sampling reconnaissance equipment. The Tu-85, which started flight testing in early 1951, had an unrefueled combat range of 7,400 miles; however, production of the plane was canceled in favor of jet-powered bombers.⁵⁰¹

Aircraft and submarine reconnaissance would not have been necessary to determine what the Americans were doing in the Pacific. Even if the Soviets did not conduct "on site" radiological sample collection at the PPG, stratospheric fallout from U.S. tests could "rain out" and be collected just as easily over Soviet territory as it could anywhere else along its path through the stratosphere.

A revised CASTLE security plan, issued after the series had started, concluded that it would be to the Soviets' advantage not to interfere with the test series:

a. Summary of Situation: (subject to continual revision)

- (1) The world political situation as of 15 June 1954 shows a general picture of

⁵⁰⁰Zaloga, pp. 79, 80.

⁵⁰¹Operation Order No. 1-53, Headquarters, Joint Task Force Seven, 20 June 1953, Annex C, pp. C-1, C-2; "The Billion Dollar Bomber," William Green, AIR ENTHUSIAST, Vol. I No. 4, September 1971, p. 217; Operation Plan CJTF No. 3-53, Headquarters, Joint Task Force Seven, 10 February 1954, Annex C, pp. C-1, C-2.

widespread political unrest.

- (2) Strictly speaking, the world is at peace but it is actually in an indefinite state known as "cold war" in which the forces of the East (Communists) are lined up against those of the West (Anti-Communist) and oppose each other openly by every means short of a declaration of war. In maintaining this state of cold war, the scientific discovery and application of atomic energy has played an important part in restraining the potential enemy.

The fear of the destructive force of atomic weapons has undoubtedly proved a deterrent to the aggressive attitude of any enemy government. They are striving by every possible means to increase their technical knowledge and application of atomic energy and thereby increase their potential military force.

- (3) It may be expected that the enemy will make a determined effort to gain knowledge of all atomic experiments and to interfere with them by every means possible, short of operations which would precipitate hostilities.

b. Enemy Capabilities: The capabilities listed below make it possible for the enemy (considered herein as the Soviets) to take the following action against the operational (CASTLE) area:

- (1) Submarine reconnaissance
- (2) Air reconnaissance
- (3) Ground reconnaissance or sabotage by landing parties from (a) submarine
- (4) Raids from submarines or surface ships
- (5) Raids by aircraft.

c. Conclusions: Capability (b) (1) is considered the most likely to occur under conditions existing today.

Capability (b) (2) is a possibility but is less likely and capabilities (b) (3), (4), and (5) most likely would be exercised in the event of open hostilities between the United States and the USSR. It is believed that information as to the operations of JTF Seven and the results thereof would be of more practical value to the Soviets than hindering or impeding the CASTLE operation.⁵⁰²

To counter possible submarine probes, or other enemy surface, subsurface, or air action, Joint Task Force 132 was to be provided by the Navy with four destroyer escorts, nine P2V patrol planes, and four F4U fighters. The underwater detection unit at Eniwetok was also to be manned by naval personnel.⁵⁰³

The sailing of the CURTISS to Eniwetok was conducted under radio silence with ships blacked out at night and on a route well away from normal shipping lanes. The CURTISS arrived at Eniwetok on January 24 (local time) where she unloaded the SHRIMP, RUNT, JUGHEAD, and ALARM CLOCK devices; she arrived at Bikini on January 27 and unloaded the MORGENSTERN device at Eninman island.⁵⁰⁴ The alternate seaborne command post, the USS ESTES, sailed from San Diego on January 18 and arrived at Eniwetok on February 3.

Certain other special materials for devices which could not be readied in time for the CURTISS's departure were later airlifted to Eniwetok.

Project experimenters arrived in the forward area in January and February 1954. TU 4 operations got underway in the first week of January when personnel arrived. All the LASL test devices, with the exception of ZOMBIE and RUNT II, were shipped to Eniwetok aboard the CURTISS (the latter two devices were flown to Eniwetok after CASTLE began).

Typical pre-shot activities during nuclear tests included handling, cab installation, and arming of an exercise

⁵⁰²Annex B to Operation Plan No. 4-5, Intelligence, Security, and Public Information, Headquarters, Task Group 7.4 Provisional, Kirtland Air Force Base, New Mexico, 15 June 1954, pp. B-1, B-2.

⁵⁰³Draft Operation Order No. 1-53, Headquarters, Joint Task Force Seven, Washington, D.C., 1 May 1953, p. F-6.

⁵⁰⁴Memorandum dated 4 December 1953 to Commander, Joint Task Force 7, from P. L. Hooper, J-3, Plans and Operations, JTF 7, Subject: Major Military Support Phasing.

("dummy") model of the test device or weapon. The dummy was a full-size physical and electrically-functional replica of the actual nuclear device, without high explosives or nuclear materials.

This exercise often included a test of the firing circuit; on several shot rehearsals, banks of photo-flash lamps mounted atop the shot cab were fired at the rehearsal "zero hour" to align and test cameras and light-sensing experimental equipment.

Some slight revisions of facilities were required as well as some modifications to the device installations. Trial runs with dummy units were started as soon as practicable, involving the movement of the SHRIMP dummy unit to shot location at Bikini, movement of the TX-14 ALARM CLOCK dummy, on its barge, to final Bikini anchorage, and trial loadings of dummies onto each barge.

Between January 20 and 21, the dummy SHRIMP device was transported by an LST from Eniwetok to Bilini, where it was transferred to an LCU off Eninman island and then taken to its ground zero site. A barge loaded with a dummy ALARM CLOCK was transferred between Eniwetok and Bikini atolls aboard the BELLE GROVE between January 28 and 29 and placed in its planned surface zero site in the vicinity of Aomoen island. The dummy ALARM CLOCK was returned to Eniwetok between February 2 and 3.⁵⁰⁵

Since original CASTLE plans called for detonation of four LASL devices on barges and one on land, all on a fairly tight schedule, the majority of TU 4 personnel were assigned to work on specific devices. The overall TU 4 responsibility for each device was given to the LASL project leader who had followed the design and fabrication of the device. TU 4 activities were coordinated closely with TU 3 on the assembly and test fillings of the EC16 JUGHEAD cryogenic bomb.

By January 25 (local time), the SHRIMP, JUGHEAD, RUNT and ALARM CLOCK devices had been unloaded from the CURTISS at Eniwetok. Final assembly of the SHRIMP device was completed on February 17; the device was shipped by LST to Bikini the following day, loaded aboard a truck, and driven to the shot cab near Namu on February 20, where installation in the cab was completed by February 22. A general task force rehearsal on the following day revealed several technical problems that would have been sufficient to preclude detonation.⁵⁰⁶

⁵⁰⁵Joint Task Force Seven, Commander Task Group 7.3, Final Report, Operation CASTLE, ca, June 14, 1954, p. 12b-1.

⁵⁰⁶Hewlett and Holl, pp. 171, 172.

The last components were loaded into the device and detonators installed on February 28. As the first test of a "dry" thermonuclear device, the SHRIMP had a special significance: its performance would not only dictate the remaining CASTLE agenda, but it could also conceivably change the future of the entire U.S. thermonuclear weapons program.

Figure IV-23 and **Figure IV-24** show the SHRIMP device in its shot cab near Namu. The device was supported from below, with diagnostic light pipes near the primary and at opposite ends of the secondary, the latter to measure radiation transit time. These pipes conducted light to the mirror array outside the cab.

The arming team proceeded from the firing bunker at Enyu to the shot cab at the end of the causeway on Namu and began the arming process. This consisted of mechanically altering the device to create a firing circuit. This circuit was still not complete, however, as a gap in the circuit existed in the firing bunker on Enyu.

After work on the device was completed, the arming party and the military police detailed to guard the device withdrew by boat to Enyu. The fleet evacuated the lagoon before the arming process was completed, thus ensuring that no task force personnel except for the firing party were in the atoll lagoon while the device could be detonated. Final links in the firing circuit were then closed, and the SHRIMP was fired at 0645 AM, March 1, 1954 (February 28 in the continental U.S.).

The Bravo Shot

Operation CASTLE began on an inauspicious and tragic note. Because of experimental inexperience with solid-fueled thermonuclear devices, scientists at Los Alamos had seriously underestimated the affinity of ordinary lithium-7 for neutrons: the yield of the SHRIMP device was 2 1/2 times its most probable estimate.

The results of this miscalculation were spectacular:

Suddenly the skies in the west lighted up and a great flare of whitish yellow light splashed against the clouds and illuminated the water. The startled seaman grasped the rough wood of the cabin with his hands and gazed in awe at the spectacle in the west.

It seemed like minutes though it was for seconds that he was transfixed by the dazzling light. It changed to a yellow red

and then to a flaming orange red before Suzuki came to his senses and dashed back to the cabin to tell his mates what he had seen. As he entered the cabin, Takagi, a cabinmate, was humming a song. Suzuki blurted out, "The sun rises in the west!"⁵⁰⁷

Within a second, a fireball nearly four miles in diameter was formed, and a crater more than 6,000 feet wide and over 200 feet deep was gouged from the reef.⁵⁰⁸ At ground zero, station 20, the shot cab, was obliterated, taking with it 61 tons of steel and 18 1/2 tons of wood, along with nearly 81 million cubic yards of coral sand and rock.⁵⁰⁹

The light from the fireball was visible for nearly a minute on Rongerik atoll, 155 miles to the east, and was observed clearly by military personnel at Kwajalein island, more than 250 miles southeast of Bikini.

On Rongerik, a weather station observer stated:

He came out with a towel and the light was so bright that he held the towel just to

⁵⁰⁷THE VOYAGE OF THE LUCKY DRAGON, Ralph E. Lapp, Harper & Brothers, New York, New York, 1957, 1958, p. 28. This book details the ordeal of 23 Japanese fishermen dusted aboard their boat by fallout from Bravo. The underestimated yield, a shift in the wind from the south to the west, and the failure of Navy patrol planes to spot the boat (which would have delayed the shot until the Japanese were out of the area) combined to cause an international incident when the fishermen returned to Japan on March 14. Their radioactive cargo of fish and the subsequent death of the ship's radio operator from hepatitis he contracted while hospitalized by radiation sickness did not help matters.

⁵⁰⁸One estimate of Bravo fireball radius was 8,000 feet. (Minutes of the 40th Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 27-29, 1954, p. 22.)

⁵⁰⁹Letter dated June 29, 1959 to Joe B. Sanders, Director, Pacific Operations, USAEC, from S. P. Howell, Manager, Engineering and Construction, Holmes & Narver, subject: Estimated Weight of Construction Materials, Operations CASTLE, REDWING, and HARDTACK I. An earlier estimate calculated CASTLE Bravo crater mass displacement at just over a billion tons. (Memorandum to Dr. John von Neumann from Lt. Col. Norair M. Lulejian, Atomic Operations Division, 23 February 1955.)

the bottom edge of his eyes. He did not have to cover his eyes but was on the verge of doing so because of the exceptionally bright light in the sky. He said that about 15 minutes after the flash, the noise and shock arrived which were very noticeable and shook and rattled the buildings. He became very frightened later when he saw the high (radiation) readings on his instruments. (He) reported that he did not see particles falling as had been reported but a haze closed in like a cloud and a dust was deposited on buildings and flat surfaces.⁵¹⁰

Closer to ground zero, the effects at the firing bunker on Enyu were even more terrifying:

I looked at the clock and calculated that the shock wave wouldn't arrive for another forty-five seconds at least, so I ignored the warning and turned to talk to Cochrane.

I started to tell him of my plans to go home, but as I talked his attention wandered. His look of flushed pleasure had slowly changed to one of puzzlement; then his expression changed as his face turned white with concern. I felt it too as I talked, the words coming more and more slowly and finally stopping as all thoughts of home quickly vanished.

Something was wrong!

Grier spoke the words first, as he reached out to steady himself at the workbench.

"Is this building moving or am I getting dizzy?" he asked.

"My God, it is. It's moving!"

"The whole building is moving!"

Grier reached for the bench to steady himself as I stood bewildered in the center of the room. The whole building was moving,

⁵¹⁰Memorandum for Record dated 6 March 1954 from Maj. Gen. P. W. Clarkson, Commander, Joint Task Force Seven, Subject: Visit to Kwajalein, Friday, 5 March, p. 4.

definitely now, not shaking or shuddering as it would from the shock wave that had not arrived yet, but with a slow, perceptible rolling motion like a ship's roll; I began to feel a nausea akin to seasickness.

I was completely unable to get it through my head that the building was moving. The building is made of concrete, I told myself. The walls are three feet thick. It's anchored like a rock on this island. Besides, the shock waves can't be here yet.

But it was moving! The motion was unmistakable as it built up. Objects were beginning to slide on the workbench. My initial period of astonishment must have lasted about five seconds before I could move over to the bench to hold on and steady myself against the strong rolling motion. Subconsciously, I pushed back a screwdriver, which was rolling back and forth in a narrow arc on the bench. Then I realized what it was.

"It's the ground shock," Clark and I yelled simultaneously.

I had never felt it before, but it was due to the fact that the shock wave from the blast traveled much faster through the ground than it did through the air. Generally, the ground shock was never felt, as it died off more rapidly than did the shock wave through the air; the fact that this one was evident at all was an indication that the explosion had been one of tremendous force. A force that could move this heavy concrete building so many miles away so that it felt like it was resting on a bowl of jelly must be wreaking tremendous havoc with this whole Pacific atoll. From the way the building felt at this moment, it might just as well be floating in the middle of the Pacific Ocean. For all I knew maybe it was!⁵¹¹

⁵¹¹O'Keefe, pp. 178-179. The ground shock from Bravo was felt throughout the entire Bikini atoll. Mr. O'Keefe and his firing party, twenty miles from Bravo ground zero, did not slide into the sea, but they were trapped by fallout and

Within a minute, the fireball from the SHRIMP had risen to 47,000 feet and 2,000,000 tons of pulverized coral from the crater was pulled up into a cloud that was already more than seven miles across with a stem 2,000 feet wide.⁵¹² In this same minute, the blast wave from the explosion moved outward from the burst point, stripping the nearby islands of vegetation and animals.

Figure IV-25 shows the Bravo fireball enveloped by three "ice caps" and two condensation clouds.

Ten minutes after zero hour, the mushroom cloud was 70 miles across and rising swiftly; a minute later, windows in buildings at Rongerik atoll rattled as the blast wave hit. Several temporary buildings on Eninman island at the south end of Bikini atoll were nearly destroyed; a fire started in one of them, caused by a short-circuit in damaged electrical equipment. In a concrete bunker a mile and a half from Bravo ground zero, the blast was powerful enough to rip a 20-ton door from its hinges and slam it 15 feet through the bunker.⁵¹³

The SHRIMP device explosion vaporized its shot cab and concrete substructures and sent tons of sand and coral upward with the atomic cloud which penetrated the tropopause and reached an altitude of 114,000 feet⁵¹⁴ and spread out to a diameter of 350,000 feet, just over 66 miles.⁵¹⁵

lethally-high radiation for several hours before they could be rescued. This personal story of the Bravo shot is probably one of the best (and most frightening) accounts ever written about a thermonuclear test gone awry.

⁵¹²Minutes of the 40th Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 27-29, 1954, p. 22.

⁵¹³"We Were Trapped by Radioactive Fallout," Dr. John C. Clark, THE SATURDAY EVENING POST, July 20, 1957.

⁵¹⁴Another report lists the Bravo cloud top at only 90,000 feet, with an estimated yield of only 11 MT. (Memorandum dated March 3, 1954 to K. E. Fields, Director, Division of Military Application, USAEC, from John C. Bugher, MD, Director, Division of Biology and Medicine, USAEC, Subject: Meteorological Data on First Shot - 2/28/54 - 1845 Zulu. Bugher cites the U.S. Weather Bureau as the source for his figures.)

⁵¹⁵"A Theory of Fall-Out From Nuclear Weapons," J. M. McCampbell, Weapons Capabilities Branch, Military Evaluation Group, U. S. Naval Radiological Defense Laboratory, undated,

Unfortunately, winds at various altitudes through which the shot cloud passed were blowing in somewhat different directions than those at altitudes the lower-yield cloud was expected to reach. Winds at lower heights were expected to be from the south, which would have carried fallout in a fan-shaped pattern from northwest to northeast over the empty ocean.

In addition, the higher-than-expected yield resulted in much more highly-radioactive debris being carried up into the atmosphere; this debris was diffused over a much larger area than was previously thought possible (with only results of the IVY Mike shot to guide planners. The higher than predicted Bravo yield also resulted in the debris being two to three times more radioactive than expected.

Before Bravo, fallout planning was based on the idea that the radioactive debris cloud was almost a "point source" of contamination, with about a 15-mile radius. Bravo showed that this cloud was no "point source" but was instead an area about 100 miles in diameter.

A post-Bravo evaluation also noted that the most probable yield from the SHRIMP was expected to be three to five megatons, or one-half the IVY Mike shot yield. The upper limit of the SHRIMP shot yield was considered to be on the order of eight megatons. The actual yield of the Bravo shot was 15 megatons, three to five times the most probable yield, and nearly twice the anticipated upper limit.

As a result, more debris was carried up into the atmosphere and diffused over a much larger area than was originally thought possible. In addition, since the radioactivity of the fallout was proportional to the yield, the SHRIMP shot debris was much more radioactive and of a much larger volume than expected.

Upper altitude wind predictions 48 hours before Bravo shot time had been for easterly winds below 15,000 feet with winds of a "southerly component" above. A 24-hour wind prediction starting 18 hours after shot time was unfavorable, with northwest winds between 10,000 and 20,000 feet.

Shot day wind forecast was for easterly winds from 15 to 20 knots at surface level; 10 to 15 knot easterly winds at 10,000 feet; easterly 15 to 20 knot winds at 20,000 feet; southeasterly 15 to 25 knot winds at 30,000 feet; southerly 15 to 25 knot winds at 40,000; southwesterly 10 to 20 knot winds at 50,000 feet; and southeasterly 5 to 15 knot winds at

60,000. The radiation safety outlook was "very favorable."⁵¹⁶

A forecast 36 hours before shot time was for easterly winds 15 to 20 knots from the surface to 15,000 feet; easterly 10 to 15 knots between 15,000 and 25,000 feet; southerly 5 to 10 knots between 25,000 and 30,000 feet; southwesterly at 20 knots between 30,000 and 50,000 feet; and northeasterly 10 to 15 knots at 60,000 feet.

The 10 megaton IVY Mike blast had pushed its cloud to 20 miles; Bravo's four to eight megaton yield was expected to result in a lower cloud top. At the time of the CASTLE Bravo shot, the algorithm used for predicting shot cloud heights was $H = 8.3 + 23.3 (\log_{10} W)$, where W was yield in kilotons and H was cloud top height in thousands of feet above burst point.

Using this formula, the cloud from a 4 MT shot should reach about 92,300 feet; for 6 MT, 96,331 feet; for 8 MT, 99,242 feet; and for 10 MT, 101,500 feet.⁵¹⁷

At midnight of February 28 (local time), winds at 20,000 feet were forecast to be from the west, blowing toward Rongelap and Rongerik atolls; however, the scientists did not expect that significant quantities of highly-radioactive fallout would be carried by the winds aloft far enough to reach those sites. Fallout was expected only from low altitudes and it would be subject to light winds which would limit its travel from Bikini. At Bikini, both Enyu and Eninman islands were expected to be heavily contaminated.

The most probable cloud track was expected to be in a narrow cone on an east-northeasterly bearing, north of Rongelap and Rongerik atolls.⁵¹⁸ This situation was virtually unchanged four and a half hours later (two and a quarter hours before shot time).

Figure IV-26 shows both the forecasted and actual winds directions at Bravo zero hour.

⁵¹⁶Memorandum for Record dated 12 April 1954 from Alvin C. Graves, Scientific Director, JTF 7, and Maj. Gen. P. W. Clarkson, USA, Commander, JTF 7, Subject: BRAVO Shot, Operation CASTLE, pp. 2, 3, 4.

⁵¹⁷Outline of Material to be Presented by AFSWP Personnel at Tripartite Conference on Atomic Weapon Effects, 15, 16, and 19 February 1954, p. 5.

⁵¹⁸Radsafe Narrative Sequence of Events, Tab B to Memorandum for Record dated 12 April 1954 from Alvin C. Graves, Scientific Director, JTF 7, and Maj. Gen. P. W. Clarkson, USA, Commander, JTF 7, Subject: BRAVO Shot, Operation CASTLE.

Official Weather Bureau winds aloft data gathered after Bravo showed that winds from the ocean surface to an altitude of 5,000 feet were generally from the east. Between 5,000 and 8,000 feet, the air was calm. From 8,000 to 55,000 feet, winds were westerly or southwesterly, with a maximum speed of 35 to 40 knots at 40,000 feet. Between 55,000 and 90,000 feet, winds were variable in lower levels, changing to easterly with a velocity of 35 knots at 90,000 feet.⁵¹⁹

Another report listed shot-time winds from the west-southwest at 35 knots between zero and 55,000 feet, and from the east at 40 knots between 65,000 and 90,000 feet. This sharp reversal in direction caused heavy fallout from those portions of the shot cloud stem and mushroom below 65,000 feet.⁵²⁰

By 6:45 PM on March 1, 12 hours after Bravo zero hour, winds at 20,000 feet, which had previously been from the west-southwest, were discovered to be actually blowing from the northwest and depositing fallout on the atolls east and south of Bikini. Fallout began raining down on Rongerik atoll, 133 nautical miles from Bikini, about seven hours after zero hour.⁵²¹

Winds at still higher altitudes were out of the west around shot time, causing the fallout to blow eastward, over many of the task force vessels, several inhabited atolls east and south of Bikini, and an undetected 100-ton Japanese fishing boat just outside the restricted area.

Due to in part to predicted fallout patterns, pre-shot airborne radar searches for transient surface shipping were conducted mainly to the northwest of Bikini, so the tiny Japanese boat immediately east of the atoll was not

⁵¹⁹Memorandum dated March 3, 1954 to K. E. Fields, Director, Division of Military Application, USAEC, from John C. Bugher, MD, Director, Division of Biology and Medicine, USAEC, Subject: Meteorological Data on First Shot - 2/28/54 - 1845 Zulu. Bugher cites the U.S. Weather Bureau as the source for his figures.

⁵²⁰Minutes of the 40th Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 27-29, 1954, p. 22.

⁵²¹Radsafe Narrative Sequence of Events, Tab B to Memorandum for Record dated 12 April 1954 from Alvin C. Graves, Scientific Director, JTF 7, and Maj. Gen. P. W. Clarkson, USA, Commander, JTF 7, Subject: BRAVO Shot, Operation CASTLE.

detected.⁵²² Another Japanese fishing boat had been discovered on February 17 inside the danger area, 26 nautical miles northeast of Eniwetok atoll; the boat was escorted out of the area.⁵²³

An hour after the shot, considerable contamination was found on the Bikini island airstrip and on decks of the task force ships downwind of the blast. The extensive fallout of the Bravo shot spread out east and west for hundreds of miles, contaminating the sea, Bikini, and nearby downwind atolls.

Figure IV-27 shows the CASTLE Bravo shot fallout "footprint."

A U.S. Navy gasoline tanker, the USS PATAPSCO (AOG-1), approximately 180 to 195 nautical miles northeast of Bravo's ground zero, was dusted on March 2 by radioactive fallout as she sailed from Eniwetok to Pearl Harbor in Hawaii.⁵²⁴ Detectable levels of airborne radiation from Bravo eventually reached far to the west of Eniwetok atoll, more than 200 miles west of Bikini.

After the Bravo shot, a teletype message dated March 9 from LASL to the AEC stated that "It would appear that the weather played a dirty trick as we have seen it do in the past." In the fall of 1953, one meteorological study had warned that fallout from Bikini might blow to the east:

... the easterly winds (over Bikini between January and March) will vary in direction with height, becoming strong westerlies above 30,000 feet and reaching (speeds) as high as 100 knots at 45,000 feet on occasion. These winds are associated with a weather system aloft which can become quite intense and persist for periods in excess of a week. This situation is the one to be most

⁵²²Teletype dated March 17, 1954 to Chairman, USAEC, from JTF 7, Eniwetok listing all aerial searches between Bravo-2 to Bravo day; these searches ranged out several hundred miles on the "forecast significant cloud travel" line, mostly to the northwest and northeast of Bikini, and never to the east.

⁵²³Memorandum for Record dated 10 March 1954 from Lt. Col. R. A. House, Chief, Tech Branch, J-3, Subject: Protection of Transient Shipping during Operation CASTLE.

⁵²⁴DNA 6035F, pp. 218, 230, 232; teletype dated 13 March 1954 from Commander in Chief Pacific Fleet to Chief of Naval Operations. The PATAPSCO was decontaminated at Pearl Harbor at Oahu in the Hawaiian islands.

wary of during this period of operations.⁵²⁵

An October 1953 AEC study of the dispersion of fallout from the IVY Mike shot noted:

The wind structure over the Marshall Islands area during the (fall 1952) test period can be considered in three broad layers: the trade wind layer of steady east-northeast winds extending from the surface to about 25,000 feet; the upper troposphere, from 25,000 to about 55,000 feet, which consists of a series of moving large-scale vortices; and the lower stratosphere, from 55,000 to 130,000 feet, containing variable east or west winds with speeds decreasing from values as low as 50 knots near the base of the layer to light (and uncertain) values above.

Unfortunately, wind data over the Pacific is almost completely lacking at heights above 100,000 feet and only sporadic soundings are available in the remainder of the lower stratosphere. Even climatological mean winds are not available at these great heights. The paucity of data is critical in view of the great height reached by the Mike cloud and the fact that the bulk of the airborne debris is probably contained in the mushroom (cloud).⁵²⁶

Another problem with Pacific testing was the paucity of surface weather reporting stations. In an area the size of the continental U.S., there were only 32 stations. By comparison, the same area in the U.S. was covered by more than 300 stations. For this reason, weather forecasting depended very heavily on airborne and seaborne weather reconnaissance, and on the 32 island-based stations.⁵²⁷

⁵²⁵Excerpts from a paper entitled "Climatology of the Eniwetok-Bikini Area for the Months of January, February, March, and April," published in the fall of 1953.

⁵²⁶A METEOROLOGICAL ANALYSIS OF THE TRANSPORT OF DEBRIS FROM OPERATION IVY, USAEC NYO-4555(Del.), Lester F. Hubert, et. al., Weather Bureau, October 30, 1953, p. 2.

⁵²⁷Letter dated 25 November 1953 to Lt. Col. Mahlon Hammond, TG 7.4, Kirtland AFB, New Mexico, from Lt. Col. Herschel H.

Air Force weather forecasters in the Pacific were unhappy with the way that they were being made scapegoats for the Bravo fallout:

Colonel Slater and I are concerned over comments in the press regarding the weather forecast for the Bravo (shot). We believe the Air Force has been unjustifiably placed in a bad light with the public due to releases made by nationwide press services. Many of these articles have indicated or implied that "fall out" over inhabited areas (subsequent to) Bravo was due to a poor forecast.

The same (press) releases talk about the shot (yield) being two to three times as large as predicted. Furthermore, no reference has been made to the lack of experience with fallout on the IVY test of a high-yield devices. Apparently the statement by Mr. Strauss has been amplified to the discredit of the Air Force Weather Service.

General Clarkson and Dr. Graves have both told Col. Slater they consider that a good weather forecast was made for Bravo. It was my impression that you thought the forecast was at least adequate. Cols. Slater and Bonnat both feel that it was as good as could be expected considering the limitations of data in this area and the science of meteorology. Perfect forecasts are practically unattainable and greater discrepancies could easily occur (in the future).

The Air Force noted that their forecasters were being adversely affected by unjustified press comments. These men had both military and civilian careers to consider, and their reputations were at stake.

Accompanying this memorandum was a list of unfavorable press citations regarding the weather forecast for Bravo shot:

Known newspaper and magazine articles

Slater, USAF. At the time of Operation IVY, there were only nine weather stations in the central South Pacific. (Moore and Bechanan, p. 204.)

regarding the tests, which have thrown the Air Force weather forecasting in a bad light with the general public. Each of these releases had wide circulation through nationwide press services.

1. Statement by AEC chairman Lewis Strauss quoted in an Associated Press dispatch dated 31 March:

"Winds failed to follow predictions but shifted south of that line and the little islands of Rongelap, Rongerik, and Uterik were in the path of the fallout."

2. Column by the Alsop brothers, Washington Post, 24 March 1954:

"The unfortunate Japanese were the victims of a chance 'fallout' of radioactive material from the bomb cloud, which resulted from a combination of bad luck and bad meteorological forecasting. Ordinarily, the cloud (rises) into the stratosphere and is harmlessly dissipated."

3. International News Service release, 19 March:

"Representative Chet Holifield, Democrat of California, revealed, in a report on why 264 persons on Kwajalein (sic) and nearby islands suffered radiation exposure, that the test blast was 'larger than expected' and that unpredicted high altitude winds played a part in the incident."

4. United Press release, March 11th, referring to a statement by Representative Carl Hinshaw, Republican of California:

"He said that the upper winds were not forecast correctly for the recent test and the 'fallout' was blown over the islands."

5. TIME magazine on 12 April:

"Then, last month, came the U.S. explosion that Strauss describe as being twice the estimated size. It became famous prematurely because an unexpected wind shift showered a Japanese fishing boat with

radioactive ash."⁵²⁸

A subsequent Air Force memorandum comparing forecasted winds for Bravo four hours before shot time with observed winds at Rongerik at zero hour indicated that the most significant differences were at altitudes of 15,000, 20,000, and 55,000 feet. The H-4 forecast for 15,000 feet was for winds of 18 knots from a bearing of 250°, or approximately west-southwest. Observed winds at Rongerik at this altitude at shot time were 13 knots from 320°, the northwest.

At 20,000 feet, the forecast was for 12 knot winds from the west; H-hour winds were 19 knots from approximately the west-northwest. At 55,000 feet, the forecast was for 10 knot winds from northeast; shot time winds at this elevation were instead six knots from the north-northwest.⁵²⁹ The resultant "footprint" of fallout from Bravo was caused by the radioactive materials -- primarily pulverized coral and coral ash -- falling from great height through winds at different directions at different altitudes.

In general, the more that winds came from the east and north, the greater the likelihood of fallout on the inhabited atolls to the east and south of Bikini.

The radioactive contamination from Bravo became so serious and widespread that on March 3, two days after the shot, Rongerik, Rongelap, Ailinginae and Utirik atolls east of Bikini were evacuated. Ailinginae was 79 nautical miles from Bravo ground zero; Rongelap, 100 nautical miles; Rongerik, 133 nautical miles; and Utirik was 270 nautical miles from surface zero.⁵³⁰

Although Ailinginae atoll was not inhabited in the sense that islanders lived on it permanently, it was regularly used and coconuts, fish, and other crops were harvested from the atoll by the islanders on Rongelap, who depended upon these

⁵²⁸Memorandum for Commander, Task Group 7.4, Provisional dated 13 April 1954 from Lt. Col. Mahlon B. Hammond, USAF, subject: Weather Forecast for Bravo. Emphasis in original.

⁵²⁹Undated memorandum and attachments to Commander, Air Weather Service, Andrews AFB, from Brig. Gen. Howell M. Estes, Jr., Commander, TG 7.4 Provisional, subject: Weather Forecasts for JTF 7.

⁵³⁰Memorandum for Record dated 12 April 1954 from Alvin C. Graves, Scientific Director, JTF 7, and Maj. Gen. P. W. Clarkson, USA, Commander, JTF 7, Subject: BRAVO Shot, Operation CASTLE, p. 1.

foodstuffs for their continued sustenance. The removal of Ailinginae from the economic orbit of Rongelap resulted in critically lowered living standards for the Marshallese.⁵³¹

The AEC ordered a blackout on news of the aftereffects of the Bravo shot. The AEC advised that no press release would be issued about either the extensive fallout or the evacuations of the islands unless they were forced to do so by a "leak" or other circumstances.⁵³²

However, the Bravo shot did not go entirely unnoticed: the flash from the blast was seen in Okinawa, just over 2,600 miles to the west-northwest of Bikini, and the United States Geological Service earthquake monitoring station in Berkeley, California, where Teller had "observed" the IVY Mike shot, picked up a strong seismic signal from the Bravo test.⁵³³

The first public notice in the press of the extensive Bravo contamination came as the result of two letters a Marine corporal temporarily stationed at Kwajalein atoll wrote to his mother in Cincinnati, Ohio and which she brought to the attention of a local newspaper. Unlike the case of letters subsequent to the IVY Mike shot, here was an example of draconian AEC security plans thwarted by an individual who was not associated with the CASTLE task force.⁵³⁴

One newspaper account of the Marine's letters said:

... a nuclear explosion in late February at Eniwetok (sic) had seared natives 75 miles away and shook buildings 300 miles away. "I was walking back to the barracks from chow and it was just getting (to be) daylight when all of a sudden the sky lighted up, a

⁵³¹Letter dated February 5, 1953 to James P. Davis, Director, Office of Territories, Department of the Interior, from Elbert D. Thomas, High Commissioner, Trust Territory of the Pacific Islands.

⁵³²Teletype dated March 4, 1954 to Admiral Clarke, Commander in Chief Pacific Fleet, from General Clarkson, Commander, JTF 7.

⁵³³Teletype dated March 4, 1954 to Gen. K. E. Fields from Alvin Graves.

⁵³⁴Letter dated March 26, 1954 to Thomas E. Murray, Acting Chairman, USAEC, from E. C. Stakman, Chairman, Advisory Committee for Biology and Medicine, USAEC, Appendix "C" to AEC 141/25, Atomic Energy Commission, Use of the Nevada Proving Grounds, Note by the Secretary, June 24, 1954.

bright orange, and remained that way for what seemed like a couple of minutes. About 10 to 15 minutes later, back at the barracks, we heard a very loud rumbling that sounded like thunder. Then the whole barracks began shaking as if there had been an earthquake. This was followed by a very high wind."

"There were two destroyers that pulled in here today bearing natives of one of the Marshall Islands that was within 75 miles of the blast. They were suffering from various burns and radioactivity (sic)."⁵³⁵

As a result of the story, the AEC announced on March 11 that 28 Americans and 236 Bikini natives had been "unexpectedly" subjected to "some radiation" during the "course of a routine atomic test in the Marshall Islands."⁵³⁶ In reality, the Bravo shot had been anything but routine, and "some radiation" considerably understated the extent of the exposure of the Rongelapese to fallout.⁵³⁷ (On March 1, the AEC had announced that "JTF Seven has detonated an atomic device at the AEC's Pacific Proving Grounds in the Marshall Islands.")

Three days later, on March 14, the FUKURYU MARU fishing vessel reached Japan and reported its 23 crewmen ill with radiation sickness. The boat had been just beyond the eastern edge of the 80-mile radius of the danger zone, and its misfortunes caused an international uproar.⁵³⁸

⁵³⁵Teletype dated March 15, 1954 to CJTF 7 from Brig. Gen. K. E. Fields, Director of Military Application, USAEC quoting Cincinnati, Ohio newspaper accounts of the letters dated March 3 and 4, 1954. Fields asked Clarkson if disciplinary action was "indicated." The sound of Bravo should have taken about 20 minutes to reach Kwajalein atoll. The Marine corporal was not a member of JTF; nor was he stationed at Kwajalein. He was merely passing through the Marshalls at the time of Bravo. No action was taken against him. (Letter dated May 13, 1954 to Representative Sterling Cole, JCAE, from K. D. Nichols, General Manager, USAEC.)

⁵³⁶News Roundup, BULLETIN OF THE ATOMIC SCIENTISTS, 1954, p. 141.

⁵³⁷For a good account of AEC dissemblance after Bravo, see "Writing the History of a Controversial Program: Radiation Safety, the AEC, and Nuclear Weapon Testing," Barton C. Hacker, THE PUBLIC HISTORIAN, Vol. 14 No. 1, Winter 1992, p. 51.

Two other Japanese fishing vessels, the KOEI MARU and the MYOJIMM MARU, were also contaminated when they were brushed by the edge of the Bravo cloud.⁵³⁹

One radiological monitor concluded at the time that "Operation CASTLE must be reported as the nightmare of radiological safety test operations." A report noted that "everyone and everything in the northern Marshall Islands had become radiologically contaminated to some extent."

One estimate placed Bravo fission yield at eight megatons, nearly half its yield; much of this rained out or fell out over the Pacific and the eastern Marshall islands.⁵⁴⁰ (Actual Bravo fission yield was 10 megatons, two-thirds of its total yield.)

At Rongelap atoll, 110 miles east of Bikini, fallout density ranged from five to 61 megacuries per square mile in a strip about 20 miles wide. Fallout had begun dropping as a fine sand or fluffy powder about 12 hours after the Bravo shot. The drinking water at Rongelap was heavily contaminated.⁵⁴¹

The AEC later prepared a secret map (see **Figure IV-28**) that showed what might happen if a comparable 15 MT bomb were dropped on Washington, D.C. and winds carried fallout to the northeast. The map was created by superimposing the Bravo fallout pattern over the northeastern U.S. with Washington, D.C. as ground zero.

Lethal fallout would have covered what was left of D.C., as well as Baltimore, Philadelphia, and part of New York City, over an area of 6,000 square miles. Dangerous fallout would have reached nearly to Boston, with fringes extending through

⁵³⁸AF ATOMIC ENERGY PROGRAM, p. 299. Subsequent articles in Japanese newspapers in March and April 1954 stated that the U.S. had tested a lithium-fueled device; within weeks, Japanese physicists had deduced that the SHRIMP had contained a uranium-238 jacket around its fusion stage. (Hewlett and Holl, p. 176; Lapp, pp. 148-156.)

⁵³⁹Memorandum to John C. Bugher, Division of Biology and Medicine, USAEC, from Merrill Eisenbud, subject: Contamination of the Fukuryu Maru and Associated Problems in Japan: Preliminary Report, p. 8.

⁵⁴⁰"The 'Humanitarian' H-Bomb," Ralph E. Lapp, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 11, 1956, p. 262.

⁵⁴¹Minutes of Thirty-Ninth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 31-April 2, 1954, pp. 13, 14.

New England almost to the U.S.- Canadian border.

Fallout would have begun dropping onto D.C. from 15 to 45 minutes after the blast; on Baltimore within 45 to 90 minutes after the explosion; within five to seven hours on Philadelphia; within nine to 11 hours on New York; and within 12 to 14 hours on Boston.⁵⁴² Bravo had shown that a large yield surface detonation could produce extremely serious radiological contamination over a distance more than 120 miles downwind and dangerous contamination about 250 miles downwind.

A number of prominent U.S. politicians and lesser-known anonymous sources, all of whom should have known better, soon made inaccurate and premature statements concerning both the Bravo test and its significance to the U.S. These sources also disclosed accurate information that was highly classified at the time:

Representative (Carl T.) Durham, ranking Democrat on the Joint Committee on Atomic Energy, said today the United States "exploded this month" a thermonuclear weapon many times more powerful than the explosion of November 1, 1952.

That 1952 explosion itself was "a success," Durham told a reporter. "This more recent one is just much greater."

Representative Sterling Cole, chairman of the Senate-House atomic group (Author's note: the JCAE), had announced last night that the United States has the hydrogen bomb -- thermonuclear is the scientific adjective for this new weapon -- and can deliver it anywhere in the world.

Cole also said it is a fair assumption that Russia, too, has the H-bomb.

....

Durham said the explosion this month was in the testing grounds of the Pacific Ocean.

While Durham went on the record, even higher estimates of destructive power were

⁵⁴²Hewlett and Holl, pp. 181, 182; letter to USAEC from Col. Robert N. Isbell, Chief, Atomic Operations Division, Deputy Commander/Technical Operations, Headquarters, Air Research and Development Command, USAF (date illegible).

assigned to the new explosion by others in a position to know, but who asked not to be quoted by name.

These sources said a "warm up" explosion of a hydrogen device in the Pacific March 1 surprised American scientists with a blast at least 600 times more powerful than the Hiroshima atomic bomb. They said it was three or four times stronger than had been expected.

The Hiroshima bomb, which claimed 60,000 victims, was measured at 20,000 tons of TNT. The March 1 hydrogen explosion was measured at between 12 and 14 million tons of TNT, it was said. These sources also said it was five times greater than the measured force of what was believed to be the first H-bomb test, November 1, 1952. Other comparisons with the 1952 tests varied.

Those who reported the 600-times Hiroshima effect also said of the March 1 blast:

The blast was set off from a 150-foot tower and shot a nuclear cloud more than 17 miles into the air.

The shock was felt at Kwajalein atoll, 176 miles away.

...

And, it was added, scientists now believe they can build hydrogen weapons with an explosive power of 100 million tons of TNT, as compared with the 12-14 million ton estimate given for the March 1 blast.

...

As destructive as this test was, the first of a series, others are planned that will be even more powerful, the sources reported. One blast set now for the latter half of April may run as high as 45,000,000 to 50,000,000 tons, the sources said.

The reports came in the wake of official disclosure that this country now has a deliverable hydrogen bomb and the planes to carry it to any target in the world. An unofficial source added that crude but usable H-bombs are already being

stockpiled.

According to congressional sources, the test at Bikini did not involve an H-bomb as such, but rather a hydrogen "device." Estimates of its destructive power were put at 12,000,000 to 14,000,000 tons of TNT.

This would make it more than twice the power of the world's first hydrogen explosion, set off by this country at Eniwetok November 1, 1952. This first blast has been estimated unofficially at around 5,000,000 tons TNT equivalent. (Author's note: this was about half the actual yield of the IVY Mike test, which was still classified in the spring of 1954.)⁵⁴³

When President Eisenhower announced that he would answer questions about current nuclear testing at a press conference on March 24, the AEC suggested that he include in his comments the following statement of justification of continued testing:

The routine tests now underway at the Pacific Proving Grounds are concerned with the most recent developments in hydrogen explosives. Results to date have been of great value and significance, substantially adding to our knowledge of atomic weapons and to our readiness.

The explosive yield from a device recently tested has been greater than that of any previously detonated. This is in accord with the record of previous test series, each of which has regularly surpassed the results of earlier ones. Our constant aim is to obtain the maximum efficiency possible from nuclear material, not only in order that our armed forces may be equipped with the most effective weapons, but also because such research is a positive way of learning what a possible enemy might be capable of directing against us.⁵⁴⁴

⁵⁴³Teletype message dated March 17, 1954 to Lewis L. Strauss, Chairman, AEC, from K. D. Nichols, General Manager, AEC.

⁵⁴⁴Teletype dated 19 March 1954 from Commander Joint Task Force Seven to Kenneth Nichols, USAEC.

Later in March, after the second CASTLE shot, in a draft of his report to President Eisenhower, AEC Chairman Lewis Strauss defended his agency and the need for testing in the Pacific:

.. The results of the tests are important in the following respects:

(1) We now know that a considerably lighter, less complicated, an less expensive weapon can go into the stockpile. [deleted]

(2) [deleted]

... The results of these two findings will mean a saving in the costs which we had budgeted for the next fiscal year to meet the JCS requirements. This saving will be material ($\pm$ \$100 million) but what is most important is that we will be able to meet the JCS requirements much sooner and the DOD costs for delivery and logistics will be very much diminished (savings in several hundred millions of dollars).

With respect to the public relations aspect of the tests, the facts are as follows. The first shot was fired on March 1st. It was fired on a so-called island. These islands are little spits of sand and coral, 200 to 300 yards across, many dozen of which compose one atoll. Subsequent large shots will be fired on barges and there will be no destruction of "real estate."

The first shot was computed in advance by the scientists at Los Alamos to have a minimum yield of four megatons and a maximum yield of perhaps eight. Its actual yield as presently computed was 15 megatons. The scientists are not yet able to account for the difference which, while important, was far from "out of control."

The second shot which was computed to have a yield from two to seven megatons apparently had a yield of nine. There were no untoward results from the second shot. The first shot, however, was followed by incidents resulting from the fall-out (sic) of sand, coral, and water carried up by the blast, due to the fact that the meteorologists had no been able to accurately predict the wind direction.

A sudden shift in the wind caused the fall-out to occur approximately due east of the test site and this occasioned exposure to the U.S. personnel who manned the weather stations on some of these atolls, involving about 28 people, and to the native population, involving about 236 people. All were promptly evacuated to Kwajalein by the task force. ...

The newspaper stories which I have been shown since my return to the effect that the (first) test "got out of hand" and was 'devastating' are, of course, simply untrue. The exaggerations have provided fuel for those who were opposed from the beginning to any attempt on the part of our Government (sic) to be armed with this weapon.

The relative ease and economy with which we have now discovered it can be made surely point up the fact that it is something which the enemy can in time possess if he does not already have it. This is information which it would have been a national disaster for us not to know.⁵⁴⁵

Post-Bravo Operations

Bravo's 15 MT yield was half again that of the IVY Mike shot, and approximately 1,000 times that of the LITTLE BOY bomb dropped over Hiroshima in August 1945. The unexpectedly high yield of the shot and the heavy contamination and damage to structures on Bikini that ensued brought about radical changes in the CASTLE operational concept and in the schedule of the remaining shots. Bravo was the only CASTLE shot to be fired on its original schedule.

Afterwards, all test personnel at Bikini lived aboard ship, traveled to and from their stations by helicopter or boat, and subsequent firings at Bikini for shots Romeo, Koon, Union and Yankee was accomplished by means of a radio link from the USS ESTES to the evacuated control station on Enyu.

In addition, stringent new weather criteria were imposed to guarantee that fallout would not drop onto inhabited areas again (these new criteria brought on many delays in the shot schedule; Romeo, for example, was postponed seventeen

⁵⁴⁵"Notes for Report to The President on My Return from the Pacific Proving grounds," draft memo by Lewis Strauss dated 30 March 1954.

times).⁵⁴⁶

Another significant change was a substantial expansion of the "danger area" surrounding the Bikini-Eniwetok area. The original CASTLE danger area was rectangular and enclosed approximately 67,645 square statute miles centered approximately halfway between Bikini and Eniwetok atolls. The enlarged area, defined by a 450 nautical mile radius semicircular arc centered on Bikini, encompassed over a half-million square miles, enclosing a large area to the north, east, and west of Bikini.⁵⁴⁷ (See **Figure IV-29.**)

This vast enlargement of the restricted area drew a protest from the Japanese government. Approximately 150 Japanese tuna fishing boats were in and around the newly-proclaimed zone. The catch from the area accounted for only about 1% of the total tuna caught by the Japanese fishing industry. The largest fishing areas lay east and southeast of the Eniwetok-Bikini area, around the Gilbert, Solomon, and Samoan islands.

About 300 fishing boats, ranging in size from 100 to 600 tons, operated in those waters and caught about 65% of all tuna caught by tuna boats in those areas. These vessels usually passed through the enlarged danger zone on their way to and from the fishing grounds, and the expanded danger zone not only made fishing within it dangerous, it also forced ships operating in the Gilbert-Solomon-Samoan area to make a two or three-day detour to skirt it.⁵⁴⁸

The high yield of the SHRIMP also removed any lingering doubts that LASL may have had about how the ALARM CLOCK and the RUNT would perform. Following Bravo, the predicted yields of all subsequent CASTLE shots were raised significantly. The thermonuclear fuel reactions concerning device components were now too uncertain to make high-probability yield predictions. However, the surprising performance of

⁵⁴⁶HISTORY OF AIR FORCE ATOMIC CLOUD SAMPLING, Narrative, AFSC Historical Publications Series 61-142-1, January 1963, p. 106.

⁵⁴⁷Teletype message dated March 23, 1954 to USAEC, Washington, D. C. from CJTF-7, Eniwetok, Marshall Islands; "Answers to State Department 'List of Possible Questions,'" attachment to letter dated June 9, 1954 to Honorable David McKay, Assistant Secretary of State, from K. D. Nichols, General Manager, USAEC; letter dated 29 December 1955 to Lewis L. Strauss, Chairman, USAEC, from RADm. B. Hall Hanlon, USN, Commander, JTF 7.

⁵⁴⁸Teletype dated March 31, 1954 to JCTF7, Eniwetok, from USAEC, Washington, D.C.

unenriched lithium made it much more likely that devices containing lithium enriched to varying degrees in Li-6 would also "burn" much more efficiently and completely than previously expected.

A comparison between pre- and post-Bravo yield predictions is shown in Table 3-5; most likely yields are shown in parentheses (there was still a great deal of uncertainty). The Yankee shot yield prediction was changed when the RUNT II was substituted for the JUGHEAD:⁵⁴⁹

Table 3-5 CASTLE Yield Revisions

<u>Shot</u>	<u>Pre-Bravo Prediction</u>	<u>Post-Bravo Prediction</u>
Romeo	4 to 8 MT (6 MT) (RUNT device; 7.5% Li-6)	1.5 to 15 MT (8 MT)
Union	1 to 6 MT (3-4 MT) (ALARM CLOCK; 95% Li-6)	1 to 18 MT (5-10 MT)
Koon	0.33 to 2.5 MT (MORGENSTERN device)	0.33 to 4 MT (1.5 MT)
Yankee	6 to 10 MT (8 MT) (JUGHEAD device)	7.5 to 15 MT (9.5 MT) (RUNT II device; 40% Li-6)
Nectar	1 to 2.5 MT (1.8 MT) (ZOMBIE device)	1 to 5 MT (2-3 MT)

The unexpected yield of the SHRIMP device also caused a significant rearrangement of the shot sequence. The base camp structures at Eninman island at Bikini had suffered severe blast damage; it was obvious that scientific test installations on the island for the Koon shot would be subjected to further damage by intervening shots if Koon were not rescheduled.

The Union, Yankee, Nectar, and Romeo detonations scheduled to follow Bravo were barge shots, located closer to Eninman than Bravo had been, and it was now possible that Union might give a yield as large as Bravo.

It appeared that Koon would have to be fired as soon as possible; however, since the MORGENSTERN device could not be readied in time, the RUNT surface zero was relocated to the new Bravo crater to position it as far as possible from Eninman. Romeo, the RUNT test, was rescheduled for March 13.

⁵⁴⁹OPERATION CASTLE, Report of the Manager, Santa Fe Operations, Pacific Proving Ground, Spring 1954, p. 4.

The Romeo shot was chosen to be next in the series because JTF 7 scientists believed that, of those test devices which could be made ready in the short time available, the unenriched lithium-fueled RUNT devices was least likely to produce a yield of more than 15 megatons, thus minimizing the chances of further damaging the Koon installations.⁵⁵⁰

Substitutions and changes were made during the operation to other test devices as the series progressed. There were also repeated delays due to unfavorable weather or winds which forced many revisions and last-minute improvisations in many experiments. The revisions in the shot sequence and locations resulting from effects of the first and subsequent shots had little effect on the facilities required beyond an expansion of the barge shot program.

The AEC had been misled by results of the IVY Mike shot which seemed to suggest that radioactivity from high-yield detonations would not be carried hundreds of miles at intensities sufficient to be hazardous to health, as happened during CASTLE Bravo (only about 5% of the total IVY Mike fallout was ever located).⁵⁵¹ After Bravo, more restrictive weather criteria for testing were established, which led to even more delays and prolonged the operation.⁵⁵²

As of March 6, the new schedule for the remaining shots was to fire Romeo on March 13; Koon, March 22; Union, March 29; Nectar, April 8; Yankee, April 17; and Echo on April 24. Echo was moved to the end of the schedule because UCRL workers assigned to assemble the RAMROD were now required to ready the Koon (MORGENSTERN) device first.

⁵⁵⁰Semiannual Historical Report, Headquarters, Field Command, The Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1954 - 30 June 1954, John Wendell Bailey, Lt. Col., QMC, Field Command Historian, 30 June 1954, p. 335.

⁵⁵¹Memorandum for Record dated 12 April 1954 from Alvin C. Graves, Scientific Director, JTF 7, and Maj. Gen. P. W. Clarkson, USA, Commander, JTF 7, Subject: BRAVO Shot, Operation CASTLE, p. 2.

⁵⁵²In a letter dated April 22, 1954 to Lt. Col. Herschel H. Slater, USAF, from Col. B. G. Holzman, then Acting Chief of Staff of the Air Force Special Weapons Center at Kirtland AFB, the latter notes that after SANDSTONE, he had recommended against use of the Marshall Islands for further testing because of wind and weather problems; he closed by saying, "But, then I suppose if one waits long enough, there are occasional situations when the winds are suitable."

Assembly of the ALARM CLOCK was finished on March 2; assembly of the RUNT device was complete two days later. RUNT was installed on a barge the next day and taken to Bikini via LSD on March 7. That day, the barge was anchored over the new crater dug at Bikini by shot Bravo a little over a half-mile southwest of Namu island. **Figure IV-30** shows the RUNT; the Yankee shot device was externally identical.

Detonators and final components, including a new gas booster cylinder, were installed by March 12, and the device was finally fired at 0630 hours on March 27 following several delays caused by unfavorable upper altitude winds.

Romeo had been scheduled for firing on March 13, 15, 19, and 21; on each occasion, instrumentation had to be prepared and timing devices reset. These repeated delays led to many disruptions of planned experiments; in addition, personnel at Bikini were constantly exposed to Bravo fallout radiation which accumulated in their bodies (almost the entire atoll was contaminated by this time).

The many delays attendant to the Romeo shot inspired Dr. Alvin Graves of LASL to Shakespearean poetry:

What night is this? We wish we knew.
We think it is a minus two,
But since it could be a minus four,
Or, heaven help us, minus more,
We guess there's nothing we can do
But sit and wait the long nights through.
Until some dawn, with brilliant light,
Our Romeo doth end the night.
Then we will know, come what may come,
The night before was minus one.

A minute after firing, the Romeo cloud had climbed 44,000 feet and reached a diameter of over six miles; ten minutes after zero hour, the cloud had expanded to a diameter of nearly sixty miles. It finally ascended to 110,000 feet.

Soon after Romeo, a civilian aircraft on a military charter flight to Tokyo passed through "an unusual cloud" between Wake and Marcus islands, several hundred miles north of Bikini. While passing through the cloud, the cockpit windows were pelted with what looked like particles of coral dust. This deposit did not wash off, even after the plane later flew through heavy rain squalls.

Examination of the plane by Geiger counter showed no radioactive contamination except for slight readings from dust on the cockpit window glass. The aircraft was washed down before returning to the U.S., and the entire incident was classified "secret." Considerable effort was made to keep the news from Tokyo papers, as the Japanese were then very

alarmed over the "Lucky Dragon" accident and the AEC did not want any more unfavorable publicity.⁵⁵³

At the beginning of the operation, seven devices were scheduled for testing. Five of these were designed by LASL and two by UCRL. During CASTLE, on-site diagnostic measurements were made which caused changes in the device designs and even removed two from the schedule. The first two devices fired, LASL's SHRIMP and RUNT, gave yields considerably above those predicted and led to the conclusion that a solid-fueled weapon was practical for military stockpiling purposes.

Since this type of weapon was appreciably simpler to use than a liquid deuterium-fueled weapon, the Los Alamos JUGHEAD (EC16) device was removed from the schedule on April 1 and a second, modified RUNT device, called RUNT II, inserted in its place for the Yankee shot. The EC16 device shot was canceled after the liquid-fueled weapon was readied at Parry island.⁵⁵⁴

By April 2, not only had the JUGHEAD shot been canceled, but the EC16 was to be eliminated from the nation's nuclear weapons stockpile.⁵⁵⁵

Following the Romeo detonation, the shot sequence was again altered. Because of long delays due to unfavorable winds or weather, a capability of firing at either Eniwetok (for small yield devices) or Bikini had to be established. If weather were bad at Bikini and good at Eniwetok, for example, then much time could be saved if a shot could be fired at either atoll. Two tests, Echo and Nectar, were now assigned to Eniwetok and the remaining three, Koon, Union, and Yankee, were assigned to Bikini.

The first UCRL device test at CASTLE, shot Koon, would test fire the MORGENSTERN device. The unclassified components of both the MORGENSTERN and RAMROD devices were shipped to the PPG in December 1953; the classified components were sent aboard the USS CURTISS in January 1954. A portion of the RAMROD was flown to the PPG early in March 1954 aboard an Air Force C-124 aircraft.

⁵⁵³Overseas National Airways teletype memorandum dated 2 April 1954 to Mr. Tompkins from Mr. Larsen.

⁵⁵⁴OPERATION CASTLE, Joint Task Force Seven, Commander Task Group 7.3, Final Report, 14 June 1954, p. A-1.

⁵⁵⁵Minutes of Thirty-Ninth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 31-April 2, 1954, p. 31.

Assembly of the complex, liquid-fueled refrigerated UCRL RAMROD device was done in a shot cab on Eberiru island at Eniwetok, starting early in January 1954 and was completed, including necessary cryogenic testing, by March 6, except for installation of the primary fission stage.

One difficulty during assembly was the development of a frozen air plug in one of the deuterium lines similar to the problem in October 1952 with the IVY Mike device. This plug caused a nonexplosive bursting of several of the cryostat components and approximately a week's effort was required to return the system to operating condition.

The abandonment of the Bikini island bases after Bravo and the revision of the shot schedule placed a big workload on the BELLE GROVE and other Navy transport ships. There was several million dollars worth of equipment on Eninman island which was originally slated to remain at Bikini until after the final shot. The new schedule meant that it had to be returned to Eniwetok before the third shot. Fourteen LST trips were made between March 1 and April 7; 2,762 tons of equipment were evacuated.

The solid-fueled MORGENSTERN device was constructed in a shot cab on Eninman island at Bikini during the latter part of January and early February. **Figure IV-31** shows the Koon shot cab and support camp at the western end of Eninman island.

Except for the installation of its primary stage and high explosive detonators, assembly of the MORGENSTERN was substantially complete before Bravo shot. The bomb was undamaged by shot Bravo, and was detonated by radio command as the Koon shot at 0625 hours local time, April 7. The explosion completed the destruction of the Bravo-damaged base camp facilities at Eninman and resulted in a temporary closure of the Bikini lagoon anchorage.⁵⁵⁶

The Morning Star That Didn't Rise

UCRL was also cursed by bad scientific estimates: the MORGENSTERN exploded with only a small fraction of its predicted force. Its design had failed to yield enough energy to cause an appreciable compression and "burn" of the lithium deuteride fusion fuel charge in its secondary stage: there was an extremely long time interval between primary and secondary reactions, suggesting poor propagation of radiation from the primary stage to the secondary.

Observers aboard ships south of Bikini atoll saw only a faint

⁵⁵⁶Teletype dated April 7, 1954 to Chairman, USAEC, from Commander, JTF 7.

glow beneath low-lying rain squall clouds, and nothing further.⁵⁵⁷ At Eniwetok and Parry islands at Eniwetok atoll, approximately 175 nautical miles from Koon ground zero, each of the preceding CASTLE tests had bathed the islands in a brighter-than-day brilliance for several seconds. The light from Koon shot was like the flare of a photographic flash bulb, and faded almost immediately. It was instantly obvious that something had gone seriously wrong.

Of its total 110 KT yield, only 10 KT of the MORGENSTERN's force was generated by thermonuclear reactions.⁵⁵⁸ The failure of the device scattered uranium and plutonium near the burst point.⁵⁵⁹ The MORGENSTERN "fizzle" allegedly resulted from a simple design flaw engendered by the novelty of the technology and the weapons-design inexperience of Livermore scientists.⁵⁶⁰

⁵⁵⁷Semiannual Historical Report, Headquarters, Field Command, The Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1954 - 30 June 1954, John Wendell Bailey, Lt. Col., QMC, Field Command Historian, 30 June 1954, p. 337.

⁵⁵⁸Memorandum to Distribution dated April 7, 1954 from Vern Denton, Subject: Test News Letter.

⁵⁵⁹OPERATION REDWING 1956, DNA 6037F, Defense Nuclear Agency, Washington, D. C., 1 August 1982, p. 45.

⁵⁶⁰York, MAKING WEAPONS, TALKING PEACE, p. 78. Dr. York states that UCRL always pushed the "state of the art" of nuclear weaponry:

Our working philosophy, which I set out at the very beginning (of the laboratory) and which everyone readily accepted, called for always pushing at the technological extremes. We did not wait for higher government or military authorities to tell us what they wanted and only then seek to supply it.

Instead, we set out from the start to construct nuclear explosive devices that had the smallest diameter, the lightest weight, the least investment in rare materials, or the highest yield-to-weight ratio or that otherwise carried the state of the art beyond the currently explored frontiers. We were completely confident that the military would find a use for our product after we proved it and that did

The device name, MORGENSTERN, which means "morning star" in German, had both intended and unintended significance. Had the device worked properly, the early morning shot would indeed have looked like a "morning star."

The chief reason for the name was due to a "crazily" shaped secondary, which featured spikes radiating from a central sphere. This was a remarkable likeness to the spiked ball at the end of the short chain attached to a club which comprised the medieval European war device also called MORGENSTERN.

These spikes may have been an early effort by Teller and his colleagues to compress and ignite a central thermonuclear fuel region via implosive jets; this concept, if successful, might have led to small, very high-yield spherical secondary configurations in lieu of more conventional cylindrical designs.

According to sources, the MORGENSTERN contained a LASL-furnished primary which generated a much stronger neutron flux than anticipated by UCRL. This resulted in considerable preheating of the spiky secondary, in effect "burning up" the charge before it could be properly ignited. The primary

indeed usually turned out to be true.
(MAKING WEAPONS, TALKING PEACE, p. 75).

So much for the question of whether DOD or the weapons labs drove the nuclear arms race. A few of Livermore's more daring early schemes ended in embarrassingly-public test failures at Nevada and in the Pacific.

Dr. Norris Bradbury later evidenced some confusion over the UCRL CASTLE test devices:

When LRL (sic) was established in 1952, all of the prior and current work of the Los Alamos Scientific Laboratory was, of course, available to it. They were naturally aware of the direction of our thinking and of our general plans. Presumably not wishing to duplicate these, they elected to go along a line (pushed by Teller but about which Los Alamos was highly skeptical) then known as RAMROD. The first example of this was called Morgenstern tested in 1954. It was a failure, and thereafter LRL returned to more conventional geometries with which they have since done good work. (Letter dated August 30, 1961 to Honorable Clinton P. Anderson from Norris Bradbury, Director, LASL.)

essentially "cooked" the secondary instead of allowing it to be ignited with its full force.

According to one source, the MORGENSTERN device was the prototype of "a crucial enabling technology for (an) alternate approach to thermonuclear explosives which (Dr. Edward) Teller had been pursuing since 1942" and it "failed apparently very narrowly, and effectively suspended this line of development for another two decades." The MORGENSTERN test "pointed toward highly efficient, very large yield thermonuclear weapons of the general flavor which were subsequently favored by the Soviet Union."⁵⁶¹

The surprisingly low yield of the UCRL device, combined with the success and high credibility of diagnostic measurements associated with the shot, and the hydrodynamic similarity of the MORGENSTERN and the RAMROD (both shared a common LASL primary design), led UCRL scientists to conclude that the RAMROD device slated for the Echo shot would also fail and the shot was canceled on April 12.⁵⁶²

The failure of MORGENSTERN and the cancellation of the RAMROD test were bitter disappointments to Livermore. Following had on the heels of the UPSHOT-KNOTHOLE "fizzles," these events occasioned even more scorn and derision from LASL.

The RAMROD device was disassembled and its parts stored in the Camco building in the CMR area on Parry at Eniwetok. At

⁵⁶¹"The Development of Nuclear Explosives," Lowell Wood and John Nuckolls, article in ENERGY IN PHYSICS, WAR AND PEACE, Hans Mark and Lowell Wood, eds., Kluwer Academic Publishers, Norwell, Massachusetts, 1988, pp. 315, 316. According to Dr. Herbert York, the MORGENSTERN was the last Livermore device designed by Edward Teller.

⁵⁶²York, THE ADVISERS, p. 136; letter dated April 26, 1954 to Honorable Robert LeBaron, Chairman, MLC, from Lewis L. Strauss, Chairman, USAEC. Both the RAMROD and MORGENSTERN devices shared a common primary or radiation channel design. According to "Timetables of Weapon Events," the UCRL TX-22 device was canceled at the end of April 1954. The TX-22 was the first UCRL device to receive an AEC "TX" designation (see Volume VI). Wood and Nuckolls claim, in the article cited above, that contrary to the Strauss letter cited, the RAMROD was sufficiently different from the MORGENSTERN in several critical respects that a test of the RAMROD probably would have succeeded. (ENERGY IN PHYSICS, WAR AND PEACE, p. 316; Wood and Nuckolls imply that Edward Teller never had a bad idea in his life.) Dr. York has stated that the RAMROD test "was (to have been) of far greater significance than the (MORGENSTERN test)." ("30 Years of Technical Excellence, 1952-1982," Lawrence Livermore National Laboratory, 1982, p. 6.)

the conclusion of CASTLE, the RAMROD components were returned to the continental U.S. for reclamation.

Ironically, at the 39th GAC meeting on April 9, Dr. Willard Libby had wondered aloud whether Los Alamos should be criticized for being "too conservative." He felt that the LASL weapons development program was dominated by theoretical physicists, and that bolder experimentation, such as UCRL was doing, was in order.

When Libby -- in a surprising show of ignorance of the history of the thermonuclear weapons program -- asked if the "emergency capability" program might have been achieved sooner and better if the TX-14 had been fired in lieu of the SAUSAGE during Operation IVY in November 1952, Dr. John von Neumann replied that the design of the TX-14 depended upon results of the SAUSAGE test. Libby also overlooked the fact that insufficient Li⁶D was on hand in late 1952 to fire the solid-fueled TX-14.⁵⁶³

Von Neumann remarked about the boldness of the Livermore program; he said that if their tests were successful, they would continue to be bold and if the tests were unsuccessful, "no one would dare to be quite so bold."⁵⁶⁴ In lieu of the failed MORGENSTERN and canceled RAMROD tests, UCRL might not have been feeling so bold.

On the day of the Koon shot, following the Bravo and Romeo tests, AEC Chairman Lewis Strauss wrote to the chairman of the JCAE to inform him of the breathtaking progress being made in the Pacific:

For your information, as a consequence of the recent SHRIMP and RUNT device tests at Castle, the Commission has determined that the thermonuclear portion of the JCS requirements should be met with dry weapons utilizing enriched and unenriched lithium.

Accordingly, liquid-fueled EC16 (JUGHEAD) weapon production has been canceled. [deleted] weapons will be provided [deleted] toward meeting Emergency

⁵⁶³Letter dated August 30, 1961 to Honorable Clinton P. Anderson from Norris Bradbury, Director, LASL ("... the special material required for SHRIMP was not available at the time Mike was detonated").

⁵⁶⁴Minutes of Thirty-Ninth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 31-April 2, 1954, p. 5.

Capability and July 1, 1955 large-yield requirements. Some EC14 (ALARM CLOCK) weapons will contain lithium-6 to the extent that lithium separation capacity permits.

Additional EC17 (RUNT) weapons utilizing unenriched lithium will be produced toward meeting the remainder of the number required. A small number of EC24 weapons are also being provided as part of the Emergency Capability program.

Relative to the matter above is the decision also made by the Commission to eliminate the EC16 (JUGHEAD) shot in the current CASTLE series. Present intentions are to substitute in its place the test of an additional RUNT device with different thermonuclear booster components to determine the feasibility of [deleted] in this weapon.⁵⁶⁵

In the aftermath of international uproar over the widespread Bravo shot fallout, the AEC was forced to offer good reasons to continue Pacific testing in case a movement arose in Congress to stop the CASTLE operation. The top-secret justifications put forth were almost entirely military in nature and were not intended for public disclosure.

One of the reasons given for the requirement for high-yield weapons was to compensate for bombing errors:

1. Because we have a requirement established by the Joint Chiefs of Staff for a range of high-yield weapons.
2. Because these are proof-tests of some weapons which are already scheduled to enter the stockpile.
3. Because all our operations are directed to the end that we make (the) most effective and economic use of available materials for defense purposes.
4. Because we are entering a relatively new, unexplored field whose potential

⁵⁶⁵Letter dated April 7, 1954 to Honorable Sterling Cole, Chairman, JCAE, from Lewis L. Strauss, Chairman, USAEC.

can only be determined by full-scale tests.

5. Because the only previous high-yield (weapon) proto-type (sic) utilized a liquid fuel which caused almost unacceptable operating difficulties. Therefore, (the) search for effective and reasonably economic dry fuels must continue.
6. Because from an operational standpoint, every effort must be made to reduce weapon size and weight to enhance deliverability capabilities.
7. Because another Joint Chiefs of Staff requirement is for a weapon of even higher yield than those previously tested, which is intended to obviate bombing errors while still producing maximum destructive effectiveness.
8. Because there are still newer theories as yet untried which must be furthered if we are to retain our leadership in the thermonuclear and atomic fields.
9. Because we have not unduly or intentionally hazarded (sic) anyone and we have no intention of so doing.
10. Because we have no assurance that other powers are not engaged in similar research.
11. Because no one raised any "hue or cry" (sic) after the Russian (nuclear) tests. We seem to be the victims of a campaign directed to relinquishing our leadership in this field.
12. Because there are other related fields about which information vital to our defenses must be obtained. Radiological safety, blast effect, and related phenomena concerning high-yield weapons are of (the) utmost importance to defense planning.
13. Because we need the best weapons we can get. The adages "Walk softly but carry a big stick" or "Keep your powder dry"

have never been refuted.⁵⁶⁶

After the Echo shot was canceled, a revised CASTLE shot schedule was released on April 13. Union would be fired three days later; Nectar on April 20; and Yankee on April 27.

The Herrick L. Johnston and Camco personnel left Eniwetok in April after cancellation of Echo.⁵⁶⁷ Roll-up of all TU 3 equipment and supplies was completed on April 16, and materials returned soon afterwards to the U.S.

In the meantime, CASTLE rolled on. The ALARM CLOCK, first assembled on March 2 (it was to have been fired on March 11 as the second shot in the test series), was later disassembled when the Union shot was postponed. The device was reassembled on April 8 and loaded on a barge. The barge was taken to Bikini via LSD the next day; anchoring was completed and firing cables checked out a day later. Detonators were installed on April 15, but the scheduled shot date of April 16 was postponed again, this time by a combination of bad weather and by an unusual request for a delay from AEC headquarters.⁵⁶⁸

The AEC request resulted in an "indefinite" delay in shot date, with instructions to the task force to be ready to fire on 24-hours notice.⁵⁶⁹ After being rescheduled and canceled on

⁵⁶⁶Memorandum dated April 7, 1954 to McKay Donkin, Special Assistant to the Chairman (of the USAEC) from Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC. It is not clear whether this information was ever used by Strauss; it may have been offered to the Chairman of the JCAE. Item 7, about high yield offsetting bombing errors, was also applicable to early ICBMs with their relatively-poor accuracies and large CEPs.

⁵⁶⁷DNA 6035F, pp. 56, 405.

⁵⁶⁸Teletype dated April 16, 1954 to Commander, JTF 7, Eniwetok, from K. D. Nichols, General Manager, USAEC, Washington, D.C. The reasons for this request are not included in this message, although later Pacific tests were occasionally delayed for political reasons, such as the imminence of Japanese elections where leftist and Communist candidates might, to their advantage, criticize U. S. testing.

⁵⁶⁹Semiannual Historical Report, Headquarters, Field Command, The Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1954 - 30 June 1954, John Wendell Bailey, Lt. Col., QMC, Field Command Historian, 30 June 1954, p. 338. The reasons for this curious request for a delay are not yet known.

April 21, 23, 24, and 25, the ALARM CLOCK was finally detonated at 0610 hours on April 26 in Bikini lagoon off Yurochi island.

Within a minute, the fireball rose to 35,000 feet; it was surrounded by a cloud nearly five miles in diameter. At ten minutes after zero hour, the cloud had a diameter of nearly 24 miles. The cloud top rose to 94,000 feet.

Although the shot schedule had been changed by this time to fire Nectar immediately after Union, poor weather conditions at Eniwetok made it necessary to fire Yankee next and Nectar afterwards.

The substitute RUNT II device, added to the schedule after the Los Alamos EC16 test was canceled following the hugely-successful Bravo and Romeo shots, was airlifted by a Strategic Air Command B-36 from the U.S. to Eniwetok and arrived on April 16. The device was accompanied by a B-2 bomb lift shipped to the Pacific aboard a C-124.

The modified RUNT device contained no tritium or boron-10.⁵⁷⁰ Final assembly was completed ten days later; it was loaded on a barge on the next day, taken aboard an LSD to Bikini on April 30, and anchored in Bikini lagoon over the Union shot crater on May 2. Detonators were installed on May 4, and the device was fired at 0610 hours the following morning.⁵⁷¹

The shot cloud soared to 34,000 feet within a minute of zero hour; it swelled to a diameter of six and a half miles in the same minute. At zero plus ten minutes, the cloud was 51 miles in diameter and rising to an eventual height of 110,000 feet.

The light flash from the Yankee shot was observed by the crew and passengers aboard a transpacific Pan American Airways flight enroute to Wake and Guam islands from Hawaii.⁵⁷² Closer in, the crew aboard a B-36 effects aircraft observed a

⁵⁷⁰Minutes of the 39th Meeting of the General Advisory Committee to the USAEC, March 31, April 1 and 2, 1954, pp. 12, 13, 31.

⁵⁷¹According to NWD 86-2 (p. 16), the Yankee shot was a test of an EC17. There is no confirmation of this statement at this time. According to NWD 86-2, the Romeo shot was a test of an EC14; however, AF ATOMIC ENERGY PROGRAM, Vol. IV, pp. 211-224, repeatedly identifies the TX-14 as the ALARM CLOCK, and the ALARM CLOCK tested at CASTLE was fired during the Union shot.

⁵⁷²Teletype dated May 6, 1954 from Commander Joint Task Force Seven to Chairman, USAEC.

distinct two-pulse red glow through asbestos cabin window curtains; considerable smoke was produced in the rear gunners' compartment, as had been the case for the same type of aircraft during the Bravo shot.

The Yankee shot heat pulse was sufficiently strong to activate fire warning sensors on four of the plane's six reciprocating engines for four seconds.

After the Echo shot was canceled at Eniwetok, the Nectar shot of the Los Alamos ZOMBIE device was relocated from Bikini to Eniwetok. The bomb was flown into Eniwetok on March 5, and final assembly completed on April 14 following several postponements. After being loaded on a barge, it loaded on a barge which was towed on April 16 to the IVY Mike crater (where Elugelab had been before November 1, 1952) at the north end of Eniwetok lagoon. After the barge was securely anchored, the firing cables were checked out.

On April 17, the support jacket around the weapon was removed and dummy lugs attached to the case. Ten days later, nuclear arming was completed; the shot was scheduled for firing on May 4, 5, and 11, but the bomb was not detonated until 0620 hours on May 14 following more delays due to unfavorable winds aloft.⁵⁷³

The Nectar fireball ascended to 25,000 feet a minute after zero hour; it was wreathed in a cloud over three and a half miles in diameter. At zero plus ten minutes, the cloud was nearly 28 miles in diameter. The cloud top finally ascended to 72,000 feet.

Thus Operation CASTLE concluded with the firing of six shots.⁵⁷⁴ The final shot, Nectar, was fired 29 days after the originally-scheduled date for the sixth shot in the series (see Table 3-4). Five of the original seven devices scheduled to be fired (SHRIMP, ALARM CLOCK, ZOMBIE, RUNT, and MORGENSTERN) were detonated; two (JUGHEAD and RAMROD) were canceled; and one device (RUNT II) not on the original schedule was substituted later.

The technique of using barges as shot platforms was practical and finally essential to CASTLE. It allowed great flexibility, especially when the location of all the remaining Bikini shots had to be changed after the heavy

⁵⁷³Table 2, Actual Shot Schedule, OPERATION CASTLE, Report of the Manager, Santa Fe Operations, Pacific Proving Ground, Spring 1954, p. 5.

⁵⁷⁴OPERATION CASTLE, Report of the Manager, Santa Fe Operations, Pacific Proving Ground, Spring 1954, p. 19.

contamination following shot Bravo. The ALARM CLOCK was fired ten miles from its original planned position and the Nectar shot was fired at Eniwetok instead of Bikini. Neither change would have been possible with fixed land positions.⁵⁷⁵

The CASTLE test devices were relatively "dirty," that is, a large proportion of their total yield came from fission of plutonium, uranium-235, and uranium-238. Over 63% of the total yield of the CASTLE tests was from fission, a very high proportion.⁵⁷⁶ Table 3-6 lists the fission and fusion yields for all CASTLE tests.

Table 3-6 CASTLE Fission-Fusion Yields⁵⁷⁷

<u>Shot</u>	<u>Fission Yield</u>	<u>Fusion Yield</u>	<u>Total Yield</u>	<u>% Fission Yield</u>
Bravo	10.0 MT	5.0 MT	15 MT	66.66
Romeo	7.0 MT	4.0 MT	11 MT	63.63
Koon	0.1 MT	0.01 MT	0.11 MT	90.90
Union	5.0 MT	1.9 MT	6.9 MT	72.46
Yankee	7.0 MT	6.5 MT	13.5 MT	51.85
Nectar	1.35 MT	0.34 MT	1.69 MT	79.88
Total CASTLE Fission Yield: 30.45 MT				
Total CASTLE Fusion Yield: 17.75 MT				
Total CASTLE Fission + Fusion Yield: 48.2 MT				

⁵⁷⁵OPERATION CASTLE, Report of the Manager, Santa Fe Operations, Pacific Proving Ground, Spring 1954, p. 20.

⁵⁷⁶DNA 6035F, p. 34. According to "Known U.S. Nuclear Tests," NWD 86-2 (p. 16), the combined fission yield of all U.S. nuclear tests at the Nevada Proving Ground and the Pacific Proving Grounds between 1952 and 1954 was 37 megatons (this figure is based on a statement by an AEC spokesman in "Fallout from Nuclear Weapons Tests," Hearings before the Special Subcommittee on Radiation of the Joint Committee on Atomic Energy, U.S. Congress, Volume I, May 5-8, 1959, p. 23). These series included Operations TUMBLER-SNAPPER and UPSHOT-KNOTHOLE in Nevada, and IVY and CASTLE in the Pacific.

⁵⁷⁷Table I, attachment to letter dated December 21, 1954 to Dr. Bugher, Division of Biology & Medicine, USAEC, Washington, D.C. from Col. Robert N. Isbell, USAF, Chief, Atomic Operations Division, Air Research & Development Command, USAF.

Total CASTLE Fission Yield Percentage (all shots):
63.17
Average CASTLE Fission Yield Percentage (per shot):
70.9
Average CASTLE Fission Yield Percentage
(for shots above one megaton total yield): 66.9

The wide range in predicted yields incident to novel experimental devices made it difficult to set recording equipment in the proper range, causing the loss of much information. One set of experiments suffered loss of nearly all data during the Bravo shot: although the recording equipment functioned satisfactorily, the reinforced concrete scientific station it was in had not been designed for as high a yield as SHRIMP delivered, and airborne radioactively-contaminated particles entered the blockhouse and fogged the photographic film.

Conversely, because of the unexpectedly-low yield of the MORGENSTERN device and a rainstorm at Eninman island at shot time that obscured visibility, much data for Koon shot was not obtained.

As was the case at Eniwetok during IVY, several of the CASTLE shots substantially rearranged some of the Bikini atoll and lagoon geography. The 15 MT Bravo shot, fired on a man-made island at the end of a causeway over the reef, dug a crater 6,000 feet in diameter and 240 feet deep; the 110 KT Koon shot, fired on Eninman island, blasted a hole 990 feet in diameter and 75 feet deep; and the 6.9 MT Union shot, fired on a barge over 160 feet of water, made an excavation 3,000 feet in diameter and 90 feet deep in the bottom of Eniwetok lagoon.⁵⁷⁸

Figure IV-32 is a contour map of the Koon shot camp on Eninman before the test; **Figure IV-33** shows the contours of the same area after the shot.

A large number of weapons effects tests were also conducted. One interesting experiment was the effect of high-yield nuclear explosions on underwater naval mines, primarily to determine the utility of nuclear explosives in clearing minefields before amphibious landings. A large subsurface minefield, consisting of 123 inert unarmed plaster-filled U.S. and Soviet naval mines of various types, with detonators installed, was set 180 feet deep in seven rows at distances

⁵⁷⁸OPERATION CASTLE, Summary Report of the Commander, Task Unit 13, Military Effects, Programs 1-9, Pacific Proving Grounds, March-May 1954, WT-934 (EX), Headquarters Field Command, Armed Forces Special Weapons Project, Sandia Base, Albuquerque, New Mexico, January 30, 1959, p. 65.

from 2,000 feet to 15,000 feet from surface zero of the Union shot, and produced data on the extent of neutralization of these mines by the detonation.

Damage ranged from 0% at 15,000 feet to 100% at 2,000 feet from surface zero. Forty-eight of the mines were returned to Pearl Harbor, Hawaii for further study. **Figure IV-34** shows the damage to a Russian RIA-type naval mine after the Union shot.

The yield of the Bravo shot was approximately 25% greater than the positioning yield used for the effects studies of aircraft in flight. A weapons effects program B-36D (the same plane used at Operation IVY) was exposed to an overpressure which caused mechanical damage that required replacement of the bomb bay doors, aft lower plexiglas scanners' blisters, and the radar antenna radome; the undersides of all six engine nacelles and the landing gear and lower aft gun turret bay doors were also damaged.

Thermal effects on this airplane during the 13.5 MT Yankee shot were also considerable: a 322⁰ F thermal pulse buckled the elevator skin in four places and blistered and peeled a large portion of the paint on the lower surfaces of the horizontal stabilizer and elevators. The plane's undersides were painted with white enamel over a protective coat of clear lacquer to improve reflecting properties of the skin. This thermal pulse was accompanied by a 20-second duration radiation pulse.

During the thermal pulse, the crew of this aircraft reported seeing a bright red glow through aluminized asbestos shields placed over the aircraft's cabin windows. In the aft gunner's compartment, a great deal of smoke was produced by the burning of the sponge rubbers pads around the rims of both lower scanner's sighting blisters.

This damage led to the conclusion that "for the relatively slow-speed, propellor-driven aircraft such as the B-36, the effects of the blast wave resulting from the detonation of a nuclear weapon may very well limit the operational capabilities of the aircraft during a strategic mission..."⁵⁷⁹

⁵⁷⁹OPERATION CASTLE, Summary Report of the Commander, Task Unit 13, Military Effects, Programs 1-9, Pacific Proving Ground, March-May 1954, WT-934 (EX), Headquarters Field Command, Armed Forces Special Weapons Project, Sandia Base, Albuquerque, New Mexico, January 30, 1959, pp. 6, 7, 8, 76; OPERATION CASTLE, Pacific Proving Grounds, March-May 1954, WT-925, Project 6.2a, Blast and Thermal Effects on B-36 Aircraft in Flight, Headquarters Field Command, Armed Forces Special Weapons Project, Sandia Base, Albuquerque, New

Only high-speed, jet-powered bombers would be suitable for delivery of high-yield, unretarded free-fall thermonuclear bombs.

At the conclusion of the CASTLE test series, the hydrogen liquefaction plant at Eniwetok was mothballed, as it probably would not be needed for future weapons tests. The plant was shut down and placed in standby condition.

The total cost for Operation CASTLE between January 1, 1953 and May 31, 1954 was \$96,097,649. Operation IVY had cost \$65,933,163 during a similar period of planning and operations between July 1951 and December 1952.⁵⁸⁰

On May 13 (in the U.S.; May 14 in the Pacific), the day after the Nectar shot, AEC Chairman Lewis Strauss issued the following cryptic statement to the press:

The 1954 series of thermonuclear tests at the Atomic Energy Commission's Pacific Proving Ground has been completed.

The tests were successful in the development of thermonuclear weapons. They were essential to our national interest and have contributed materially to the security of the United States and the free world.

The tests being concluded, within a few days sea and air traffic may be safely resumed within the "warning area" which was set up for safety purposes for the time when the tests were taking place. Official notice to airmen and mariners will be published.⁵⁸¹

Mexico, June 1956, pp. 58, 59, 74, 123, 124.

⁵⁸⁰Moore and Bechanan, p. 325.

⁵⁸¹Joint DOD-AEC press release dated May 13, 1954, Subject: Statement by L. L. Strauss, Chairman, AEC, and Charles E. Wilson, Secretary, DOD. A longer internal draft of this statement dated about three weeks earlier on April 23, 1954, said:

The series of thermonuclear tests by the United States at the Atomic Energy Commission's Pacific Proving Ground has been completed. The series included six detonations.

The tests were highly successful in the

Results of CASTLE

CASTLE achieved several scientifically and militarily useful results. A number of devices, with a variety of weights, compositions, and yields, were now available for "weaponization." The tests established "emergency capability" on two weapons, the EC14 and EC17. Stockpiling of the EC14 had started just before Operation CASTLE; a handful of EC17s were being stockpiled as CASTLE concluded.

The practicability of using solid-fueled lithium-6 deuteride and lithium-7 hydride weapons was amply demonstrated; neither large quantities of tritium nor liquid deuterium were required as thermonuclear weapon components.⁵⁸² Small amounts

development of thermonuclear weapons. They have contributed materially to the security of the United States and the free world against aggression.

New knowledge obtained in this testing program will lead to manifold multiplication of the explosive potential of our resources of fissionable materials. This is the result of years of scientific and technological research and could not have been accomplished without the large investments of manpower, money, materials, and time this nation has thus far made in its program for development and production of nuclear weapons.

The rumors that the initial detonation was "out of control" were without foundation in fact. ...

The nation and the free world owe for their increased security a debt of gratitude to the men of Joint Task Force Seven and associated laboratories who conducted the tests for the Atomic Energy Commission and the Department of Defense. Special mention for their part in this enterprise should go to the management and staff members of the Los Alamos Scientific Laboratory as the agency primarily responsible for these thermonuclear developments.

⁵⁸²Atomic Energy Commission, Program Status Report, December 31, 1954, Part III - Weapons Program, pp. 5, 6. Hewlett and Holl (p. 181) characterize the CASTLE test devices as "evidence of exceptional scientific ingenuity and

of tritium were later used to boost primaries in thermonuclear weapons.

In addition, CASTLE furnished valuable experimental information that could be used in the design of lighter thermonuclear weapons; gave highly significant effects information on high-yield explosions; provided experimental evaluation of "boosted" fission primaries; and significantly reduced the national requirements for tritium production, allowing the AEC to use its new reactors to produce more plutonium, rather than tritium.⁵⁸³

CASTLE also established that low-enriched and unenriched lithium were suitable thermonuclear fuels, which resulted in the production of weapons unlimited by a shortage of lithium highly enriched in the lithium-6 isotope. The use of Li-7 also meant that production requirements for large quantities of Li-6 could be reduced significantly.

Just as Operation SANDSTONE in 1948 had demonstrated many promising new fission weapon principles, CASTLE showed the way to many fusion weapon design improvements: with a few notable exceptions, every new design principle incorporated in the CASTLE devices seemed to work, often beyond the wildest expectations of its inventors.

Just after the Bravo shot, the AEC moved rapidly to cancel the cryogenic weapons and accelerate production and deployment of solid-fueled devices:

As discussed by the Commission at meeting number 970, we have taken the following actions to realign the thermonuclear weapons program toward elimination of the [deleted] and acceleration of the [deleted] development and production.

- A. Instructions have been issued to the San Francisco Operations Office to curtail development and terminate production of [deleted] San Francisco Operations has notified Camco of this

imagination."

⁵⁸³Hewlett and Holl, pp. 179, 180; OPERATION CASTLE, Report of the Manager, Santa Fe Operations, Pacific Proving Ground, Spring 1954, pp. 20, 21. By April 1954, the new requirement for tritium was, at most, half of the previous requirement, and most of this would be used in boosted weapons and primaries. (Minutes of the 39th Meeting of the General Advisory Committee to the USAEC, March 31, April 1 and 2, 1954, p. 18.)

and a meeting is scheduled for the week of April 5 to discuss contract cancellation.

- B. San Francisco Operations has been requested to consult with LASL and Livermore and submit recommendations as soon as possible as to future AEC support of (the NBS) Boulder Laboratory.
- C. The Emergency Capability (EC) and FY 1955 thermonuclear weapon production has been rescheduled. Elimination of the TX-16 will cause a reduction of [deleted] in the EC stockpile during the months of April and May (1954), but this will be made up by July and the original schedule will be surpassed by a small number of weapons by the end of (the) EC (program) in September.

It is expected that the FY 1955 JCS requirements will be met with little difficulty by production of TX-17 weapons [deleted] Greater difficulties were envisioned for a TX-16 production program and it is doubtful whether JCS requirements would have been met with this weapon.

- D. Instructions have been issued to General Clarkson and Los Alamos to eliminate the TX-16 test shot from the current CASTLE series. The Commission will be consulted on the substitution of a test [deleted] The MLC has given verbal agreement to this change.
- E. Production Division has taken steps to accelerate production of [deleted] in keeping with the required numbers of normal TX-17's for EC and FY 1955 (requirements). To meet the [deleted] schedule in FY 1955, it may be necessary to have industry perform a portion of the lithium deuteride compounding.
- F. (The) MLC and GAC have been notified of the test and production program

changes.⁵⁸⁴

By the time CASTLE was over, the U.S. had a choice of "emergency capability" weapons, all of them solid-fueled. Shortly after CASTLE ended, the AEC canceled all of its contracts for cryogenic weapons research.⁵⁸⁵

A particularly significant result of the CASTLE tests was the discovery that scientists at Los Alamos had seriously underestimated the cross-section of ordinary lithium (Li-7) for tritium production. Only after the tests started did LASL begin experiments on fast neutron interactions with lithium.⁵⁸⁶

This misjudgment showed up most dramatically in the differences between the pre-shot yield estimates and the actual performance of devices during the tests. These variations are summarized in Table 3-7, which also shows the final shot sequence.

The CASTLE tests had shown that not only did lithium-7 generate tritium with moderated (slowed) 14 MeV neutrons, it also produced a slow neutron of its own while it generated tritium; this slow neutron then went on to generate tritium from any lithium-6 that was present:

"They really didn't know," Harold Agnew explains, "that with lithium-7 there was a (n,2n) reaction. They missed it entirely. That's why Shrimp went like gangbusters."

"When the two neutrons come out," says Agnew, "then you have lithium-6 and it went like regular lithium-6. Shrimp was so much bigger than it was supposed to be because

⁵⁸⁴Memorandum dated April 5, 1954 to K. D. Nichols, General Manager, USAEC, from Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC, subject: Revision of FY 1954 and 1955 Thermonuclear Weapon Program. The mention of LiD compounding by private industry refers to the process of chemically compounding lithium salt and deuterium gas, a process normally conducted only at the AEC's Oak Ridge facility.

⁵⁸⁵Hewlett and Holl, pp. 178, 179.

⁵⁸⁶Minutes of the Fortieth Meeting of the General Advisory Committee to the USAEC, May 27-29, 1954, p. 18; "Nuclear Data: The Numbers Needed to Design the Bombs," Ben C. Diven, John H. Manley, and Richard F. Taschek, LOS ALAMOS SCIENCE, Volume 4 No. 7, Winter/Spring 1983, pp. 121, 122.

we were wrong about the cross section." ⁵⁸⁷

Table 3-7 CASTLE Yield Variations⁵⁸⁸

<u>Shot Name</u>	<u>Device Name</u>	<u>Predicted Yield Range</u>	<u>Most Probable Yield Estimate</u>	<u>Actual Yield</u>	<u>%age Err.</u>
Bravo	SHRIMP	4-8 MT	6 MT	15 MT	150
Romeo	RUNT I	1.5-7 MT	4 MT	11 MT	175
Koon	MORGENSTERN	1/3-2.5 MT	1 MT	110 KT	-89
Union	ALARM CLOCK	1-6 MT	3-4 MT	6.9 MT	100
Yankee	RUNT II	6-10 MT	8 MT	13.5 MT	68
Nectar	ZOMBIE	1-2.5 MT	1.8 MT	1.69 MT	-6
Echo	RAMROD	65-275 KT	125 KT	Canceled after Koon "fizzle"	

The majority of neutrons from the fissioning "sparkplugs" and U-238 fusion tampers in the CASTLE test devices had kinetic energies below the 2.5 MeV reaction threshold for tritium production in Li-7, so at least some Li-6 was required, in addition to Li-7, to properly utilize these lower-energy neutrons.⁵⁸⁹ Dr. Bethe's assumption in May 1953 that Li-7 would not be useful was shown to have been spectacularly incorrect.

⁵⁸⁷Rhodes, DARK SUN, p. 541.

⁵⁸⁸Sources: Shot names and actual yields -- NVO-209, p. 4; device names (except JUGHEAD) -- OPERATION CASTLE, Pacific Proving Grounds, March - May 1954, Report of Commander, Task Group 7.1, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, WT-940 (EX), June 1954, pp. 8, 25, 28, 47, 62, 66, 67, 72 (JUGHEAD is from Hines, PROVING GROUND, pp. 161, 162, 183); estimated yield ranges and most probable yield estimate -- OPERATION CASTLE, Operation Plan No. 1-53, Task Group 7.1, 8 December 1953, pp. A-1, A-2. Percentage error calculated by computing (actual yield / most probable yield - 1) x 100. Negative values represent yields below predictions. The most probable yield estimates and predicted yield values are original estimates. After shot Bravo, all of these were revised upwards. The original pre-Bravo yield estimate figures are used to emphasize how badly the yields were underestimated. See "Weapons Physics" in Volume I for more details of the "tritium bonus."

⁵⁸⁹Letter dated 21 October 1989 to author from Lee Barker.

The recalculation of tritium cross-sections meant that less enriched lithium would be required for a given yield, and that the AEC's ADP facility at Oak Ridge would not be overburdened; the facility was already producing enriched lithium in quantities beyond its design specifications.⁵⁹⁰

Based on CASTLE results, the AEC estimated that revised requirements for tritium production were, at most, half the previous requirement. It would not be necessary to enrich all the fuel slugs at the Savannah River plant reactors with U-235, or as many at Hanford as previously planned. If no tritium were required for thermonuclear weapons beyond that required to boost primaries, then the Savannah River reactors would not be required for tritium production after 1956.⁵⁹¹

It might even be possible that no tritium at all would be required. In addition, the new hydrogen weapons would be incredibly inexpensive: whereas in 1954 a ton of TNT cost in the neighborhood of \$500, the amount of fusionable materials required to yield a ton of TNT-equivalent "bang" cost only 10 cents.⁵⁹² This factor of 5,000 would become important for not only military budgeting, but also for plans for future peaceful uses of thermonuclear explosives.⁵⁹³

Just before the CASTLE Bravo shot, the first test in the series, a handful of EC14 warheads had entered the national nuclear weapons stockpile on an "emergency" basis. In March, April and May 1954, concurrent with the CASTLE tests, the EC16, EC17 and EC24 weapons were also produced in small numbers, providing the long-planned "emergency capability" to

⁵⁹⁰Hewlett and Holl, p. 179.

⁵⁹¹Minutes of Thirty-Ninth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 31-April 2, 1954, p. 18.

⁵⁹²Memorandum to the Files dated March 31, 1954 from Ken Mansfield, subject: The Cost of Hydrogen Weapons. If one pound of Li^6D yielded 29,000 tons of TNT-equivalent blast energy, then the cost of producing one pound of lithium-6 deuteride at this time was around \$3,000.

⁵⁹³In "Sloop," a June 1967 study of the use of nuclear explosives for copper mining, a table of explosive charges ranging in yield from 10 KT to 2 MT were listed at prices between \$350,000 and \$600,000; these costs included nuclear materials, device fabrication and assembly, and arming and firing services. At these prices, the cost of a ton of TNT-equivalent energy ranged from \$35 down to 30 cents.

deter possible Soviet thermonuclear threats.⁵⁹⁴ (The Air Force expected in early 1955 to have at least 15 thermonuclear weapons ready for strikes against the Soviet Union.)⁵⁹⁵

The major military result of the CASTLE test series was the rapid adaption (weaponization) and certification by the AEC and DOD of the MKs 14, 15, 17, and 24 aerial bombs for immediate production and deployment with the Air Force and Navy. The EC14, EC16, EC17, and EC24 were canceled after CASTLE; the handful of EC14, EC16, EC17, and EC24 weapons were withdrawn from the stockpile and dismantled,⁵⁹⁶ and the EC14, EC17, and EC24 bombs were replaced in the stockpile by the respectively Mark-numbered weapons. The MK 14 was the Air Force's first deliverable hydrogen bomb.⁵⁹⁷

Solid-fueled thermonuclear weapons, of relatively light weight and high yields in the megaton range were practical and usable, which immediately made ICBM and IRBM warheads feasible. The elaborate and expensive plans and arrangements made by the armed services for the deployment, servicing, and delivery of liquid cryogenically-fueled bombs were no longer necessary. The CASTLE tests had proven that even ordinary lithium hydride -- unenriched in the rare and expensive lithium-6 and deuterium isotopes -- could be used as a hydrogen bomb fuel.

Among results of CASTLE reported to President Eisenhower were the following:

In the recent CASTLE tests at the Pacific Proving grounds, yields up to 15 megatons TNT equivalent were attained. More important, these tests demonstrated that thermonuclear weapons can be produced without using liquid deuterium or tritium, two materials which appeared essential in our earlier work on this type of weapon. Instead, we now know that lithium deuteride is a better fuel than liquid deuterium, which must be stored and handled at

⁵⁹⁴NUCLEAR WEAPONS DATABOOK, Vol. II, p. 16.

⁵⁹⁵"A Smoking Radiating Ruin at the End of Two Hours: Documents on American Plans for Nuclear War with the Soviet Union," Dr. David Alan Rosenberg, INTERNATIONAL SECURITY, Vol. 6 No. 3, Winter 1981/1982, p. 31.

⁵⁹⁶NUCLEAR WEAPONS DATABOOK, Vol. II, pp. 16, 17.

⁵⁹⁷Correction to caption of MK 17 photo on p. 161 of LOS ALAMOS SCIENCE, Volume 4 No. 7, Winter/Spring 1983.

extremely low temperatures.

Reduction in the requirement for tritium makes unnecessary the previously planned allocation of a substantial portion of our reactor capacity for the production of this material. The significance of this development lies in the fact that theoretically a reactor can produce [deleted] of plutonium in place of [deleted] of tritium.

These results indicate the feasibility of providing during calendar year 1955 a family of thermonuclear weapon types ranging in weight from [deleted]⁵⁹⁸

CASTLE had opened the way to a whole spectrum of thermonuclear weapons in a range of yields from a few to many megatons. The ZOMBIE, in particular, made many high-yield and heavy stockpiled fission bombs obsolete. Norris Bradbury of Los Alamos proposed that the weapons labs concentrate their efforts on types of weapons that would be required in large numbers and develop the best possible weapons with optimum characteristics.⁵⁹⁹

Another unique sidelight to CASTLE was the first collection, with U.S. permission, by the United Kingdom's Royal Air Force of airborne radiological samples of debris from a U.S. thermonuclear test. During British atomic tests in Woomera, Australia in October 1953, (the "Totem" test series), U.S. nuclear weapons scientists were allowed to monitor the operation.⁶⁰⁰ The U.S. reciprocated for this courtesy during CASTLE by permitting a British intelligence team to be based at Kwajalein.⁶⁰¹

Two English Electra Canberra bombers fitted with wingtip and under-belly sample collectors were flown from England to Kwajalein atoll by way of Australia. Sampling missions were

⁵⁹⁸"Summary of Program Status Report," undated, but probably ca. August 1954. Emphasis in original.

⁵⁹⁹Hewlett and Holl, p. 180.

⁶⁰⁰Memorandum for: The Secretary of State, The Secretary of Defense, and The Director of Central Intelligence dated 23 December 1953 from Lewis Strauss, Chairman, USAEC.

⁶⁰¹Minutes of Thirty-Ninth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 31-April 2, 1954, p. 6.

flown during shots Bravo, Romeo, Koon, Yankee, and Nectar. To maintain their anonymity, the RAF markings on the planes were painted out and the pilots identified themselves only as "Air Force" during their radio calls. Collected samples were flown by RAF transports to Australia then on to England, where they were analyzed.⁶⁰²

The GAC also seized upon results of CASTLE to rebut critics, both domestic and foreign, of the U.S. nuclear weapons testing program:

The Committee wishes to reiterate its opinion that our test program is one of the most important elements in our whole weapons program. Without it, we could not prove-out the advances which are being made in the weapons laboratories to increase the magnitude, the efficiency, and the variety of weapons in our stockpile. We therefore believe that if any international arrangement were made to curtail or abolish these tests, it should only be made under such arrangements that we obtain an equivalent value, or greater, for the security of the United States.⁶⁰³

Security and Operation CASTLE

A highly undesirable result of the CASTLE series, at least from the AEC's point of view, was the unexpected disclosure to the world of one of the United States' most-closely held nuclear weapon secrets, namely, the highly-classified operating principles of solid-fueled thermonuclear weapons.

This revelation resulted from yield miscalculation, capricious weather, imperfect naval air surveillance, and the efforts of a Japanese physicist. This time, the "leak" could not be attributed to unauthorized sailors' letters published in the U.S. during CASTLE (of which there were none).

⁶⁰²Memorandum for Record: For Commander's Notebook, 8 October 1954, Headquarters, Joint Task Force Seven, Subject: Summary of British Participation in Operation CASTLE. If British physicists were as skilled and knowledgeable as their Japanese counterparts in analyzing these samples, then the U.S. may have intentionally provided to England at this time many of the important characteristics and likely design principles of U.S. thermonuclear weapons.

⁶⁰³Letter dated May 29, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC.

The unexpected rise of the Bravo shot cloud into a region of unfavorable winds aloft over Bikini deposited a large amount of fallout on a Japanese fishing boat FUKURYU MARU ("Lucky Dragon"), which was located at sea just beyond the eastern edge of restricted waters around Bikini atoll. Had the boat been discovered by naval patrol planes, she would have been escorted out of the area, as was done with another Japanese fishing boat before Romeo shot.⁶⁰⁴

Missed by prowling planes, the boat, her crew, and her cargo of fish were heavily contaminated by radioactive ash from the passing Bravo cloud.

When the "Lucky Dragon" returned to Japan two weeks later and it was discovered that she had been dusted by the test, fallout samples collected from her decks were analyzed by Japanese scientists in order to determine to which radioactive materials the crew might have been exposed and the maximum radiation dosage they might have received. Among materials found were large amounts of uranium-237 which resulted from bombardment of uranium-238 by fusion-spawned high-energy neutrons.

In addition, concentrations of neptunium-239 resulting from neutron capture by uranium-238 led the Japanese to conclude that a mantle of normally unfissionable uranium-238 had been wrapped around the fusion package of the Bravo device, resulting in a fission-fusion-fission explosion.⁶⁰⁵

This was precisely the design principle of all CASTLE devices; while this particular configuration was known to be theoretically possible and had been published in unclassified books and articles, it was not known to be feasible or practical until after the IVY Mike shot, where it was first tested, and then again in solid-fueled devices after the CASTLE tests, and the AEC was deeply concerned and especially sensitive about any evidence that might suggest the success of a "dry" thermonuclear weapon.

The AEC had conducted its own analysis of Bravo fallout by the end of March; samples had shown large quantities of products resulting from fissions caused by a high "fast" neutron flux. The presence of these radioactive isotopes gave some idea of the kind of device tested, so the U.S. would not

⁶⁰⁴DNA 6035F, p. 252.

⁶⁰⁵"Preliminary Report on the Chinese Nuclear Test of June 26, 1973," R. W. Perkins and Associates, Battelle Pacific Northwest Laboratories, Richland, Washington, July 13, 1973, p. 1; "5th Chinese Nuclear Test Held 'Dirty'," Los Angeles Times, January 4, 1967.

provide this isotopic analytical information to Japan.⁶⁰⁶

The Japanese also complained that the U.S. was hindering medical treatment of the crew of the "Lucky Dragon" by withholding information on the constituents of the Bravo device; speculation was rampant: an early Japanese news report listed calcium carbonate as an important bomb component, based on preliminary fallout analysis.⁶⁰⁷ The calcium carbonate actually came from coral blasted into the stratosphere from the reef below the SHRIMP device.

The AEC was apoplectic over the disclosure of the so-called "3F" or "fission-fusion-fission" concept of thermonuclear weapons design; this information was widely disseminated in the U.S. and around the world. JCAE Chairman W. Sterling Cole made comments implying that the "Lucky Dragon" had been spying on the tests. AEC Chairman Lewis Strauss went even further, privately characterizing the "Lucky Dragon" as "a Red spy ship".⁶⁰⁸

In a public statement on March 31, Strauss mislocated the boat's position, claiming it was closer to Bikini than it actually was at the time of the March 1 blast. Strauss implied that the "Lucky Dragon" had been only 74 miles east of Bikini, inside the AEC's publicly-proclaimed "warning area;" the boat had actually been about 86 miles east of Bikini, outside the original CASTLE danger area.⁶⁰⁹ The Japanese government formally protested Strauss' March 31 statement.⁶¹⁰

A CIA analysis at the end of April concluded that Strauss' assertion was incorrect:

... that there is no evidence, direct or otherwise, that the ship was in or near

⁶⁰⁶Memorandum dated March 31, 1954 to Dr. John C. Bugher, Director of Biology and Medicine, USAEC, from Col. Vincent G. Huston, USAF, Acting Director of Military Application, USAEC, Subject: Analyses of Fission Products.

⁶⁰⁷Teletype message dated March 12, 1954 to K. D. Nichols, USAEC, from JTF Liaison Office, Hickam AFB, Hawaii, Subject: Contamination of Japanese Fishing Vessel.

⁶⁰⁸Hewlett and Holl, p. 177.

⁶⁰⁹Lapp, VOYAGE OF THE LUCKY DRAGON, p. 32.

⁶¹⁰Aide-Memoire delivered to the U.S. Department of State in Washington, D.C. by the Japanese Ambassador to the U.S. on April 12, 1954.

the danger area for the purpose of observing or recording the explosion; rather, that the ship and its crew were neither equipped nor qualified for such activities.

Secondly, it has been generally determined that there was no evidence, direct or otherwise, of a propaganda purpose; rather, that the evidence indicates that the ship had a legitimate fishing mission and its presence in the area was not unusual.⁶¹¹

If a Japanese fishing boat had inadvertently collected informative fallout samples, a Soviet trawler or submarine could easily have done the same intentionally.⁶¹² In 1954, during some of the darkest days of the Cold War, revelation of thermonuclear weapon design principles and of experimental confirmation of those principles was far more serious and potentially much more damaging to national security than the details of the IVY Mike shot "leaked" in November 1952 by careless sailors.

This was not the only disclosure of CASTLE data by the Japanese: in December 1954, at a time when the total number of CASTLE shots and their firing dates were still secret, the Japanese magazine Kagaku Asahi published an article in its December 1954 issue that accurately listed the firing dates and times of four of the six tests (the Japanese did not detect the relatively low-yield Koon and Nectar shots). The tests were discovered at long range by means of acoustic waves and ionospheric disturbances.⁶¹³

⁶¹¹Memorandum for the Record dated 28 April 1954, Subject: CIA Investigation of Fuku Ryu (sic) Maru (Fortunate Dragon) to Hydrogen Bomb Test, attachment to letter dated 29 April 1954 to the Honorable Lewis L. Strauss, Chairman, USAEC, from Frank G. Wisner.

⁶¹²A good account of the analysis of the fallout by the Japanese is on pages 148 to 156 of Lapp's VOYAGE OF THE LUCKY DRAGON; Cole's comments are noted on p. 119 of that book.

⁶¹³Memorandum dated December 28, 1954 to Dr. John C. Bugher, Director, Division of Biology & Medicine, USAEC, from Merrill Eisenbud, Manager, New York Operations Office, USAEC; "Rain from the South and Snow from the North," translation of article by Yasuo Miyake in Kagaku Asahi, December 1954, attached to Memorandum to Those Listed Below dated February 9, 1955 from Merrill Eisenbud, Manager, New York Operations Office, USAEC, Subject: Translation of Japanese Articles. As

In contrast, three years later, by May 1957, only the first three CASTLE shots had been announced publicly by the AEC.⁶¹⁴

Another alarming instance, to the AEC, at least, of classified information appearing in the public domain during CASTLE occurred when the April 12, 1954 issue of TIME magazine featured an essay entitled "The Making of the H-Bomb;" the article contained facts and accurate speculation about the fuels and mechanisms of deuterium-tritium gas-boosted weapons and "wet" and "dry" thermonuclear devices, probable thermonuclear reactions and products, results of U.S. atomic tests, and the cost of tritium in terms of lost plutonium production.⁶¹⁵ The story also unknowingly revealed the "secret" of pre-ignition compression of the main charge of fusion fuels.

Because of its early publication, the discourse did not include information about the "fission-fusion-fission" design revealed by the ashes of the Bravo shot. Nonetheless, the AEC was greatly displeased by the publication of so much of this type of information in one place; however, to have prosecuted after publication or even commented publicly on the accuracy of the TIME article would have compromised security even further.⁶¹⁶

Another article in the same issue of TIME aroused the ire of Norris Bradbury at LASL. The article implied again, as had the May 1953 FORTUNE article published by the same company that produced TIME and LIFE, that the current series of CASTLE tests, as well as the November 1952 IVY test, were

of February 1, 1956, only three of the CASTLE tests -- Bravo, Romeo, and Koon -- had been publicly announced by the AEC. ("Publicly Announced Nuclear Detonations of the United States, USSR and United Kingdom, 1945-1955," compiled February 1, 1956, by Public Information Service, U.S. Atomic Energy Commission; "News Roundup," BULLETIN OF THE ATOMIC SCIENTISTS, 1954, p. 234.)

⁶¹⁴AEC Press Releases dated March 1, March 30, April 7, and May 13, 1954; "Background Information on Nevada Nuclear Tests," Office of Test Information, Nevada Test Organization, Nevada Test Site, Mercury, Nevada, May 1, 1957, pp. x, 4.

⁶¹⁵"The Making of the H-Bomb," TIME, April 12, 1954, pp. 72, 73.

⁶¹⁶In a letter dated April 15, 1954, JCAE Chairman Sterling Cole "invited" AEC Chairman Lewis Strauss to testify before the JCAE on the article. The AEC went so far as to have the FBI investigate the publication of the article. (Excerpt from Transcript of Hearing before JCAE on May 3, 1954.)

entirely the work of Edward Teller and the University of California Radiation Laboratory at Livermore, California (Teller at this time was no longer affiliated directly with LASL).

This TIME article also insinuated that the Soviets had conducted nuclear tests on the inhabitants of several eastern European countries; stated that Dr. J. Robert Oppenheimer opposed work on the hydrogen bomb; that "in a phenomenally short time, (Teller) found short cuts through Oppenheimer's technical objections (to the Super);" and that the Soviet test in August 1953 was of a thermonuclear device "far less complex, far more economical, and far more 'transportable' than (the IVY Mike device)."

The TIME article failed to point out that the Soviet test device was not a true multistage hydrogen bomb, and that it yielded only a few hundred kilotons, not anywhere near the yield of the LASL SAUSAGE device; these facts were well known within some U.S. government circles. TIME also claimed that a future CASTLE test was to be of a device named JUGHEAD that would yield 45 megatons.⁶¹⁷

Bradbury Reviews the H-Bomb Program

Bradbury was quite upset by this thinly-veiled attack on Los Alamos and he urged the AEC to take action to counter the possibly misleading and negative impression it gave. For the benefit and knowledge of the AEC's Director of Military Application, Bradbury reviewed the history of the American hydrogen bomb. He was scrupulously fair in his description and attribution of credit to Dr. Teller's role, despite their deep personal political and technical differences over what LASL's immediate postwar priorities should have been:

⁶¹⁷"The Atom: The Road Beyond Elugelab," National Affairs, TIME Magazine, April 12, 1954, pp. 22, 23. J. Robert Oppenheimer was a frequent villain of Henry Luce's magazines; this issue of TIME showed a photo of a scowling Oppenheimer alongside that of a hopeful-looking Teller, the hero of the TIME-LIFE-FORTUNE publisher. The ultraconservative editorial viewpoints of all three magazines were very far to the right during the McCarthy era, in which this article appeared. FORTUNE and TIME also had the behind-the-scenes support of then-AEC Commissioner Lewis Strauss in preparing these defamatory articles. (Hewlett and Holl, pp. 47, 48, 57.) During a September 24, 1954 press conference, when asked about the veracity of the 45 MT blast estimate, LASL director Norris Bradbury, citing security reasons, refused to answer directly, but implied strongly that it was untrue. (Transcript of Press Conference with Norris Bradbury at Los Alamos, New Mexico, September 24, 1954, p. 7.)

In the April 12 issue of TIME magazine there appears an extended discussion of the "hydrogen bomb" in which it is apparent that the editors have a rather extraordinary misconception regarding the roles of AEC laboratories presumably involved in this development.

This, of course, is not the first time that the Luce publications have erred along these lines (e.g., the May 1953 issue of FORTUNE magazine), and the history of their misinformation in this field goes back to 1946. It has never been obvious whether corrective measures were either appropriate or possible.

The present instance is, of course, so fantastically in error, particularly with respect to the relative roles and accomplishments of the Livermore and Los Alamos laboratories, that a non-trivial morale problem is arising at both laboratories.

The persistence of the uncorrected Livermore legend in the press is now raising the question at Los Alamos as to whether or not the course of fusion weapon development as it has occurred over the last ten years and (as it is now occurring) is really understood in all appropriate Washington areas, particularly (in) the Congress.

On the other hand, Livermore personnel can only be embarrassed, as honest scientists, by the attribution of accomplishments to them which are not theirs, and also are concerned by the easily detectable exasperation of many of their Los Alamos colleagues with whom (they have had up to now) as amiable relations as one could expect between two institutions whose activities were expected to be competitive in the weapon field.

The purpose of this memorandum, therefore, is to set forth as specifically as possible the situation in the thermonuclear field over the last few years as it has actually occurred at Los Alamos. All of this is no doubt well known to you and the (Atomic

Energy) Commission, but it may be useful to summarize the situation as it has developed.

As you are aware, the thermonuclear field is a very old one, dating back to the early 1930s. The development of the atomic weapon seemed to provide the first practicable means of attaining the necessary temperatures, and during the war years, Edward Teller, while a member of the Los Alamos Scientific Laboratory, devoted a considerable portion of his time to the direction of a small group of theoretical physicists in the exploration of this problem.

At that time, the thermonuclear weapon was only known in the abstract form of the classical "super" [deleted] and there was a fundamental question, for which calculational techniques did not then exist, as to whether such a device if ignited would actually burn with a useful efficiency. (This question is now almost universally believed to be answered in the negative.) (Author's note: Edward Teller and a few other UCRL physicists insist even now, nearly fifty years later, that the "classical Super" could have been made to work.)

At any rate, strong interest in this problem continued at Los Alamos after the war, as the records of the various "super" conferences will indicate. However, it was imperative to devote our major efforts to the atomic weapon, both from the practical point of view of the national stockpile (particularly the effective use of U-235) but also because it was apparent that any advance in the thermonuclear field could only follow a high degree of technical skill with fission systems. The accomplishments of the Los Alamos Scientific Laboratory in this field during the period 1946-1949 are well known to you.

Teller left Los Alamos in 1946 to return to academic work but remained as a consultant to the Laboratory and spent most of his summers there. He returned to Los Alamos on a full time basis in July 1949 on a year's leave of absence from the University of

Chicago. With the occurrence of the first Russian atomic explosion we, of course, urged Teller to extend his stay at Los Alamos not only because interest in the potentialities of the thermonuclear field was at a high point, but (also because) the technical skills which had been acquired (by) Los Alamos in the fission field made the prospect of real progress seem more likely. An extension of this leave of absence was ultimately obtained with the assistance of the Commission.

The first significant and large scale accomplishment in this (thermonuclear) field occurred (during the GREENHOUSE George test of the CYLINDER device). As you will recall, this was the first instance in which a thermonuclear reaction had been made to take place external to a fission system.

It was designed as a first step toward the initiator (in the design of the "classical super") and Teller was instrumental in the design and theoretical interpretation of this system. Shortly after its design, it was realized that the technique could be extended to the principle we now know as radiation implosion.

In this connection, full credit should also be given to Teller for his fundamental idea of using the radiation from a fission weapon as an effective means of compression of a thermonuclear system. The possibility of using one bomb to affect another fission or fusion system was, of course, a common concern to many people, and indeed, a closely related idea was first proposed at Los Alamos by Dr. Ulam who suggested the shock (wave) from a fission weapon for this purpose. Actually, this suggestion led Teller to the idea of radiation implosion and both ideas were written up jointly and simultaneously by Ulam and Teller.

Following the [George] shot, it was obvious that the next step should be a full scale test of a real thermonuclear system. During the course of the summer of 1951, it was decided that this should be [deleted] At a scheduling meeting of the Technical Board of the Los Alamos Scientific Laboratory in

September 1951, the tentative date for the MIKE shot was picked for the Fall (sic) of 1952.

At this meeting, Teller became extremely upset and asserted that if the date (obviously a year in advance) were not set to be no later than the Summer (sic) of 1952, he would resign from the (Los Alamos) Laboratory.

Although the actual incident was trivial, underlying tensions had grown up primarily between Teller and myself, but (were) paralleled by similar reactions from my entire senior technical staff, regarding the most effective manner in which the thermonuclear program could be administered and accomplished, and (tensions) were such as to make Teller's separation from the Los Alamos Scientific Laboratory probably inevitable.

A number of sincere efforts were made by myself and other people to persuade Teller to change his mind and remain, but these efforts were unsuccessful.

In all frankness, Teller's volatility and sensitiveness (sic) tended to detract from the value of his theoretical intuition and ability and there was a general feeling, which I shared, that our progress (toward a thermonuclear weapon) would not be less rapid in his absence. He left Los Alamos in a reasonably friendly spirit; his relations with me personally since then have been friendly and agreeable, although I presume we both feel some degree of personal disappointment in the other. (Author's note: Dr. Edward Teller had testified before the Oppenheimer security clearance review hearing board on April 28; Bradbury may or may not have seen Teller's very damaging testimony by the time he wrote this memo on May 3. Bradbury had testified on April 22.)

The theoretical design, development, and test of the MIKE shot was (sic) entirely an operation of the Los Alamos Scientific Laboratory, its consultants and subcontractors, and Teller's contact with

it was completely incidental and casual. He was not present at Eniwetok at the time of its firing (Author's note: Teller witnessed the shot at a seismograph at the University of California in Berkeley), but his congratulations to the Los Alamos Scientific Laboratory, I believe, were thoroughly sincere and were received in (a spirit of) complete friendliness.

The [ALARM CLOCK] and other shots of the present (CASTLE test) program are, as you know, weapons (or weapons prototypes) also designed and developed entirely by the Los Alamos Scientific Laboratory, its consultants and subcontractors. It is my opinion that extraordinary credit is due our theoretical division for the ingenuity, skill, and persistence which they have displayed in this task.

Bradbury also outlined the genesis of the Livermore lab, and detailed how relations between the labs had improved, although he felt that LASL was still "wet nursing" UCRL:

Following his departure from Los Alamos in 1951, Teller intensified his campaign (which he had been conducting quietly for some time) for a second weapons laboratory. He returned to the University of Chicago and sought to have it established there. The Commission ultimately decided in the late Spring (sic) of 1952 to establish such a laboratory under the guidance of the Radiation Laboratory of the University of California, and located at Livermore, California. (Author's note: Teller had a lot of support from the Air Force and some members of Congress, who were unhappy with LASL, in getting the lab established.)

Relations between the Los Alamos Scientific Laboratory and Livermore have been good. They have given us excellent assistance at CASTLE in certain measurements although they indicate that they do not wish to do this in the future. On the other hand, we have furnished them with copies of all our reports and documents, our personnel have been involved (at times seemingly endlessly) in the indoctrination of theirs; we have performed experiments for them, made primary bombs for them, and are now engaged in an extensive program of high

explosives fabrication experiments for them. Teller has recently expressed the intent to spend several weeks at Los Alamos this summer.⁶¹⁸

Bradbury also made a few remarks concerning the GAC's 1949 recommendation about the hydrogen bomb, and how LASL's role in it had been distorted by the May 1953 FORTUNE article:

There is one final matter which I should like to be certain is clearly understood. In 1950, as you well recall, there was extensive public debate (but not by Los Alamos) regarding the wisdom or morality of proceeding with hydrogen weapon development. Teller, of course, conducted a strong, personal, and somewhat publicized campaign along these lines.

To the surprise and dismay of the Los Alamos Scientific Laboratory, the GAC recommended against the program. This was essentially the first time in which we found ourselves opposed to a point of view of so eminent a group of people.

However, the LASL opinion -- both mine, my senior staff, and as far as I am aware, that of all technical personnel, was to the effect that no decision as to the wisdom or morality of stockpiling or using thermonuclear weapons could possibly be made in ignorance of the facts; and that the facts of such weapons were of fundamental importance and must be known by this country as soon as possible. It was our task to ascertain those facts as rapidly and accurately as possible.

Accordingly, we did not and, in my opinion, properly did not enter into the public debates as to whether this course was sound or moral. We did insist to the Commission that the investigative, experimental, and test program must proceed at the highest

⁶¹⁸Starting later in 1954, UCRL conducted non-nuclear explosive implosion tests at the Nevada Test Site. ("Background Information on Nevada Nuclear Tests," Office of Test Information, Nevada Test Organization, Nevada Test Site, Mercury, Nevada, May 1, 1957, p. 22.) UCRL's own local HE test area, Site 300, did not open near Livermore until 1958.

priority to provide the relevant facts with which to make future decisions.

An impression has apparently grown up in some quarters that the LASL was opposed to this (thermonuclear) program. On the contrary, we were not only most strongly in favor of its exploration, but had been, were, and are deeply and persistently involved in it.

Not even a negative presidential decision, I believe, could have stopped the calculational, theoretical, or experimental effort, although it would have denied actual full scale tests. As a laboratory, we privately deplore the (May 1953) FORTUNE statement and its exaggeration of a situation in which the LASL played a role almost exactly contrary to that ascribed to it by inference.

To a degree, the Oppenheimer security clearance review hearings being conducted at this time in Washington, D.C. were also a slap at LASL, which Oppenheimer had help found and which he had directed during World War II. By attacking Oppenheimer as a security risk who deliberately slowed hydrogen bomb development, LASL and its staff were also slandered, if only by inference, as FORTUNE and TIME were doing.

The director of LASL closed his memo with a request that the AEC help clarify LASL's history with the Congress. Bradbury attributed blame for some public misunderstanding on the mere existence of the new Livermore lab; not only had he stated previously that it was overly-dependent upon LASL for help, but he was now noting, without saying so directly, that UCRL was doing little to set the record straight and may, in fact, have been trading upon its unearned "legend" to enhance its chances at winning future weapons development contracts.

Had the public and Congress known the details of UCRL's test failures during UPSHOT-KNOTHOLE and CASTLE, the new lab's reputation might have shrunk considerably. Unfortunately, secrecy hid the results of these tests:

May I make it clear in closing that I am not making a specific request for public clarification of the role of Los Alamos in the events of the last three years. Official denials of those parts of the TIME story which are incorrect is not likely to be very impressive or to undo much of the damage which has been done.

A conscious attention on the part of the Commission to more positive public statements in the future may be helpful, although I am far from sanguine that the problem is soluble and (I) fear that it is inextricably related to the fact that two weapon laboratories do, indeed, exist.

My major purpose in describing this situation in some detail is to provide you with the basic facts in case such an account may be useful. Those scientists, whose opinions the (Los Alamos) Laboratory values, know the accomplishments of the Laboratory and are unlikely to be seriously impressed by TIME, FORTUNE, or the speculations of the press services. The DOD appears to have been under some misapprehensions (between 1950 and 1951), but it is unlikely that they still retain them at the present time.

The clarity of understanding of the Joint Committee (on Atomic Energy) of the Congress (and, indeed, congressmen in general) is far less certain and may be greatly influenced by what they see in print. It is to the Commission that we must look to carry out the responsibility of making sure that no errors in understanding occur in those lofty areas, and hence my concern that there be as factual an account as possible in your hands.⁶¹⁹

To a large extent, it was the fault of the AEC and Presidents Truman and Eisenhower that misleading accounts like those in TIME appeared. The White House and the Commission had been, and still were, hysterically obsessed with secrecy that was

⁶¹⁹Memorandum dated May 3, 1954 to Brig. Gen. K. E. Fields, AEC DMA, from N. E. Bradbury, Subject: Thermonuclear History. Emphasis in original. A few minor editorial changes have been made to clarify the original text; Bradbury liked long paragraphs with few commas. Fortunately for historians, Norris Bradbury seldom wrote a short letter nor did he ever apparently hesitate to express himself on any number of subjects, especially when the good reputation of LASL was being denigrated. Teller's Livermore lab would not have a successful full-scale thermonuclear test until the summer of 1956, still two years in the future at the time of Operation CASTLE.

frequently totally unnecessary.

Public speculation about CASTLE was rampant, and reliable information, both historical and technical, was unconfirmed by authoritative sources. While this may have kept potential adversaries off balance, it also served to exacerbate problems and misunderstandings not only between the two weapons labs, but also between the public and the government and between branches of the government.

Not until the release of the Oppenheimer hearing transcripts later in 1954 would the American public have much idea of what had been going on at LASL since the end of World War II.⁶²⁰ The information in Bradbury's memo was highly classified in 1954; the concept of radiation implosion was kept classified within the U.S. government until 1980, although it was well known inside many academic and private fusion energy research laboratories by that time.

Despite Bradbury's statement "that I am not making a specific request for public clarification of the role of Los Alamos in the events of the last three years," within just a few months, he and the AEC would be forced to defend the laboratory against an even more vicious and defamatory attack by journalists.

The government's publication of the Oppenheimer hearing transcripts would also cast much light on the postwar activities of the New Mexico weapons laboratory, although this was not a primary reason for release of the transcripts.

Underscoring LASL's primacy in thermonuclear weaponry, on August 31, 1954, the TX-15 was scheduled to enter the U.S. nuclear weapons stockpile in March 1955. This was the end of "emergency capability" and the beginning of normal thermonuclear bomb deployment.

⁶²⁰Hewlett and Holl, p. 110. For an early reconstruction of the U.S. hydrogen program, based on the Oppenheimer hearing transcript, see "The H-Bomb Chronology," Roland Sawyer, BULLETIN OF THE ATOMIC SCIENTISTS, 1954, pp. 287-290, 300.

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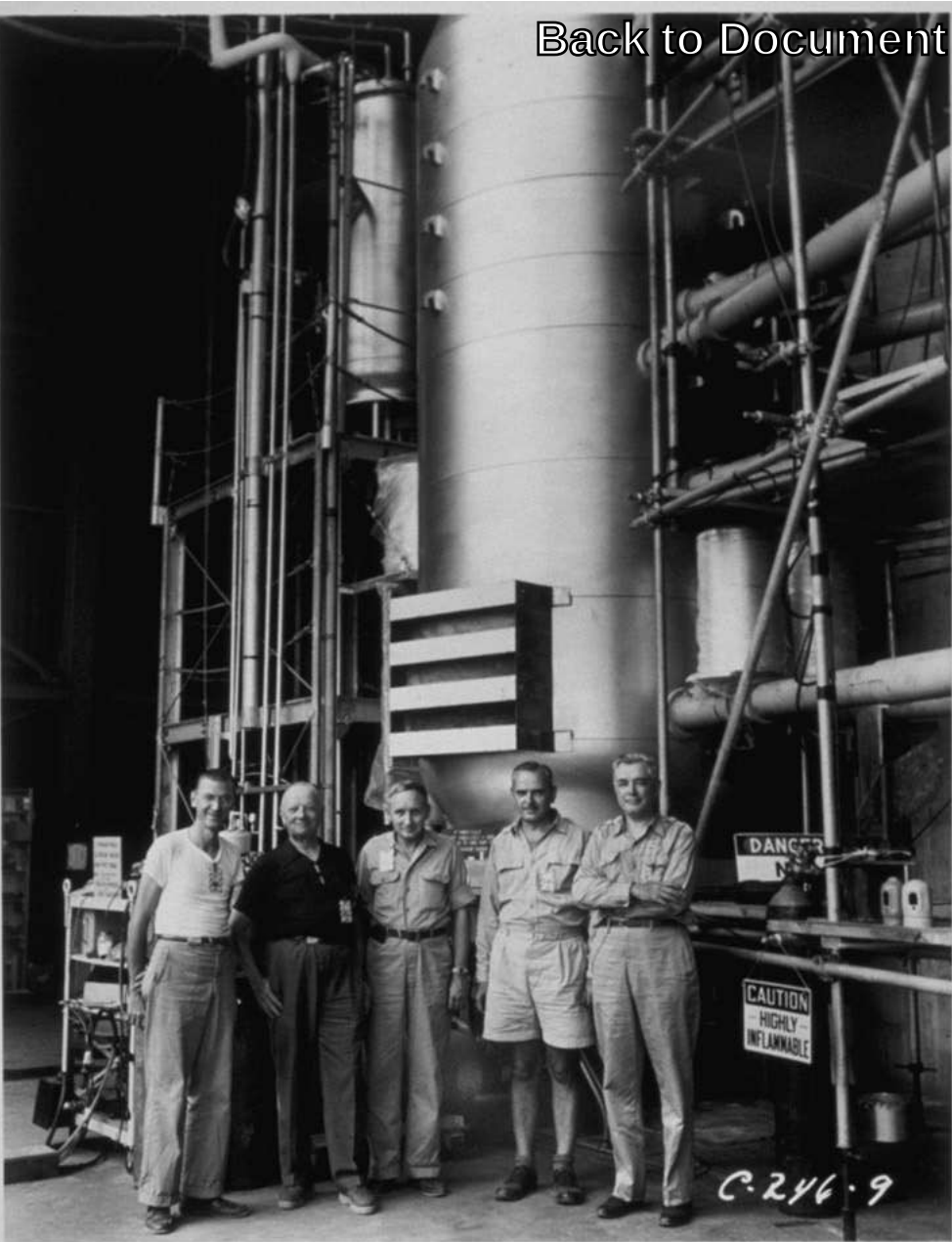
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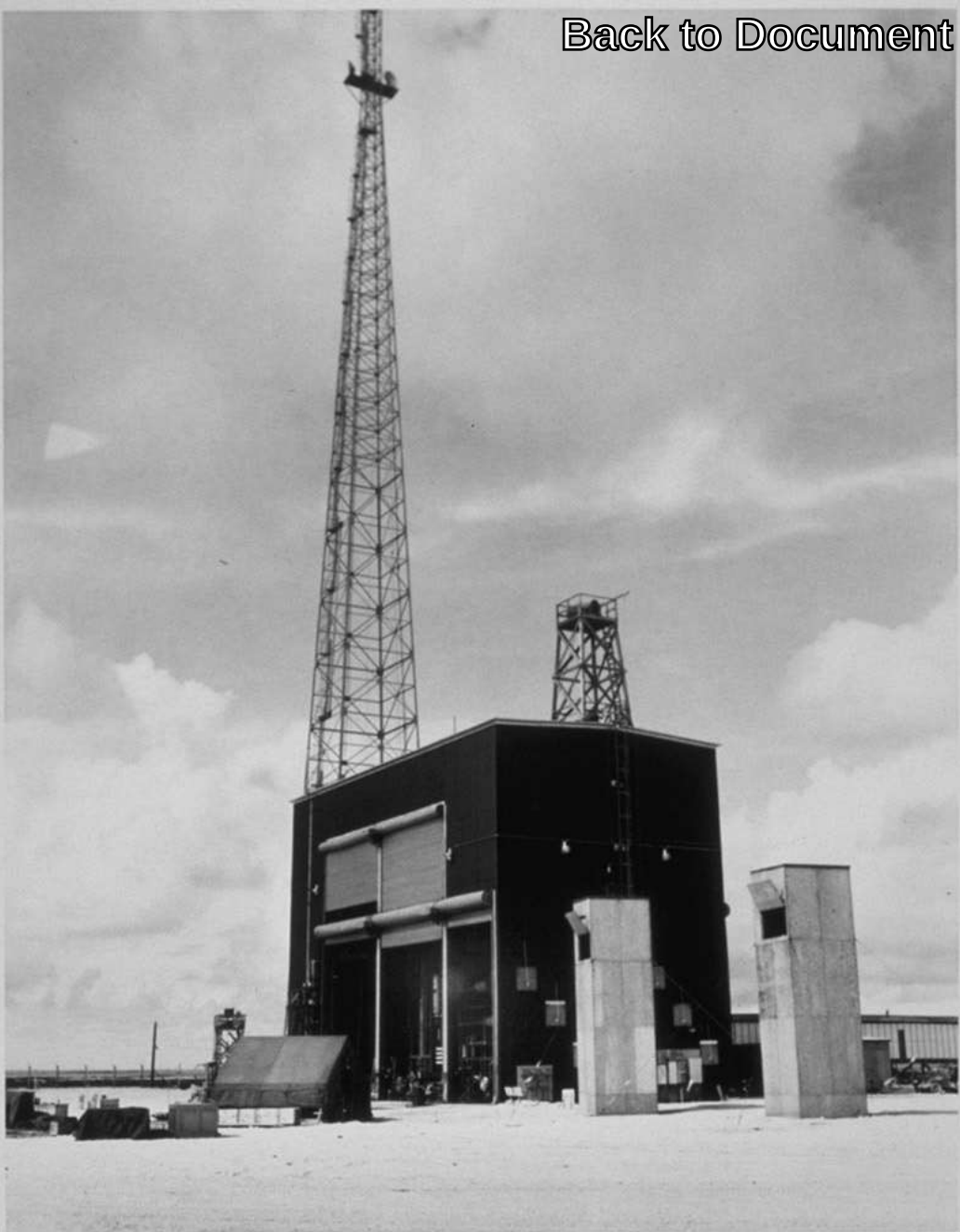
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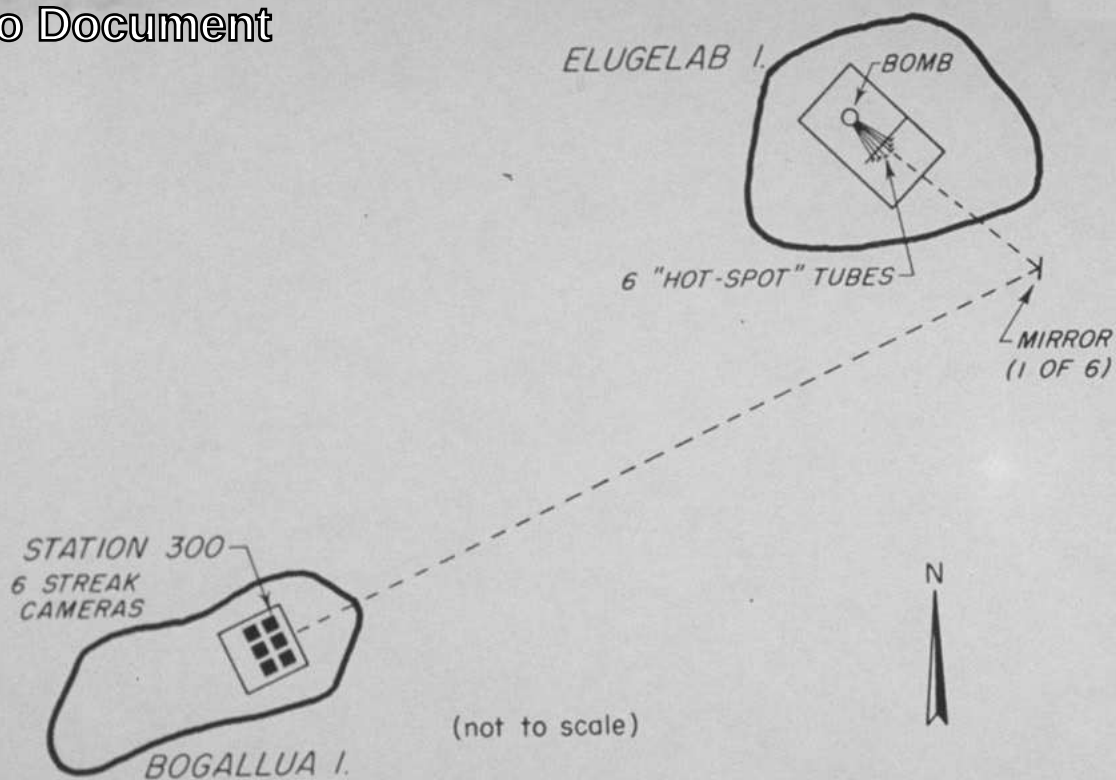
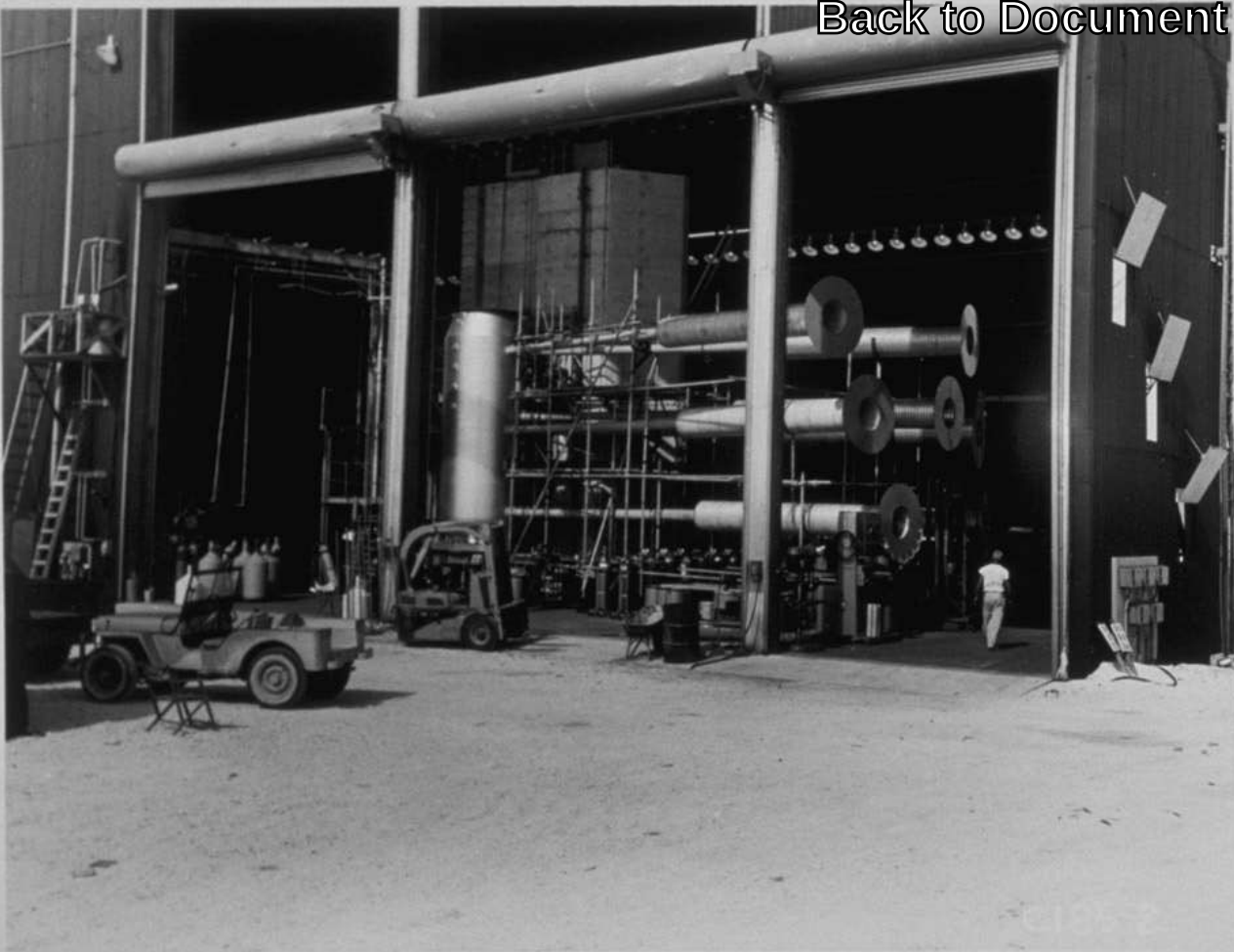


Fig. 3.3.1

Mike Shot Hot-Spot Observation

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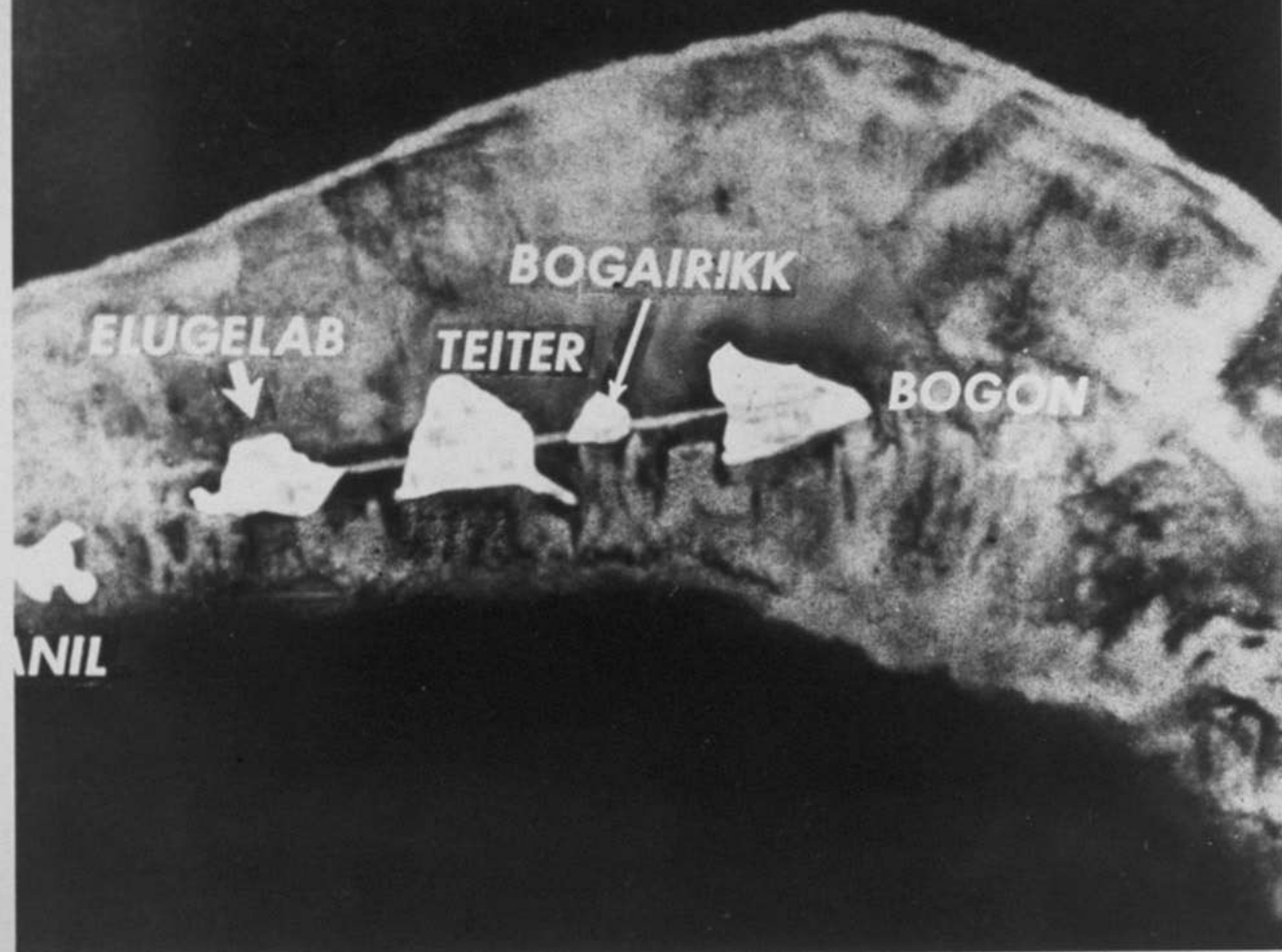


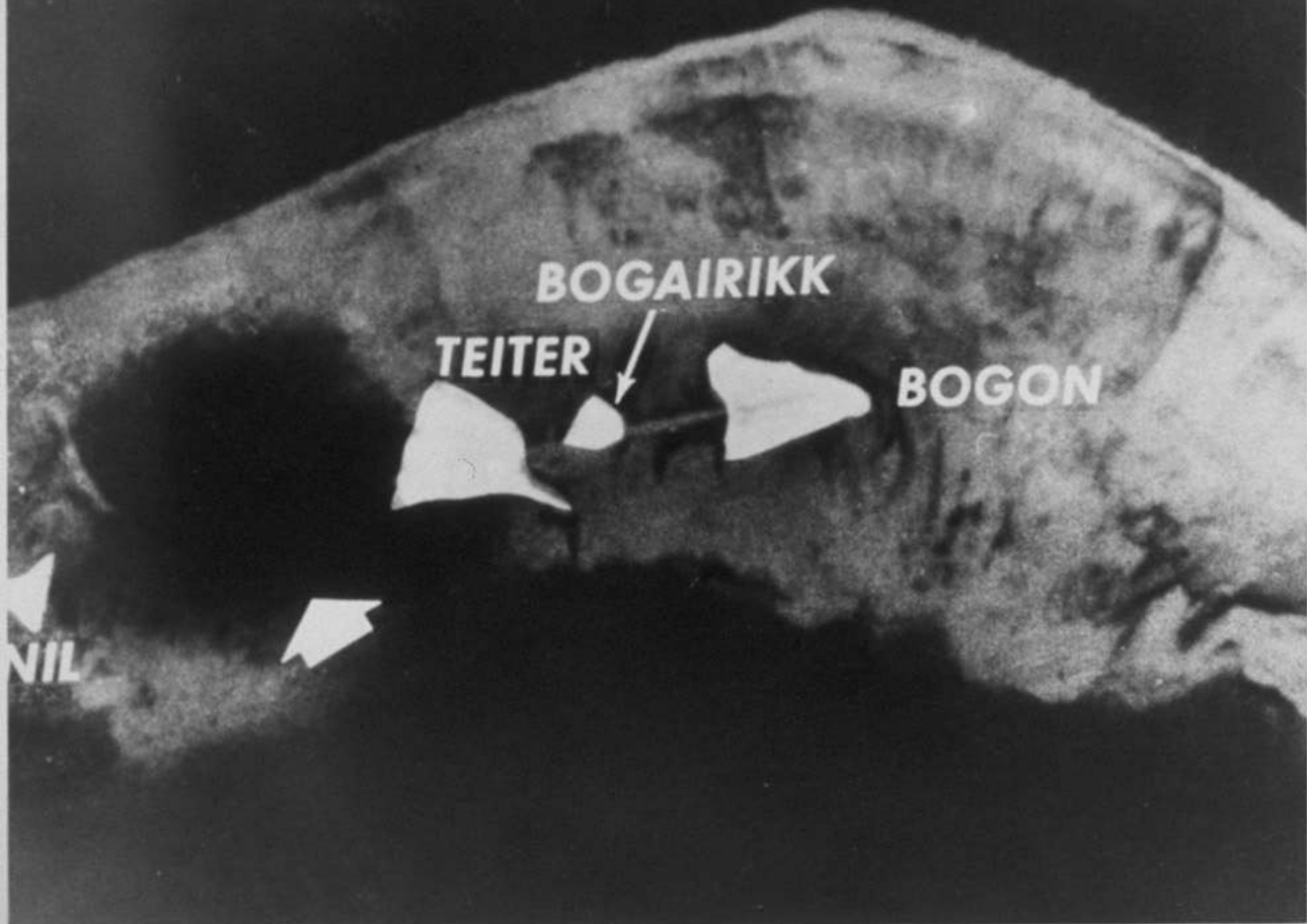
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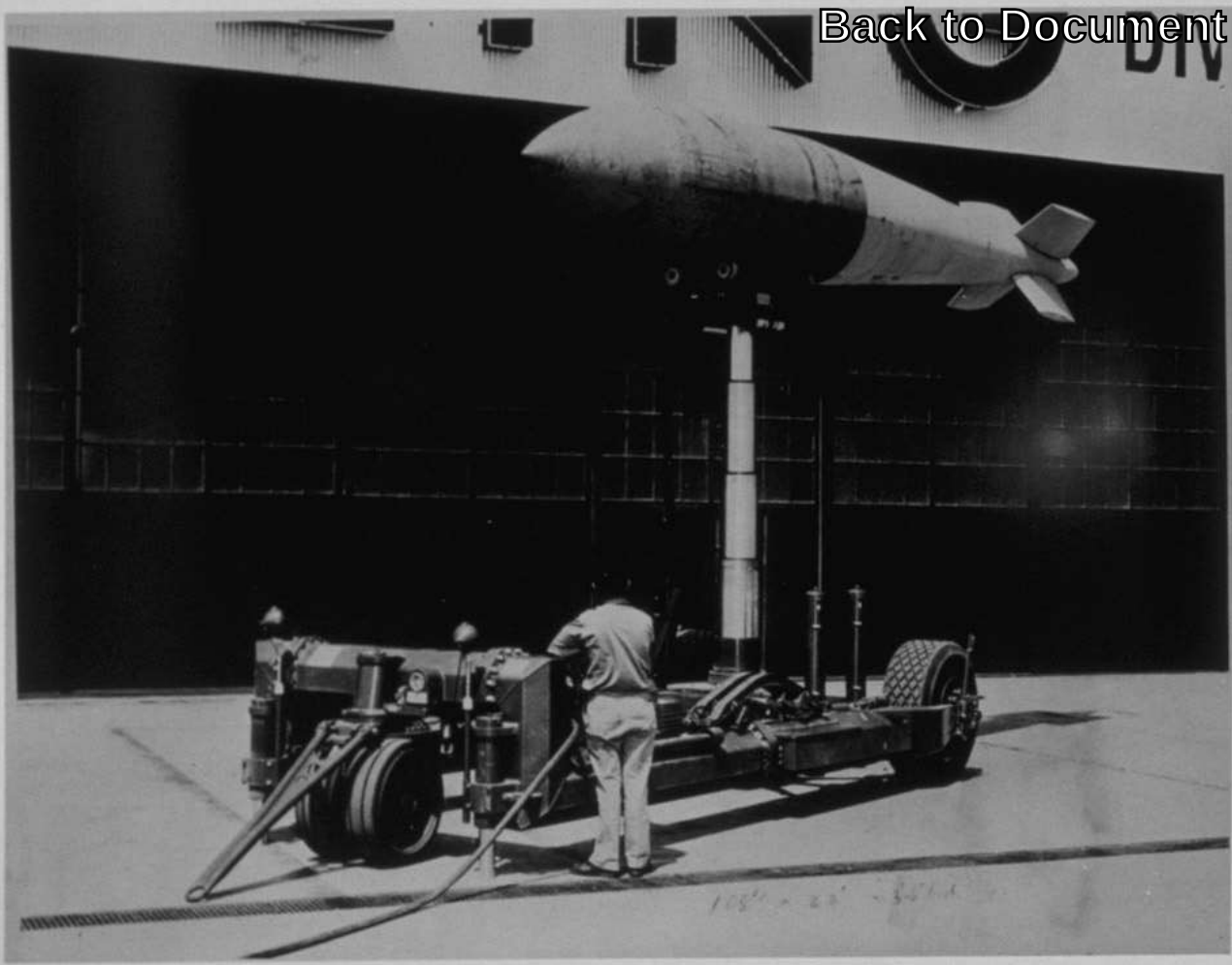


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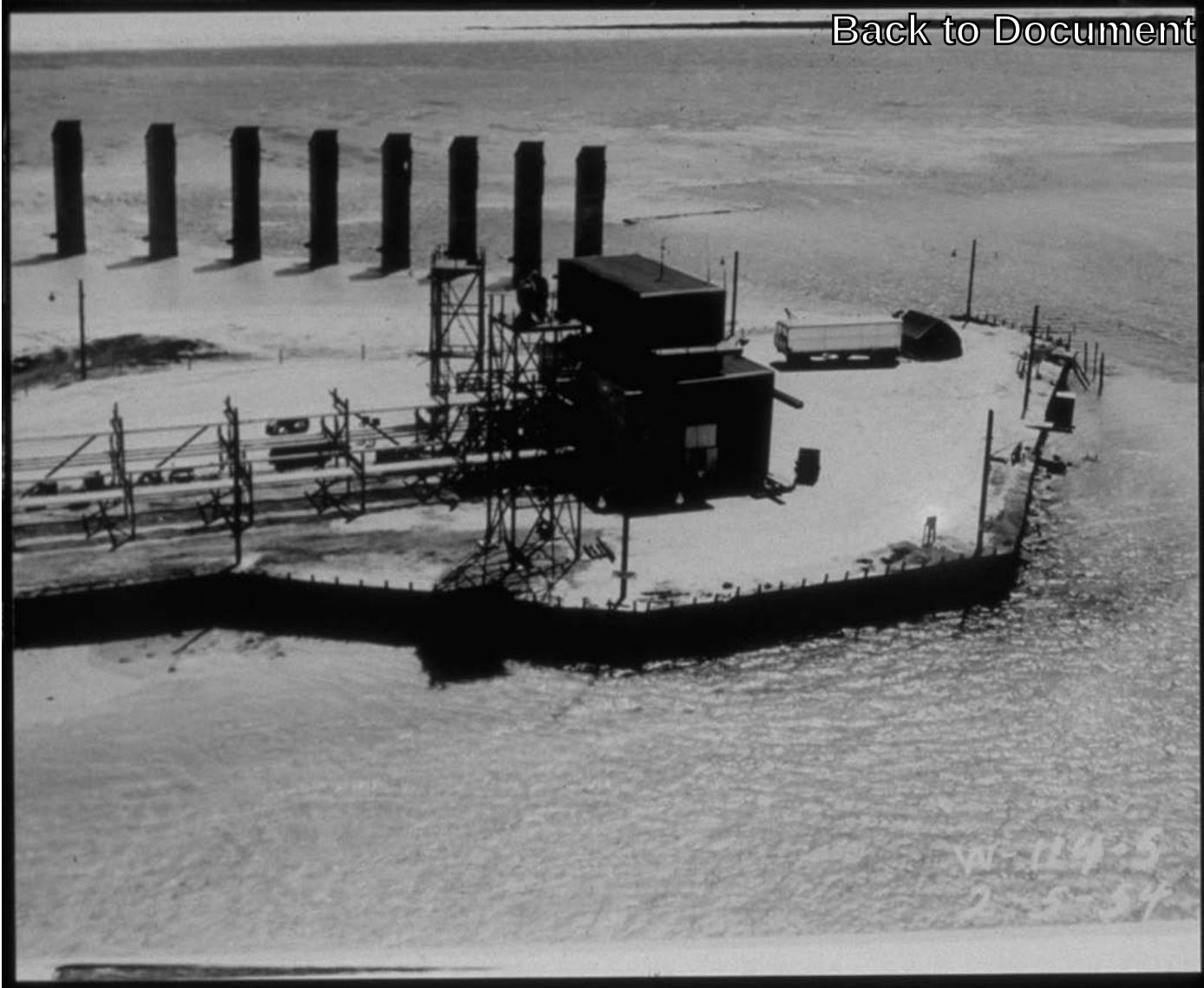


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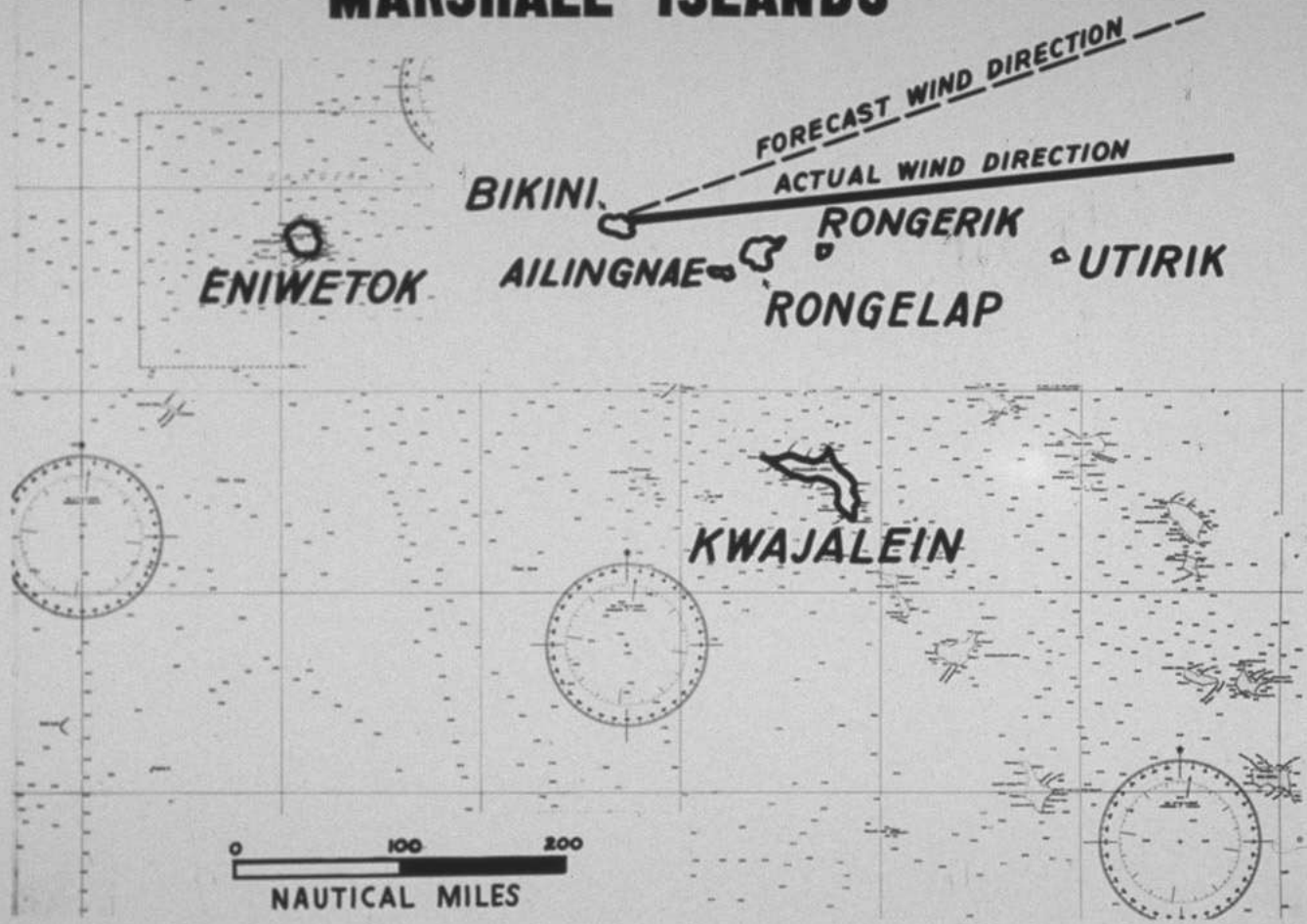


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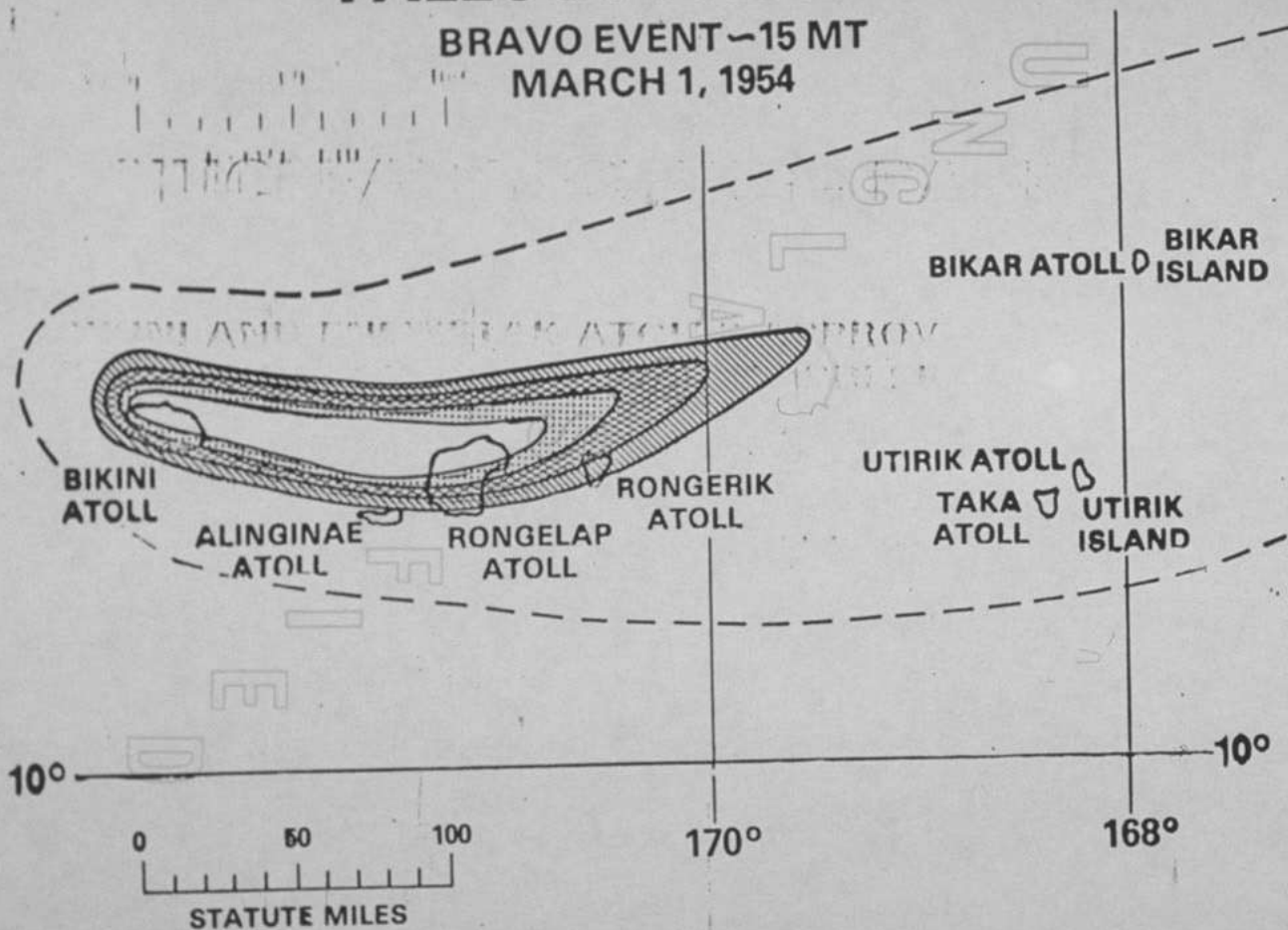


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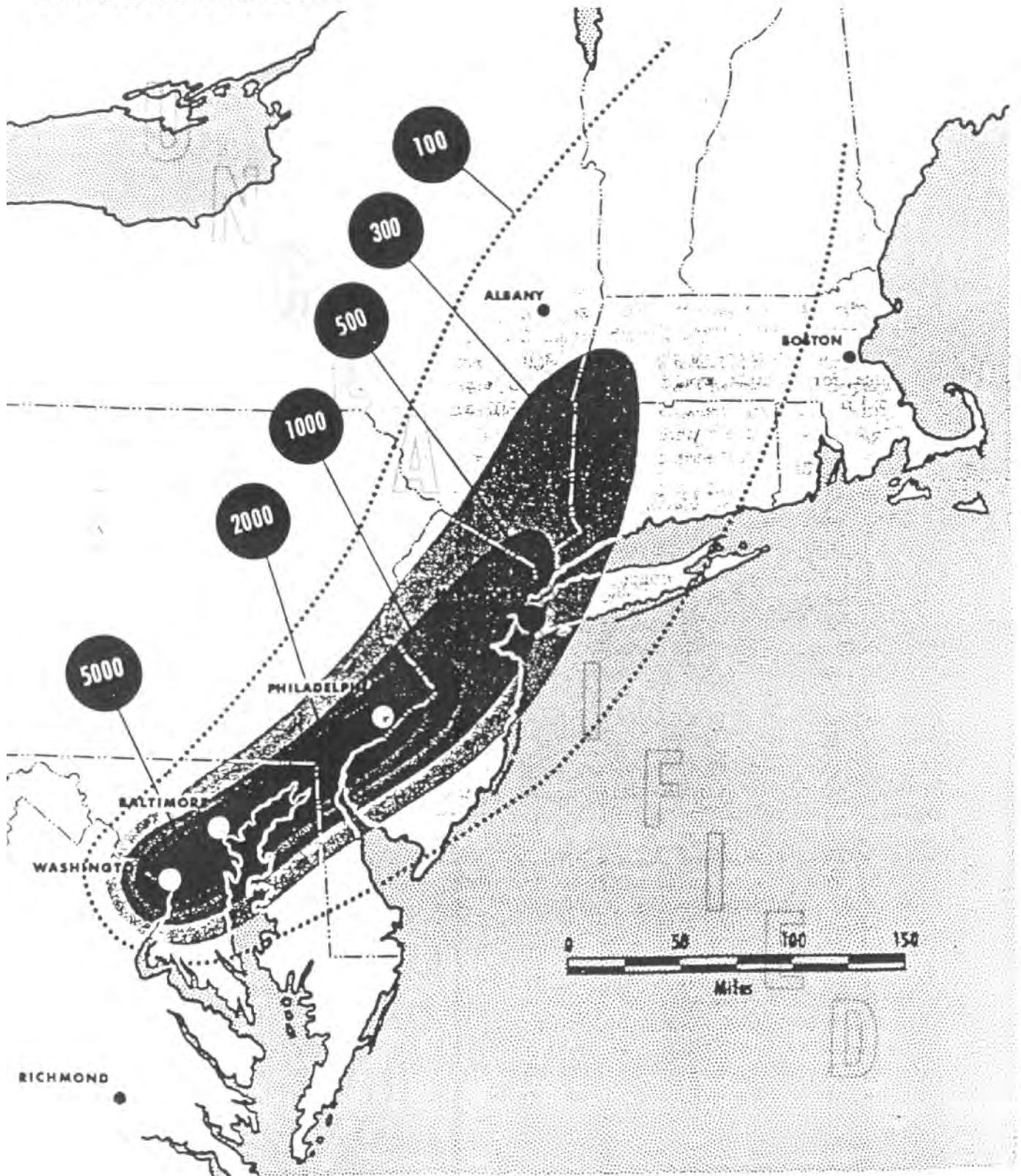


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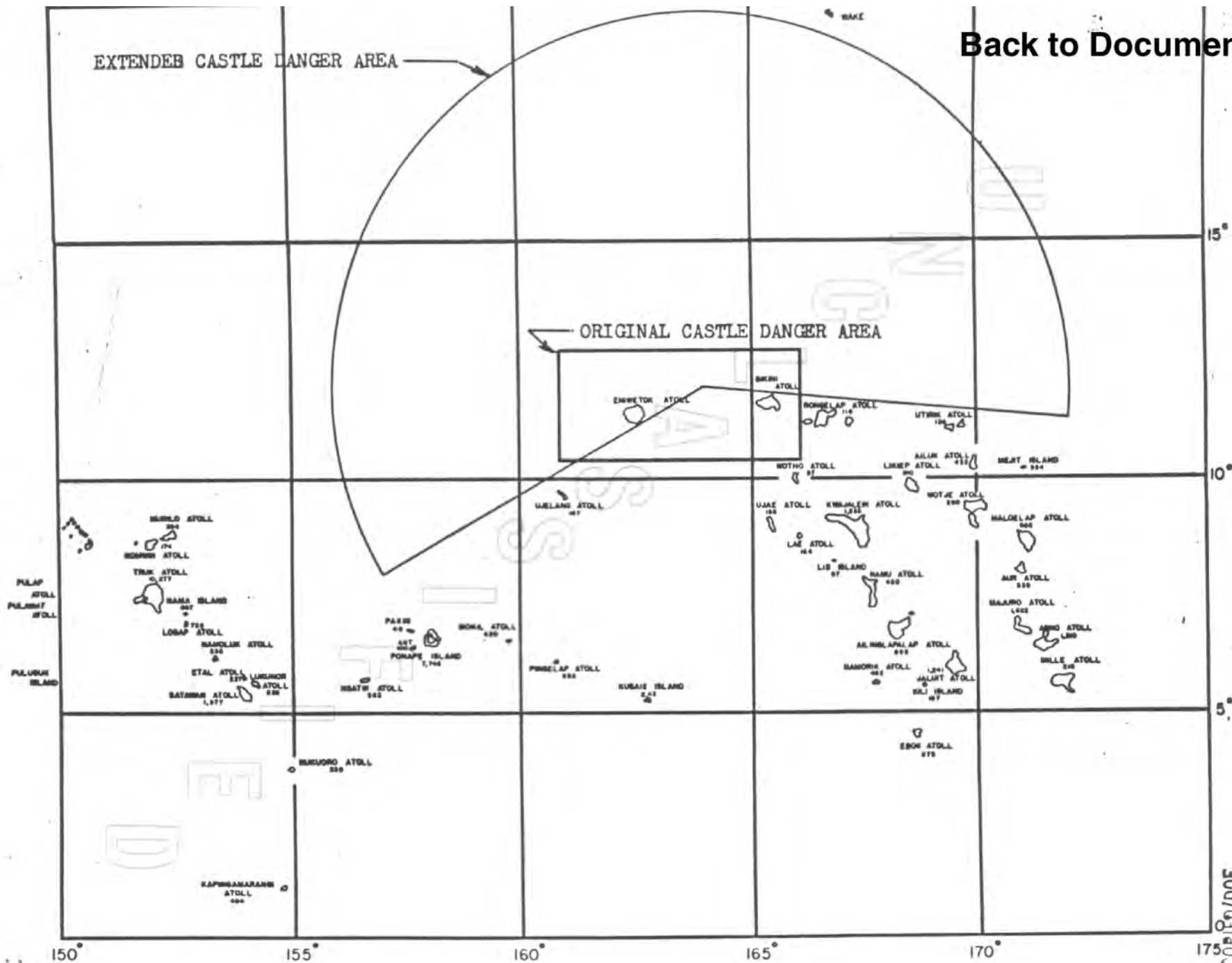
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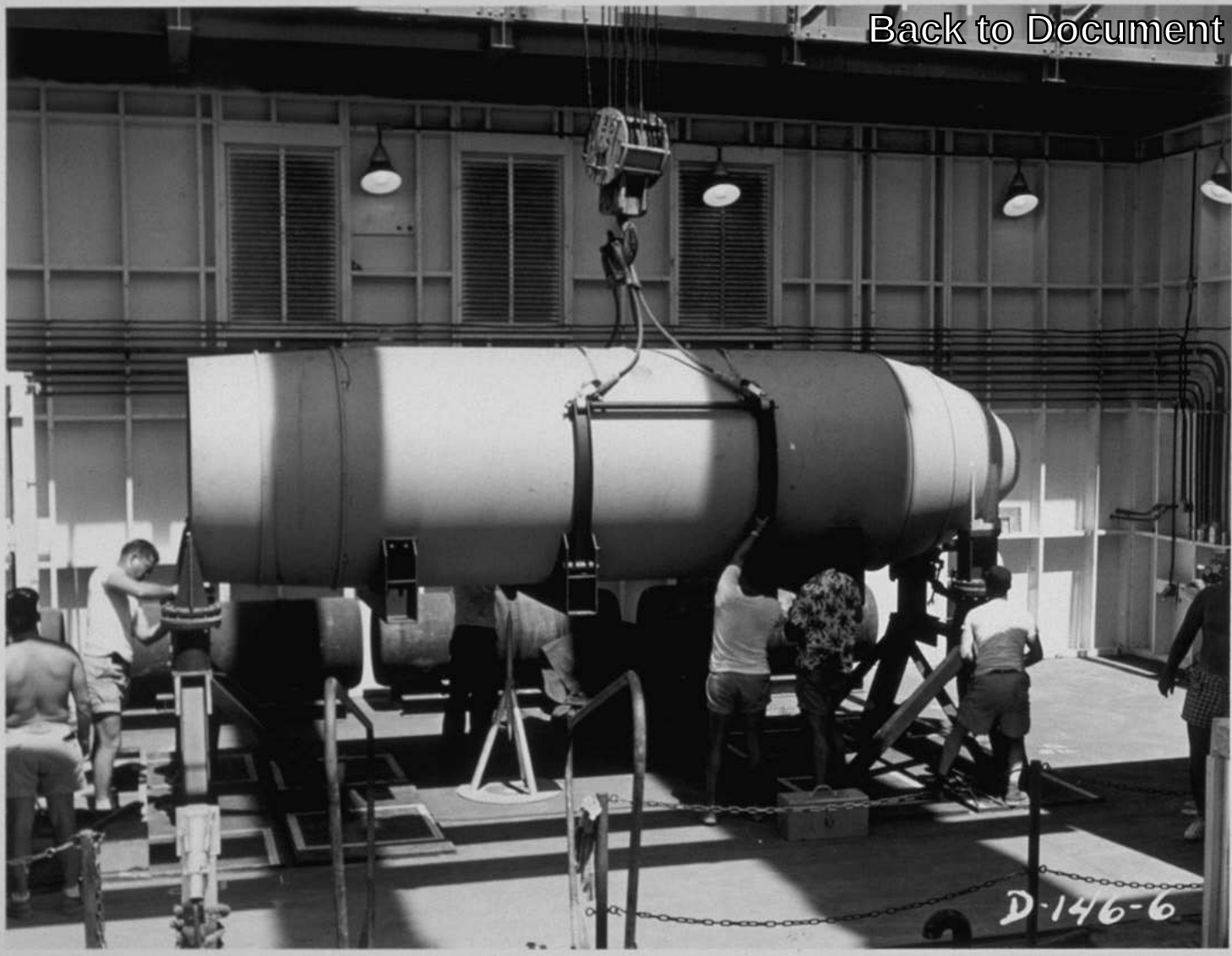


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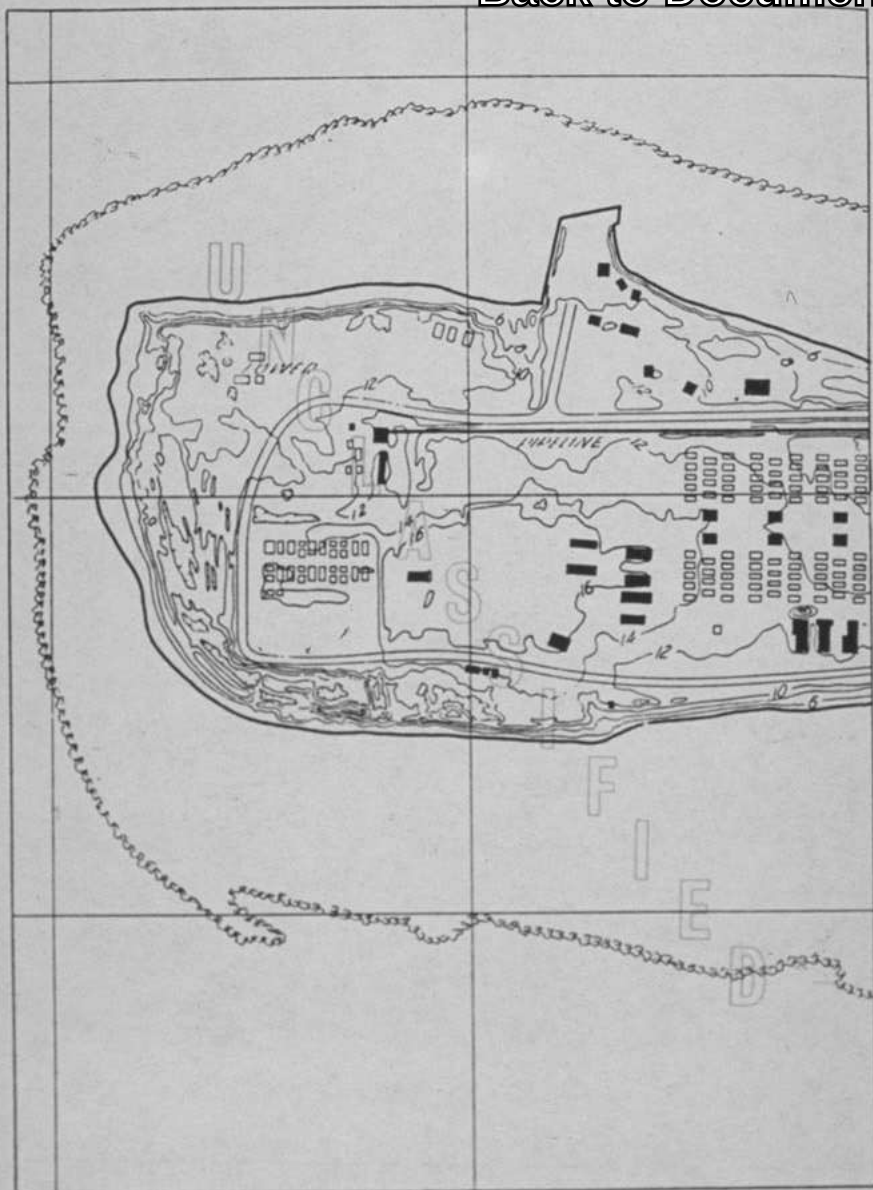


Fig. 3.8 Preshot 3 Contours

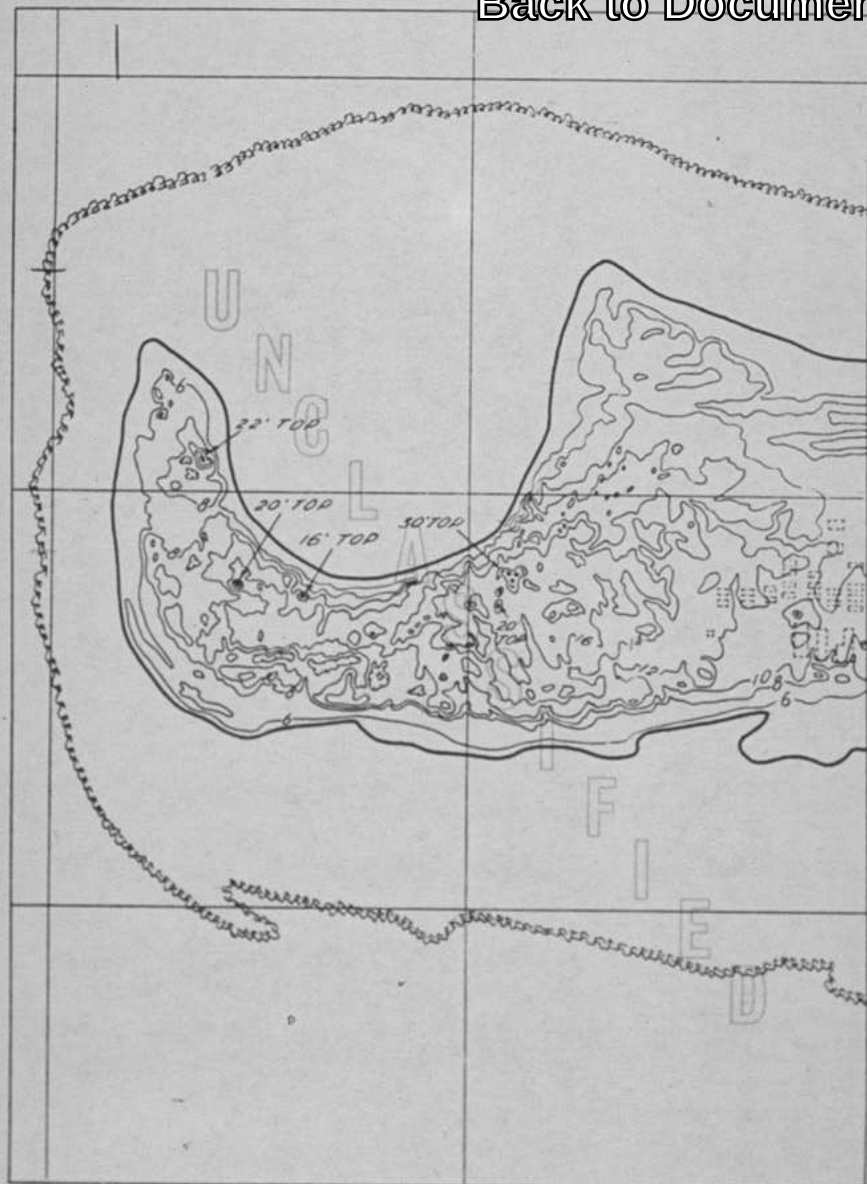


Fig. 3.9 Postshot 3 Contours

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